

MAHARASHTRA UNIVERSITY OF HEALTH SCIENCES NASHIK



NAME OF THE INSTITUTE WITH LOGO

KRIYA SHARIR HUMAN PHYSIOLOGY

Name of the student:

Seat number:

Academic year:

NAME OF THE INSTITUTE WITH LOGO

KRIYA SHARIR
HUMAN PHYSIOLOGY

(Subject code: AyUG-KS)

DEPARTMENT OF KRIYA SHARIR

NAME OF THE COLLEGE

(Approved by - National Commission for Indian System of Medicines, New Delhi)

(Approved by - Maharashtra University of Health Sciences Nashik)

DEPARTMENT OF KRIYA SHARIR

Batch – 20.....-20.....

CERTIFICATE

This is to certify that, Mr. / Ms., has satisfactorily completed the course of practical in the department of Kriya Sharir prescribed by the Maharashtra University of Health Sciences Nashik as a part of the first year B.A.M.S. course summer/winter during the year 20.....-20..... having university exam. seat number.....

Sign. of Teacher

Sign.of H.O.D.

Sign. of Internal Examiner

Sign. of External Examiner

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Teacher's signature

Approach for Using Kriya Sharir Practical Record Book

Dear Students,

Welcome to your journey through the **Kriya Sharir Practical Record Book**! This record is carefully crafted to guide you through 23 essential physiology practicals, aligned with the latest syllabus and examination patterns. To make the most of this resource, please adhere to the following guidelines:

Follow Instructions: Pay close attention to the instructions provided in the record book and refer to your textbook for additional insights before starting any experiment.

Record Observations: While documenting your observations, ensure you explain any deviations from the expected results.

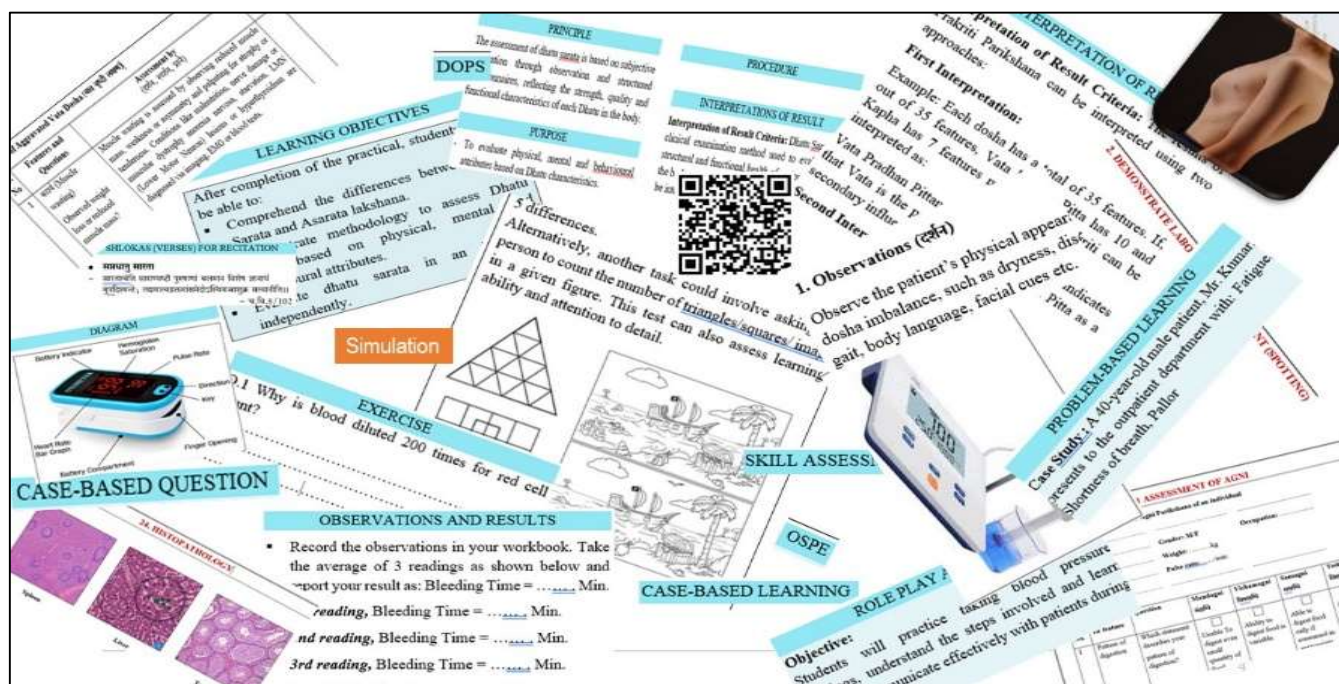
Comprehensive Structure: The record book is enriched with diverse sections, including: Proforma, OSPE (Objective Structured Practical Examination), CBL (Case-Based Learning), PBL (Problem-Based Learning)

Role Play Activities, QR Codes for Interactive Content, Shlokas for Recitation

Visuals and Calculations: Essential diagrams and calculations are thoughtfully incorporated within the relevant sections. Ensure your entries are clear and precise.

This record book is designed not just to assist you in learning but also to inspire a deeper understanding of physiology. Engage with each section, explore the connections, and make your practical sessions both productive and enjoyable.

Best wishes on your academic journey!



Demo



Syllabus

1. DHATU SARATA PARIKSHANA

LEARNING OBJECTIVES

After completion of the practical, students should be able to:

- Comprehend the differences between Dhatu Sarata and Asarata lakshana.
- Demonstrate methodology to assess Dhatu sarata based on physical, mental and behavioural attributes.
- Evaluate dhatu sarata in an individual independently.

PRINCIPLE

The assessment of dhatu sarata is based on subjective evaluation through observation and structured questionnaires, reflecting the strength, quality and functional characteristics of each Dhatu in the body.

PURPOSE

- To evaluate physical, mental and behavioural attributes based on Dhatu characteristics.
- To aid in understanding an individual's constitution and potential health predispositions.
- To provide a diagnostic and preventive approach to maintaining holistic health.

PROCEDURE

1. Preparation: Familiarize yourself with the proforma and the qualities (Lakshana) of each Dhatu Sarata. Refer to the provided material on behavioural attributes to understand key indicators. Ensure a respectful and confidential environment for the participant.

2. Conducting the Assessment: Physical Observations (ex.- hair, nail texture) Behavioural and Mental Attributes: Inquire about mental resilience, enthusiasm, memory and other behavioural traits.

3. Recording Responses: Mark responses for each attribute as YES or NO based on the participant's condition. If YES: The attribute aligns with Dhatu

Sarata Lakshana and assigns a score of 1 and If NO: The attribute aligns with Dhatu Asarata Lakshana and assigns a score of 0. Tally the scores for each Dhatu Sarata type and calculate the percentage for each Dhatu Sarata. Maintain participant confidentiality throughout the process. Ensure unbiased questioning during the assessment. Perform the procedure in a calm and quiet environment. Avoid misinterpretation of subjective responses.

INTERPRETATIONS OF RESULT

Interpretation of Result Criteria: Dhatu Sarata is a clinical examination method used to evaluate the structural and functional health of various tissues in the body. The results of Dhatu Sarata Parikshana can be interpreted using two approaches:

First Interpretation: Based on the calculated percentage, evaluate the quality of each Dhatu Sarata:

>70% :Pravar (High-quality) Dhatu Sarata

30-70%:Madhyam (Moderate-quality) Dhatu Sarata

<30% :Avar (Poor-quality) Dhatu Sarata

On the basis of above score enlist name of Dhatu(s) having Pravar, Madhyam and Avar Dhatu Sarata

Second Interpretation: Depends on how many Pravar Dhatu Sarata an individual possesses. Sarva Sara (optimal tissue health), Madhya Sara (average tissue health) and Asara (suboptimal tissue health). (Charak Samhita)

Sarva Sara: Possess all 8 kinds of Sarata as Pravar.

Madhya Sara: Possess Sarata in 1-7 Dhatus as Pravar.

Avar Sara: Possess no Sarata as Pravar.

Report the findings clearly in the Dhatu Sarata Examination Form, ensuring consistency with observed characteristic

SHLOKAS (VERSES) FOR RECITATION

■ सप्तधातु सारता

- सारतश्चेति साराण्यष्टौ पुरुषाणां बलमान विशेष ज्ञानार्थं मुपदिश्यन्ते; तद्यथात्वग्रक्तमांसमेदोऽस्थिमज्जाशुक्र सत्वानीति॥
- च.वि.8/102

- सारतश्चेत्यादौ सारशब्देन विशुद्धतरो धातुरूच्यते।- चक्रपाणि
- त्वगसृङ्मांससमेदोऽस्थिमज्जाशुक्राणि धातवः।
ओजः सत्त्वं च सर्वं च तत्सारं तु निबोध मे॥ - का.सू.28
9 वा ओजसार यह काश्यपसंहिता का वैशिष्ट्य है।

■ रससार

तत्र स्निग्धश्लक्ष्णमृदुप्रसन्नसूक्ष्माल्पगम्भीर सुकुमारलोमा सप्रभेव च त्वक् त्वक्सारानाम्। सा सारता सुखसौभाग्यैश्चर्योपभोग बुद्धिविद्यारोग्य प्रहर्षणान्यायुष्यत्वं चाचष्टे॥ - च.वि.8/103

■ रक्तसार

कर्णाक्षिमुखजिह्वानासौष्ठपाणिपादतलनखललाट मेहनस्निग्धक्तवर्णं श्रीमद् भ्राजिष्णु रक्तसारानाम्। सा सारतासुखमुद्धतां मेधां मनस्वित्वं सौकुमार्यमनतिबलमक्लेश-सहिष्णुत्व मुष्णासहिष्णुत्वं चाचष्टे॥ - च.वि.8/104

■ मांससार

शंखललाटकृकाटिकाक्षिगण्डहनुग्रीवास्कन्धोदरकक्षवक्षः पाणिपादसन्धः स्थिरगुरुशुभमांसोपचितामांससारानाम्। सा सारता क्षमाधृतिमलौल्यं वित्तं विद्यां सुखमार्जवमारोग्यं बलमायुश्चदीर्घमाचष्टे॥ - च.वि.8/105

■ मेदसार

वर्णस्वरनेत्रकेशलोमनखदन्तौष्ठमूत्रपुरीषेषुविशेषतः स्नेहोमेदः साराणाम्। सा सारता वित्तैश्वर्यसुखोपभोगप्रदानान्यार्जवं सुकुमारोपचारतां चाचष्टे॥ - च.वि.8/106

■ अस्थिसार

पार्श्विगुल्फजान्वरत्निजत्रुचिबुकशिरः पर्वस्थूलाः स्थूलास्थिनखदन्ताश्चास्थिसाराः। ते महोत्साहाः क्रियावन्तः क्लेशसहाः सारस्थिरशरीरा भवन्त्यायुष्मन्तश्च॥-च.वि.8/107

■ मज्जासार

मृद्वांगबलवन्तः स्निग्धवर्णस्वराः स्थूलदीर्घवृत्तसन्धयश्च मज्जासाराः। ते दीर्घायुषो बलवन्तः श्रुतवित्तविज्ञानापत्य संमानभाजश्च भवति॥ - च.वि.8/108

■ शुक्रसार

सौम्याः सौम्यप्रेक्षिणः क्षीरपूर्णलोचना इव प्रहर्षबहुलाः स्निग्धवृत्तसार समसंहतशिखरदशनाः प्रसन्नस्निग्धवर्णस्वरा भ्राजिष्णवो महास्फिचश्च शुक्रसाराः। ते स्त्रीप्रियोपभोगा बलवन्तः सुखैश्वर्यारोग्यवित्तसंमानापत्य भाजश्च भवन्ति॥

■ सत्वसार

स्मृतिमन्तो भक्तिमन्तः कृतज्ञः प्राज्ञाः शुचयो महोत्साहा दक्षा धीराः समरविक्रान्तयोधिनस्त्यक्तविषादाः सुव्यवस्थितगति गम्भीरबुद्धिचेष्टाः कल्याणाभिनिवेशिनश्च सत्वसाराः तेषां स्वलक्षणेरेव गुणा व्याख्याताः। - च.वि.8/108

■ सर्वसार

तत्र सर्वः साररूपेतः पुरुषा भवन्त्यतिबलाः परमसुखयुक्ताः क्लेशसहाः सवरिम्भेष्वात्मनि जातप्रत्ययाः कल्याणाभि निवेशिनः स्थिरसमाहितशरीराः सुसमाहितगतयः सानुनाद स्निग्धगम्भीर महास्वराः सुखैश्वर्यवित्तोपभोगसंमानभाजो मन्दजरसो मन्दविकाराः प्रायस्तुल्यगुणा विस्तीर्णापत्याश्चिरजीविनश्च॥
- च.वि.8/109

■ सप्तधातु सारता (सुश्रुत)

- रससार-सुप्रसन्नमृदुत्वग्रोमाणं त्वक्सारं विद्यादिति सौभाग्य।
- रक्तसार-स्निग्धताग्रनखनयनतालुजिह्वौष्ठपाणिपादतलरक्तेना
- मांससार-अच्छिद्रगात्रं गूढास्थिसन्धिं मांसोपचितं च मांसेन॥
- मेदसार-स्निग्धमूत्रस्वेदस्वरं बृहत् शरीरमायासासहिष्णुं मेदसा।
- अस्थिसार-महाशिरः स्कन्धं दृढदन्तहन्वस्थि नखमस्थिभिः॥
- मज्जासार-अकृशमुत्तमबलं स्निग्धगम्भीरस्वरं सौभाग्योपपन्नं महानेत्रं मज्जा॥
- शुक्रसार-स्निग्धसंहतेश्वेतास्थिदन्तनखं बहुलकामप्रजं शुक्रेण॥
- सु.सू.35/16

■ काश्यप

- रससार-त्वग्रोगरहितो भोगी प्रसन्नव्यञ्जनच्छविः।
- सद्यः क्षतप्ररोहश्च त्वक्सारः सुतनूरूहः॥
- रक्तसार-रक्तसारोऽरूणाभासः.....॥ - का.सू.28

1.1 DHATU SARATA PARIKSHANA

Objective : To assess the Dhatu Sarata of an individual

Name: :
Age : **Gender:** M/F **Occupation:**
Body temperature : **Weight:**kg **Desha:**.....
Blood pressure : **Pulse rate:**...../min **Rutu/Kal:**.....
Address :
.....

1. रससारता (Rasasarata)	Response (YES/ NO)	Score
a. Skin		
1. स्निग्ध (Unctuous skin) Does your skin remain oily throughout year?		
2. श्लक्ष्ण (Smooth skin) Does your skin feel smooth throughout the year?		
3. मृदू (Soft) Does your skin feel soft throughout the year?		
4. प्रसन्न (Healthy) Does your skin remain healthy?		
b. Hair		
5. सूक्ष्म (Fine) Does your hair feel fine?		
6. अल्प (Less) Does your hair count is minimum (null)?		
7. गंभीर (Deep-rooted) Do your skin hairs fall easily on rubbing?		
8. सुकुमार (Delicate) Does your skin hair feel soft?		
c. Mental/ Economic/ Education		
9. सुख (Happy) Do you feel happy about small things?		
10. सौभाग्य (Good fortune) Do you feel you get good fortune?		
11. ऐश्वर्य (Fame/prosperity) - Do you feel you have/will have fame?		
12. उपभोग (Enjoyment) Do you enjoy/ celebrate every small thing?		
13. बुद्धि (Intelligence) Do you always find a way to get out of situations?		
14. विद्या (Knowledge) What field do you master in?		
15. आरोग्य (Healthy) How often do you get ill in a year?		

16. प्रहर्ष (Exhilarated) Do your acts give happiness to others?		
17. सद्यःक्षतप्ररोहण (Easy wound healing) Does your wound heal too early or take time?		
18. प्रसन्नव्यञ्जनछवि (Good looking) Does your look good? (usually mates compliment)		
19. त्वग्रोगरहीतो (without skin disease) How often do you get skin disease in a year?		
रससारता x 100 = % 19		
2. रक्तसारता (Raktasarata)	Response (YES/ NO)	Score
1. स्निग्ध (कर्ण/अक्षि/मुख/जिह्वा/नासा/ओष्ठ/पाणिपादतल/नख/ललाट/मेहन)(unctuous at - Ears, eyes, face, tongue, nose, lips, palm sole, nails, forehead, genitals)		
2. रक्तवर्ण (कर्ण/अक्षि/मुख/जिह्वा/नासा/ओष्ठ/पाणिपादतल/नख/ललाट/मेहन) (Redness of Ears, eyes, face, tongue, nose, lips, palm sole, nails, forehead, genitals)		
Mental/ Educational		
3. सुख (Happiness) - Do you remain positive towards bad/worst situations?		
4. मेधा (Intellect) - Do you remain to withstand your decision throughout?		
5. मनस्विता (Controlled mind) While in group discussion, do you admire other's opinions or always have self-opinion?		
6. सुकुमारता (Delicate) Does your skin get frequently infected/ allergic to foreign substances?		
7. अनतिबलम् (Moderate strength) Do you tolerate hard work or get easily fatigued during work?		
8. अक्लेशसहिष्णूत्वं (Less pain bearer) Does your body tolerate pain or painful stimuli?		
9. उष्णसहिष्णूत्वं (Intolerant to heat) - Do you take hot drinks (tea/coffee) as soon as they are served or after a few minutes?		
रक्तसारता x 100 = % 9		
3. मांससारता (Mamsasarata)	Response (YES/ NO)	Score
1. स्थिर (Firmness) ललाट, कृकाटिका, नेत्र, कपोल, हनु, ग्रीवा (गर्दन), स्कंध (कन्धा), उदर, कक्षा, वक्ष, पानी, पायू, संधि (Firm- forehead, nape, cheek, jaws, neck, shoulders, abdomen, axilla, chest, limb & joints)		
2. गुरु (Heavyness) ललाट, कृकाटिका, नेत्र, कपोल, हनु, ग्रीवा (गर्दन), स्कंध (कन्धा), उदर, कक्षा, वक्ष, पानी, पायू, संधि		
3. स्थूल (Compact) ललाट, कृकाटिका, नेत्र, कपोल, हनु, ग्रीवा (गर्दन), स्कंध (कन्धा), उदर, कक्षा, वक्ष, पानी, पायू, संधि		

4. मांसपचित (Well covered with muscles) ललाट, कृकाटिका, नेत्र, कपोल, हनु, ग्रीवा (गर्दन), स्कंध (कन्धा), उदर, कक्षा, वक्ष, पानी, पायू, संधि		
Mental/Educational		
5. क्षमावान(Forgiveness) Do you easily forgive someone's did? Or keep angry for a long time?		
6. धीरता (Patience) Do you have the patience to do something afterward/as a result of done work?		
7. अलौल्यता (Honest) Do you remain honest towards a given task/ work or take shortcuts to complete it?		
8. धनी (Wealthy) Do you believe health is wealth?		
9. विद्यायूक्त (Knowledgeable) Do you like to learn new things?		
10. सुखी (Happy) Do you feel happy about having good strength and fitness?		
11. आरोग्य (Healthy) Do you easily recover from physical psychic trauma?		
12. बलयुक्त (Having strength) Do you perform heavy exercise/strenuous work?		
13. अच्छिद्रगात्रा (Well-developed muscles) Does the body improve (muscular) pain after long-term exercise/eating well?		
14. गूढ अस्थिसन्धि (Deep seated bones & joint) Is your body fatty/fleshy from the beginning/starting phase of life?		
माससारता x 100 = % 14		
4. मेदसारता (Medasarata)	Response (YES/ NO)	Score
विशेषतः स्निग्ध		
1. वर्ण (Complexion) Does your skin remain oily irrespective of season (in a dry/cold)?		
2. स्वर (Voice) Does your voice get cracked frequently?		
3. नेत्र (Eyes) Do you often feel your eyes dry/watery?		
4. केश (Hairs) Do you often feel your scalp oily/dry?		
5. लोम (Body hairs) Do your limb hairs feel soft/rough?		
6. नख (Nails) Does your nails feel shiny/thick/ brittle?		
7. दंत (teeth) - Does your tooth have luster?		
8. ओष्ठ (Lips) Do your lips remain smooth/soft throughout the year?		
9. मूत्र (Urine) Do you strain/effort during micturition?		

10. पुरिष (Stool) - Do you pass feces without strain/effort?		
11. स्वेदन (Profused sweating) - Do you often sweat even after minimal work?		
12. सर्वशरीर (Unctuous body) - Do your body parts feel soft/unctuous?		
13. आयास असहिष्णू (Intolerance to exertion) Despite your huge physical appearance do you get easily fatigued even after minimal work?		
14. बृहत्शरीर (Big muscular body exertion) Does the physical appearance of a person seem/feel huge or lean?		
15. वित्त (Wealthy) What kind of work do you believe in earning? Sedentary/hard work?		
16. सुखोपभोग (Happiness/Enjoyment) Do you find more happiness in a sedentary lifestyle?		
17. सुकुमारोपचारता (Responds to small dosage) Despite of huge body, do you tolerate high doses of medicine?		
18. ऐश्वर्य उपभोग (Prosperity) Do you enjoy having a comfortable/luxurious lifestyle?		
19. आर्जव (Humble) - Are you humble and respectful towards other's work?		
मेदसारता x 100 = % 19		
5. अस्थिसारता (Asthisarata)	Response (YES/ NO)	Score
1. स्थूल- पाष्णी, गूल्फ, जानु, अरत्नि, जत्रु, चिबूक, शिर, पर्व, अस्थि, नख, दंत (Prominent-heal, ankle, knee, elbow, collar bone, chin, head, interphalangeal joint, bone, nail and teeth) (Compare with others)		
2. महाशिर- (Prominent head) - दर्शन		
3. महास्कंध (Prominent shoulder) - दर्शन		
4. महोत्साह (Super enthusiastic) Do you always have enthusiasm/excitement for the work you do?		
5. क्लेशसहत्व (Enduring) Does your body remain tolerant to pain or do you sustain hard work/stimulus?		
6. स्थिरशरीर (Stable body weight) Do you think/compare the difficult/physical work to your body build?		
7. क्रियावंत (Very active) Do you feel low when there is nothing to do with your busy work?		
8. सारशरीर-सुगठित (Firm and strong body) - दर्शन		
9. आयुष्मंत (Long-lived) - family history		
अस्थिसारता x 100 = % 9		
6. मज्जासारता (Majjasarata)	Response (YES/ NO)	Score
1. मृदुअंग (Softness of organ) Do your body parts often feel soft & smooth?		
2. बलवंत (Powerful) Does physical & strenuous work are easy going for you?		

3. स्निग्धस्वर (Sweet voice) Do people around you often find your voice smooth & soft?		
4. स्निग्धवर्ण (Unctuous complexion) Do you find your complexion soft & unctuous?		
5. संधी - स्थूल/दीर्घ/वृत्त (Broad, long, rounded, prominent joints) Have you ever noticed that your joint is larger, bigger & rounded than usual? In comparison to others?		
6. महानेत्रं (Big eyes) Have you ever noticed that your eyes are larger than others?		
7. गंभीरस्वर (High pitch voice) Do people around you find your voice high pitch/ melodies as compared to others?		
8. अकृशं (Masculine body) Did you notice lately that your physique remains constant/maintained throughout the year?		
9. उत्तम बलं (Full strength) Do you often feel that you have enough strength to perform strenuous/physical activity?		
10. श्रुतभाजं (Good listener) Do people around you find you as a good listener or are you a good listener?		
11. विज्ञानभाज (Endowed with knowledge) Are you naturally endowed with a curiosity for scientific knowledge, or do you cultivate your understanding through independent study?		
12. वित्तभाजं (Wealthy) Do you think you have enough money to fulfill your needs/desires/luxury?		
मज्जासारता x 100 = % 12		
7. शूक्रसारता (Shukrasarata)	Response (YES/ NO)	Score
1. सौम्य (Gentle) Do you feel that your nature is calm & composed towards others?		
2. सौम्यप्रेक्षिण (Looking with charm) Did you often notice that people around you get happy after meeting you?		
3. क्षीरपूर्णनेत्र (Milky eyes) - Does your eyes are whiter (milky) in appearance?		
4. प्रहर्षबहूलत्व (High libido) Do you often feel like you are having excessive libido/sexual desires?		
5. स्निग्धदंत (Shinning teeth) Is your teeth appearance shiny/smooth?		
6. वृत्तदंत (Round shape teeth) Does your teeth are rounded in nature or shaped?		
7. सारदंत (Strong teeth) Do you feel that your teeth are strong and you can easily crack hard objects/ food items or they are brittle?		
8. संहतदंत (Symmetrical teeth) Do you have gaps in your teeth or they are symmetrical?		

9. समदंत (Ubroken teeth) Are all your teeth complete in number or have some of them fallen in the past?		
10. शिखरदशन (Aligned teeth) Are your teeth in proper rows or scattered in nature?		
11. प्रसन्नस्वर (Pleasant voice) Did you often notice that people around you find your voice delightful?		
12. स्निग्धस्वर (Sweet voice) Did you often notice that people around you find your voice soft?		
13. प्रसन्नवर्ण (Pleasant complexion) Do you often notice your skin is shiny or smooth or it is dull or rough?		
14. स्निग्धवर्ण (Unctuous complexion) - Is your facial appearance oily?		
15. भ्राजिष्णू (Bright complexion)- Do you think, sometimes when you meet new people they find you personally glorious or majestic?		
16. महास्फिक (Promonant buttocks)-Does your pelvic region usually appear large?		
17. स्निग्ध/संहत/श्वेत - नख/दंत Do your nails & teeth appear lustrous, white and properly aligned?		
18. स्त्रीप्रीय (Liked by women) Did you often find attraction for the opposite gender around you?		
19. स्त्रीउपभोगी (Lovable for women) Do you have more sexual desires than usual?		
20. बलवान (Strong) Do you often found easy to perform strenuous physical activity or lift weights?		
21. सुखभाज (Happy) Do you manage to remain happy with all the ups and downs in your life or get depressed in some situations?		
22. ऐश्वर्यभाज (Supremacy) Do you feel you have all the opulence & luxury you desire in life?		
23. आरोग्यभाज (Healthy) Does your mental & physical health remain good mostly or your sick frequency?		
24. सम्मानभाज (To be honored) Did you notice that people around you respect you & keep faith in your words?		
25. वित्तभाज (Wealthy) Do you think you have enough wealth to fulfill all your needs or desires?		
26. अपत्यभाज (Progeny) How many children do you have or desire to have in the future?		
शुक्रसारता x 100 = % 26		
8. सत्वसार (Satvasarata)	Response (YES/ NO)	Score
1. स्मृतिमंत (Having good memory) Do you think you are a quick learner?		
2. भक्तिमंत (Devoted/Spiritual) - Do you remain dedicated for a long period to achieve something important to you?		

3. कृतज्ञ (Grateful) - Do you outlook your life you lead till the date? How do you deal with tough times or adversities you face?		
4. प्राज्ञ (Wise) - Do you always forgive people?		
5. शुची (Cleanliness) - Do you need everything neat & clean always?		
6. महोत्साहा (Very Enthusiastic) Do you celebrate/enjoy very small things or success in your day-to-day life?		
7. दक्ष (skilled) Are you able to perform multiple tasks efficiently side by side?		
8. धीर (Restrict) - Are you able to control/limit your emotions and desires?		
9. समरविक्रान्तयोद्ध (Hardihood fighter) Do you take a stand for your thoughts or things you want to achieve even in difficult situations?		
10. अविषाद (Free from anxiety) Do you remain in the state irrespective of the situation/condition?		
11. गम्भीर बुद्धी (Highly intelligent with a serious mindset) Does your activity & thought process go hand in hand?		
12. सुव्यस्थित चेष्टा (Having proper gait) Does your act always hurt someone nearer/closer to you?		
13. कल्याण अभिनिवेश (Active participation in virtuous acts) Do you always ensure that your acts/progress/goal must have morals /virtue?		
14. अतिबल (Powerful) Do you withstand physical work effortlessly?		
15. परमसुखयुक्त (Very happy) Do you remain unaffected even after an enormous amount of stress?		
सत्त्वसार x 100 = % 15		

Result:

Interpretation of particular Dhatu Sarata:

Pravar Sarata (>70%) :

Madhyam Sarata (30-70%) :

Avar Sarata (<30%) :

Interpretation of overall Dhatu Sarata: Sarva Sara/Madhya Sara/Avar Sara.



Teacher's signature

1.2 DHATU SARATA PARIKSHANA

Objective : To assess the Dhatu Sarata of an individual

Name: :

Age : **Gender:** M/F **Occupation:**

Body temperature : **Weight:**kg **Desha:**.....

Blood pressure : **Pulse rate:**...../min **Rutu/Kal:**.....

Address :

1. रससारता (Rasasarata)	Response (YES/ NO)	Score
a. Skin		
1. स्निग्ध (Unctuous skin) Does your skin remain oily throughout year?		
2. श्लक्ष्ण (Smooth skin) Does your skin feel smooth throughout the year?		
3. मृदू (Soft) Does your skin feel soft throughout the year?		
4. प्रसन्न (Healthy) Does your skin remain healthy?		
b. Hair		
5. सूक्ष्म (Fine) Does your hair feel fine?		
6. अल्प (Loss) Does your hair count is minimum (null)?		
7. गंभीर (Deep-rooted) Do your skin hairs fall easily on rubbing?		
8. सुकुमार (Delicate) Does your skin hair feel soft?		
c. Mental/ Economic/ Education		
9. सुख (Happy) Do you feel happy about small things?		
10. सौभाग्य (Good fortune) Do you feel you get good fortune?		
11. ऐश्वर्य (Fame/prosperity) - Do you feel you have/will have fame?		
12. उपभोग (Enjoyment) Do you enjoy/ celebrate every small thing?		
13. बुद्धि (Intelligence) Do you always find a way to get out of situations?		
14. विद्या (Knowledge) What field do you master in?		
15. आरोग्य (Healthy) How often do you get ill in a year?		

16. प्रहर्ष (Exhilarated) Do your acts give happiness to others?		
17. सद्यःक्षतप्ररोहण (Easy wound healing) Does your wound heal too early or take time?		
18. प्रसन्नव्यञ्जनछवि (Good looking) Does your look good? (usually mates compliment)		
19. त्वग्रोगरहीतो (without skin disease) How often do you get skin disease in a year?		
रससारता x 100 = % 19		
2. रक्तसारता (Raktasarata)	Response (YES/ NO)	Score
1. स्निग्ध (कर्ण/अक्षि/मुख/जिह्वा/नासा/ओष्ठ/पाणिपादतल/नख/ललाट/मेहन)(unctuous at - Ears, eyes, face, tongue, nose, lips, palm sole, nails, forehead, genitals)		
2. रक्तवर्ण (कर्ण/अक्षि/मुख/जिह्वा/नासा/ओष्ठ/पाणिपादतल/नख/ललाट/मेहन) (Redness of Ears, eyes, face, tongue, nose, lips, palm sole, nails, forehead, genitals)		
Mental/ Educational		
3. सुख (Happiness) - Do you remain positive towards bad/worst situations?		
4. मेधा (Intellect) - Do you remain to withstand your decision throughout?		
5. मनस्विता (Controlled mind) While in group discussion, do you admire other's opinions or always have self-opinion?		
6. सुकुमारता (Delicate) Does your skin get frequently infected/ allergic to foreign substances?		
7. अनतिबलम् (Moderate strength) Do you tolerate hard work or get easily fatigued during work?		
8. अक्लेशसहिष्णूत्वं (Less pain bearer) Does your body tolerate pain or painful stimuli?		
9. उष्णसहिष्णूत्वं (Intolerant to heat) - Do you take hot drinks (tea/coffee) as soon as they are served or after a few minutes?		
रक्तसारता x 100 = % 9		
3. मांससारता (Mamsasarata)	Response (YES/ NO)	Score
1. स्थिर (Firmness) ललाट, कृकाटिका, नेत्र, कपोल, हनु, ग्रीवा (गर्दन), स्कंध (कन्धा), उदर, कक्षा, वक्ष, पानी, पायू, संधि (Firm- forehead, nape, cheek, jaws, neck, shoulders, abdomen, axilla, chest, limb & joints)		
2. गुरू (Heavyness) ललाट, कृकाटिका, नेत्र, कपोल, हनु, ग्रीवा (गर्दन), स्कंध (कन्धा), उदर, कक्षा, वक्ष, पानी, पायू, संधि		
3. स्थूल (Compact) ललाट, कृकाटिका, नेत्र, कपोल, हनु, ग्रीवा (गर्दन), स्कंध (कन्धा), उदर, कक्षा, वक्ष, पानी, पायू, संधि		

4. मांसपचित (Well covered with muscles) ललाट, कृकाटिका, नेत्र, कपोल, हनु, ग्रीवा (गर्दन), स्कंध (कन्धा), उदर, कक्षा, वक्ष, पानी, पायू, संधि		
Mental/Educational		
5. क्षमावान(Forgiveness) Do you easily forgive someone's did? Or keep angry for a long time?		
6. धीरता (Patience) Do you have the patience to do something afterward/as a result of done work?		
7. अलौल्यता (Honest) Do you remain honest towards a given task/ work or take shortcuts to complete it?		
8. धनी (Wealthy) Do you believe health is wealth?		
9. विद्यायूक्त (Knowledgeable) Do you like to learn new things?		
10. सुखी (Happy) Do you feel happy about having good strength and fitness?		
11. आरोग्य (Healthy) Do you easily recover from physical psychic trauma?		
12. बलयुक्त (Having strength) Do you perform heavy exercise/strenuous work?		
13. अच्छिद्रगात्रा (Well-developed muscles) Does the body improve (muscular) pain after long-term exercise/eating well?		
14. गूढ अस्थिसन्धि (Deep seated bones & joint) Is your body fatty/fleshy from the beginning/starting phase of life?		
माससारता x 100 = % 14		
4. मेदसारता (Medasarata)	Response (YES/ NO)	Score
विशेषतः स्निग्ध		
1. वर्ण (Complexion) Does your skin remain oily irrespective of season (in a dry/cold)?		
2. स्वर (Voice) Does your voice get cracked frequently?		
3. नेत्र (Eyes) Do you often feel your eyes dry/watery?		
4. केश (Hairs) Do you often feel your scalp oily/dry?		
5. लोम (Body hairs) Do your limb hairs feel soft/rough?		
6. नख (Nails) Does your nails feel shiny/thick/ brittle?		
7. दंत (teeth) - Does your tooth have luster?		
8. ओष्ठ (Lips) Do your lips remain smooth/soft throughout the year?		
9. मूत्र (Urine) Do you strain/effort during micturition?		

10. पुरिष (Stool) - Do you pass feces without strain/effort?		
11. स्वेदन (Profused sweating) - Do you often sweat even after minimal work?		
12. सर्वशरीर (Unctuous body) - Do your body parts feel soft/unctuous?		
13. आयास असहिष्णू (Intolerance to exertion) Despite your huge physical appearance do you get easily fatigued even after minimal work?		
14. बृहत्शरीर (Big muscular body exertion) Does the physical appearance of a person seem/feel huge or lean?		
15. वित्त (Wealthy) What kind of work do you believe in earning? Sedentary/hard work?		
16. सुखोपभोग (Happiness/Enjoyment) Do you find more happiness in a sedentary lifestyle?		
17. सुकुमारोपचारता (Responds to small dosage) Despite of huge body, do you tolerate high doses of medicine?		
18. ऐश्वर्य उपभोग (Prosperity) Do you enjoy having a comfortable/luxurious lifestyle?		
19. आर्जव (Humble) - Are you humble and respectful towards other's work?		
मेदसारता x 100 = % 19		
5. अस्थिसारता (Asthisarata)	Response (YES/ NO)	Score
1. स्थूल- पाष्णी, गूल्फ, जानु, अरत्नि, जत्रु, चिबूक, शिर, पर्व, अस्थि, नख, दंत (Prominent-heal, ankle, knee, elbow, collar bone, chin, head, interphalangeal joint, bone, nail and teeth) (Compare with others)		
2. महाशिर- (Prominent head) - दर्शन		
3. महास्कंध (Prominent shoulder) - दर्शन		
4. महोत्साह (Super enthusiastic) Do you always have enthusiasm/excitement for the work you do?		
5. क्लेशसहत्व (Enduring) Does your body remain tolerant to pain or do you sustain hard work/stimulus?		
6. स्थिरशरीर (Stable body weight) Do you think/compare the difficult/physical work to your body build?		
7. क्रियावंत (Very active) Do you feel low when there is nothing to do with your busy work?		
8. सारशरीर-सुगठित (Firm and strong body) - दर्शन		
9. आयुष्मंत (Long-lived) - family history		
अस्थिसारता x 100 = % 9		
6. मज्जासारता (Majjasarata)	Response (YES/ NO)	Score
1. मृदुअंग (Softness of organ) Do your body parts often feel soft & smooth?		
2. बलवंत (Powerful) Does physical & strenuous work are easy going for you?		

3. स्निग्धस्वर (Sweet voice) Do people around you often find your voice smooth & soft?		
4. स्निग्धवर्ण (Unctuous complexion) Do you find your complexion soft & unctuous?		
5. संधी - स्थूल/दीर्घ/वृत्त (Broad, long, rounded, prominent joints) Have you ever noticed that your joint is larger, bigger & rounded than usual? In comparison to others?		
6. महानेत्रं (Big eyes) Have you ever noticed that your eyes are larger than others?		
7. गंभीरस्वर (High pitch voice) Do people around you find your voice high pitch/ melodies as compared to others?		
8. अकृशं (Masculine body) Did you notice lately that your physique remains constant/maintained throughout the year?		
9. उत्तम बलं (Full strength) Do you often feel that you have enough strength to perform strenuous/physical activity?		
10. श्रुतभाजं (Good listener) Do people around you find you as a good listener or are you a good listener?		
11. विज्ञानभाज (Endowed with knowledge) Do you often find interest when the topic is about science? Or are you studying science on your own?		
12. वित्तभाजं (Wealthy) Do you think you have enough money to fulfill your needs/desires/luxury?		
मज्जासारता x 100 = % 12		
7. शूक्रसारता (Shukrasarata)	Response (YES/ NO)	Score
1. सौम्य (Gentle) Do you feel that your nature is calm & composed towards others?		
2. सौम्यप्रेक्षिण (Looking with charm) Did you often notice that people around you get happy after meeting you?		
3. क्षीरपूर्णनेत्र (Milky eyes) - Does your eyes are whiter (milky) in appearance?		
4. प्रहर्षबहूलत्व (High libido) Do you often feel like you are having excessive libido/sexual desires?		
5. स्निग्धदंत (Shinning teeth) Is your teeth appearance shiny/smooth?		
6. वृत्तदंत (Round shape teeth) Does your teeth are rounded in nature or shaped?		
7. सारदंत (Strong teeth) Do you feel that your teeth are strong and you can easily crack hard objects/ food items or they are brittle?		
8. संहतदंत (Symmetrical teeth) Do you have gaps in your teeth or they are symmetrical?		

9. समदंत (Ubroken teeth) Are all your teeth complete in number or have some of them fallen in the past?		
10. शिखरदशन (Aligned teeth) Are your teeth in proper rows or scattered in nature?		
11. प्रसन्नस्वर (Pleasant voice) Did you often notice that people around you find your voice delightful?		
12. स्निग्धस्वर (Sweet voice) Did you often notice that people around you find your voice soft?		
13. प्रसन्नवर्ण (Pleasant complexion) Do you often notice your skin is shiny or smooth or it is dull or rough?		
14. स्निग्धवर्ण (Unctuous complexion) - Is your facial appearance oily?		
15. भ्राजिष्णू (Bright complexion)- Do you think, sometimes when you meet new people they find you personally glorious or majestic?		
16. महास्फिक (Promonant buttocks)-Does your pelvic region usually appear large?		
17. स्निग्ध/संहत/श्वेत - नख/दंत Do your nails & teeth appear lustrous, white and properly aligned?		
18. स्त्रीप्रीय (Liked by women) Did you often find attraction for the opposite gender around you?		
19. स्त्रीउपभोगी (Lovable for women) Do you have more sexual desires than usual?		
20. बलवान (Strong) Do you often found easy to perform strenuous physical activity or lift weights?		
21. सुखभाज (Happy) Do you manage to remain happy with all the ups and downs in your life or get depressed in some situations?		
22. ऐश्वर्यभाज (Supremacy) Do you feel you have all the opulence & luxury you desire in life?		
23. आरोग्यभाज (Healthy) Does your mental & physical health remain good mostly or your sick frequency?		
24. सम्मानभाज (To be honored) Did you notice that people around you respect you & keep faith in your words?		
25. वित्तभाज (Wealthy) Do you think you have enough wealth to fulfill all your needs or desires?		
26. अपत्यभाज (Progeny) How many children do you have or desire to have in the future?		
शुक्रसारता x 100 = % 26		
8. सत्वसार (Satvasarata)	Response (YES/ NO)	Score
1. स्मृतिमंत (Having good memory) Do you think you are a quick learner?		
2. भक्तिमंत (Devoted/Spiritual) - Do you remain dedicated for a long period to achieve something important to you?		

3. कृतज्ञ (Grateful) - Do you outlook your life you lead till the date? How do you deal with tough times or adversities you face?		
4. प्राज्ञ (Wise) - Do you always forgive people?		
5. शुची (Cleanliness) - Do you need everything neat & clean always?		
6. महोत्साहा (Very Enthusiastic) Do you celebrate/enjoy very small things or success in your day-to-day life?		
7. दक्ष (skilled) Are you able to perform multiple tasks efficiently side by side?		
8. धीर (Restrict) - Are you able to control/limit your emotions and desires?		
9. समरविक्रान्तयोद्ध (Hardihood fighter) Do you take a stand for your thoughts or things you want to achieve even in difficult situations?		
10. अविषाद (Free from anxiety) Do you remain in the state irrespective of the situation/condition?		
11. गम्भीर बुद्धी (Highly intelligent with a serious mindset) Does your activity & thought process go hand in hand?		
12. सुव्यस्थित चेष्टा (Having proper gait) Does your act always hurt someone nearer/closer to you?		
13. कल्याण अभिनिवेश (Active participation in virtuous acts) Do you always ensure that your acts/progress/goal must have morals /virtue?		
14. अतिबल (Powerful) Do you withstand physical work effortlessly?		
15. परमसुखयुक्त (Very happy) Do you remain unaffected even after an enormous amount of stress?		
सत्त्वसार x 100 = % 15		

Result:

Interpretation of particular Dhatu Sarata:

Pravar Sarata (>70%) :

Madhyam Sarata (30-70%) :

Avar Sarata (<30%) :

Interpretation of overall Dhatu Sarata: Sarva Sara/Madhya Sara/Avar Sara.



Teacher's signature

1.3 DHATU SARATA PARIKSHANA

Objective : To assess the Dhatu Sarata of an individual

Name: :

Age : **Gender:** M/F **Occupation:**

Body temperature : **Weight:**kg **Desha:**.....

Blood pressure : **Pulse rate:**...../min **Rutu/Kal:**.....

Address :

1. रससारता (Rasasarata)	Response (YES/ NO)	Score
a. Skin		
1. स्निग्ध (Unctuous skin) Does your skin remain oily throughout year?		
2. श्लक्ष्ण (Smooth skin) Does your skin feel smooth throughout the year?		
3. मृदू (Soft) Does your skin feel soft throughout the year?		
4. प्रसन्न (Healthy) Does your skin remain healthy?		
b. Hair		
5. सूक्ष्म (Fine) Does your hair feel fine?		
6. अल्प (Loss) Does your hair count is minimum (null)?		
7. गंभीर (Deep-rooted) Do your skin hairs fall easily on rubbing?		
8. सुकुमार (Delicate) Does your skin hair feel soft?		
c. Mental/ Economic/ Education		
9. सुख (Happy) Do you feel happy about small things?		
10. सौभाग्य (Good fortune) Do you feel you get good fortune?		
11. ऐश्वर्य (Fame/prosperity) - Do you feel you have/will have fame?		
12. उपभोग (Enjoyment) Do you enjoy/ celebrate every small thing?		
13. बुद्धि (Intelligence) Do you always find a way to get out of situations?		
14. विद्या (Knowledge) What field do you master in?		
15. आरोग्य (Healthy) How often do you get ill in a year?		

16. प्रहर्ष (Exhilarated) Do your acts give happiness to others?		
17. सद्यःक्षतप्ररोहण (Easy wound healing) Does your wound heal too early or take time?		
18. प्रसन्नव्यञ्जनछवि (Good looking) Does your look good? (usually mates compliment)		
19. त्वग्रोगरहीतो (without skin disease) How often do you get skin disease in a year?		
रससारता x 100 = % 19		
2. रक्तसारता (Raktasarata)	Response (YES/ NO)	Score
1. स्निग्ध (कर्ण/अक्षि/मुख/जिह्वा/नासा/ओष्ठ/पाणिपादतल/नख/ललाट/मेहन)(unctuous at - Ears, eyes, face, tongue, nose, lips, palm sole, nails, forehead, genitals)		
2. रक्तवर्ण (कर्ण/अक्षि/मुख/जिह्वा/नासा/ओष्ठ/पाणिपादतल/नख/ललाट/मेहन) (Redness of Ears, eyes, face, tongue, nose, lips, palm sole, nails, forehead, genitals)		
Mental/ Educational		
3. सुख (Happiness) - Do you remain positive towards bad/worst situations?		
4. मेधा (Intellect) - Do you remain to withstand your decision throughout?		
5. मनस्विता (Controlled mind) While in group discussion, do you admire other's opinions or always have self-opinion?		
6. सुकुमारता (Delicate) Does your skin get frequently infected/ allergic to foreign substances?		
7. अनतिबलम् (Moderate strength) Do you tolerate hard work or get easily fatigued during work?		
8. अक्लेशसहिष्णूत्वं (Less pain bearer) Does your body tolerate pain or painful stimuli?		
9. उष्णसहिष्णूत्वं (Intolerant to heat) - Do you take hot drinks (tea/coffee) as soon as they are served or after a few minutes?		
रक्तसारता x 100 = % 9		
3. मांससारता (Mamsasarata)	Response (YES/ NO)	Score
1. स्थिर (Firmness) ललाट, कृकाटिका, नेत्र, कपोल, हनु, ग्रीवा (गर्दन), स्कंध (कन्धा), उदर, कक्षा, वक्ष, पानी, पायू, संधि (Firm- forehead, nape, cheek, jaws, neck, shoulders, abdomen, axilla, chest, limb & joints)		
2. गुरु (Heavyness) ललाट, कृकाटिका, नेत्र, कपोल, हनु, ग्रीवा (गर्दन), स्कंध (कन्धा), उदर, कक्षा, वक्ष, पानी, पायू, संधि		
3. स्थूल (Compact) ललाट, कृकाटिका, नेत्र, कपोल, हनु, ग्रीवा (गर्दन), स्कंध (कन्धा), उदर, कक्षा, वक्ष, पानी, पायू, संधि		

4. मांसपचित (Well covered with muscles) ललाट, कृकाटिका, नेत्र, कपोल, हनु, ग्रीवा (गर्दन), स्कंध (कन्धा), उदर, कक्षा, वक्ष, पानी, पायू, संधि		
Mental/Educational		
5. क्षमावान(Forgiveness) Do you easily forgive someone's did? Or keep angry for a long time?		
6. धीरता (Patience) Do you have the patience to do something afterward/as a result of done work?		
7. अलौल्यता (Honest) Do you remain honest towards a given task/ work or take shortcuts to complete it?		
8. धनी (Wealthy) Do you believe health is wealth?		
9. विद्यायूक्त (Knowledgeable) Do you like to learn new things?		
10. सुखी (Happy) Do you feel happy about having good strength and fitness?		
11. आरोग्य (Healthy) Do you easily recover from physical psychic trauma?		
12. बलयुक्त (Having strength) Do you perform heavy exercise/strenuous work?		
13. अच्छिद्रगात्रा (Well-developed muscles) Does the body improve (muscular) pain after long-term exercise/eating well?		
14. गूढ अस्थिसन्धि (Deep seated bones & joint) Is your body fatty/fleshy from the beginning/starting phase of life?		
माससारता x 100 = % 14		
4. मेदसारता (Medasarata)	Response (YES/ NO)	Score
विशेषतः स्निग्ध		
1. वर्ण (Complexion) Does your skin remain oily irrespective of season (in a dry/cold)?		
2. स्वर (Voice) Does your voice get cracked frequently?		
3. नेत्र (Eyes) Do you often feel your eyes dry/watery?		
4. केश (Hairs) Do you often feel your scalp oily/dry?		
5. लोम (Body hairs) Do your limb hairs feel soft/rough?		
6. नख (Nails) Does your nails feel shiny/thick/ brittle?		
7. दंत (teeth) - Does your tooth have luster?		
8. ओष्ठ (Lips) Do your lips remain smooth/soft throughout the year?		
9. मूत्र (Urine) Do you strain/effort during micturition?		

10. पुरिष (Stool) - Do you pass feces without strain/effort?		
11. स्वेदन (Profused sweating) - Do you often sweat even after minimal work?		
12. सर्वशरीर (Unctuous body) - Do your body parts feel soft/unctuous?		
13. आयास असहिष्णू (Intolerance to exertion) Despite your huge physical appearance do you get easily fatigued even after minimal work?		
14. बृहतशरीर (Big muscular body exertion) Does the physical appearance of a person seem/feel huge or lean?		
15. वित्त (Wealthy) What kind of work do you believe in earning? Sedentary/hard work?		
16. सुखोपभोग (Happiness/Enjoyment) Do you find more happiness in a sedentary lifestyle?		
17. सुकुमारोपचारता (Responds to small dosage) Despite of huge body, do you tolerate high doses of medicine?		
18. ऐश्वर्य उपभोग (Prosperity) Do you enjoy having a comfortable/luxurious lifestyle?		
19. आर्जव (Humble) - Are you humble and respectful towards other's work?		
मेदसारता x 100 = % 19		
5. अस्थिसारता (Asthisarata)	Response (YES/ NO)	Score
1. स्थूल- पाष्णी, गूल्फ, जानु, अरत्नि, जत्रु, चिबूक, शिर, पर्व, अस्थि, नख, दंत (Prominent-heal, ankle, knee, elbow, collar bone, chin, head, interphalangeal joint, bone, nail and teeth) (Compare with others)		
2. महाशिर- (Prominent head) - दर्शन		
3. महास्कंध (Prominent shoulder) - दर्शन		
4. महोत्साह (Super enthusiastic) Do you always have enthusiasm/excitement for the work you do?		
5. क्लेशसहत्व (Enduring) Does your body remain tolerant to pain or do you sustain hard work/stimulus?		
6. स्थिरशरीर (Stable body weight) Do you think/compare the difficult/physical work to your body build?		
7. क्रियावंत (Very active) Do you feel low when there is nothing to do with your busy work?		
8. सारशरीर-सुगठित (Firm and strong body) - दर्शन		
9. आयुष्मंत (Long-lived) - family history		
अस्थिसारता x 100 = % 9		
6. मज्जासारता (Majjasarata)	Response (YES/ NO)	Score
1. मृदुअंग (Softness of organ) Do your body parts often feel soft & smooth?		
2. बलवंत (Powerful) Does physical & strenuous work are easy going for you?		

3. स्निग्धस्वर (Sweet voice) Do people around you often find your voice smooth & soft?		
4. स्निग्धवर्ण (Unctuous complexion) Do you find your complexion soft & unctuous?		
5. संधी - स्थूल/दीर्घ/वृत्त (Broad, long, rounded, prominent joints) Have you ever noticed that your joint is larger, bigger & rounded than usual? In comparison to others?		
6. महानेत्रं (Big eyes) Have you ever noticed that your eyes are larger than others?		
7. गंभीरस्वर (High pitch voice) Do people around you find your voice high pitch/ melodies as compared to others?		
8. अकृशं (Masculine body) Did you notice lately that your physique remains constant/maintained throughout the year?		
9. उत्तम बलं (Full strength) Do you often feel that you have enough strength to perform strenuous/physical activity?		
10. श्रुतभाजं (Good listener) Do people around you find you as a good listener or are you a good listener?		
11. विज्ञानभाज (Endowed with knowledge) Do you often find interest when the topic is about science? Or are you studying science on your own?		
12. वित्तभाजं (Wealthy) Do you think you have enough money to fulfill your needs/desires/luxury?		
मज्जासारता x 100 = % 12		
7. शूक्रसारता (Shukrasarata)	Response (YES/ NO)	Score
1. सौम्य (Gentle) Do you feel that your nature is calm & composed towards others?		
2. सौम्यप्रेक्षिण (Looking with charm) Did you often notice that people around you get happy after meeting you?		
3. क्षीरपूर्णनेत्र (Milky eyes) - Does your eyes are whiter (milky) in appearance?		
4. प्रहर्षबहूलत्व (High libido) Do you often feel like you are having excessive libido/sexual desires?		
5. स्निग्धदंत (Shinning teeth) Is your teeth appearance shiny/smooth?		
6. वृत्तदंत (Round shape teeth) Does your teeth are rounded in nature or shaped?		
7. सारदंत (Strong teeth) Do you feel that your teeth are strong and you can easily crack hard objects/ food items or they are brittle?		
8. संहतदंत (Symmetrical teeth) Do you have gaps in your teeth or they are symmetrical?		

9. समदंत (Ubroken teeth) Are all your teeth complete in number or have some of them fallen in the past?		
10. शिखरदशन (Aligned teeth) Are your teeth in proper rows or scattered in nature?		
11. प्रसन्नस्वर (Pleasant voice) Did you often notice that people around you find your voice delightful?		
12. स्निग्धस्वर (Sweet voice) Did you often notice that people around you find your voice soft?		
13. प्रसन्नवर्ण (Pleasant complexion) Do you often notice your skin is shiny or smooth or it is dull or rough?		
14. स्निग्धवर्ण (Unctuous complexion) - Is your facial appearance oily?		
15. भ्राजिष्णू (Bright complexion)- Do you think, sometimes when you meet new people they find you personally glorious or majestic?		
16. महास्फिक (Promonant buttocks)-Does your pelvic region usually appear large?		
17. स्निग्ध/संहत/श्वेत - नख/दंत Do your nails & teeth appear lustrous, white and properly aligned?		
18. स्त्रीप्रीय (Liked by women) Did you often find attraction for the opposite gender around you?		
19. स्त्रीउपभोगी (Lovable for women) Do you have more sexual desires than usual?		
20. बलवान (Strong) Do you often found easy to perform strenuous physical activity or lift weights?		
21. सुखभाज (Happy) Do you manage to remain happy with all the ups and downs in your life or get depressed in some situations?		
22. ऐश्वर्यभाज (Supremacy) Do you feel you have all the opulence & luxury you desire in life?		
23. आरोग्यभाज (Healthy) Does your mental & physical health remain good mostly or your sick frequency?		
24. सम्मानभाज (To be honored) Did you notice that people around you respect you & keep faith in your words?		
25. वित्तभाज (Wealthy) Do you think you have enough wealth to fulfill all your needs or desires?		
26. अपत्यभाज (Progeny) How many children do you have or desire to have in the future?		
शुक्रसारता x 100 = % 26		
8. सत्वसार (Satvasarata)	Response (YES/ NO)	Score
1. स्मृतिमंत (Having good memory) Do you think you are a quick learner?		
2. भक्तिमंत (Devoted/Spiritual) - Do you remain dedicated for a long period to achieve something important to you?		

3. कृतज्ञ (Grateful) - Do you outlook your life you lead till the date? How do you deal with tough times or adversities you face?		
4. प्राज्ञ (Wise) - Do you always forgive people?		
5. शुची (Cleanliness) - Do you need everything neat & clean always?		
6. महोत्साहा (Very Enthusiastic) Do you celebrate/enjoy very small things or success in your day-to-day life?		
7. दक्ष (skilled) Are you able to perform multiple tasks efficiently side by side?		
8. धीर (Restrict) - Are you able to control/limit your emotions and desires?		
9. समरविक्रान्तयोद्ध (Hardihood fighter) Do you take a stand for your thoughts or things you want to achieve even in difficult situations?		
10. अविषाद (Free from anxiety) Do you remain in the state irrespective of the situation/condition?		
11. गम्भीर बुद्धि (Highly intelligent with a serious mindset) Does your activity & thought process go hand in hand?		
12. सुव्यस्थित चेष्टा (Having proper gait) Does your act always hurt someone nearer/closer to you?		
13. कल्याण अभिनिवेश (Active participation in virtuous acts) Do you always ensure that your acts/progress/goal must have morals /virtue?		
14. अतिबल (Powerful) Do you withstand physical work effortlessly?		
15. परमसुखयुक्त (Very happy) Do you remain unaffected even after an enormous amount of stress?		
सत्त्वसार x 100 = %		
15		

Result:

Interpretation of particular Dhatu Sarata:

Pravar Sarata (>70%) :

Madhyam Sarata (30-70%) :

Avar Sarata (<30%) :

Interpretation of overall Dhatu Sarata: Sarva Sara/Madhya Sara/Avar Sara.



Teacher's signature

EXERCISE

This image shows a full page of primary-ruled paper. It contains ten identical horizontal rows. Each row is defined by three lines: a solid top line, a dashed midline, and a solid bottom line. The entire page is white, and there are no margins or other markings.

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OSPE-1

Station: Dhatu Sarata Assessment of Rasa Dhatu

Objective: To evaluate the quality of Rasa Dhatu based on physical and behavioural characteristics.

Station Setup: Dhatu Sarata Assessment Form (specific to Rasa Dhatu) & chair for the participant.

Procedure:

Vitals Assessment: Measure and record: Body temperature, Blood pressure, Pulse rate, Weight etc

Observation: Examine skin texture for unctuousness, smoothness and softness. Assess hair quality, skin health and nail texture visually.

Behavioural Assessment: Ask structured questions on happiness, mental resilience and intelligence as outlined in the proforma. Example: Do you feel happy about small achievements? Does your skin feel soft and smooth throughout the year?

Documentation: Mark responses as "YES" or "NO." and calculate the total score and percentage.

Checklist for Evaluation: Correct measurement of vitals. Accurate observation and recording of physical attributes by completion of all questions. Correct calculation of the Sarata percentage and interpretation of result.

(Note: The teacher may allocate marks as follows: **2 marks** for accurately writing the verses of *Rasa Dhatu*, 1 mark for proper measurement of any two vital measurements, 5 marks for covering at least 15 out of 19 features of *Rasa Dhatu*, 2 marks for correctly calculating the percentage and 3 marks for writing and interpreting the proper result. The teacher retains the discretion to modify the entire scenario of *Sarata* and the distribution of marks and may choose to assess an individual *Sarata*, a group of *Sarata* or all *Sarata* collectively, in accordance with the examination pattern and assessment criteria.)

Expected Outcomes: An appropriate assessment of Rasa Dhatu quality. Identification of its strength or deficiencies.

ROLE PLAY ACTIVITY

Role-playing involves two participants simulating a practitioner-patient scenario. It enhances communication skills, interpretation of subjective criteria and application of the scoring system.

Example: Scenario: Practitioner: A student acting as a health practitioner evaluates a patient's Dhatusarata based on the proforma.

Patient: Another student or volunteer acts as a patient with pre-defined characteristics.

Step-by-Step Activity:

Introduction:

Practitioner introduces themselves and explains the purpose of the evaluation.

Example Dialogue: *"I will ask you a few questions about your health and appearance to assess your Dhatusarata."*

Questionnaire-Based Evaluation:

The practitioner asks questions from the proforma: "Does your hair feel fine and soft?"

"Do you easily recover from wounds or physical trauma?" The patient responds based on the provided characteristics.

Physical Assessment:

The practitioner examines the patient's skin, hair or other physical features, noting relevant observations (e.g., soft skin, shiny nails).

Scoring and Feedback:

The practitioner tallies scores for each section. Provide a brief summary to the patient: *"Based on the evaluation, your Dhatusarata indicates good physical and mental health."*

Educational Outcomes:

Students learn the practical application of theoretical knowledge.

Encourages empathy and communication skills.

Prepares students for real-world patient interactions.

CASE-BASED LEARNING

Case Study Title: Evaluation of Dhatu Imbalance in a Patient

Case Scenario:

A 35-year-old male presents with complaints of frequent fatigue, low energy and dry skin. On examination, the nails appear brittle and the skin texture is rough and dry. The patient reports occasional episodes of low enthusiasm and delayed wound healing.

Problem:

Evaluate the patient's Dhatu Sarata to identify imbalances contributing to his condition.

Learning Objectives:

- Assess Dhatu Sarata systematically.
- Correlate physical and mental attributes with potential imbalances in specific Dhatus.
- Propose interventions for improving Dhatu quality.

Guiding Questions:

1. Based on the patient's symptoms, which Dhatus are likely deficient?
2. How does the condition of the nails and skin reflect the state of Rasa and Rakta Dhatus?
3. What questions would you ask to evaluate mental resilience and emotional attributes?

Step-by-Step Process:

1. **Assessment of Symptoms:** Identify key indicators of Rasa and Rakta Dhatu deficiencies (e.g., dry skin, brittle nails, low energy).
2. **Structured Questionnaire:** Ask relevant questions related to physical, mental and behavioural traits: Do your wounds take a long time to heal? Do you experience fatigue frequently even with minimal work?
3. **Analysis and Interpretation:** Correlate symptoms with Dhatu characteristics. Example:

Rough skin and brittle nails indicate potential Rasa and Rakta Dhatu imbalance.

Discussion and Recommendations:

Suggest interventions: Dietary changes (unctuous food for Rasa Dhatu, iron-rich food for Rakta Dhatu), Lifestyle modifications (regular hydration, stress management practices like yoga), Therapeutic interventions (supplements to strengthen Dhatus).

Expected Outcomes:

Identification of Rasa and Rakta Dhatu deficiencies. Practical recommendations for restoring balance and improving overall health.

Assessment of Students:

Ability to identify and correlate symptoms with Dhatu imbalances. Clear and logical recommendations based on Ayurvedic principles. Effective communication of findings and solutions.

2. DEMONSTRATE LABORATORY EQUIPMENT (SPOTTING) MICROSCOPE

PRINCIPLE

A microscope operates on the principles of magnification and resolution, allowing observation of objects or details too small for the naked eye. It uses lenses to bend light (in optical microscopes) or electrons (in electron microscopes) to produce an enlarged image. Magnification increases the apparent size of the object, while resolution ensures clarity and detail, enabling precise visualization of microscopic structures.

TYPES

Simple microscope - with a single lens

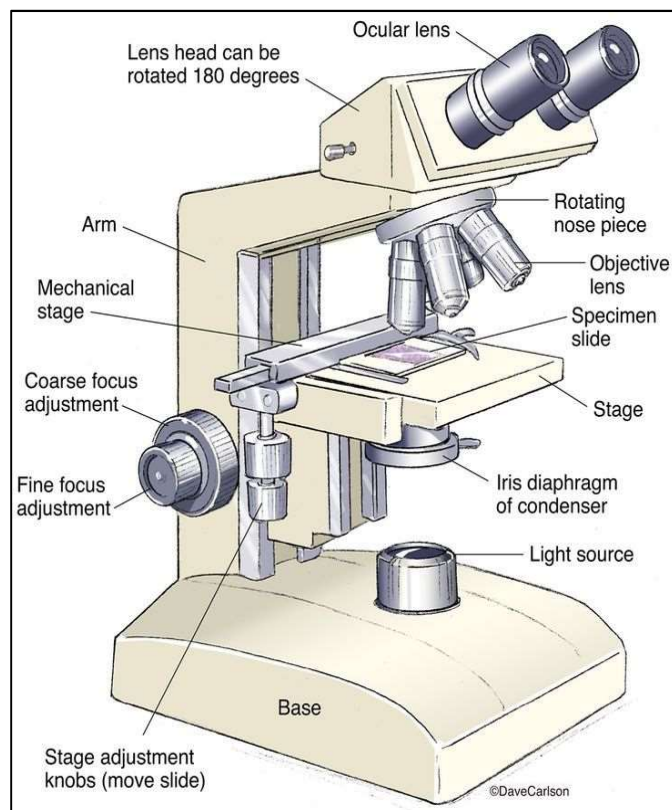
Compound microscope - with two lenses, one is objective and other one is eyepiece lens

Feature	Simple Microscope Diagram	Compound Microscope Diagram
Number of Lenses	Single lens	Multiple lenses (objective and eyepiece)
Complexity	Basic structure	Detailed structure with more parts
Light Control	Simple mirror or light source	Condenser, diaphragm and built-in light
Focus Mechanism	Basic adjustment	Coarse and fine focus knobs
Magnification	Low (up to 20x)	High (up to 1000x or more)

BINOCULAR MICROSCOPE

A binocular microscope uses two eyepieces (one for each eye) to provide a more comfortable viewing experience compared to monocular microscopes. The microscope magnifies the object by using a combination of lenses: the objective lenses and the eyepieces.

DIAGRAM



PARTS

Eyepiece (Wide-Field): The lens through which you view the specimen.

Objective Lenses (10x, 40x, 100x):
10x Objective: Used for general observation and scanning. Ex.- WBC, Sperm count

40x Objective: Used for detailed viewing of structures like cells and tissues. Ex.- RBC, Platelet count

100x Oil Immersion Objective: Requires a drop of immersion oil between the lens and the slide. The oil has a similar refractive index to glass, which enhances resolution and brightness by reducing light refraction. Ex.- DLC

Stage: Flat platform where the slide is placed.

Illuminator: Light source beneath the stage.

Condenser: Focuses light from the illuminator onto the specimen.

Focus Knobs: Coarse and fine adjustment knobs for focusing.

Diaphragm: Controls the intensity and angle of light reaching the specimen.

APPLICATIONS

Cell count: Used for counting different type of blood cells ex- RBC, WBC, Platelet

Histology: Used to study tissue samples, especially at high magnifications (40x and 100x) to view cellular details.

Microbiology: Essential for observing bacteria and other microorganisms using the 100x oil immersion objective.

Cytology: Examines cells and cell structures, helping in the diagnosis of diseases like cancer, infectious diseases, fetal abnormalities, thyroid etc.

Pathology: Pathologists use binocular microscopes to examine tissue biopsies and identify disease at the cellular level.

Clinical Research: Widely used in laboratories for studying cells, tissues and microorganisms under high magnification.

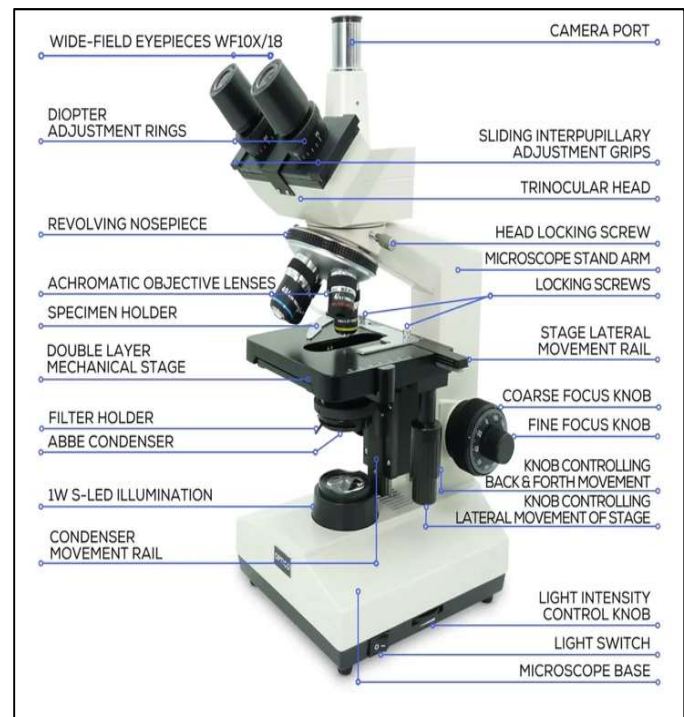
LIMITATIONS

- **Limited Depth of Field:** At higher magnifications, the depth of field decreases, meaning only a thin plane of the sample is in focus at a time.
- **Resolution Limit:** Light microscopes, including those with oil immersion, have a resolution limit of around 0.2 micrometers, which limits viewing extremely small structures.
- **Maintenance:** The spring-loaded mechanism and oil immersion lens require regular cleaning and maintenance to ensure clear viewing and prevent damage.

TRINOCULAR MICROSCOPE WITH DIGITAL CAMERA

A trinocular microscope features an additional optical path for a camera attachment, enabling simultaneous viewing through the eyepieces and digital imaging. The integrated digital camera captures high-resolution images or videos of the specimen, which can be displayed on external devices like computers, smart boards or projectors via appropriate software. This setup facilitates real-time sharing, documentation and detailed analysis of specimens, making it ideal for research, education and professional presentations.

DIAGRAM



PARTS

Trinocular Head: Allows simultaneous viewing and image capturing.

Digital Camera: Captures images/videos; specifications vary (e.g., resolution, frame rate).

Software: Interfaces with the camera for image processing, annotation and projection.

Eyepieces (Ocular Lenses): For direct viewing.

Objective Lenses: Provide primary magnification.

Trinocular Head: Houses the eyepieces and the camera port.

Camera Port: Attachment point for the digital camera.

Stage: Platform where the specimen slide is placed.

Illuminator: Light source for illuminating the specimen.

Focus Knobs: Coarse and fine adjustments for focusing.

Base and Arm: Structural support for the microscope.

APPLICATIONS

Histology: Projecting tissue samples for educational demonstrations.

Pathology: Sharing live microscopic views for consultations.

Research: Documenting and analyzing specimens collaboratively.

Education: Interactive teaching sessions with real-time specimen viewing.

LIMITATIONS

Cost: High-quality digital microscopes and cameras can be expensive.

Technical Expertise: Requires knowledge to operate and maintain the equipment.

Compatibility: Ensuring software compatibility with smart boards and other devices.

Resolution Constraints: Digital cameras may have limitations in capturing extremely fine details.

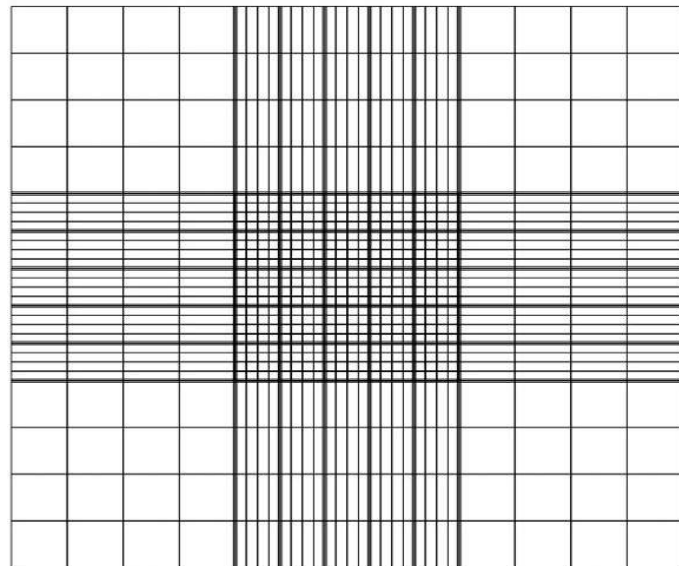
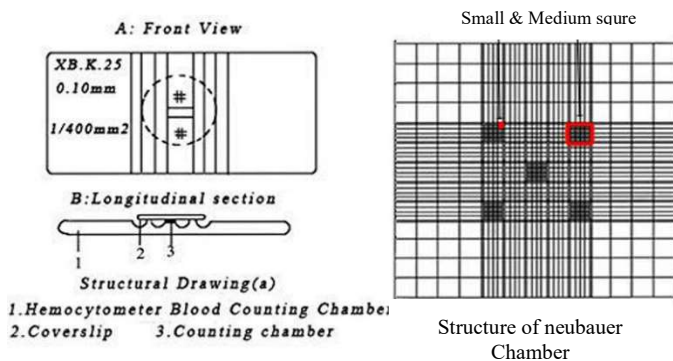


NEUBAUER CHAMBER

PRINCIPLE

The principle behind Neubauer's chamber involves counting cells within a known volume of fluid to estimate cell concentration. The chamber has a specialized grid etched onto a thick glass slide, allowing cells in a small, defined area to be counted. The chamber's specific dimensions help standardize the volume of fluid on the grid, so the concentration of cells per unit volume (usually per microlitre or millilitre) can be calculated by applying a mathematical formula based on the grid size.

DIAGRAM



PARTS

- **Large Outer Square:** This is the largest boundary in the chamber. It encompasses smaller squares, making up the counting grid.
- **Medium Square Sections:** These are smaller sections within the large square, further divided

into groups of smaller squares. Each medium square is typically used for different types of cell counts, depending on cell size and type.

- **Central Small Squares:** There are usually 25 groups of 16 small squares, which are used for precise counting of cells in samples where a high degree of accuracy is needed.

APPLICATION IN PHYSIOLOGY

- **Blood Cell Counts:** Counting red blood cells (RBCs), white blood cells (WBCs) and platelets in blood samples for hematological assessments.
- **Sperm Counting:** Determining sperm concentration in semen analysis for fertility studies.
- **Culture Cell Counting:** In cell biology labs, Neubauer's chamber is used for counting cultured cells to prepare cell suspensions of specific densities for experiments.
- **Microbial Counts:** Estimating the concentration of bacteria, fungi or yeast in a given sample.

LIMITATIONS

- **Manual Counting Error:** Human error during manual counting can lead to inaccurate results.
- **Clumping of Cells:** Cells may clump together, making it challenging to obtain an accurate count.
- **Sample Preparation Variability:** Inconsistent sample preparation, such as improper dilution or uneven distribution of cells, can skew results.
- **Time-Consuming:** Manual counting is relatively time-consuming compared to automated cell counters.



HAEMOCYTOMETER PIPETTE - RBC

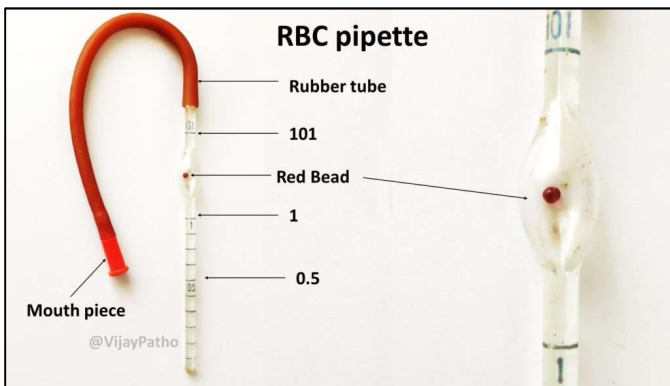
1. RBC PIPETTE

An RBC pipette is a specialized laboratory pipette used to measure and dilute blood for counting red blood cells (RBCs).

PRINCIPLE

The principle of the RBC pipette is to create a precise 1:200 dilution of blood by drawing a small, known volume of blood and then mixing it with an appropriate diluent. The pipette is marked to help measure this specific volume accurately. When the diluted sample is placed on a counting chamber, such as a Neubauer's chamber, RBCs can be counted in a specific grid area, allowing calculation of the total RBC count in the blood sample.

DIAGRAM



PARTS

- **Capillary Tip:** The narrow, pointed end of the pipette used to draw a precise amount of blood up to the 0.5 mark.
- **Stem:** marked with two graduation points, typically the 0.5 mark and the 101 mark, to measure and control the dilution accurately.
- **0.5 Mark:** This mark indicates the volume to which the blood sample should be drawn. Filling up to this mark ensures that the blood sample is precise for creating a 1:200 dilution.
- **101 Mark:** This mark is located near the top of the stem, showing the level to which the diluent

should be added. The diluent helps achieve a standardized 1:200 dilution of the blood.

- **Mixing Bulb:** Located near the upper part of the pipette, the mixing bulb often contains a glass bead to aid in mixing the blood sample thoroughly with the diluent. This ensures a homogenous mixture for accurate counting.

APPLICATION IN PHYSIOLOGY

- **Red Blood Cell Count:** The RBC pipette is used in haematology labs to create a standardized blood dilution, allowing accurate counting of red blood cells (erythrocytes) using a haemocytometer.
- **Diagnostic Assessments:** By measuring the RBC count, healthcare providers can assess conditions like anaemia, polycythaemia or other blood disorders.
- **Research Studies:** RBC counts are also used in research involving blood physiology, studying effects of various treatments or understanding disease mechanisms involving blood cell production or destruction.

LIMITATIONS

- **Manual Error:** Requires precise handling, challenging for beginners.
- **Time-Consuming:** Manual dilution takes longer than automated methods.
- **Cross-Contamination Risk:** Improper cleaning can lead to contamination.

Fixed Dilution Ratio: 1:200 ratio may not suit all applications.



HAEMOCYTOMETER PIPETTE - WBC

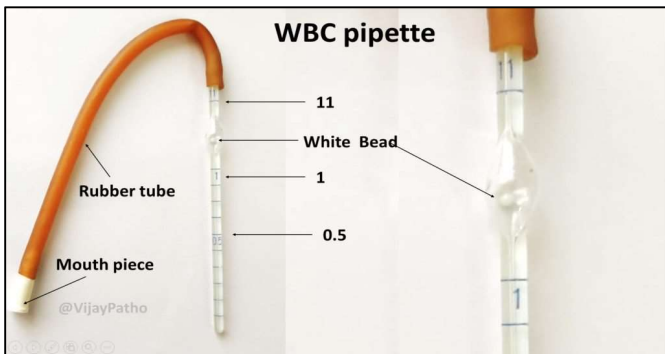
2. WBC PIPETTE

A WBC pipette is a specialized laboratory tool used to dilute blood samples for counting white blood cells (WBCs).

PRINCIPLE

The WBC pipette operates on the principle of creating a 1:20 dilution of blood for accurate counting of white blood cells. The pipette is designed with specific markings to measure a precise volume of blood and then mix it with a diluent. This dilution is then placed on a counting chamber, such as a Neubauer chamber, where WBCs can be counted within a specific grid area. This allows for an accurate calculation of WBC concentration in the original blood sample.

DIAGRAM



PARTS

- **Capillary Tip:** The narrow, pointed end of the pipette is used to draw a precise amount of blood up to the 0.5 mark.
- **Stem:** The long part of the pipette with two key markings (0.5 and 11) that indicate the specific volumes needed to achieve a 1:20 dilution.
- **0.5 Mark:** Indicates the volume to which the blood should be drawn. Drawing to this mark ensures a precise amount of blood is taken.
- **11 Mark:** Located near the upper end of the stem, this mark shows the level to which the diluent should be added, creating the standardized 1:20 dilution of the blood sample.

- **Mixing Bulb:** Situated above the stem, the mixing bulb often contains a small glass bead that helps mix the blood and diluent thoroughly, resulting in a homogenous solution for accurate counting.

APPLICATION IN PHYSIOLOGY

- **White Blood Cell Counting:** The WBC pipette is essential in haematology labs for diluting blood to a standardized concentration, allowing for accurate white blood cell counts using a hemocytometer.
- **Disease Diagnosis and Monitoring:** WBC counts are crucial in diagnosing and monitoring infections, leukemias and immune-related disorders.
- **Physiological Research:** used in research that involves studying immune responses, infection patterns and white blood cell physiology.

LIMITATIONS

- **Manual Handling Errors:** Precise handling is required to ensure accurate results.
- **Time-Consuming Process:** The manual process of preparing and diluting the blood sample is slower than automated systems.
- **Potential for Contamination:** Inadequate cleaning between samples can lead to contamination and inaccurate counts.
- **Fixed Dilution Ratio:** 1:20 ratio may not suit all applications.

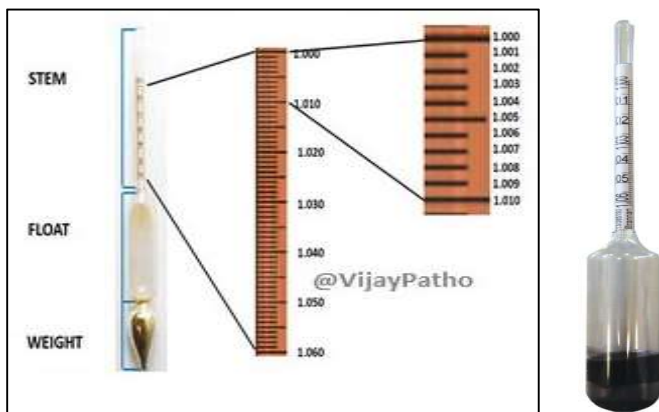


URINOMETER

PRINCIPLE

The urinometer works on the principle of buoyancy, similar to a hydrometer. When placed in urine, the urinometer will float at a certain level depending on the density of the urine. The greater the concentration of solutes in the urine, the higher its density, causing the urinometer to float higher. Conversely, less concentrated (dilute) urine will have a lower specific gravity, causing the urinometer to sink more. The scale on the urinometer stem allows for direct reading of specific gravity.

DIAGRAM



PARTS

- **Graduated Stem:** The urinometer's long, calibrated stem shows a scale corresponding to specific gravity values. The scale typically ranges from 1.000 to 1.060, the typical range for urine-specific gravity.
- **Weighted Bulb:** Located near the bottom of the urinometer, the weighted bulb ensures the device floats vertically in the liquid. It is calibrated so that the depth at which the urinometer sinks is proportional to the density of the urine.
- **Temperature Bulb:**
Some urinometers have a temperature bulb to help adjust for temperature variations, as specific gravity readings can be affected by temperature changes.

APPLICATION IN PHYSIOLOGY

- **Assessment of Hydration Status:** Higher specific gravity indicates concentrated urine, often seen in dehydration.
- **Kidney Function Analysis:** Low specific gravity may indicate conditions like chronic kidney disease, while high specific gravity can indicate dehydration.
- **Diagnosis of Metabolic Conditions:** Measuring urine specific gravity can also help diagnose metabolic conditions like diabetes mellitus,
- **Monitoring of Certain Medical Conditions:** In conditions like diabetes insipidus or adrenal gland disorders.

LIMITATIONS

- **Temperature Sensitivity:** Deviations in temperature can affect buoyancy and cause inaccurate results.
- **Requires Adequate Sample Volume:** The urinometer requires a relatively large volume of urine to float urinometer.
- **Less Accurate for Small Solutes:** It does not distinguish between different solutes. Substances like glucose and protein lead to misleading results.
- **Manual Reading Errors:**

Reading the specific gravity on the urinometer stem can be subjective, as meniscus level interpretation may vary slightly between users.

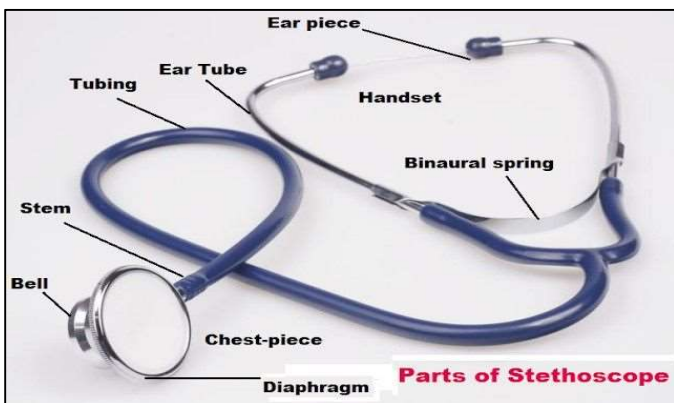


STETHOSCOPE

PRINCIPLE

The stethoscope works on the principle of sound conduction and amplification. It captures sound waves generated within the body and transmits them through hollow tubes to the listener's ears, where they can be interpreted. The diaphragm and bell of the stethoscope pick up sounds from the body, with the diaphragm capturing high-frequency sounds (like lung sounds) and the bell capturing low-frequency sounds (such as heart murmurs).

DIAGRAM



PARTS

- **Ear Tips:** Soft, comfortable tips that fit into the listener's ears, forming a seal that reduces background noise and improves sound transmission.
- **Binaurals:** The metal tubes that connect the ear tips to the tubing.
- **Tubing:** The flexible rubber or PVC tube that transmits sound from the chest piece to the binaural. It is designed to minimize sound loss and prevent external noise interference.
- **Chest Piece:** The part of the stethoscope that contacts the patient's body, containing both the diaphragm and bell.
- **Diaphragm:** The flat, larger surface on the chest piece that captures high-frequency sounds (e.g., lung and heart sounds).

- **Bell:** The smaller, hollow side of the chest piece that detects low-frequency sounds (e.g., heart murmurs and some vascular sounds). It is typically used with light pressure on the skin.

APPLICATION IN PHYSIOLOGY

- **Cardiac Assessment:** Used to listen to heart sounds, including normal heartbeats, murmurs and rhythm abnormalities, which provide information about heart function.
- **Lung Examination:** Lung sounds like breath sounds, crackles, wheezes and rales, which aid in diagnosing respiratory conditions.
- **Blood Pressure Measurement:** Stethoscope is used to listen for Korotkoff sounds in the brachial artery, enabling accurate blood pressure measurement.
- **Gastrointestinal Examination:** Bowel movements or abnormal sounds that could indicate issues like bowel obstruction.
- **Vascular Examination:** Used to assess abnormal sounds suggesting turbulent blood flow, often due to arterial blockages.

LIMITATIONS

- **Limited Sound Amplification:** A stethoscope relies on passive sound transmission and does not amplify sound electronically.
- **User Dependence:** Proper technique and training are required to interpret sounds accurately.
- **No Visual Information:** Unlike imaging devices the stethoscope provides no visual information, limiting its diagnostic scope.



SPHYGMOMANOMETER

Principle

1. Manual Sphygmomanometer:

The manual sphygmomanometer operates on the auscultatory method, which involves listening to the sounds of blood flow (Korotkoff sounds) using a stethoscope. The device uses an inflatable cuff to temporarily occlude blood flow in an artery, typically the brachial artery. As the cuff inflates, it restricts blood flow. The cuff then gradually deflates and the healthcare provider listens for the first appearance of Korotkoff sounds (indicating systolic blood pressure) and the disappearance of these sounds (indicating diastolic blood pressure).

2. Digital Sphygmomanometer:

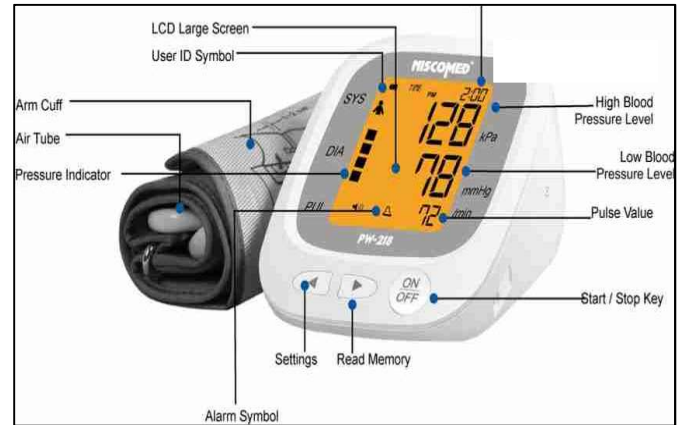
The digital sphygmomanometer utilizes the oscillometric method. It automatically inflates and deflates the cuff and uses electronic sensors to detect oscillations in the arterial wall caused by blood flow. The device's microprocessor analyzes these oscillations to determine systolic and diastolic blood pressure values, which are then displayed digitally.

DIAGRAM

1. Manual Sphygmomanometer Diagram:



2. Digital Sphygmomanometer Diagram:



PARTS

Manual Sphygmomanometer:

- **Cuff:** A fabric or nylon band that wraps around the upper arm. It inflates to apply pressure to the brachial artery, temporarily stopping blood flow.
- **Inflation Bulb:** A rubber bulb connected to the cuff via flexible tubing. It manually pumps air into the cuff to increase pressure.
- **Valve:** A small valve attached to the inflation bulb. It controls the release of air from the cuff, allowing gradual deflation.
- **Pressure Gauge (Manometer):** A calibrated dial (aneroid) or mercury column that measures cuff pressure. It display pressure in mmHg.

Digital Sphygmomanometer:

- **Cuff:** Similar to manual, wraps around the upper arm or wrist. It inflates to measure blood pressure.
- **Pump Mechanism:** An automatic pump is integrated into the device. It inflates and deflates the cuff electronically without manual pumping.
- **Sensor:** Electronic sensors located in the cuff or connected tubing. It detects oscillations in the arterial wall as the cuff deflates.
- **Digital Display:** An LCD or LED screen. It shows systolic and diastolic blood pressure readings and often heart rate.

- **Control Buttons:** Buttons for operating the device. It starts/stops measurements, adjusts settings and switches modes.
- **Microprocessor:** An electronic processing unit inside the device. It analyzes oscillometric data to calculate blood pressure values.
- **Power Source:** Batteries or rechargeable units. It gives power to the device's electronic components.

APPLICATIONS IN PHYSIOLOGY

Both Manual and Digital Sphygmomanometers:

- Determines systolic and diastolic blood pressure, a critical physiological parameter reflecting cardiovascular health.
- Identifies abnormal BP levels, such as hypertension and hypotension aiding in early diagnosis and treatment planning.
- BP readings are used to evaluate the risk of cardiovascular diseases, including heart attack and stroke.
- Measures BP changes during physical exertion or stress tests to understand the body's physiological response to stressors.
- Essential for managing chronic conditions like diabetes, kidney disease and heart failure by tracking BP over time to guide therapeutic adjustments.
- Low BP readings are crucial in assessing conditions like hemorrhagic shock, dehydration or sepsis, aiding in emergency management.
- Used in clinical trials and research to evaluate the effectiveness and safety of cardiovascular drugs, such as antihypertensive medications.
- Health screenings to identify individuals at risk for hypertension and promote preventive healthcare practices.
- Facilitates studies on vascular tone, arterial compliance and autonomic nervous system function, contributing to a deeper understanding of human physiology.

LIMITATIONS

Manual Sphygmomanometer:

- **Operator Dependent:** Accuracy relies heavily on the skill and experience of the user.
- **Requires Additional Equipment:** Needs a stethoscope for auscultatory measurements.
- **Subject to Observer Bias:** Variability in readings due to subjective interpretation.
- **Time-Consuming:** Manual inflation and deflation take more time.

Digital Sphygmomanometer:

- **Calibration and Accuracy:** This may require regular calibration to maintain accuracy.
- Generally more expensive than manual sphygmomanometers.
- Accuracy depends on the quality of oscillometric algorithms.

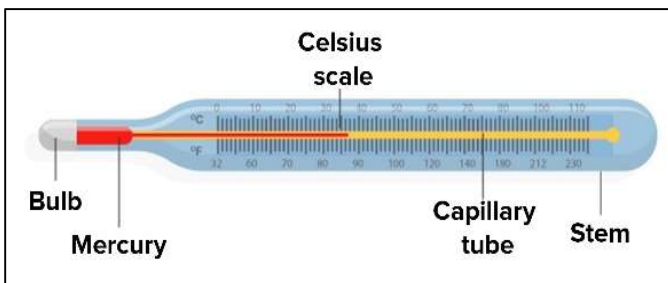


CLINICAL THERMOMETER

PRINCIPLE

The clinical thermometer works on the principle of thermal expansion. When the thermometer's bulb (filled with mercury) is placed in contact with the human body, the liquid expands with rising temperature and moves up the capillary tube. The temperature reading is taken at the point where the liquid level stabilizes, typically indicated in degrees Celsius ($^{\circ}\text{C}$) or Fahrenheit ($^{\circ}\text{F}$).

DIAGRAM



PARTS

- **Bulb:** It contains a temperature-sensitive liquid (typically mercury) that expands when exposed to heat, indicating body temperature.
- **Capillary Tube:** A very narrow glass tube allows the liquid to move up as it expands. The narrow diameter ensures precise measurements by limiting the rapid movement of the liquid.
- **Scale:** Calibrated markings along thermometer tube, typically in Celsius ($35\text{--}42^{\circ}\text{C}$) or Fahrenheit ($94\text{--}108^{\circ}\text{F}$). It provides an accurate body temperature by correlating the liquid's position with specific temperature values.
- **Constriction Point (Kink):** It prevents the liquid from immediately falling back down once the thermometer is removed.
- **Stem:** The outer glass structure of the thermometer, protects the capillary tube, scale and bulb. Provides a safe, easy-to-handle exterior.

APPLICATIONS IN PHYSIOLOGY

- **Detecting Fever:** Clinical thermometers are primarily used to measure body temperature to detect fever, which can indicate infections or other health conditions.
- **Temperature Monitoring in Illness:** Useful for monitoring temperature changes in patients with illnesses such as the flu, COVID-19 and other infections where temperature tracking is crucial.
- **Assessing Metabolic and Hormonal Activity:** Body temperature fluctuations can indicate metabolic or hormonal imbalances, making thermometers useful in diagnostic assessments.
- **Post-Surgery Monitoring:** Used in clinical settings to monitor temperature after surgery, as fever can indicate post-operative infections.
- **Home Healthcare:** Many households use clinical thermometers for regular health monitoring, providing a quick and accessible way to assess health changes.

LIMITATIONS

- **Breakable Material:** Glass thermometers are fragile and prone to breaking if mishandled.
- **Slow Response Time:** Clinical thermometers typically take a minute or more to provide an accurate reading, which can be inconvenient compared to digital thermometers.
- **Reading Difficulty:** Requires precise reading of the liquid level within the narrow capillary tube, which can be challenging for users with vision impairments.

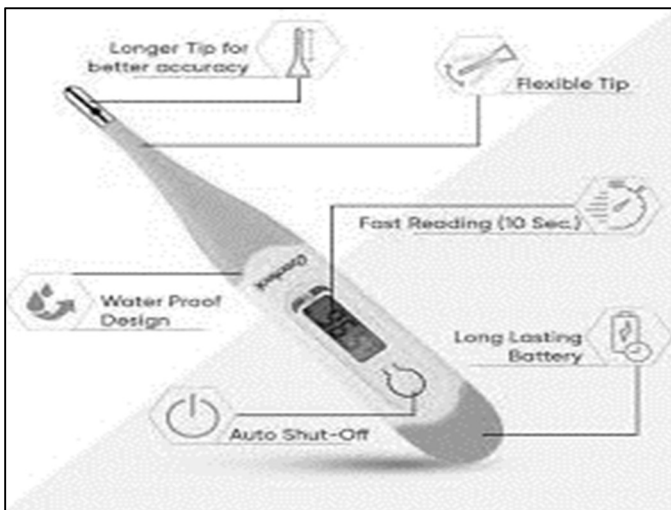


DIGITAL THERMOMETER

PRINCIPLE

A digital clinical thermometer operates based on the principle of thermal sensing and electrical signal conversion. It uses a thermistor, a temperature-sensitive resistor, whose resistance changes with temperature variations. The thermistor is connected to a microcontroller that converts these changes in resistance into a digital temperature reading. The output is displayed on a screen, typically in degrees Celsius ($^{\circ}\text{C}$) or Fahrenheit ($^{\circ}\text{F}$). This technology ensures accurate, fast and reliable temperature measurements.

DIAGRAM



PARTS

- **Probe:** The tip of the thermometer that comes in contact with the body part (e.g. oral, rectal or underarm) to measure temperature.
- **Thermistor:** A temperature-sensitive resistor located inside the probe that detects temperature changes.
- **Microcontroller:** A small processing unit that interprets the resistance changes from the thermistor and converts them into a temperature reading.
- **LCD Screen:** Displays the temperature reading in a digital format.

- **Battery:** Provides power for the thermometer's operation.
- **Buttons:** Typically includes an On/Off button and sometimes additional buttons for switching between Celsius and Fahrenheit or storing previous readings.

LIMITATIONS

- **Battery Dependency:** Requires a functioning battery to operate, which can limit its usability in emergencies if not charged or replaced.
- **Accuracy in Certain Conditions:** Environmental factors, such as extreme cold or heat, can affect the accuracy of the readings.
- **Delay in Response:** While generally fast, the thermometer may take slightly longer in comparison to infrared thermometers.
- **Limited Lifespan:** The electronic components and batteries may degrade over time, requiring replacement.
- **Sensor Damage:** The thermistor can be sensitive to physical damage, impacting accuracy.
- **Calibration Needs:** May require periodic calibration to maintain accuracy, particularly in professional or research settings.
- **Surface Limitations:** Less accurate if not used correctly (e.g., improper placement or insufficient contact with the body).
- **Cost:** Higher cost compared to traditional mercury thermometers.



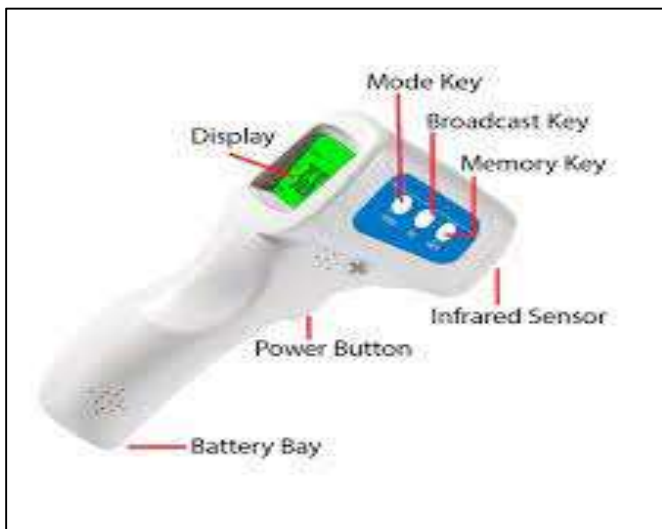
NON-CONTACT INFRARED THERMOMETER

PRINCIPLE

The non-contact infrared thermometer operates on the principle of infrared radiation detection. The human body emits infrared radiation as heat, which is proportional to the surface temperature. The thermometer has an infrared sensor that detects this radiation and converts it into an electrical signal. This signal is processed by an internal microprocessor to calculate the temperature, which is displayed digitally on the screen.

The thermometer typically measures temperature from the forehead or another accessible body part. It assumes that the surface temperature correlates well with core body temperature under stable conditions.

DIAGRAM



PARTS

- **Infrared Sensor:** Captures the infrared radiation emitted by the body surface.
- **Microprocessor:** Converts the infrared signal into a temperature reading.
- **Display Screen:** Shows the measured temperature digitally.
- **Battery Compartment:** Powers the device, usually with AA or AAA batteries.

- **Control Buttons:** Allow user interactions such as switching modes, turning the device on/off and initiating measurements.
- **Memory Function (Optional):** Some models can store previous readings for tracking purposes.
- **Calibration Settings (Optional):** Enable adjustments for enhanced accuracy.

LIMITATIONS

- **Surface Temperature Dependency:** Measures surface temperature, which may not always correlate accurately with core body temperature, especially in cold or hot environments.
- **Environmental Interference:** External factors like wind, ambient temperature or humidity can affect readings.
- **Limited Accuracy:** Less precise than invasive methods like rectal or oral thermometers in certain clinical conditions.
- **Calibration Requirements:** Needs regular calibration to maintain accuracy.
- **Limited Range:** Effectiveness reduces if the device is held too far from the body.
- **Battery Dependency:** Requires charged batteries and low battery levels can affect performance.
- **High Initial Cost:** Generally more expensive than contact thermometers.
- **Not Suitable for Core Temperature Monitoring:** Cannot replace invasive methods for precise core temperature measurements.



CLINICAL HAMMER

PRINCIPLE

The clinical hammer operates on the principle of eliciting the stretch reflex, which is an involuntary contraction of a muscle in response to a quick stretch. When the hammer strikes a tendon, it momentarily stretches the muscle attached to that tendon. This stimulates sensory receptors called muscle spindles, which send signals through sensory nerves to the spinal cord. The spinal cord, in turn, sends a signal back through motor nerves, causing the muscle to contract. This reflex action is a key indicator of nerve health and function.

DIAGRAM



PARTS

- **Head:** The striking part of the hammer, typically made of rubber and designed in various shapes (triangular, round or cylindrical). It is used to tap tendons to elicit reflexes.
- **Neck:** The short part between the head and handle.
- **Handle:** Allows the examiner to hold the hammer securely and control the force of the strike.
- **Pointed Tip (Some Models):** A pointed end is often located at the handle's base in certain hammer models. It is used for checking plantar reflex (Babinski response) or for light sensory touch assessments.

APPLICATIONS IN PHYSIOLOGY

▪ Reflex Testing in Neurological Exams:

It is essential for checking deep tendon reflexes, which help assess the integrity of the spinal cord and peripheral nervous system. Common reflexes tested include the knee (patellar), ankle (Achilles) and elbow (biceps) reflexes.

▪ Evaluating Nervous System Function:

Reflex responses provide insight into nerve transmission and spinal cord health. Absent, exaggerated or asymmetric reflexes can indicate neurological issues such as nerve damage, lesions or neuropathy.

▪ Assessing Muscle Tone and Reflex Pathways:

Reflex testing can help detect conditions like hyperreflexia or hyporeflexia, which are often associated with upper and lower motor neuron disorders, respectively.

▪ Monitoring Disease Progression or Recovery:

Reflex testing is a quick, non-invasive way to monitor changes in neurological conditions, assess the effectiveness of treatment or detect signs of recovery after nerve injuries.

LIMITATIONS

- **Limited Information on Central Nervous System (CNS):** It provides little direct information about higher brain functions.
- **Subjectivity in Interpretation:** Reflex responses are often graded based on subjective observation, which can lead to variability between different examiners.



PH METER

PRINCIPLE

A pH meter operates on the principle of electrochemical measurement. It consists of a glass electrode sensitive to hydrogen ions and a reference electrode. When the electrode is immersed in a solution, hydrogen ions interact with the glass membrane, creating a voltage difference between the glass and reference electrodes. This voltage is then measured and converted to a pH value by the meter, as the voltage is directly proportional to the hydrogen ion concentration.

DIAGRAM



PARTS

- **Glass Electrode:** It detects the hydrogen ion concentration by generating a voltage proportional to the pH level.
- **Reference Electrode:** Contains a stable solution that provides a constant reference voltage.
- **Probe:** A combination of the glass and reference electrodes encased in a cylindrical tube.
- **Display Screen:** It displays the pH reading as a numeric value.
- **Control Panel:** Typically includes buttons or dials for calibration and adjustments.
- **Temperature Sensor (in some models):** for temperature-related changes in pH.

APPLICATIONS IN PHYSIOLOGY

▪ Blood pH Monitoring:

Maintaining a pH of around 7.4 is crucial for human blood and deviations can indicate metabolic or respiratory disorders. The pH meter is used to measure blood pH in clinical labs.

▪ Urine and Saliva Testing:

Used to assess the pH of urine and saliva, helping diagnose conditions related to kidney function, hydration status and acid-base balance.

▪ Gastric Acidity Measurement:

pH meters help measure gastric juice acidity, which is important in diagnosing conditions like acid reflux or ulcers.

▪ Cell Culture and Biochemistry Labs:

Monitoring the pH of culture media is essential for maintaining the viability of cells, as well as for biochemical assays sensitive to pH.

▪ Nutrition and Metabolism Studies:

pH is used in studies related to diet, metabolic processes and acid-base homeostasis in the body.

LIMITATIONS

- **Calibration Required:** Regular calibration with standard buffer solutions is needed for accurate readings.
- **Fragile Electrode:** The glass electrode is delicate and can break easily, making it unsuitable for high-impact or rough environments.



DIGITAL WEIGHING SCALE

PRINCIPLE

The digital weighing machine operates on the principle of strain gauge technology. A load cell within the machine converts mechanical force (weight) into an electrical signal. When a person steps on the machine, their weight applies force on the load cell, causing it to deform slightly. This deformation changes the electrical resistance of the strain gauge, which is converted into a digital reading by an analog-to-digital converter (ADC). The result is displayed as the person's weight in the chosen unit (kilograms or pounds).

DIAGRAM



Digital



Analog

PARTS

- **Top Platform:** The flat surface where the user stands to measure weight.
- **Load Cell:** Key sensor that measures weight by detecting deformation caused by the applied load.
- **Analog-to-Digital Converter (ADC):** analog signal from the load cell into a digital format.
- **Microcontroller:** Processes the digital signal and translates it into readable weight data.
- **Display Unit:** A digital screen (LED or LCD) that shows the measured weight.
- **Battery/Power Source:** Provides power to the electronic components of the machine.
- **Housing/Enclosure:** Protects internal components and provides aesthetic appeal

APPLICATIONS IN PHYSIOLOGY

- **Body weight measurement & monitoring:** used for precise body weight measurement and monitoring.
- **BMI Calculation:** Along with height, weight measurement helps calculate Body Mass Index, an important indicator of overall health.
- **Drug Dosage Calculations:** Some medications require precise weight-based dosing.
- **Tracking weight changes** in chronic conditions like fluid retention or loss related to heart or kidney diseases.
- **Research and Sports Physiology:** Assists athletes in monitoring their weight for performance optimization.
- **Undernutrition or Overnutrition:** Regular weight measurements in children are essential for assessing normal growth and development. Helps identify undernutrition or overnutrition.
- **Pregnancy:** Tracks weight gain during pregnancy to ensure healthy fetal development.
- **Clinical Research:** Used in studies to correlate weight with various health parameters.

LIMITATIONS

- **Calibration Needs:** Digital scales may require periodic calibration to maintain accuracy, especially in clinical and research settings.
- **Power Dependence:** Requires batteries or electricity; cannot function without power.
- **Environmental Sensitivity:** Extreme temperature or high humidity may affect the performance of electronic components.

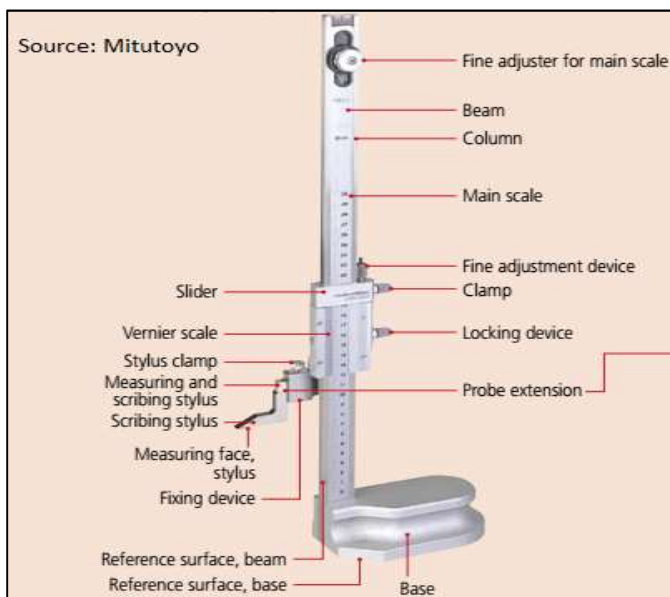


HEIGHT MEASUREMENT SCALE

PRINCIPLE

The height measurement scale works on the principle of a vertical measuring rod or tape. The scale has a movable headpiece or sliding bar that adjusts to rest on the top of the individual's head. The position of the headpiece is then read against the calibrated markings on the vertical rod or tape to determine the person's height. This simple mechanical design provides an accurate, reliable reading by measuring the vertical distance from the floor to the headpiece.

DIAGRAM



PARTS

- **Headpiece:** A movable, flat, horizontal bar that slides up or down the measuring rod.
- **Measuring Rod (Scale):** Provides an accurate measure of height.
- **Base / Platform:** for standing
- **Leveling Knob (in some models):** Helps to stabilize the stadiometer and keeps it level on uneven floors for accuracy.
- **Locking Mechanism (some models):** Lock the headpiece in place to prevent movement and allows accurate reading of the measurement.

APPLICATIONS IN PHYSIOLOGY

- **Growth and Development Assessment:** Regular height measurements are crucial in pediatrics for tracking growth patterns, helping identify potential developmental issues or growth disorders.
- **BMI Calculation:** Height measurements are combined with weight to calculate Body Mass Index (BMI), which is used to assess body fat, health risks and nutritional status.
- **Nutritional Assessment:** Height measurement is an important indicator of malnutrition or stunted growth, especially in children, as it can reflect long-term nutritional health.
- **Assessment of Adult Health:** Height is sometimes measured in adults for general health records, as changes in height over time (e.g., reduction due to spinal compression) can be indicative of aging or conditions like osteoporosis.
- **Sports and Physical Fitness Assessments:** Height is an important metric in athletic and fitness settings, particularly in sports where height affects performance (e.g., basketball, volleyball).

LIMITATIONS

- **Inaccuracy Due to Positioning Errors:** If the individual doesn't stand completely upright (bedridden patients) or if the headpiece isn't positioned correctly, the reading can be inaccurate.
- **Dependence on a Flat Surface:** uneven floors can affect accuracy.



GLUCOMETER

PRINCIPLE

The basic principle behind the glucometer is electrochemical detection. It measures the electrical current produced by the reaction of glucose with specific enzymes (like glucose oxidase or glucose dehydrogenase) on a test strip. When a drop of blood is placed on the test strip, the enzyme reacts with the glucose in the blood to form gluconic acid. This reaction generates electrons, creating an electrical current. The glucometer measures the strength of this current, which is directly proportional to the glucose concentration in the blood.

DIAGRAM



PARTS

- **Test Strip Slot & Test Strip:** A disposable strip coated with enzymes (e.g., glucose oxidase) that reacts with the glucose in the blood sample.
- **Display Screen:** The screen shows the blood glucose reading, typically in mg/dL or mmol/L.
- **Control Buttons:** used for finding previous readings and setting date and time.
- **Lancing Device:** A small, pen-like tool used to prick the skin (usually on the fingertip) by lancet to obtain a drop of blood.

APPLICATIONS IN PHYSIOLOGY

▪ Diabetes Management:

Glucometers are vital for people with diabetes to monitor their blood sugar levels. Regular monitoring helps in adjusting medication, diet and exercise to maintain optimal blood glucose level.

▪ Immediate Feedback:

Glucometers provide real-time results, helping patients make quick decisions about insulin dosing or dietary changes.

▪ Diagnostic Tool:

Healthcare professionals may use the data for treatment adjustments.

LIMITATIONS

- **Accuracy Issues:** Glucometer readings can sometimes vary from laboratory glucose testing due to environmental factors (like temperature and humidity), the quality of the test strips or the meter's calibration.
- **Interference from Substances:** Certain substances like vitamin C, acetaminophen or high levels of lipids in the blood may interfere with glucometer readings and lead to inaccurate results.
- **No Continuous Monitoring:** Unlike continuous glucose monitors (CGMs), glucometers provide only a snapshot of blood glucose levels at a single point in time, requiring frequent testing.

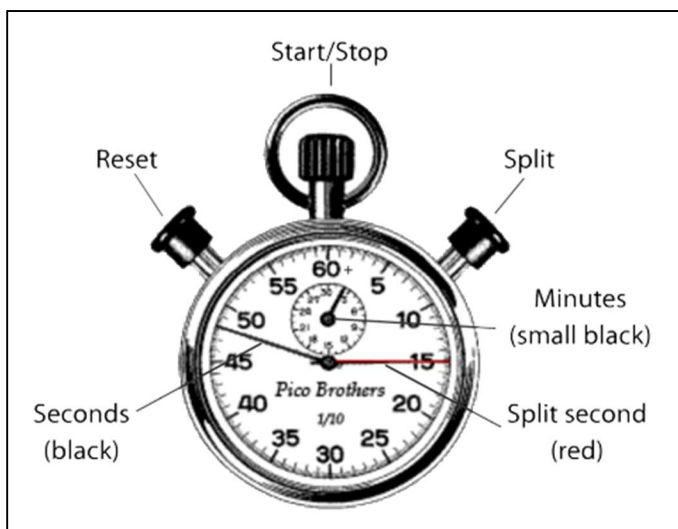


STOPWATCH

PRINCIPLE

The principle of a stopwatch is based on the measurement of time intervals from the moment the user starts the device to the moment they stop it. Most modern stopwatches use a quartz crystal oscillator, which vibrates at a precise frequency, ensuring accurate timekeeping. When the start button is pressed, the stopwatch begins counting from zero and when the stop button is pressed, it displays the elapsed time. Some advanced stopwatches can also measure laps, splits or multiple intervals.

DIAGRAM



PARTS

- **Start/Stop Button:** Initiates or stops the counting of time.
- **Reset Button:** Resets the time to zero after measurement.
- **Digital/Analog Display:** Shows the time elapsed, either in digits (for digital stopwatches) or using a dial and hands (for analog stopwatches).
- **Lap Button:** Records intermediate times or lap times in certain stopwatches.
- **Battery Compartment:** Stores the power source (for digital stopwatches).

APPLICATIONS IN PHYSIOLOGY

- **Reaction Time Measurement:** Stopwatches are used in experiments to measure human reaction times ex.- BTCT & provide time-related data on sensory and motor responses.
- **Heart Rate Measurement:** They are used to measure heartbeats and heart rate.
- **Muscle Endurance Testing:** In physiology labs or clinics, stopwatches help measure how long a subject can maintain muscle contraction during endurance tests.
- **Exercise Testing:** Stopwatches track time during endurance, speed or time trials in various physiological assessments (e.g., VO2 max tests).

LIMITATIONS

- **Human Error:** The start and stop of the stopwatch rely on manual operation, which can introduce human error, particularly in response timing.
- **Limited Precision:** While stopwatches can measure milliseconds, their accuracy depends on the reaction speed of the operator in starting and stopping the timer.
- **Limited to Time Measurement:** A stopwatch can only measure time and cannot provide additional physiological data, such as distance or force.



BT/CT KIT

PRINCIPLE

Bleeding Time (BT): This test measures the time it takes for bleeding to stop after a small standardized puncture is made on the skin. It evaluates platelet function, blood vessel integrity and the initial phase of clot formation. A filter paper or similar material is used to blot the blood at regular intervals until the bleeding stops.

Clotting Time (CT): This test measures the time it takes for blood to clot in a glass capillary tube or on a clean surface. It assesses the coagulation pathways (intrinsic and common pathways) that lead to fibrin clot formation. A stopwatch is used to monitor the process.

DIAGRAM



PARTS

- **Lancet/Blade:** Used to make a small, standardized puncture or incision on the skin to initiate bleeding.
- **Stopwatch:** Measures the time taken for bleeding to stop or for the blood to clot, essential for both BT and CT tests.
- **Filter Paper:** Used to blot the blood at regular intervals in the bleeding time test to determine when the bleeding stops.
- **Capillary Tubes:** used in the clotting time test to collect a blood sample

APPLICATIONS IN PHYSIOLOGY

Bleeding Time (BT)

- Assesses platelet function, vascular integrity and capillary fragility.
- It is useful in detecting platelet disorders (e.g., thrombocytopenia, von Willebrand disease).

Clotting Time (CT):

- Evaluates the blood's ability to form clots and the functionality of clotting factors.
- It helps diagnose clotting factor deficiencies like haemophilia or other coagulation disorders.

LIMITATIONS

Bleeding Time Test:

- It can be affected by operator technique, as incision size and depth can vary.
- Not always a reliable measure of platelet function or small vessel health in certain bleeding disorders.
- Has largely been replaced by more sensitive tests like platelet function analyzer (PFA-100).

Clotting Time Test:

- It provides a general overview but does not indicate specific factor deficiencies.
- Results can be affected by temperature, sample handling and technique, leading to variability.
- Not as precise as more modern tests like prothrombin time (PT) or activated partial thromboplastin time (APTT).



EVACUATED BLOOD COLLECTION TUBES

PRINCIPLE

Evacuated blood collection tubes use a vacuum to draw a precise blood volume into a sterile, sealed tube when a vein is punctured. The vacuum ensures consistency and prevents contamination. Tubes may contain additives like anticoagulants or clot activators, tailored for specific tests. This system maintains sample integrity and enhances safety for diagnostic and clinical use.

DIAGRAM



Color	Anticoagulant	Uses
	No anticoagulant	Serological examination in biochemistry
	Sodium Fluoride	Glucose estimation
	EDTA	Hematological examination like complete hemogram, ESR
	3.2% sodium citrate	Coagulation studies like PT, APTT
	Heparin	Bone marrow studies
	Citrate	Blood culture
	(K2)EDTA	Blood Bank tests. Blood typing. ABO grouping etc

PARTS

- **Tube Cap:** Color-coded for identification of additives.
- **Tube Body:** Made of glass or plastic, designed to hold a specific volume.
- **Additives:** Chemical substances to preserve the blood or aid in testing.
- **Label:** Provides information for identification.
- **Markings:** Indicate volume capacity.

APPLICATIONS IN PHYSIOLOGY

- **Hematology:** Tubes with EDTA for complete blood counts (CBC).
- **Biochemistry:** Serum separator tubes for testing electrolytes, glucose, and enzymes.
- **Coagulation Studies:** Tubes with sodium citrate for PT and APTT tests.
- **Microbiology:** Blood culture bottles for detecting infections.
- **Specialized Tests:** Tubes for DNA, RNA, or hormone analysis.

LIMITATIONS

Additive Interference: Some additives may interact with specific tests, causing errors.

Volume Issues: Over- or under-filling can affect test accuracy.

Cross-Contamination: Improper sealing or handling may compromise sample integrity.

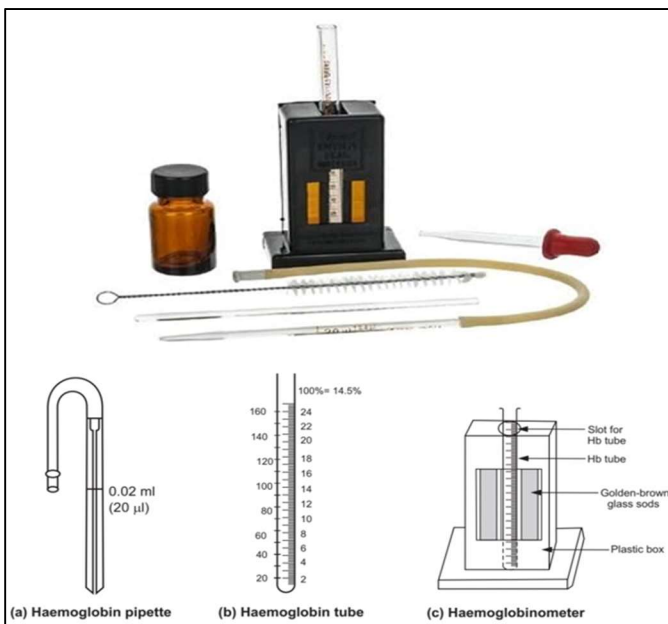


SAHLI'S HAEMOGLOBINOMETER

PRINCIPLE

Sahli's Haemoglobinometer is used to estimate hemoglobin concentration in blood based on colour comparison. It utilizes acid hematin, which is formed by mixing a blood sample with hydrochloric acid (HCl). The blood hemoglobin is converted into brownish-yellow acid hematin, which is then matched against a calibrated colour scale. The hemoglobin concentration is determined by matching the colour intensity of the acid hematin solution to a reference standard.

DIAGRAM



PARTS

- **Hemoglobin Tube (Graduated Tube):** used to hold the acid hematin solution, with graduations marked in grams per deciliter (g/dL) to measure hemoglobin concentration.
- **Comparator Tubes:** These tubes provide a reference for standard colour comparison to estimate hemoglobin levels.
- **Hemoglobin Pipette:** A capillary pipette marked to draw a precise volume of blood (usually 20 microliters) for the test. It ensures accuracy in blood sample measurement.

- **Mixing Stirrer/Glass Rod:** Used to mix the blood and hydrochloric acid (HCl) solution within the hemoglobin tube.
- **Diluting Solution (HCl):** A 0.1 N hydrochloric acid solution is used to lyse the red blood cells and convert hemoglobin into acid hematin.

APPLICATIONS IN PHYSIOLOGY

- **Clinical Hemoglobin Measurement:** Sahli's Haemoglobinometer is widely used in clinical settings to estimate hemoglobin levels in blood, which is essential for monitoring hemoglobin levels in various health conditions.
- **Public Health Screenings:** Used in public health settings and rural healthcare facilities as a simple and cost-effective tool to screen for anaemia, particularly in regions with limited resources.
- **Educational Tool:** Commonly used in educational laboratories for teaching and demonstration purposes, helping students understand hemoglobin concentration analysis.

LIMITATIONS

- **Subjective Colour Matching:** this can lead to variations based on individual perception and ambient lighting.
- **Limited Sensitivity:** The device is less sensitive and may not detect minor variations in hemoglobin concentration.
- **Old Technology:** is used less frequently for more accurate and faster results.

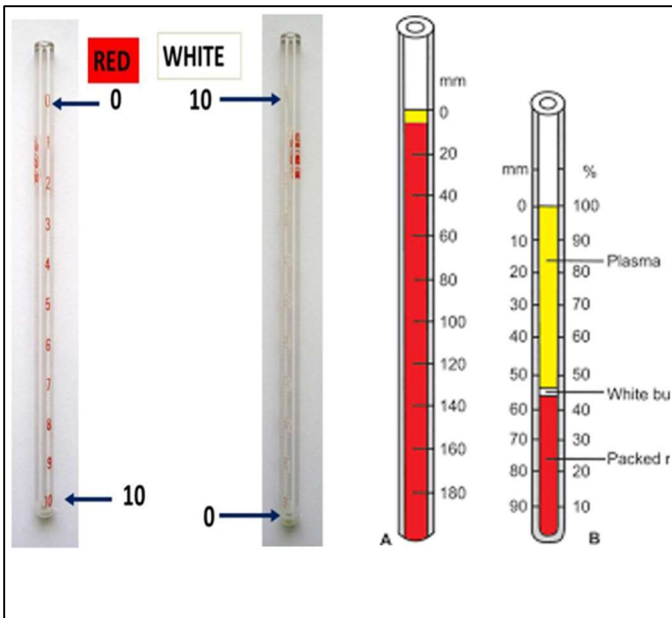


WINTROBES TUBE

PRINCIPLE

Wintrobe's tube is used to determine the erythrocyte sedimentation rate (ESR), which reflects inflammation or disease processes in the body. The principle is based on the rate at which red blood cells (RBCs) settle to the bottom of a vertical tube when blood is allowed to stand undisturbed. The rate is influenced by factors like plasma proteins, RBC shape and the presence of inflammatory markers.

DIAGRAM



PARTS

- **Tube:** A narrow, graduated glass tube, typically 1 cm in diameter and 10 cm long.
- **Blood Sample:** The blood is drawn into the tube and left to stand.
- **Plasma:** The upper part of the tube, where the liquid portion of the blood remains after the RBCs settle.
- **RBC Layer:** The red blood cells form a sediment at the bottom of the tube.
- **Graduation Markings:** Used to measure the distance the RBCs have settled, giving the ESR in mm/hour.

APPLICATIONS IN PHYSIOLOGY

- **Erythrocyte Sedimentation Rate (ESR):** Primarily used to detect inflammatory conditions, infections, autoimmune diseases (like rheumatoid arthritis) and certain cancers.
- **Indicator of Disease Activity** ESR can help assess disease activity in conditions like tuberculosis or systemic lupus erythematosus.

LIMITATIONS

- **Non-specific:** ESR is an indirect marker and does not pinpoint the exact cause of inflammation.
- **Influenced by Various Factors:** Age, gender, anaemia, pregnancy and even posture can affect ESR readings.
- **Slow:** Requires approximately 1 hour to get results, limiting its use for rapid diagnosis.

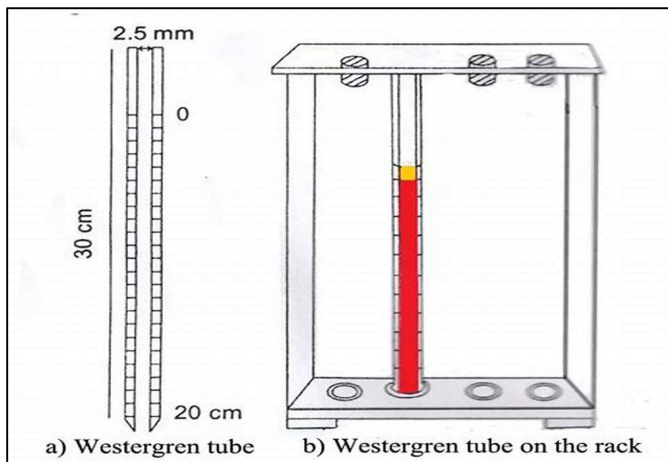


WESTERGREN TUBE

PRINCIPLE

The principle of the Westergren Tube involves the collection of small blood samples in a capillary tube coated with an anticoagulant, generally lithium heparin. This anticoagulant prevents blood clotting, allowing accurate measurement and separation of plasma or serum from cells. When centrifuged, the blood components separate into layers, enabling analysis based on cellular density.

DIAGRAM



PARTS

▪ Capillary Tube:

A small, thin glass or plastic tube, typically around 1 mm in diameter and several centimeters (30 cm) long. It is used to hold a precise amount of blood, facilitating the measurement of small volumes.

▪ Anticoagulant Coating (Lithium Heparin):

Coated inside the tube, lithium heparin prevents blood clotting by neutralizing thrombin and other clotting factors.

▪ Sealing Wax/Plug:

One end of the tube is sealed after blood collection, using wax or plug, to prevent blood from leaking during the centrifugation process.

APPLICATION IN PHYSIOLOGY

- **Hematocrit Measurement:** Used to determine the percentage of red blood cells (RBCs) in a blood sample. After centrifuging, the packed cell volume (PCV) can be calculated by measuring the RBC layer in relation to the total length of the sample.
- **Blood Gas Analysis:** Enables the analysis of oxygen, carbon dioxide and pH levels in the blood, which is crucial for assessing respiratory and metabolic health.
- **Electrolyte and Enzyme Studies:** Useful in monitoring various electrolyte and enzyme levels in the blood, aiding in diagnosing and managing conditions affecting metabolic and electrolyte balance.

LIMITATIONS

- **Sample Size Limitation:** Only small volumes can be obtained, limiting analysis options for larger panels or tests.
- **Risk of Haemolysis:** Improper handling and processing can sometimes lead to cell rupture, which may distort results, especially for potassium and enzyme measurements.
- **Short Storage Time:** Blood in heparin-coated tubes must be processed quickly; otherwise, cellular changes may affect the test results.

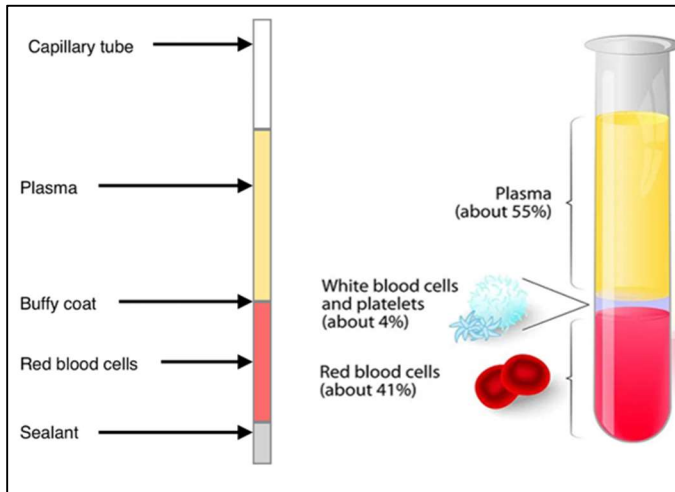


PCV TUBE

PRINCIPLE

PCV tube, also known as a microhematocrit tube, is used to determine the packed cell volume of blood, which is the proportion of blood volume that is occupied by red blood cells (RBCs). The test helps diagnose conditions like anaemia or polycythemia.

DIAGRAM



PARTS

- **Capillary tube:** Holds the blood sample
- **Blood sample layers:** Plasma, buffy coat and red blood cells
- **Clay plug:** Seals one end of the tube
- **Centrifuge rotor:** Used to spin the tube
- **Measurement scale:** To determine the percentage of packed cells

APPLICATIONS IN PHYSIOLOGY

- **Hematocrit Measurement:** The primary use of the PCV tube is to measure the hematocrit or packed cell volume, which indicates the proportion of red blood cells in the blood. It's useful for diagnosing anaemia (low PCV) or polycythemia (high PCV).
- **Dehydration Diagnosis:** A high PCV may indicate dehydration, as plasma volume

decreases in such conditions, leaving a higher proportion of RBCs.

- **Blood Doping Detection:** Used in sports medicine to detect blood doping or erythropoietin (EPO) misuse by measuring abnormally high PCV levels.
- **Monitoring Hematologic Disorders:**

Helpful in monitoring patients with conditions like sickle cell disease, where abnormal red blood cell levels can be critical.

LIMITATIONS

- **Small Sample Volume:** The small volume of blood used in a PCV tube limits the ability to perform multiple or more complex tests on the same sample.
- **Crude Measurement:** PCV measures the volume of RBCs, but does not give information about the shape, size or quality of the cells, which may be important in diagnosing certain conditions (like thalassemia).
- **Buffy Coat Overlap:** In some conditions, the buffy coat layer can be difficult to distinguish clearly, potentially affecting the accuracy of the reading.
- **Requires Centrifugation:** The need for a centrifuge means that the test may not be readily performed in remote or resource-limited settings without the right equipment.

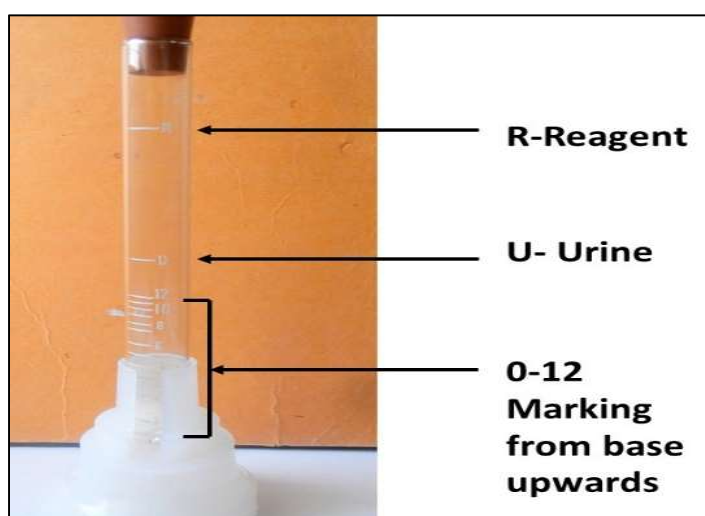


ALBUMINOMETER

PRINCIPLE

The principle of an albuminometer is based on the precipitation of albumin. Albumin is precipitated by adding a reagent like a strong acid, typically sulfosalicylic acid or trichloroacetic acid, to the sample. The precipitated albumin forms visible cloudiness or sediment, which can be measured either qualitatively or quantitatively. The extent of turbidity or the volume of the precipitate is directly proportional to the concentration of albumin in the sample.

DIAGRAM



PARTS

- **Graduated glass tube:** The gradations help in measuring the volume of the albumin precipitate.
- **Sample chamber:** The area inside the tube where the sample (urine or serum) is placed.
- **Reagent chamber:** The part of the tube where the precipitating agent (sulfosalicylic acid, trichloroacetic acid) is added to the sample to form a reaction.
- **Cork or stopper:** Used to seal the tube during the reaction, loss of material
- **Precipitate zone:** After the reaction, this is where the precipitate settles and its volume is measured against the gradations on the tube.

APPLICATIONS IN PHYSIOLOGY

- **Proteinuria Detection:** In clinical settings, albuminometers are primarily used to detect proteinuria (the presence of excess albumin in urine). This is often a sign of kidney dysfunction or disease.
- **Liver Function Assessment:** Measuring albumin in serum helps in assessing liver function because albumin production decreases in liver diseases like cirrhosis.
- **Monitoring of Kidney Diseases:** Quantifying albumin in urine helps monitor the progression of kidney diseases, such as glomerulonephritis and diabetic nephropathy.
- **Diagnosis of Nephrotic Syndrome:** Increased albumin in urine can be a sign of nephrotic syndrome and albuminometers can help in its detection.

LIMITATIONS

- **Sensitivity:** May not detect low albumin levels, limiting early diagnosis.
- **Accuracy:** Results can vary; modern methods are more precise.
- **Time-consuming:** Requires multiple steps, less convenient than automated methods.
- **Interference:** Other substances may cause false positives or inaccuracies.



NADI TARANGINI

PRINCIPLE

The principle behind Nadi Tarangini is to measure the radial pulse at three specific locations on the wrist i.e. Vata, Pitta and Kapha points. These pulse readings are based on the Ayurvedic concept that the body's physiological and pathological states are reflected in the quality of the pulse. The system uses pressure sensors to capture the pulse waveform and the collected data is then analysed through machine learning algorithms to provide insights into a person's health, based on Ayurvedic principles. The system translates the pulse data into digital signals, which are then processed to detect imbalances in the doshas (Vata, Pitta, Kapha), which according to Ayurveda, are responsible for physiological functions in the body.

DIAGRAM



PARTS

- **Pulse Sensor:** Captures pulse data from the wrist.
- **Main Processing Unit:** Processes the captured pulse signals.
- **Digital Display:** Shows the output as graphical waveforms representing the different types of pulses (Vata, Pitta, Kapha) for easy interpretation.

APPLICATIONS IN PHYSIOLOGY

- **Dosha Analysis:** Helps in determining the balance of the three doshas (Vata, Pitta, Kapha) in a person's body, which is key to Ayurvedic diagnosis.
- **Chronic Disease Monitoring:** Nadi Tarangini can track physiological imbalances over time, making it useful in managing chronic diseases, lifestyle disorders and monitoring general well-being.
- **Preventive Healthcare:** By detecting subtle imbalances in the body's energy flow, the device can help in early diagnosis and prevention of diseases before they manifest physically.
- **Personalized Ayurvedic Treatment:** Aids in providing personalized treatment plans according to an individual's unique dosha constitution, thus tailoring lifestyle and dietary recommendations.

LIMITATIONS

- **Reliability of Data:** While the device converts traditional pulse reading into a digital format, the accuracy of the data in clinical settings compared to traditional practitioners may vary.
- **Dependence on Software Interpretation:** The diagnosis relies heavily on algorithms and software interpretation, which may not always capture the nuances of human pulse variability that an experienced Ayurvedic practitioner might feel manually.



PULSE OXIMETER

PRINCIPLE

The pulse oximeter works on the principle of spectrophotometry. It uses light to measure oxygen saturation levels in the blood by emitting two wavelengths of light (typically red and infrared) through a body part, usually, a fingertip or ear lobe. Oxygenated haemoglobin absorbs more infrared light and allows more red light to pass through, whereas deoxygenated haemoglobin absorbs more red light and less infrared light. The oximeter calculates the ratio of absorbed red and infrared light to estimate the percentage of haemoglobin that is saturated with oxygen (SpO_2). The heart rate is also measured from the pulsatile blood flow.

DIAGRAM



PARTS

- **Light Emitting Diode (LED):** One that emits red light and one that emits infrared light.
- **Photodetector:** Receives the light after it passes through the body part.
- **Probe/Clip:** attaches to finger/other body parts.
- **Display Unit:** Readings of SpO_2 and heart rate.
- **Microprocessor:** Processes the signals and computes the oxygen saturation.
- **Power Source:** Typically a battery compartment for portable oximeters

APPLICATIONS IN PHYSIOLOGY

- **Monitoring Oxygen Saturation:** used to monitor SpO_2 levels in patients with respiratory or cardiovascular issues, such as COPD, asthma, pneumonia or heart failure, to ensure sufficient oxygenation.
- **Assessing Hypoxia:** Pulse oximeters can help detect early signs of hypoxia, which is crucial in situations like sleep apnea, altitude sickness or during intensive care.
- **Fitness and Sports Physiology:** Athletes and fitness enthusiasts use pulse oximeters to monitor heart rate and blood oxygen levels during workouts, especially at high altitudes.
- **COVID-19 Monitoring:** Pulse oximeters became widely used during the COVID-19 pandemic to monitor oxygen saturation in patients at risk of developing severe respiratory distress.

LIMITATIONS

- **Accuracy Affected by Movement:** the patient moves during measurement or if the sensor isn't properly attached.
- **Limited Accuracy in Low Perfusion States:** In cases where blood flow to the extremities is poor (e.g., in shock, hypothermia or severe peripheral vasoconstriction), the device may not give accurate readings.
- **Interference from External Factors:** Nail polish, artificial nails, cold extremities or bright external light can interfere with the oximeter's readings.



CENTRIFUGE MACHINE

PRINCIPLE

The centrifuge works on the principle of centrifugal force. When a sample is spun at high speeds, the denser components move outward, away from the axis of rotation, while less dense components stay closer to the center. This separation is proportional to the mass, rotational speed and the radius of the rotor.

DIAGRAM



PARTS

- **Rotor Head:** The part that holds the sample tubes. Different rotors can accommodate different tube sizes.
- **Digital Display/Regulator:** Displays the current speed (RPM) and allows the operator to set the desired speed and time.
- **Motor:** Powers the rotor and ensures the spinning motion. Higher speeds generate more centrifugal force.
- **Tube Holders/Slots:** Designed to hold the tubes at specific angles, allowing for efficient separation of the components.
- **Base/Chassis:** A sturdy foundation that absorbs vibration and ensures the machine stays stable during high-speed operation.

APPLICATION IN PHYSIOLOGY

- **Hematocrit Measurement:** A small centrifuge is used to determine the ratio of red blood cells to plasma in a blood sample, known as hematocrit.
- **Separation of Blood Components:** Used in laboratories to separate blood into plasma, white blood cells (buffy coat) and red blood cells for testing or analysis.
- **Urine Sediment Analysis:** Helps in sedimenting solid particles in urine for microscopic analysis, useful in diagnosing kidney disorders.
- **Cell Fractionation:** Used for separating different cellular components such as nuclei, mitochondria and cytoplasm for biochemical analysis.
- **Protein Isolation:** In biological research, centrifuges are used to isolate proteins from samples for further biochemical studies.

LIMITATIONS

- **Sample Capacity:** Limited number of tubes can be processed at a time, so large sample volumes may require multiple rounds.
- **Heat Generation:** High-speed centrifuges can generate significant heat, which might affect sensitive biological samples. Some models include refrigeration to mitigate this.
- **Risk of Imbalance:** If the rotor is not properly balanced, it can cause damage to the machine or inaccurate results.

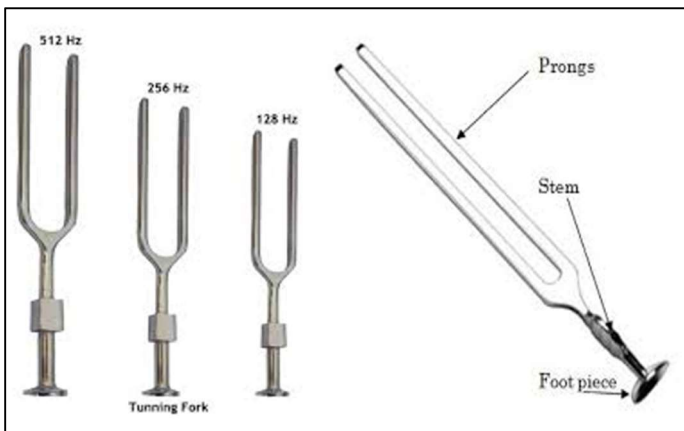


TUNING FORK

PRINCIPLE

A tuning fork operates on the principle of resonance. When struck, its tines (the two prongs) vibrate at a specific frequency, producing a pure tone. This vibration causes surrounding air molecules to compress and rarefy, creating sound waves that propagate outward. The frequency of the sound produced depends on the length, mass and material of the tines. In physiology, tuning forks are often used for vibratory sensation testing and auditory assessments. The pure tone generated helps clinicians assess hearing function or detect sensory deficits.

DIAGRAM



PARTS

- **Tines (Prongs):** The U-shaped arms of the tuning fork that are responsible for producing the sound. When the tines are struck, they vibrate at a set frequency, which is determined by their length and material.
- **Stem (Handle):** The central, non-vibrating part of the tuning fork. It is used for holding the tuning fork. The stem transfers vibrations to the tested area during physiological exams, especially in bone conduction tests.
- **Base of Stem:** The lower end of the stem is sometimes referred to as the base, which is used to rest on hard surfaces or bones during auditory or vibratory sensation tests.

APPLICATIONS IN PHYSIOLOGY

▪ Hearing test

Rinne's Test: Compares air conduction (AC) to bone conduction (BC) of sound. Normally timing AC is greater than BC

Weber's Test: Assesses unilateral hearing loss by placing the vibrating tuning fork at the center of the forehead.

▪ Neurological Examination:

Loss of vibratory sense can be indicative of peripheral neuropathy, spinal cord lesions or other neurological disorders.

LIMITATIONS

- **Subjective Nature:** Tuning fork tests, especially in auditory assessments, rely on the patient's response, making them subjective and prone to error.
- **Limited Frequency Range:** Tuning forks produce a single, fixed frequency (e.g., 128 Hz or 512 Hz), which limits their ability to assess a wide range of auditory or vibratory functions compared to more comprehensive instruments like audiometers.
- **Sensitivity to External Noise:** In a noisy environment, tuning fork assessments, especially hearing tests, can be difficult to perform accurately due to interference from ambient sounds.



NASAL SPECULUM

PRINCIPLE

The nasal speculum works on the principle of mechanical retraction, where it gently separates the nostrils to allow better visualization of the nasal cavity. By applying pressure to the handles, the blades open, expanding the nasal opening for examination, diagnosis or minor procedures. This allows the practitioner to inspect the nasal septum, turbinates and internal structures.

DIAGRAM



PARTS

▪ **Blades:**

The blades are two flat or slightly curved ends that go inside the nostrils. The size of the blades varies according to the patient's age.

▪ **Handles:**

The ergonomic design helps in applying the right amount of pressure without causing discomfort to the patient.

▪ **Hinge (or Pivot Point):**

It allows the opening and closing movement of the blades.

▪ **Spring Mechanism:**

Some nasal speculums come with a spring-loaded mechanism that helps to automatically retract the blades when the handles are released.

APPLICATIONS IN PHYSIOLOGY

- **Nasal Examination:** It is commonly used for inspecting the nasal cavity for blockages, infections, polyps, septum deviation or any foreign bodies.
- **Diagnostic Procedures:** Used in conjunction with other diagnostic tools like endoscopes or otoscopes to examine the internal nasal structures more deeply.
- **Surgical Procedures:** During minor nasal surgeries, such as cauterization of nosebleeds or removal of small polyps, the nasal speculum provides clear access.
- **Physiological Studies:** Useful in studies involving nasal airflow, olfactory assessments and conditions like rhinitis or sinusitis where nasal obstruction is a concern.

LIMITATIONS

- **Limited Field of View:** The nasal speculum provides a restricted view of the nasal passages, limiting visualization of the deeper sinuses or posterior nasal cavity.
- **Inflexibility for Deep Nasal Structures:** While it allows for superficial examination, it cannot provide access to the deeper regions of the nasal cavity like endoscopes can.
- **Risk of Trauma:** If improperly used, it can cause trauma to the sensitive mucosa of the nose, leading to bleeding or injury.

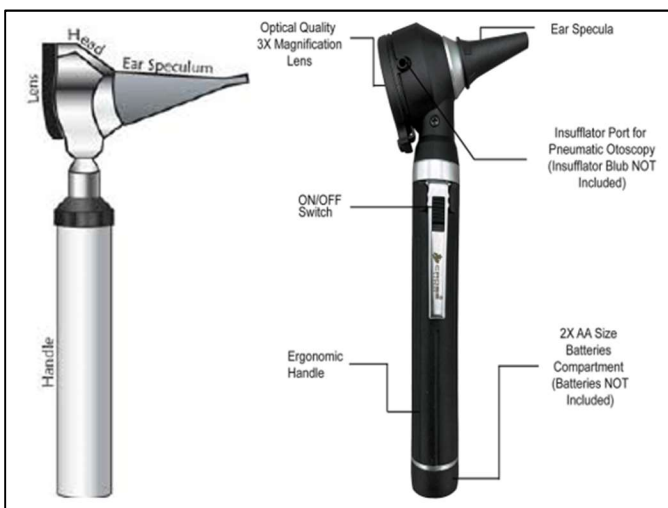


EAR SPECULUM

PRINCIPLE

The principle behind an ear speculum is mechanical dilation, allowing visualization of the ear canal by gently widening the entrance of the external auditory canal. The ear speculum helps guide the otoscope or other diagnostic instruments into the ear canal without causing damage to the ear structures. By increasing the opening, the physician can observe the ear canal and the eardrum and identify infections, obstructions or any other abnormalities.

DIAGRAM



PARTS

- **Cone-shaped Tip:** The narrow part of the ear speculum is the tip that is inserted into the ear canal. It is designed in a conical shape to fit comfortably in the external auditory canal.
- **Base:** The broader part of the speculum is the base, which allows it to be attached to an otoscope.
- **Flanged Edges:** providing a safety stop to prevent the instrument from being inserted too deeply into the ear canal, reducing the risk of injury.
- **Light Conduction:** In combination with an otoscope, the speculum allows light to be conducted through its hollow center, illuminating the ear canal and eardrum for clear visibility.

APPLICATIONS IN PHYSIOLOGY

- **Ear Canal Examination:** The ear speculum is used to widen the external auditory canal, allowing the physician to inspect for infections, earwax buildup, foreign bodies or structural abnormalities.
- **Tympanic Membrane Inspection:** It facilitates the examination of the tympanic membrane (eardrum) to check for perforations, infections or signs of middle ear issues (otitis media).
- **Diagnostic Tool in Otology:** Often used with an otoscope, the ear speculum helps in diagnosing conditions like ear infections, hearing loss, tinnitus and fluid buildup in the middle ear.

LIMITATIONS

- **Limited View of Middle Ear:** The ear speculum only allows visualization of the external ear canal and eardrum. It does not provide access to the middle or inner ear, limiting its use for deeper ear conditions.
- **Discomfort:** If not sized properly, the insertion of the speculum may cause discomfort patients.
- **Inability to Visualize Deep Obstructions:** While the ear speculum allows for viewing the outer part of the ear canal, deep obstructions or issues may not be visible without advanced tools like endoscopes or imaging.
- **Dependent on Light Source:** The ear speculum alone cannot provide visualization; it requires an otoscope or a similar tool to shine light into the ear.



TONOMETER

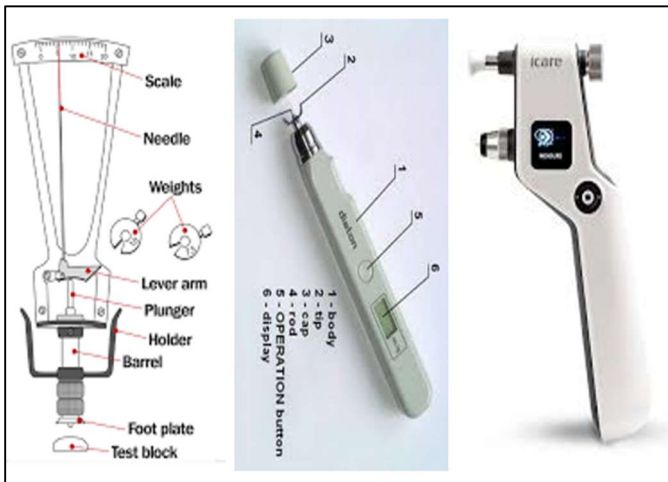
PRINCIPLE

The basic principle of tonometry relies on applanation or indentation of the cornea to estimate intraocular pressure.

Applanation tonometry: The force required to flatten a fixed area of the cornea is measured and this force correlates with intraocular pressure (IOP).

Indentation tonometry: Measures how much the cornea indents under a given force. The amount of indentation is inversely proportional to the IOP.

DIAGRAM



PARTS

- **Tonometer Head/Probe:** The part that comes into contact with the cornea. In applanation tonometers, it is usually a prism that flattens a small part of the cornea.
- **Pressure Sensor:** Found in electronic tonometers, this sensor measures the force required to either flatten or indent the cornea.
- **Dial or Gauge:** The device that displays the intraocular pressure reading. In mechanical tonometers, it may be a dial, while electronic ones have digital displays.
- **Supporting Arm:** Holds the tonometer in place during measurement and allows the operator to adjust the device properly.

APPLICATIONS IN PHYSIOLOGY

- **Intraocular Pressure Measurement:** The primary use of a tonometer is to measure IOP. This is vital in diagnosing and managing conditions like glaucoma, which can lead to blindness if untreated.
- **Glaucoma Screening:** Since elevated IOP is a risk factor for glaucoma, tonometry is routinely used in eye examinations for early detection.
- **Corneal Rigidity Studies:** Tonometers can also help in studies related to corneal biomechanics, as corneal thickness and rigidity affect IOP readings.
- **Post-Surgical Monitoring:** After eye surgeries, such as cataract surgery, monitoring IOP is important to prevent complications.

LIMITATIONS

- **Corneal Thickness Influence:** The accuracy of tonometry readings can be affected by corneal thickness. A thicker cornea can lead to overestimated IOP, while a thinner cornea can result in underestimation.
- **Patient Discomfort:** Applanation and indentation tonometers require direct contact with the eye, which can be uncomfortable for some patients and may require anesthetic drops.
- **Operator Skill Dependency:** Accurate readings with manual tonometers (like Goldmann) depend heavily on the skill and experience of the examiner.



ECG MACHINE

PRINCIPLE

The Electrocardiograph (ECG) machine records the electrical activity of the heart over time. The heart generates electrical impulses that cause it to contract, pumping blood throughout the body. These electrical signals spread through the heart muscle, creating voltage changes, which the ECG machine detects using electrodes placed on the skin. The machine amplifies and records these signals as a waveform, corresponding to the depolarization and repolarization of the heart during each cardiac cycle. Each part of the waveform represents different phases of the heart's activity: the P-wave indicates atrial depolarization, the QRS complex represents ventricular depolarization and the T-wave shows ventricular repolarization.

DIAGRAM



PARTS

- **Electrodes and Leads:** The standard 12-lead system includes 3 standard limb leads, 3 augmented leads and 6 precordial leads to measure electrical potential differences.
- **Galvanometer:** Converts electrical signals into mechanical movement or digital signals for processing.
- **Graph Paper/Screen:** Outputs the ECG trace either on graph paper or a screen, with the horizontal axis representing time and the vertical axis showing the amplitude of electrical activity.

▪ APPLICATIONS IN PHYSIOLOGY

▪ **Diagnosis of Arrhythmias:**

ECG helps in identifying abnormal heart rhythms such as atrial fibrillation, ventricular tachycardia and bradycardia.

▪ **Ischemic Heart Disease:**

Changes in the ST segment and T-wave on an ECG are used to detect myocardial ischemia (reduced blood flow to the heart) and infarction (heart attack).

▪ **Heart Rate Monitoring:**

An ECG provides real-time monitoring of the heart rate and rhythm, which is essential during surgeries, in ICUs or in patients with cardiac issues.

▪ **Electrolyte Imbalance Detection:**

Abnormal levels of potassium, calcium or magnesium affect the heart's electrical activity, which can be seen on the ECG.

▪ **Assessment of Heart Chamber Enlargement:**

ECG can suggest enlargement of the atria or ventricles by characteristic changes in the waveform.

LIMITATIONS

- **Limited Scope:** Records only electrical activity and not mechanical function, missing short events unless continuous monitoring (e.g., Holter) is used.
- **Subject to Interpretation Errors:** The accuracy of ECG interpretation depends on the skill and experience of the physician or technician and errors can lead to misdiagnosis.



DIGITAL SPIROMETER

PRINCIPLE

The Digital Spirometer is a pulmonary function testing device that measures the volume of air inhaled and exhaled by the lungs, as well as the speed (flow) of the breath. It works on the principle of flow sensing and volume displacement.

When a patient breathes into the mouthpiece, the spirometer's sensors detect changes in airflow and volume. These signals are converted into digital data, which is processed by the spirometer's software to generate graphs such as the flow-volume loop and volume-time curve. The digital spirometer uses pneumotachometers or ultrasonic flow sensors to measure airflow and lung volume accurately.

DIAGRAM



PARTS

- **Mouthpiece:** for breathing into the device.
- **Flow Sensor (Pneumotachometer or Ultrasonic Sensor):** Measures the rate of airflow.
- **Digital Display/Screen:** Displays real-time spirometry results, including graphs and data.
- **Control Panel:** Used to input patient data and control the spirometry test.
- **USB/PC Connection (optional):** Connects to a computer for software analysis and data storage.

APPLICATIONS IN PHYSIOLOGY

- **Diagnosis of Respiratory Diseases:** Spirometry assesses the extent of airflow obstruction in COPD, Asthama, Pre-operative Evaluation of patients.
- **Monitoring Lung Health:** Used for routine screening in smokers, individuals exposed to environmental pollutants or patients with chronic respiratory conditions.
- **Occupational Health Screening:** Detects early lung damage in workers exposed to dust, chemicals or respiratory irritants through regular testing.
- **Exercise Testing:** Evaluates lung performance during physical activity, aiding in diagnosing exercise-induced asthma or other respiratory issues.

LIMITATIONS

- **Patient Effort Dependent:** Accurate spirometry results require maximal effort from the patient during testing, especially for forced expiratory maneuvers. Inadequate effort can lead to inaccurate results.
- **Not Suitable for All Patients:** Spirometry may be difficult for very young children, elderly patients or individuals with certain neuromuscular disorders who cannot perform the required breathing maneuvers correctly.
- **Interference from External Factors:** External factors like poor technique, suboptimal environmental conditions (e.g., humidity or temperature) or mechanical faults (e.g., leaks in the mouthpiece) can affect test accuracy.



3. DEMONSTRATE BLOOD COLLECTION

LEARNING OBJECTIVES

After completion of the practical, students should be able to:

- Describe the purpose of collecting a blood sample.
- Indicate how to attain and maintain asepsis when collecting a blood sample.
- Collect capillary blood under guidance of teacher from a finger-prick and precautions to be taken during a skin-prick.
- Indicate the steps for obtaining a blood sample by venepuncture.

PURPOSE

- **Diagnosis of Health Conditions:** Blood samples are commonly used to detect diseases, infections and health conditions. Blood samples can assess the functioning of organs like the liver, kidneys, heart and lungs etc.
- **Checking for Nutritional Deficiencies:** Blood tests can reveal deficiencies in essential nutrients, such as iron, vitamin D or B12.
- **Other Functions:** Assessing Immune Function, Blood testing can serve as a preventive measure to screen for diseases before symptoms appear, such as cancer markers, metabolic conditions or genetic disorders.

ASEPSIS

Asepsis refers to the condition of being free from septic or infectious material - bacteria, viruses, etc.

- **All the instruments to be used for collecting blood:** syringes, needles, lancets and cotton and gauze swabs should preferably be sterilized in an autoclave. Irradiated and sealed, single-use syringes, needles, lancets and blades are now freely available and are in common use.
- **Cleaning/Sterilization of Skin:** At least 2–3 sterile cotton/gauze swabs soaked in 70% alcohol, methylated spirit or ether should be used to clean and scrub the area.

COLLECTION OF VENOUS BLOOD

PRINCIPLE

Venipuncture involves the insertion of a sterile needle into a vein, typically in the arm, to collect blood. The principle is based on creating a negative pressure inside the collection device (syringe or vacuum tube), which facilitates blood flow from the vein into the device. Proper vein selection ensures minimal discomfort and efficient blood collection.

APPARATUS AND MATERIALS

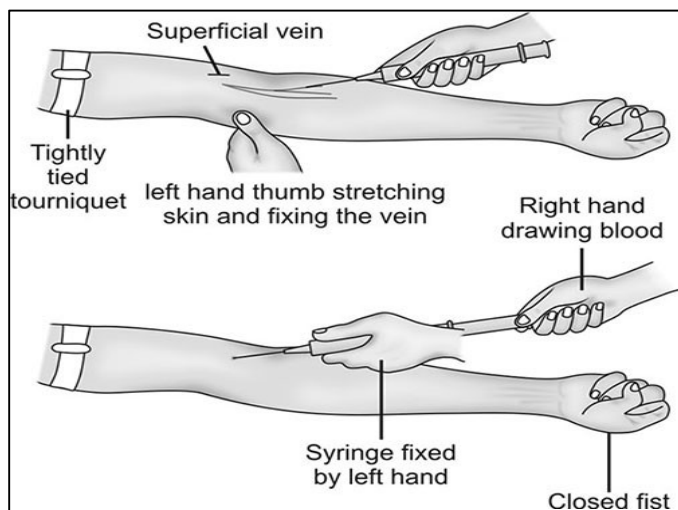
- Disposable gloves.
- Sterile gauze pieces moist with 70% alcohol/methylated spirit.
- Tourniquet.
- Sterile, disposable, one-time use, 5 or 10 ml syringe. Two 22-gauge needles with short bevels.
- 10 ml test tubes or vials, with or without anticoagulant.

PROCEDURE

1. Let the subject seat comfortably on a chair with an arm rest or near a table. Reassure the subject by your approach and conversation.
2. Examine both arms in front of the elbows to locate a suitable vein. Ask your assistant to compress the upper arm with his hands to make the veins prominent.
3. The antecubital vein is usually sufficiently large to take a wide-bore needle.



4. If the vein is neither visible nor palpable, try the other arm. (You should avoid superficial veins because they are notoriously slippery. Veins above the wrist or on the back of the hand may have to be used).
5. Once a suitable vein has been selected, support the subject's arm over the edge of the table.
6. Apply the tourniquet about 2-3 cm above the elbow to obstruct the venous return. The subject may open and close her fist to increase the venous return and make the veins engorged (filled) with blood. If the vein is still not sufficiently prominent, a few 'slaps' with your fingers over the region may do so.
7. Clean the skin over the selected vein with gauze and alcohol and allow it to dry.
8. With the fingers of your left hand supporting and steadying the elbow from behind, stretch the skin over the vein downward with your left thumb placed about 4 cm below the vein. This traction fixes the vein and prevents its slipping when it is punctured.



9. Hold the syringe between your fingers and thumb of the right hand.
10. With the first finger placed near the butt of the needle, puncture the skin and push in the needle under the skin with a firm and smooth thrust.
11. At an angle of 15-20° to the skin. Slightly pull the plunger back with your thumb and little finger

to produce a little negative pressure in the syringe. Advance the needle gently along the vein and puncture it from the side, a few mm ahead of the skin puncture.

12. As the vein is punctured, all resistance will suddenly cease and blood will start to enter the syringe. With the needle still in the vein and supporting the syringe with your left hand, gently pull the plunger back with the thumb and fingers of your right hand.
13. Do not withdraw blood faster than the punctured veins which are filling too much pressure applied to the plunger is likely to cause mechanical injury and haemolysis of red cells. The subject may open and close the fist to enhance venous return.
14. When enough blood has been collected, release the tourniquet and press a fresh swab over the skin puncture. Withdraw the needle gently but keep the swab in position.
15. Ask the subject to flex the arm and keep it so to maintain pressure on the puncture site till the bleeding stops.
16. Expel the blood gently (by removing needle) into the container; do not apply force as it may cause mechanical injury to red cells. Gently shake or swirl the container between your palms so that the anticoagulant (if used) mixes well with the blood without frothing.

PRECAUTIONS

- All aseptic precautions must be observed and disposable gloves, syringes and needles must be used.
- The tourniquet must be removed before taking the needle out of the vein to avoid the formation of a hematoma.
- The blood from the syringe should be transferred to the container without delay to prevent clotting.
- Ask the subject to keep the swab in position till the bleeding from the puncture site stops.

COLLECTION OF CAPILLARY BLOOD

PRINCIPLE

Capillary blood collection involves obtaining a small blood sample from capillaries by puncturing the skin, usually on a fingertip, earlobe or heel (in infants). The principle is based on stimulating blood flow from the superficial capillaries through a small incision or puncture, enabling collection using micro collection tubes or glass slides. This method is ideal for small sample volumes and point-of-care testing.

APPARATUS AND MATERIALS

1. Blood Lancet/Pricking needle.
2. Sterile gauze/cotton, moist with 70% alcohol/methylated spirit.
3. Glass slides, pipettes, etc. according to requirements.

Selection of Site for Skin Prick

- In adults and older children, capillary blood is generally obtained from a skin puncture made on the tip of the ring finger or middle or on the lobe of the ear.
- In infants and young children in whom the fingers are too small for a prick, the medial or lateral side of the pad of the big toe or heel is used.

Important

The thumb and little finger are never pricked because the underlying palmar fasciae (venous bursae) from these digits are continuous with those of the forearms. Any accidental injury to these fasciae may cause the infection to spread into the forearm.

PROCEDURE

1. All aseptic precautions must be taken.
2. Clean and vigorously rub the ball of the finger with the spirit swab, followed by a final cleaning

with dry gauze. (Scrubbing increases local blood flow).

3. Allow the alcohol to dry by evaporation.
4. Steadying the finger to be pricked in your left hand, apply a gentle pressure on the sides of the ball of the finger with your thumb and forefinger to raise a thick, broad ridge of skin. (Do not touch the pricking area).
5. Hold the lancet between the thumb and fingers of your right hand. prick the skin with a sharp and quick vertical stab to a depth of 3–4 mm and release the pressure.
6. The blood should start to flow slowly, spontaneously and freely (without any squeezing) - if a good prick has been given.
7. Wipe away the first 2 drops of blood with dry, sterile gauze as it may be contaminated not only with tissue fluid, but also with epithelial and endothelial cells which will appear as artifacts in the blood film.
8. Allow a fresh drop of blood of sufficiently large size (about 3–4 mm diameter) to well up from the wound and make a blood smear or fill a pipette as the case may be.
9. Clean the area of the prick with a fresh swab and ask the subject to keep the swab pressed on the wound with his/her thumb till the bleeding stops, which occurs in a minute or so.

PRECAUTIONS

- The selected site should be clean, free from infection, edema or skin disease.
- The site should be vigorously cleaned and Scrubbing increases local blood flow.
- The puncture should be deep enough to give free-flowing blood but not so very deep that it takes inordinately long time for the bleeding to stop.
- Do not press or squeeze the finger to increase the blood flow from the skin-prick, though the arm or the hand may be milked towards the fingers.

EXERCISE

Q.1 What measures are taken to prevent infection during venepuncture and skin-pricking?

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Q.2 What are the main differences between venous blood and capillary blood?

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Q.3 What is the difference between plasma and serum?

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Q.3 Why is patient consent important before collecting blood?

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OSPE-1

OSPE Station: Venous Blood Collection and Aseptic Technique

Learning Objectives

1. Demonstrate understanding of aseptic techniques.
2. Identify and prepare the necessary materials and apparatus for blood collection.
3. Perform venous blood collection following proper procedural and safety guidelines.
4. Demonstrate adherence to precautions to prevent contamination and ensure patient safety.

Station Setup

- Displayed items: Disposable gloves, sterile gauze with spirit, tourniquet, sterile syringes, needles, test tubes/vials.
- Patient (simulated or mannequin) seated with arm exposed for blood collection.

Tasks

Each student will perform and/or explain the following steps in 15 minutes. Each points having 1 or 2 marks each.

Checklist

Preparation of Equipment and Knowledge of Aseptic Technique (3 points):

- Identify and confirm the necessary equipment (disposable gloves, alcohol swabs, sterile syringe, needles and test tube). (Y/N)
- Explain how to ensure aseptic conditions by using sterile, single-use equipment. (Y/N)
- Demonstrate the process of sterilizing the skin with a spirit swab before venipuncture. (Y/N)

Patient Comfort and Vein Selection (3 points):

- Describe how to make the patient comfortable and reassured (communication approach). (Y/N)
- Identify and palpate for a suitable vein in the antecubital area. (Y/N)
- Explain what actions to take if the vein is not immediately visible (using a tourniquet, other arm or gently tapping). (Y/N)

Venipuncture Technique (4 points):

- Apply the tourniquet above the elbow, instruct the patient to open/close their fist and select a prominent vein. (Y/N)
- Demonstrate the proper technique to clean the selected site with sterile gauze soaked in spirit and allow it to dry. (Y/N)
- Position the needle and demonstrate (or simulate) the venipuncture angle (15-20°) and gentle insertion technique to avoid haemolysis. (Y/N)
- Explain or show how to draw blood slowly and steadily. (Y/N)

Post-Collection Protocol (3 points):

- Release the tourniquet, apply pressure with a sterile swab as the needle is removed and instruct the patient to keep their arm flexed. (Y/N)
- Describe how to transfer blood into the test tube carefully, to prevent cell damage. (Y/N)
- Demonstrate gentle mixing if anticoagulant is present. (Y/N)

Precautions and Safety (2 points):

- Removal of the tourniquet before withdrawing the needle. (Y/N)
- Ensuring blood is transferred to the container immediately to avoid clotting. (Y/N)

ROLE PLAY ACTIVITY**Title:** Phlebotomy Procedure Role-Play

Objective: To practice and demonstrate effective phlebotomy techniques and communication skills in blood sample collection.

Roles: **Phlebotomist** (Teacher/Lab Technician)

Patient (Individual)

Scenario: The patient arrives at the lab for a routine blood test.

Script: Phlebotomist:

1. Greets the patient and verifies their identity.
2. Explains the procedure to ensure patient understanding.
3. Asks the patient to roll up their sleeve and relax their arm.
4. Prepares necessary equipment (gloves, tourniquet, needle, tubes).
5. Locates a vein and cleans the area with antiseptic.
6. Applies the tourniquet, inserts the needle and collects samples.
7. Removes the needle and applies pressure to the puncture site.

Patient:

8. Responds to questions and follows instructions.
9. May ask questions or show mild anxiety or discomfort (optional).

Key Points:

Patient identification and verification. Clear procedure explanation and consent. Proper hand hygiene and glove use. Sterile technique and accurate equipment handling. Communication for patient comfort. Accurate sample labeling and handling.

Debriefing Questions:

1. What were the key steps in the phlebotomy procedure?
2. How did the phlebotomist ensure patient comfort?
3. How might you handle an anxious patient?

PROBLEM-BASED LEARNING

Case Study:

A phlebotomist, John, is tasked with collecting a blood sample from a 65-year-old patient, Mrs. Smith, for a complete blood count (CBC). Upon reviewing Mrs. Smith's medical history, John notes that she has a history of difficult venipuncture due to scarred veins from previous surgeries and blood draws.

Problem:

As John attempts to collect the blood sample from Mrs. Smith's median cubital vein, he encounters difficulty locating a suitable vein. After three unsuccessful attempts, Mrs. Smith begins to show signs of distress and anxiety.

Guiding Questions:

1. What alternative veins could John consider for blood collection in this scenario?
2. What measures could John take to improve his chances of successful venipuncture on the next attempt?
3. How would you prioritize Mrs. Smith's comfort and well-being while ensuring the collection of the required blood sample?

Discussion and Resolution:

In this session, students work together to identify alternative approaches for blood collection, discuss optimal techniques for difficult venipuncture cases and brainstorm effective ways to prioritize patient comfort. Students present their responses, exploring both practical solutions and empathetic approaches to ensure Mrs. Smith's well-being.

REFERENCES

- A textbook of practical physiology, CL. Ghai, 8th edition.
- Essentials of medical physiology, K. Sembulingam and P. Sembulingam, 6th edition.



Teacher's Signature

4. ESTIMATE HAEMOGLOBIN

LEARNING OBJECTIVES

After completion of the practical, students should be able to:

- Indicate the relevance of this estimation.
- Determine the Hb level by Sahli's acid hematin method.
- Name other methods of estimation of Hb.
- Indicate normal levels of Hb in different age groups and sexes.

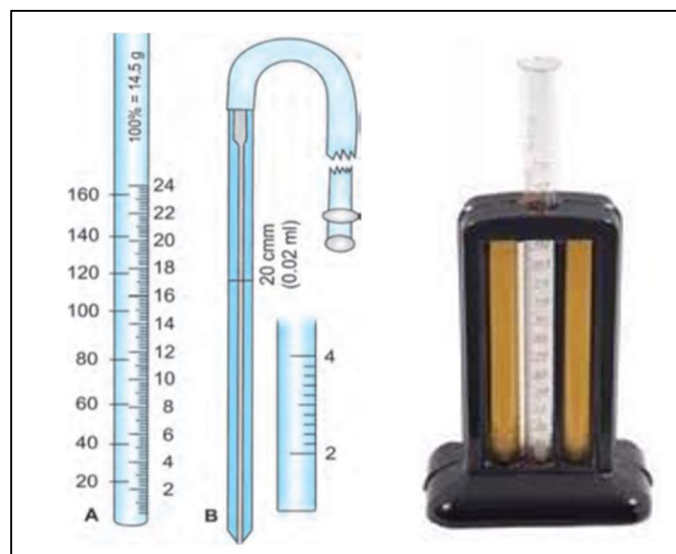
PRINCIPLE

The Hb present in a measured amount of blood is converted by dilute hydrochloric acid into acid hematin, which in dilution is golden brown colour. The intensity of colour depends on the concentration of acid hematin which, in turn, depends on the concentration of Hb. The colour of the solution (i.e. its hue and depth), after dilution with water, is matched against golden-brown tinted glass rods by direct vision. The readings are obtained in g%.

APPARATUS AND MATERIALS

1. Sahli (Sahli-Adams) Haemoglobinometer (Haemometer). The set consists of:
 - a) Comparator
 - b) Haemoglobin tube
 - c) Haemoglobin pipette
 - d) Stirrer
 - e) Pasteur pipette
 - f) Distilled water
2. Decinormal (N/10) hydrochloric acid (0.1 N HCl) solution.
3. Materials for skin prick.
 - Sterile lancet/needle.
 - Sterile gauze and cotton swabs.
 - Methylated spirit/70% alcohol.

DIAGRAM



(A) Sahli-Adam's haemoglobin tube: It has graduations in g% on one side and in percentage on the other. In this tube, 100% is equal to 14.5 g Hb per 100 ml blood. (B) Haemoglobin pipette: It has only one marking, indicating 20 cmm (0.02 ml or 20 μ l). There is no bulb (Inset: Each division represents 0.2 g of Hb). (source- A Textbook of Practical Physiology, CL Ghai)

PROCEDURES

1. Using a dropper, place 8–10 drops of N/10 HCl in the Hb tube or up to the mark 20%.
2. Get a finger prick under aseptic conditions, wipe away the first 2 drops of blood. When a large drop of free-flowing blood has formed again, draw blood up to the 20 cmm mark (0.02 ml).
3. Carefully wipe the blood sticking to the tip of the pipette with a cotton swab.
4. Immerse the tip of the pipette to the bottom of the acid solution and expel the blood gently. Rinse the pipette 3–4 times by drawing up and blowing out the clear upper part of the acid solution till all the blood has been washed out from it.
5. Withdraw the pipette from the tube, touching it to the side of the tube, thus ensuring that no mixture is carried out of the tube.
6. Mix the blood with the acid solution with the flat end of the stirrer by rotating and gently moving it up and down.

7. Put the Hb tube back in the comparator and let it stand for 6–8 minutes (or as advised by the manufacturer). During this time, the acid ruptures the red cells, releasing their Hb into the solution (haemolysis). The acid acts on the Hb and converts it into acid hematin which is deep golden brown colour.

Important

The colour of acid hematin does not develop fully immediately, but its intensity increases with time, reaching a maximum, after which it starts to decrease. An adequate time, usually 6–8 minutes, must be allowed before its dilution is started. Too little time and all Hb may not be converted into acid hematin. And, waiting too long, may result in fading of colour. In either case, the result will be falsely low.

8. Diluting and matching the colour

The next step is to dilute the acid hematin solution with distilled water (preferably buffered water, if available) till its colour matches the colour of the standard tinted glass rods in the comparator.

9. Take the Hb tube out of the comparator and add distilled water drop by drop (or larger amounts depending on the experience), stirring the mixture each time and comparing the colour with the standard.

10. Hold the comparator at eye level, away from your face, against bright but diffused light. Read the lower meniscus (lower meniscus is read in coloured transparent solutions).

OBSERVATIONS AND RESULTS

Compare your colour matching with that of your work-partner and record the observations in your workbook. Take the average of 3 readings as shown below and report your result as: Hb =g/dl.

1st reading, when the colour is slightly darker than the standard:.....g/dl.

2nd reading, when, after adding a few drops of distilled water, the colour exactly matches the standard: g/dl.

3rd reading, when, after adding some more drops, the colour becomes a little lighter than the standard:..... g/dl.

Result: Hb=g/dl.

Normal Values

Males: 14.5 g/dl (13.5–18 g/dl).

Females: 12.5 g/dl (11.5–16 g/dl).

Advantages of Sahli's Method

1. The method is simple, fairly quick and accurate.
2. It does not require any costly apparatus, since it needs only direct colour matching.
3. Its running cost is minimal and can, therefore, be used in mass surveys.

Disadvantages of Sahli's Method

1. Since the acid hematin is not in true solution, some turbidity may occur.
2. The method estimates only the oxyHb and reduced Hb, other forms, such as carboxyHb and metHb are not estimated.
3. Also, the degree of error may be high if proper precautions are not taken.

PRECAUTIONS

1. The finger should not be squeezed and there should not be any blood sticking to the outside of the pipette tip.
2. Only the recommended time should be allowed for the formation of acid hematin by the action of acid on Hb.
3. When diluting the colour of acid hematin solution, avoid over-dilution because the colour cannot be concentrated.

4. When matching the colour, the solution should be uniformly golden brown throughout the solution. The dark colour near the bottom of the tube indicates poor mixing.
5. During colour matching, 3 readings should be taken to reduce personal error.

EXERCISE

Q.1 What is the principle on which the Sahli method is based?

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Q.2 Why is Sahli's method most frequently employed as a routine test?

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Q.3 What is meant by the term 'normality' of a solution? What is N/10 HCl and how will you prepare it?

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Q.4 How would the estimation of Hb be affected if less or more than 8–10 drops of N/10 HCl is taken in the Hb tube?

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Q.5 Can tap water be used for diluting and colour matching?

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Q.6 Why is it necessary to wait for 6–8 minutes after adding blood to N/10 HCl solution?

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Q.7 What are the alternative methods for performing Estimate Haemoglobin Practicals?

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Q.8 What are the normal levels and ranges of Hb at different ages?

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Q.9 Compared to males, why are the haemoglobin levels lower in females?

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Q.10 What is the fate of haemoglobin?

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Q.11 Why is the haemoglobin level high in the newborns?

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Q.12 What are the functions of haemoglobin?

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Q.13 What is anaemia and how is it graded according to Hb concentration? What are its common causes in India?

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OSPE Station: Hemoglobin Estimation using Sahli's Acid Hematin Method

Learning Objectives

1. Indicate the relevance of hemoglobin estimation.
2. Determine the hemoglobin level by Sahli's acid hematin method.
3. Identify other methods of hemoglobin estimation.
4. State normal hemoglobin levels across different age groups and sexes.

Station Setup

Displayed items: Sahli Haemoglobinometer set (comparator, Hb tube, Hb pipette, stirrer), N/10 HCl solution, distilled water, sterile lancet/needle, sterile gauze and cotton swabs, methylated spirit/70% alcohol and waste disposal container.

Patient Setup: Simulated patient or mannequin for finger-prick demonstration.

Tasks

Each student will perform and/or explain the following steps within 15 minutes. Each point carries 1 or 2 marks.

Checklist

Preparation of Equipment and Knowledge of Principle (4 points):

- Identify and confirm the necessary equipment (Hb pipette, Hb tube, stirrer, N/10 HCl solution, distilled water). (Y/N)
- Explain the principle of the Sahli's method (conversion of Hb to acid hematin and its colour matching). (Y/N)
- Demonstrate proper handling and calibration of the Sahli Haemoglobinometer. (Y/N)
- Explain the significance of maintaining aseptic conditions during the procedure. (Y/N)

Finger Prick and Blood Collection (2 points):

- Perform finger prick under aseptic conditions (clean the site, use a sterile lancet and wipe the first two drops). (Y/N)
- Draw blood accurately up to the 20 cmm mark using the Hb pipette. (Y/N)

Preparation of Acid Hematin Solution (2 points):

- Add & mix 8–10 drops of N/10 HCl solution into the Hb tube up to the 20% mark. (Y/N)
- Allow the solution to stand for 6–8 minutes for acid hematin formation. (Y/N)

Colour Matching and Dilution (3 points):

- Dilute the acid hematin solution with distilled water drop by drop until the colour matches the standard. (Y/N)
- Hold the comparator at eye level against bright, diffused light for accurate matching. (Y/N)
- Take three readings and calculate the average Hb value. (Y/N)

Observation, Results and Precautions (2 points):

- Record the Hb value in g/dl accurately, ensuring it is within the normal range. State normal Hb ranges for males and females. (Y/N)
- List at least two precautions to minimize error during the procedure (e.g., avoid over-dilution, ensure proper mixing). (Y/N)

Additional Knowledge (2 points):

- Mention the advantages of the Sahli method (e.g., simplicity, cost-effectiveness). (Y/N)
- Name at least two other methods of Hb estimation (e.g., cyanmethemoglobin method, automated analyzers). (Y/N)

PROBLEM-BASED LEARNING

Case Study:

A 30-year-old female patient, Rohini, presents to the outpatient department with symptoms of fatigue, weakness and shortness of breath. Her medical history reveals:

- Menstrual irregularities.
- Dietary deficiencies.

Laboratory Results:

- Haemoglobin (Hb) estimation: 8 g/dL (using Sahli's method).
- Red Blood Cell (RBC) count: 3.5 million/Ml.
- Mean Corpuscular Volume (MCV): 70 fL.
- Mean Corpuscular Haemoglobin (MCH): 22 pg

Problem:

The physician suspects anaemia and requests confirmation of the haemoglobin level. However, the laboratory technician noticed:

- The haemoglobin sample was stored at room temperature for 2 hours before analysis.
- The patient's sample showed marked lipemia.

Questions:

1. How would you evaluate the reliability of the initial haemoglobin result?
2. What factors could have affected the haemoglobin estimation?
3. What corrective measures would you take for accurate haemoglobin estimation?
4. Interpret the patient's laboratory results in the context of her clinical presentation.

Learning Objectives:

1. Understand principles of haemoglobin estimation methods (Sahli's, cyanmethaemoglobin).
2. Recognize factors affecting a measurement.

3. Apply quality control measures.
4. Interpret haemoglobin results in clinical context.

Guided Discussion Questions:

1. What are the advantages and limitations of Sahli's method?
2. How does lipemia affect haemoglobin estimation?
3. What are the implications of delayed sample analysis?
4. How would you differentiate between iron-deficiency anaemia and other types?

Expected Outcomes:

1. Critical evaluation of laboratory results.
2. Understanding of factors affecting haemoglobin measurement.
3. Accurate interpretation of haemoglobin results.

Assessment:

1. A written report on the reliability of the initial haemoglobin result.
2. Group presentation on haemoglobin estimation methods.
3. Class discussion participation.

REFERENCES

- A textbook of practical physiology, CL. Ghai, 8th edition.
- Essentials of medical physiology, K. Sembulingam and P. Sembulingam, 6th edition.



Teacher's Signature

5. ESTIMATE BLEEDING TIME & CLOTTING TIME

LEARNING OBJECTIVES

After completing the practical, students should be able to:

- Indicate the clinical importance of performing BT/CT.
- Define haemostasis and describe how bleeding stops from a finger prick.
- Determine BT and CT by the routine laboratory methods and give their normal values.
- Name the conditions in which BT and CT are increased.
- Describe the physiological basis of bleeding disorders.

BLEEDING TIME (BT) & CLOTTING TIME (CT)

Bleeding Time (BT) is the time interval between the skin puncture and spontaneous, unassisted (i.e. without pressure) stoppage of bleeding. The BT test is an *in vitro* test of platelet function.

Clotting time (CT) is the time interval between the entry of blood into a glass capillary tube or a syringe and the formation of fibrin threads.

BLEEDING TIME (BT)

Bleeding time by Duke's finger-tip method

PRINCIPLE

Bleeding time (BT) measures the duration required for bleeding to stop after a standardized skin puncture. It evaluates primary hemostasis, focusing on platelet adhesion, aggregation and vascular integrity in forming a hemostatic plug. BT provides insight into platelet function and vessel health but does not assess coagulation factor activity, making it a test of primary clot formation.

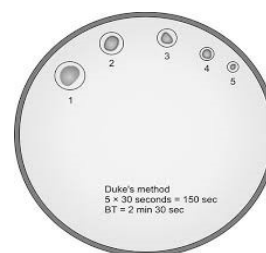
MATERIALS FOR BTCT

- Equipment for sterile finger-prick.
- Clean filter papers.
- Stopwatch.
- Capillary tubes

PROCEDURES

1. Obtain a sufficiently deep (2–3 mm) finger-prick under aseptic conditions to allow free-flowing blood. Start the stopwatch and note the time.
2. Absorb/remove the blood drops every 30 seconds by touching the puncture site with the filter paper along its edges, without pressing or squeezing the wound.
3. Note the time when the bleeding stops, i.e. when there is no trace of blood spot on the filter paper. Encircle this spot and number it as well. This is the endpoint. (Do not keep the filter paper on the table and then press your wound on it).
4. Count the number of blood spots and express your result in minutes and seconds.

Normal bleeding time = 1–5 minutes.



OBSERVATIONS AND RESULTS

- Record the observations in your workbook. Take the average of 3 readings as shown below and report your result as: Bleeding Time = Min.
1st reading, Bleeding Time = Min.
2nd reading, Bleeding Time = Min.
3rd reading, Bleeding Time = Min.
- **Result:** Bleeding Time = Min.

PRECAUTIONS

- The skin site chosen for BT should be scrubbed well with alcohol to increase the blood flow.
- The skin should be dry and the puncture should be 3–4 mm deep to give free-flowing blood. Do not squeeze.
- Do not press the filter paper on the puncture site.

CLOTTING TIME (CT)

Capillary blood clotting time by Wright's capillary glass tube method

PRINCIPLE

The principle is based on evaluating the activation of coagulation factors through intrinsic and common pathways, leading to fibrin formation. It assesses the efficiency of the coagulation cascade and helps identify disorders affecting clotting factor activity, such as hemophilia or anticoagulant use.

PROCEDURES

- Absorb the first 2 drops of blood on a separate filter paper/glass slide and allow a large drop to form.
- Now dip one end of the capillary tube in the blood; the blood rises into the tube by capillary action. This can be enhanced by keeping its open end at a lower level. (or fill the capillary directly with finger-tip blood drops)
- Record the time when blood starts to enter the tube. This is the zero time.
- Hold the capillary tube between the palms of your hands to keep the blood near body temperature (in winter, you may blow on it).
- Gently break off 1 cm bits of glass tube from one end, at intervals of 30 seconds and look for the formation of fibrin threads between the broken ends. The end-point is reached when fibrin threads span a gap of 5 mm between the broken ends ("rope formation"). Note the time.

Normal clotting time = 3–6 minutes.

OBSERVATIONS AND RESULTS

- Record the observations in your workbook. Take the average of 3 readings as shown below and report your result as: Clotting Time = Min.
1st reading, Clotting Time = Min.
2nd reading, Clotting Time = Min.
3rd reading, Clotting Time = Min.
- **Result:** Clotting Time = Min.

EXERCISE

Q.1 Name the conditions in which only the bleeding time is prolonged while the clotting time is normal.

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Q.2 What are the factors on which BT and CT depend?

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Q.3 How does BT differ from CT? What is the interrelation between them and which aspects of hemostasis are tested by them?

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Q.4 How do platelet function and vascular integrity influence bleeding time (BT)?

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Q.5 What are clinical importance of doing BT & CT?

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Q.6 What is meant by the term clotting of blood? Name the various clotting factors, their sources and their role in clotting.

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Q.9 What are the alternative methods for performing Bleeding Time (BT) and Clotting Time (CT) practicals?

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Q.8 Name the conditions where the clotting time is increased and those where it is decreased.

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OSPE-1

Aim: Determine the bleeding time of a patient by the Dukes method.

Checklist:

1. Did the candidate explain the procedure correctly? (Y/N)
2. Was the patient's finger positioned correctly? (Y/N)
3. Was the puncture made correctly? (Y/N)
4. Were filter papers applied correctly? (Y/N)
5. Was the bleeding time recorded accurately? (Y/N)

OSPE-2

Aim: Determine the clotting time of a patient (Capillary Method).

Checklist:

1. Did the candidate explain the procedure correctly? (Y/N)
2. Was the patient's finger positioned correctly? (Y/N)
3. Was the puncture made correctly? (Y/N)
4. Were capillary tubes filled and sealed correctly? (Y/N)
5. Was clotting time recorded accurately?

PROBLEM-BASED LEARNING

Case Study :

A 45-year-old male patient, Mr. Kumar, presents to the emergency department with: Severe epistaxis (nosebleed), Easy bruising, History of hypertension.

Laboratory Results: Hemoglobin (Hb): 10 g/dL, Platelet count: 150,000/ μ L, Prothrombin Time (PT): 14 sec. (normal: 11-16 sec)

Problem:

Estimate Mr. Kumar's Clotting Time (CT) and Bleeding Time (BT) to assess his coagulation status.

Guiding Questions:

1. What is the significance of estimating CT and BT in Mr. Kumar's case?
2. Describe the capillary method for estimating CT.
3. How would you perform BT estimation using Duke's method?
4. Interpret Mr. Kumar's laboratory results concerning his CT and BT.

Guided Discussion Questions:

1. What are the differences between CT and BT? (Discuss their roles in evaluating platelet function, coagulation pathways and vascular integrity.)
2. How do anticoagulants affect CT and BT? (Explore the mechanisms by which anticoagulants prolong CT but typically do not affect BT.)
3. What are the clinical implications of prolonged CT or BT? (Link prolonged times to disorders like hemophilia, von Willebrand disease or platelet dysfunction.)
4. How would you manage Mr. Kumar's epistaxis considering his coagulation status? (Discuss

immediate interventions (e.g., nasal packing, tranexamic acid) and long-term management (e.g., addressing hypertension)

Expected Outcomes:

1. Accurate estimation of CT and BT.
2. Understanding of coagulation disorders.
3. Ability to interpret laboratory results.
4. Critical thinking and problem-solving skills.

Discussion and Resolution

CT and BT Interpretation: Normal platelet count with epistaxis and bruising may suggest a vascular or mild platelet function issue rather than a significant coagulation factor defect.

Management: Addressing hypertension, using local hemostatic measures and monitoring for recurrent bleeding episodes are essential.

Assessment:

- Written report on CT and BT estimation.
- Group presentation on coagulation disorders.
- Class discussion participation.

REFERENCES

- A textbook of practical physiology, CL. Ghai, 8th edition.
- Essentials of medical physiology, K. Sembulingam and P. Sembulingam, 6th edition.



Teacher's Signature

6. ESTIMATE BLOOD GROUPING

LEARNING OBJECTIVES

After completion of the practical, students should be able to:

- Understand the principle behind blood group determination.
- Learn the procedure for blood group testing using antisera and saline.
- Identify different blood groups based on agglutination reactions.
- Explore the significance of agglutination and its clinical relevance in transfusion medicine.

PRINCIPLE

The surfaces of red cell membrane contain a variety of genetically determined antigens, called isoantigens or agglutinogens, while the plasma contains antibodies (agglutinins). To determine the blood group of a person, his/her red cells are made to react with commercially available antisera containing known agglutinins. The slide is then examined under the microscope to detect the presence or absence of clumping and hemolysis (agglutination) of red cells which occurs as a result of antigen-antibody reaction.

PURPOSE

To determine an individual's blood group using anti-A, anti-B and anti-D (Rh) sera by observing agglutination or hemolysis of red cells.

APPARATUS AND MATERIALS

1. Microscope
2. Sterile blood lancet or needle
3. Sterile cotton/gauze swabs
4. Clean, dry microscope slides
5. Anti-A serum: [contains monoclonal anti-A antibodies (against human); these antibodies are also called anti-A or alpha (α) agglutinins]

6. Anti-B serum: [contains monoclonal anti-B antibodies (against human); these antibodies are also called anti-B or beta (β) agglutinins]
7. Anti-D (anti-Rh) serum: [Contains monoclonal anti-Rh (D) antibodies (against human). These antibodies are also called anti-D agglutinins.
8. Normal saline.

PROCEDURES

1. Prepare 3 slides and divide each into two halves using a glass-marking pencil.
2. Label the slides: anti-A, anti-B, anti-D on the left halves and control on the right halves.
3. Add 8-10 drops of normal saline to the center of a 4th slide (marked 'S').
4. Obtain a finger-prick sample under aseptic conditions and add 2 drops of blood to the saline on slide 'S'.
5. Mix the blood with saline to create a red cell suspension.
6. Add one drop of anti-A serum on the first slide, anti-B serum on the second and anti-D serum on the third slide.
7. Place a drop of normal saline on the control sides of each slide.
8. Add a drop of the red cell suspension to each side (both test and control).
9. Wait for 8–10 minutes, then inspect the 3 antisera-red cell mixtures ("test" mixtures) and "control" mixtures, first with the naked eye to see whether agglutination (clumping and hemolysis of red cells) has taken place or not. Then confirm under low magnification microscope.

Preparation of red cell suspension.

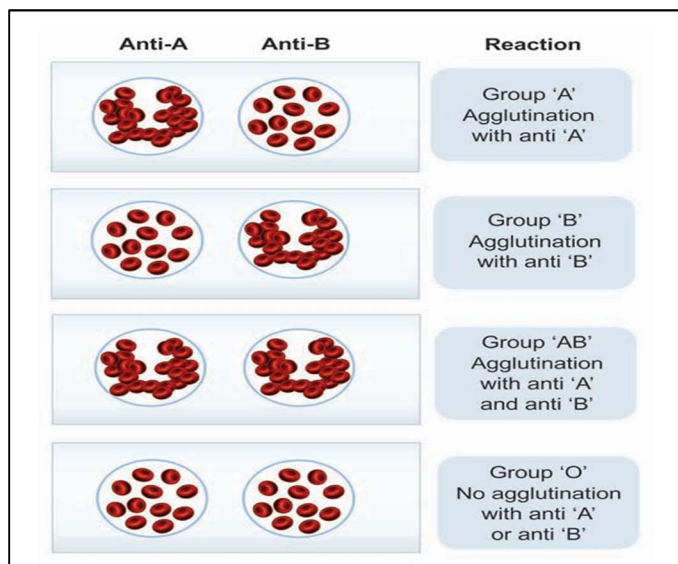
A suspension of red cells in saline should preferably be prepared and used instead of adding blood drops directly from the fingerpick to the antisera for the following reasons; Dilution of blood permits easy

detection of agglutination and hemolysis, if present. (Red cells in undiluted blood tend to form large rouleaux and masses. These may be difficult to disperse and may be mistaken for agglutination). Plasma factors likely to interfere with agglutination are eliminated.

OBSERVATIONS

1. If agglutination occurs with antiserum A: The anti serum A contains α -antibody. The agglutination occurs if the RBC contains A antigen. So, the blood group is A.
2. If agglutination occurs with antiserum B: The anti serum B contains β -antibody. The agglutination occurs if the RBC contains B antigen. So, the blood group is B.
3. If agglutination occurs with both antisera A and B: The RBC contains both A and B antigens to cause agglutination. And, the blood group is AB.
4. If agglutination does not occur either with antiserum A or antiserum B: The agglutination does not occur because RBC does not contain any antigen. The blood group is O.
5. If agglutination occurs with the anti-D serum: The individual is Rh positive (+ve). If no agglutination occurs: The individual is Rh negative (-ve).

RESULTS



Important

The red cells-saline mixture on the “control” sides of each slide will act as a control to confirm agglutination or no agglutination on the corresponding test side.

ABO BLOOD GROUPING INTERPRETATION			
ANTI SERA TYPE			BLOOD TYPE
Anti-A	Anti-B	Anti-D	
			O+ (O Positive)
			O- (O Negative)
			A+ (A Positive)
			A- (A Negative)
			B+ (B Positive)
			B- (B Negative)
			AB+ (AB Positive)
			AB- (AB Negative)

Blood Type Compatibility

		Donor Blood Type							
		O-	O+	B-	B+	A-	A+	AB-	AB+
Recipient Blood Type	AB+								
	AB-								
	A+								
	A-								
	B+								
	B-								
	O+								
	O-								

- **Result:** The given blood sample belongs to the blood group.

PRECAUTIONS

- Ensure aseptic conditions while collecting the blood sample.
- Allow enough time (8–10 minutes) for the agglutination reaction to take place.
- Use separate lancets or needles for each sample to avoid contamination.

EXERCISE

Q.1 What is a blood group system? What is the physiological and genetic basis of blood grouping?

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Q.2 What is Landsteiner Law? How does it apply to all blood groups?

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Q.3 What is cross matching?

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Q.4 Why should you wait for 8–10 minutes before checking for agglutination?

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Q.5 What is agglutination and what is its mechanism?

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Q.6 What is meant by the terms universal donor and universal recipient?

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Q.7 What is Rh factor and what is its clinical significance?

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Q.8 If an Rh –ve mother carries Rh +ve fetus, what are the complications that are likely to occur?

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Q.9 What does agglutination with anti-A serum indicate about the patient's blood group?

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OSPE-1

OSPE Station: Venous Blood Collection and Aseptic Technique

OSPE Station: Microscope, 3 slides, anti-A, anti-B, anti-D sera, normal saline, sterile lancet/needle, cotton swabs. Blood sample from a finger prick. Equipment for preparing red cell suspension.

Tasks:

Divide slides and mark with respective labels for anti-A, anti-B, anti-D and controls.

Prepare red cell suspension from the blood sample.

Add antisera to the respective slides.

Add red cell suspension to both test and control sides.

Wait and observe for agglutination.

Record observations and determine the blood group.

Checklist:

Aseptic technique in blood collection

Correct slide labeling and reagent addition

Proper observation for agglutination

Correct interpretation of results

PROBLEM-BASED LEARNING

Problem:

Determine the blood group of the patient based on the agglutination reaction with antisera A, B and D. Discuss the relevance of Rh factor in blood transfusions.

Guiding Questions:

1. What does agglutination with anti-A serum indicate about the patient's blood group?
2. Why is the Rh factor (anti-D serum) important in blood group determination?
3. What would happen if the patient received blood from an incompatible group?

Guided Discussion Questions:

1. What would be the result if agglutination occurred with both anti-A and anti-B sera?
2. How does agglutination help us identify the presence of specific antigens on the red blood cells?

Expected Outcomes:

The patient's blood group is A positive. Understanding the importance of blood group compatibility, especially the Rh factor, in transfusion and pregnancy.

Discussion and Resolution:

By identifying the agglutination patterns, it is clear that the patient has blood group A and is Rh-positive. This is critical information for ensuring safe blood transfusions and managing potential Rh incompatibility during pregnancy.



Teacher's Signature

7. PRAKRITI ASSESSMENT

LEARNING OBJECTIVES

After completion of the practical, students should be able to:

Understand the concept of Prakriti and its significance in Ayurveda.

Learn the methods of assessing Prakriti through Darshan, Sparshan and Prashna.

Accurately determine the predominant dosha(s) in an individual.

PRINCIPLE

Prakriti or an individual's constitution is determined by analyzing the dominant doshas - Vata, Pitta and Kapha. The assessment involves observation (Darshan), touch (Sparshan) and inquiry (Prashna) to evaluate physical, physiological and psychological traits.

PURPOSE

To identify the Prakriti type of an individual for personalized health management.

To enhance diagnostic skills in Ayurveda for holistic care.

INSTRUCTIONS

Instructions for Prakriti Assessment

1. **Prakriti Assessment Methods** Students must assess Prakriti using given Prakriti proforma.
2. **Dosha Characteristics** Each dosha (Vata, Pitta, Kapha) column contains 35 characteristic features.
3. **Student Responsibility:** Assess the characteristic features in the examinee and tick (✓) the respective column if the feature aligns with the examinee's traits.

4. **Guidelines for Marking;** Mark (✓) based on occurrence frequency:

- Rarely Occurs: Do not mark.
- Sometimes Occurs: Mark cautiously.
- Often Occurs: Confidently mark

Allocate 1 mark only when certain that the feature aligns often. Refer to provided clues under each point if needed.

5. **Dosha Characteristic Calculation:** Count the total number of features marked (✓) in each dosha column.

PROCEDURE

1. Explain the process and objective of the assessment to the participant.
2. Record basic details such as name, age, gender and weight.
3. Conduct Darshan (visual examination): Observe anatomical traits like body build, height, skin texture, hair type etc.
4. Perform Sparshan (tactile examination): Assess skin temperature and texture, pulse, joint movements etc.
5. Use Prashna (inquiry) for physiological and psychological traits: Ask about sleep patterns, appetite, dreams, decision-making tendencies etc.
6. Tick characteristics observed under respective doshas in the Prakriti assessment form.
7. Calculate scores for Vata, Pitta and Kapha characteristics.

PRECAUTIONS

Ensure accurate measurement of physical parameters. Avoid leading questions during inquiry.

Consider external factors (e.g., age, environmental conditions) affecting the participant's traits.

INTERPRETATION OF RESULT

Interpretation of Result Criteria: The results of Prakriti Parikshana can be interpreted using two approaches:

First Interpretation:

Example: Each dosha has a total of 35 features. If, out of 35 features, Vata has 20, Pitta has 10 and Kapha has 7 features marked, the Prakriti can be interpreted as:

Vata Pradhan Pittanubandhi Prakriti. This indicates that Vata is the predominant dosha with Pitta as a secondary influence.

Second Interpretation:

Ekdoshaja Prakriti: If more than 75% (i.e., more than 26 out of 35) characteristics of a single dosha are marked, it signifies the predominance of that specific dosha.

Dwidoshaja Prakriti: If 40%-60% of the characteristics of two doshas are marked, it indicates the simultaneous balanced traits of two doshas.

Sama Prakriti: If 30%-35% of the characteristics of all three doshas are marked, it suggests an equal predominance trait of all three doshas.

Report the findings clearly in the Prakriti Examination Form, ensuring alignment with the observed characteristics

SHLOKAS (VERSES) FOR RECITATION

प्रकृति

- प्रकृति तु स्वभाव। - चक्रदत्त
- जन्ममरणान्तरालभाविनि अविकारिणी दोषस्थिति प्रकृति।
- रसवैशेषिक सूत्र
- शुक्रशोणितसंयोगे यो भवेत दोषउत्कटः प्रकृतिजार्यते तेन तस्या मे लक्षणं शृणु॥ - सु.शा.4/62
- प्रकृति शरीर स्वरूपम्। - अरुणदत्त
- प्रकार-7, हारित-4

■ वातप्रकृति

वातस्तु रूक्षलघुचलबहुशीघ्रशीतपुरुषविषदः।
तस्य रौक्ष्याद्वातला रूक्षपचिताल्पशरीराः प्रतत
रूक्षक्षामसन्नसक्त जर्जरस्वरा जागरूकाश्च भवन्ति।
लघुत्वान्नलघुचलपलितचेष्टाहारव्याहाराः।
चलत्वात्अनवस्थितसन्ध्यक्षिभ्रूहन्वोष्ठजिह्वाशिरःस्कन्धपाणिपादा
बहुत्वाद्बहुप्रलापकण्डरासिराप्रतानाः
शीघ्रत्वाच्छ्रीघ्रसमारम्भक्षोभविकाराः
शीघ्रनासरागविरागाः श्रुतग्राहिणोऽल्पस्मृतयश्च।
शैत्याच्छीतासहिष्णवः प्रततशीतकोद्वेष स्तम्भाः।
पारूष्यात् पुरुषकेशश्मश्रुमनखदशनवदनपाणिपादाः
वैशद्यात् स्फुटिताङ्गावयवाः सततसन्धिश्चक्षुर्गामिनश्च भवन्ति।
त एवं गुणयोगाद्वातलाः प्रायेणल्पबलाश्चाल्पायुष्यश्चाल्पापत्या
श्चाल्प साधनाश्चाल्पधनाश्च भवन्ति॥
- च.वि.8/98

■ पित्तप्रकृति

पित्तमुष्णं तीक्ष्णं द्रवं विस्त्रमम्लं कटूकं च।
तस्य औक्षण्यात् पित्तला भवन्त्युष्णासहा, उष्णमुखाः सुकुमारा
वदातगात्राः प्रभुतपिप्लुव्यङ्गतिलकालकाः क्षिप्रवलीपलित
खालित्यदोषाः प्रायोमृदुअल्प कपिलश्मश्रुलोमकेशाश्च।
तैक्ष्ण्यात् तीक्ष्णपराक्रमाः, तीक्ष्णानाग्रेयः, प्रभूताशनपानाः
क्लेशासहिष्णवो, दन्दशूकाः।
द्रवत्वाच्छिथिलमृदुसन्धिमांसाः प्रभूतसृष्टस्वेदमूत्रपुरीषाश्चः
विस्त्रत्वात् प्रभूतपूतिकक्षास्यशिरः शरीरगन्धाः,
कटुवम्लत्वादल्पशुक्रव्यवायापत्याः।
त एवंगुणयोगात् पित्तला मध्यबला मध्यायुषो मध्यज्ञानविज्ञान
वित्तोपकरणवन्तश्च भवन्ति।
- च.वि.8/97

■ कफप्रकृति

स्निग्धश्लक्ष्णमृदु मधुरसारसान्द्र मन्दस्तिमितगुरु शीतविज्जलाच्छः। तस्य स्नेहात् श्लेष्मलाः स्निग्धाङ्गाः, श्लक्ष्णत्वात् श्लक्ष्णाङ्गाः, मृदुत्वात् दृष्टीमुखसुकुमारावदातगात्राः, माधुर्यात् प्रभुतशुक्र व्यवायापत्यः, सारत्वात् सारसहतास्थिरशरीराः, साद्रत्वात् उपचिपरिपुर्णसर्वगात्राः, मन्दत्वात् मन्दचेष्टाहारव्याहाराः, स्तैमित्यात् शीघ्रारंभाल्पक्षोभविकाराः, गुरुत्वात् साराधिष्ठिता वस्थितगतयः शैत्याद् अल्पक्षुत्तृष्णासंतापस्वेद दोषा विज्जलत्वात् सुश्लिष्टसारसंधिबंधना, तथाऽच्छवात् प्रसन्न दर्शनाननाः प्रसन्नस्निग्धवर्णस्वराश्च भवन्ति। त एवं गुणयोगात् श्लेष्मल बलवन्तो वसुमन्तो विद्यावन्त ओजस्विनः शान्ताआयुष्यन्तश्च भवन्ति।

- च.वि.8/96

■ भौतिक प्रकृति (5,3,31)

विषजातो यथा कीटो न विषेण विपद्यते तद्वत्प्रकृतयो मर्त्ये शक्नुवन्ति न बाधितुम्। प्रकृतिमिह नराणां भौतिकीं केचिदाहुः पवनदहनतौः कीर्तितास्तास्तुति स्थिरविपुलशरीरः पार्थिवश्च क्षमावान्, शुचिरथचिरजीवी नाभसः खैर्म। - सु.शा.4/79 पार्थिव प्रकृति - स्थिरविपुलशरीर क्षमावान् नाभस प्रकृति - शुचिरथ चिरजीवी खैर्महाद्भि

■ जात्यादी प्रकृति

(चरक -6)(वाग्भट सप्तप्रकृति-बल)

तत्र प्रकृतिर्जातिप्रसक्ता च कुलप्रसक्ता च देशानुपातिनी च कालानुपातिनी च वयोऽनुपातिनी च प्रत्यात्मनियता चेति जातिकुलदेशकालवयः प्रत्यात्मनियता हि तेषां तेषां पुरुषाणां ते ते भाव शेषा भवन्ति।

- च.इ.1/5, अ.सं.शा.4/20

■ प्रकृति निर्मापक भाव

तद्यथा- शुक्रशोणितप्रकृति कालगर्भाशयप्रकृतिं आतुराहाराविहाराप्रकृतिं महाभूतविकारप्रकृतिं च गर्भशरीरम् पेक्षते एतानि हि येन येन दोषेणाधिकेनैकेनानेकेन वा समनुबध्यन्ते तेन तेन दोषेणा गर्भोऽनुबध्यते ततः सा सा दोषप्रकृतिरुच्यते मनुष्याणां गर्भादिप्रवृत्ता तस्माच्छ्लेष्मलाः प्रकृत्या केचित् पित्तलाः केचित् वातलाः केचित् संसृष्टाः केचित् समाधातवः केचिद्भवन्ति तेषां हि लक्षणानि व्याख्यास्यामः।

- च.वि.8/95

प्रकृति - सुश्रुत एवं अष्टांग हृदय

■ वातप्रकृति

- सु.शा.4/63: प्रजागरूकः शीतद्वेषी दुर्भगः स्तेनो गान्धर्वचित्त क्राथी दन्तनखादि धमनीततः। अनुकत्व-अज गोमायु (शृगाल) शश आखु उष्ट्र गृध्र काक खर श्व (कुत्ता)
- अ.हं.शा.3/85-89: स्फूर्तितधूसर केशगात्राः, बहुभूजः मधुरसात्म्य कटू उष्ण सात्म्य कृष्णदीर्घ आकृति, वृत्त अचार मृतोपसानि नेत्र, प्रोद्बद्धपिण्डिका।

■ पित्तप्रकृति

- सु.शा.4/67: शिथिलाङ्गता दुर्भग बहुभुग् समितिदुर्निवारवीर्य (युद्ध पराक्रम) स्वप्न-कनक पलाश कणिकार, हुताश (अग्नि), विद्युत, उल्का व्यथितास्यगति (कष्ट से बोलना) अनुकत्व-भूजंग, उल्लू, गंधर्व, यक्ष, मार्जार, कपि, व्याघ्र, ऋक्ष, नकुल।
- अ.ह.शा.3/90-95: शूरोमानी, अश्रितवत्सल, घर्मद्वेषी पिङ्गकेशो, अल्परोमा शुचि, मेधावी, प्रशिथिलसंधिबंधमासो, नारीणाभिमतो अल्पशुक्रकामः। आवासः, भुर्युच्चार, तनु अल्प पक्ष्माणि, पण्डित।

■ कफप्रकृति

- सु.शा.4/71: बलवान् स्थिरग्राहीदृष्टवैरश्च। रक्तान्तनेत्र सुविभक्तगात्र सत्वगुणोपपन्नः गुरुमानकरच्च। मधुरप्रिय, सुभग वर्ण - इन्दीवर (कमल), निखिंश आर्द्रारिष्ट, शरकाण्ड शब्द - जलमृद्गसिंहघोषः। अनुकत्व - ब्रह्मा रूद्र इन्द्र वरूण सिंह अश्व गज गो वृषे ताक्ष्य हंस सुश्रुत ने हर प्रकृति का वर्णन 9 भागों में किया है।
- अ.ह.शा.3/97,103: सात्विक सत्यसंसर्ग, प्रलम्बबाहुः पृथुपीनवक्षा महाललाटो, घननीलकेशः। चारुवर्ष्मा दृढं वैरम्। दीर्घदर्शी दीर्घसूत्री। प्रियंगु, दुर्वा, गोरोचन, पद्मसुवर्णवतवर्ण।

प्रकृति - शारंगधर- शां.पू.6/20

■ वातज प्रकृति

अल्पकेशः कृशो रूक्षो वाचालश्चमानसः। आकाशचारीः स्वप्नेषु वातप्रकृतियो नरः॥

■ पित्तज प्रकृति

अकालपलितैर्व्याप्तो धीमान्स्वेदी च रोषणः। स्वप्नेषु ज्योतिषां द्रष्टा पित्तप्रकृतियो नरः॥

■ कफज प्रकृति

गम्भीरबुद्धिः स्थूलाङ्गः स्निग्धकेशो महाबलः। स्वप्ने जलाशयलोकी श्लेष्मप्रकृतियो नरः॥

7.1 PRAKRITI ASSESSMENT

Objective : To assess the Prakriti of an individual

Name: :

Age : **Gender:** M/F **Occupation:**

Body temperature : **Weight:**kg

Blood pressure : **Pulse rate:**...../min

Address :

Physical Traits		Prakriti assessment preform	Score		
Sr No	Domain	Feature	Vataj	Pittaj	Kaphaj
Anatomical Traits					
1	Built	अपचित: (Thin built)	1	0	0
		उपचित: (Well built) Decide with the help of BMI (<18.5 = Vata, >25 = Kapha) or Measure the width of the four fingers of the hand together to calculate one finger's breadth in cm. Calculate height as per using self-anguli (finger) measurement, which is 84 angul. Whether any confounder which is affecting the weight of the participant is present or not?	0	0	1
2	Height	ह्रस्वाकृति (Short height)	0	0	1
		दीर्घाकृति (Tall height) Short or tall decide as compared with others. (may considered below 5 feet & above 6 feet)	1	0	0
3	Appearance	बहु कण्डरा सिरा प्रतान: (Prominent tendons and veins)	1	0	0
		सुकुमार गात्र (Delicate body) Are you sensitive to excessive heat/summer season/hot food items especially spicy food items/ harsh medicine?	0	1	0

		सुकुमार गात्र (Delicate body) Whether the person has a delicate appearance?	0	0	1
		महाललाट (Broad forehead) (The forehead is broader than 4 anguli)	0	0	1
4	Skin	Colour			
		गौर (Lotus/straw/gold colour)	0	0	1
		गौर- पीतांग (Fair with yellowish tinge)	0	1	0
		धुसर/कृष्ण (Dusky/dark)	1	0	0
		Texture			
		रूक्षांग / स्फुटितांग: (Dry skin/cracking of skin) (Non lustrous skin)	1	0	0
		प्रसन्न स्निग्धांग: (Unctuous skin/ smooth clear skin without moles, freckles & dryness etc.)	0	0	1
		प्रभुततिलव्यंगपिप्लु: (Presence of moles, pimple, freckles)	0	1	0
		क्षिप्रवली/शीघ्रवली: (Early appearance of wrinkles, may be seen under the eyes first)	0	1	0
		Temperature			
		उष्माङ्गा (Warm feeling on touch at dorsal surface of the hand and forehead)	0	1	0
		अल्पसंताप (Body temperature towards the lower side of normalcy) Ask for the normal temperature of the body or Measure the temperature with the thermometer to decide the range of temperature.	0	0	1
		उष्णमुख: (Tendency for frequent ulceration of mouth) Do you feel hotness in mouth / suffer from ulceration of mouth?)	0	1	0
5	Joints	अनवस्थित सन्धिपाद (Undue frequent movement of different body parts like i.e joints, eyes, eyebrows, jaws, lips, tongue, head, shoulder, hands and legs (sandhi, akshi, bhru, hanu, osth, jihva, shira, skandha, paani and paad - physician's observations)	1	0	0

		सशब्द सन्धिगामि (Joints sound) Is there any sound (at times) on normal movement especially in knee/ ankle /toes or shoulder?	1	0	0
6	Muscle	शिथिल मृदु मांसः (Lax muscles) Observe the muscle tone while keeping in mind confounding factors like ageing.	0	1	0
		उद्भट्ट पिण्डिका (Prominent/hard calf muscles)	1	0	0
7	Eyes	Size			
		तनु लोचनः (Small eyes)	1	0	0
		विशालाक्षः (Big eyes) (after opening the eyes, at the lower portion of the cornea a little part of the white sclera is also visible)	0	0	1
		Appearance/Texture			
		रूक्ष/खर नेत्रः (Dry/rough eyes) (often having feelings of dryness/roughness/irritation/eye fatigue)	1	0	0
		Colour of eyes			
		धुसर नेत्र (Dull white sclera)	1	0	0
		ताम्रनयन (Coppery, Reddish eyes)	0	1	0
		शुक्लाक्ष/व्यक्त सित असिताक्षः (milky white / clearly distinguished sclera) रक्तान्त नेत्र (Reddish canthus)	0	0	1
		Eye lashes			
		अल्पपक्ष्म/तनुपक्ष्मः (Less in number/thin eye lashes)	0	1	0
		पक्ष्मलाक्षः (Thick/dense eye lashes)	0	0	1
		Eye movements			
		चलदृष्टीः (Unsteady gaze) (Unable to hold attention/ frequent movement of eyes observed by assessor)	1	0	0
		Special feature of eyes			
		क्रोधेन/मद्येन/ रवेश्च रागमः (Eyes redden during anger/ after alcohol consumption/after exposure to sunlight)	0	1	0

8	Body hair	Texture			
		परुष/रूक्ष/स्फुटितकेश: (Hard/ dry/spilt hair ends)	1	0	0
		मृदु: (Soft)/ अल्प केश: (Scanty hair)/ कपिल केश: (Reddish brown hair)	0	1	0
		स्निग्ध: (Unctuous, oily hair)	0	0	1
		Appearance			
		कुटिल केश: (Curly hair)/ घन केश (Dense hair) श्याम केश:(Black hair)/	0	0	1
		कपिल केश: (Reddish brown hair)	0	1	0
		धूसर केश: (Dusky hair)	1	0	0
		Tendency of hair fall & Greying			
		अकाल पलित: (Premature greying) or शीघ्र खालित्य: (Early hair loss/baldness) Whether the person is having grey hair (approx. more than 20%) before the age of 35 years? Whether the person has baldness/is facing any significant hair fall (more than 50% before the age of 50 years)?	0	1	0
9	Teeth	Texture			
		रूक्ष दन्त/परुष दशन: (Dry teeth /rough teeth)	1	0	0
		Size			
		सूक्ष्म दन्त/अति दन्त: (Small teeth/big teeth)	1	0	0
10	Nails	Size			
		अल्प नख/तनु नख:(Small nails/thin nails)	1	0	0
		दीर्घ नख:(Long nails)	0	0	1
		Texture			
		रूक्ष नख/परुष नख/स्तब्ध नख (Dry nails/rough nails)	1	0	0
		Colour			
		ताम्र नख:(Coppery coloured nails)	0	1	0
		Nature			
		नख: वृद्धि (Excessive nail growth)	1	0	0

Physiological Traits

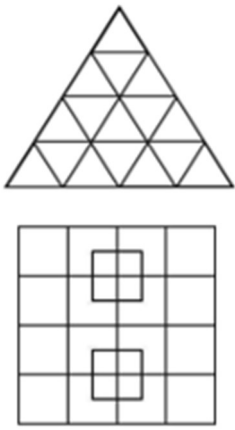
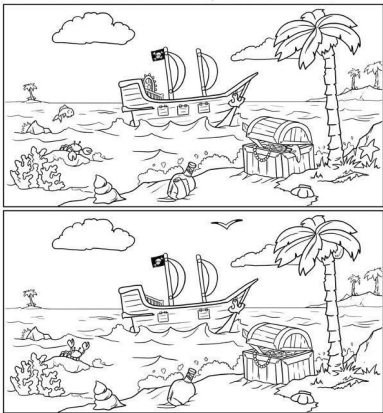
11	Locomotion	Gait and Locomotion			
		द्रुतगति/चपलगति/लघुगति: Speedy/ quick, Swift gait/Small steps The person walks very fast or he/she starts walking instantly/ immediately?	1	0	0
		सारगति/अधिष्ठित गति: (Consistent /straight unswerving way) Complete way i.e. heel strike first followed by toes touching the ground completely.	0	0	1
		Initiation of Action			
		लघु चेष्टा/शीघ्र समारम्भः:(Quick initiation of action) If a task is given to you do you have a tendency to initiate it immediately?	1	0	0
		मन्द चेष्टा/अशीघ्रारम्भः:(Delayed initiation)	0	0	1
12	Speech/ Voice	Voice Quality			
		सन्न स्वर/रूक्ष स्वर/जर्जर स्वर/प्रतत स्वर/बद्ध स्वर/चलवाच/ स्तब्ध स्वर/तनुस्वर: The assessor has to analyse the voice quality as per Feeble voice/ Unpleasant/Shattered, Broken voice/ Unclear speech / Stammering/ Fast, shaking voice/Hoarseness of voice/Low speech voice. Whether any confounder which is affecting the voice of the participant is present or not?	1	0	0
		प्रसन्न स्वर/गंभीर स्निग्ध स्वर (Clear voice/Deep, Pleasant voice)	0	0	1
13	Appetite	Quantity & Frequency			
		शीघ्रक्षुत/दन्दशुक : (Frequent intake of more food) प्रभुताशन (Excessive intake/ Excess Quantity)	0	1	0
		अल्पक्षुत/अल्पअशन: (Less hunger/Less intake, less frequency)	0	0	1

		How many times in a day do you take main meals and how many times refreshments? 2 main meals and 3 or more refreshments or 3 main meals and 2 or more refreshments or None of the above			
		बहुभुज/अनल्पाशन (Frequent but less intake) How much quantity of food you take in a day? Less than average (4 chapati, little rice, 1 bowl dal etc.) or Average (6-10 chapattis, rice, dal, sabji, fruits, salads etc.) or More than average	1	0	0
		Tolerance to hunger			
		क्षुत सहिष्णु (Can tolerate hunger)	0	0	1
		तीक्ष्ण बुभुक्षा: (Intense feeling of hunger) Suppose you skip 1 meal do you tolerate it easily? Can tolerate (Kapha) or Cannot tolerate (Pitta) or It varies (None)	0	1	0
		Eating habit			
		लघु आहार/चपल आहार: (Fast eating habit)	1	0	0
14	Thirst	मंद आहार: (Slow eating habit) Suppose you are having lunch with your friends / family members. Usually you finish your food - Before your friends / family members (Vata) or At par / with your friends / family members (Kapha)	0	0	1
		Quantity & Intensity			
		शीघ्र पिपासा: (Frequently thirsty) How frequently you take water/fluids in a day? (<4 or 4-7 or >7 times/day)	0	1	0
		प्रभूतपान/तीक्ष्ण तृष्णा: (Drink more water/Intense thirst)	0	1	0
		अल्प पान/अल्पतृष्णा: (Drink less water/Less thirst) When you feel thirsty and water is not instantly available how you react? Immediately search for water or wait for sometime. How much water / liquids do you take per day approximately? (<1 or 1-2 or >2 lit/day)	0	0	1

15	Stool	प्रभूत-सृष्ट पुरीषः मूत्रः (Easy & satisfactory bowel evacuation) (Easy & adequate flow of urine) The quantity of stool passed by you in comparison to food intake is: Adequate/ Bulky or Less Do you usually have easy, quick and effortless bowel evacuation without constipation or use of any laxative?	0	1	0
16	Sweat	प्रभूत.सृष्ट स्वेदः (Profuse sweating)	0	1	0
		अल्प स्वेदः (Scanty sweating) How much do you sweat on exposure to sun rays / exercise / physical activity? Profuse (Pitta) Scanty (Kapha)	0	0	1
		कक्ष-आस्य-शिर प्रभूत पूतिगंध/स्वेदोदुर्गन्धः (Putrid, foul odour especially from axilla, scalp, mouth & head / Bad odour of sweat) Have any of your family members / friends ever told you about your strong body odour?	0	1	0
17	Sleep	जागरूक/अल्पनिद्रा (Easy awakening) (Less sleep) While sleeping are you awakened by a little noise? How many hours do you sleep in 24 hours? (<6 or 7-8 or > 8 hr.)	1	0	0
		निद्रा प्रियः (Like sleeping) Whether you like to sleep if given a chance or in free time?	0	0	1
18	Nature Dreams	आकाशचारी स्वप्नेषुः (Dreaming of Flying in the sky, walking mountains, dried water bodies, trees etc.)	1	0	0
		स्वप्नेषु ज्योतिषां दृष्टाः (Dreaming of falling of meteors, fire, Lightning, Gold etc.)	0	1	0
		स्वप्नेषु जलाशय लोकीः (Dreaming of water bodies, Lotus, water birds like swans, ducks etc.) Do you often see dreams? if ans is No – then skip the question	0	0	1

Psychological Traits

19	Indecisive-ness	अनवस्थित आत्मा/अव्यवस्थितमति: (Indecisiveness often change in decisions) After taking a decision how often you feel the need to change your decisions?	1	0	0
20	Comprehe-nsion	श्रुतग्राही: (Quick comprehension/ Grasping)	1	0	0
		चिरग्राही: (Delayed comprehension) If the participant is able to grasp new information quickly, he/ she may be considered as Shrutagrahi. Else, if his/her grasping is delayed he/she may be considered as Chiragrahi. Tell a short story or passage and ask the person to summarize it. The participant who answers 60 % or above (able to answer 6 or more out of 10) correctly may be taken as Shrutagrahi.	0	0	1
21	Memory	स्मृति मान: (Having good & long-term memory) To assess good memory, ask the person to recall details of a meaningful past event, like a childhood story and evaluate their ability to retrieve vivid, accurate and sequential details. Alternatively, give them a list of 5 single words and 5 paired words to memorize. Test recall after 5 minutes for short-term memory and after 1-2 hours for long-term memory. If they recall 4-5 single words and 3 or more pairs, they likely have good long-term memory.	0	0	1
		अल्प स्मृति/चल स्मृति: (Less memory/Unstable memory)	1	0	0
22	Relationship	मंद मित्र/अदृढ सौहृद्/शीघ्र सौहृद्: (Few friends/ Fickle friendship/ Make friends easily, quickly)	1	0	0
		स्थिर सौहृद् : (Stable cordial friendship)/ Do your relations with your friends / relatives / neighbours last for a long time?	0	0	1

23	Intelligence	निपुण मति (Learning ability)/ मेधावी (Intelligent) A picture puzzle may be presented to a person to identify a set number of differences (e.g., 15 differences) between two otherwise similar images within 3 minutes. If the person successfully identifies 12–15 differences within the time limit, they may be considered to have good learning ability or intelligence. Total triangles are 27, squares are 40 and 5 differences. Alternatively, another task could involve asking the person to count the number of triangles/squares/ image in a given figure. This test can also assess learning ability and attention to detail.  	0	1	0
24	Temperament	क्रोधी/क्षिप्र कोप-क्षिप्र प्रसाद (Short tempered/ Becomes angry quickly & gets pleased quickly) The assessor may ask the individual the following question: "What is your usual reaction if someone breaks a queue in front of you?" A. Ignore it, remain calm or politely request them to step back – overall calm demeanor. B. Start an argument, shout at them or lose your temper – overall short-tempered behaviour. C. Get angry or irritated quickly but calm down shortly afterward – overall volatile temper. (when an answer is B or C then score 1 for pitta) (when an answer is A then score 1 for Kapha) अल्प क्रोधः (Calm & patient, Less anger) Ask the question - What is the opinion of your family members/friends/relatives about you? (calm, short tempered, get angry)	0	1	0
			0	0	1

Behavioural Appearance

25	Nature of Speech/ Talking	वाचाल (talkative or talking irrelevantly) The Assessor will observe the talking style nature of speech of an individual. The nature of speech may be confirmed by friends or associates to check/observe talkativeness or way of manners. The assessor may be asked the following questions. What is your future plan? How much do you serve your parents? How do you judge your personality? etc. If the person seems to be talkative or talking irrelevant, may be considered as talkative & give a score 1 for vata	1	0	0
		मितवाक्, वदान्य/ विनित (Polite /humble) Do you often remain polite soft & submissive even in stressful/unwanted/anxious conditions? If a person speaks limited, mannered way, soft-spoken may be considered as polite/humble. If the answer is yes then score 1 for Kapha	0	0	1
		प्रगल्भ वक्ता (profound orator) Are you often able to talk effectively at length on a particular topic of your choice eg. Pollution, humanity, women power, agriculture, corruption etc? If the answer is yes then score 1 for pitta	0	1	0
		समितिषु दुर्निवारवीर्य (Dominant speaker) Are you often able to dominate during the conversation & establish your views If the answer is yes then score 1 for pitta			
		तीक्ष्ण पराक्रम/शुर/न भयात् प्रणम्याति: (Very brave, valour/ never surrender) Are you often able to stand for your faith /values/morals despite what others say & you face extreme adverse situations and challenges bravely till the goal is achieved? If the answer is yes then score 1 for pitta	0	1	0
26	Likings	शीताभिलाषी (like cold climate & food items)	0	1	0
		उष्ण अन्नपानाकांक्षा: (Desirous of hot food & drinks) What type of food items you like the most in moderate season?	1	0	0

		अतिस्वेदन /विमर्दनेन सौख्यम् गच्छति: (like sudation & massage)	1	0	0
		यात्राप्रियदयित माल्य विलेपन मंडन (Fond of cosmetics)	0	1	0
		शास्त्रप्रिय (Fond of reading/listening classical texts)	0	0	1
27	Tolerance	हीनसत्त्व/शीघ्रत्रास (Having less tolerances & get frightened easily) How do you usually react when you face a stressful situation?	1	0	0
28	Bravery	तीक्ष्ण पराक्रम/शूर न भयात् प्रणम्यति:। (Very brave/ never surrender in fear) Do you consider yourself someone who stands firm in the face of fear? Why or why not?	0	1	0
29	Affectionate	आश्रित वात्सल्य (affectionate to dependents) How do you typically show affection to those who depend on you?	0	1	0
30	Competitive spirit	भूरिईर्ष्या: (competitive spirit to friends/ relatives/ neighbours) What motivates you to compete with others? Is it the desire to win, improve or be ahead of others?	0	1	0
31	Jealous	मात्सर्यः (Jealous of friends/Relatives /Neighbours) How do you feel when someone close to you achieves something significant, like a promotion, a new car or a special recognition?	1	0	0
32	Egoist	अभिमानि (Having superiority complex) Have you ever dismissed someone's advice or opinions because you felt they were not qualified to give it?	0	1	0
33	Forgiveness & solemnity	गम्भीर व क्षमावान (Stable mind forgiveness to other) Do you believe in giving people second chances? Why or why not?	0	0	1
34	Rigid	अनतेषु/अमृदुः (Harsh towards non-loyal) How do you handle conflicts or disagreements with people you perceive as disloyal?	0	1	0
35	Respectfulness & Humanity	गुरुमानयिता व विनीत (Respect to teachers/ elders & polite/Humble) Scenario based questions- How is your attitude towards house help, waiters, etc?	0	0	1

36	Enmity	दृढवैरः (Having strong enmity) Do you find it difficult to forgive those who have wronged you in the past? Why or why not? Clue - Have you any friends /relatives/colleagues with whom you had differences & never patched up again because you never wanted to patch up or your friends, but you did not accept?	0	0	1
37	Compliment freak	स्तुतिप्रिय (Fond of exaggerated praise) How often do you seek feedback from others on your work? What kind of feedback do you value the most?	0	1	0
38	Modesty	सुलज्जौः (Feel shy while expressing) Do you feel awkward when someone praises you for your achievements?	0	0	1
30	Generosity	परिगन्य चिरात् प्रददातिबहुः (Denotes generosity & Judiciously) Do you feel comfortable saying no to donation requests when necessary?	0	0	1
40	Planner	अभियोगावानः (Systemic & ideal planner) How do you set goals for yourself, both short-term & long-term?	0	1	0
41	Physical activity	व्यायामशीलः (Doing regular physical exercise) How often do you exercise each week?	0	0	1

Result:

Predominance of Doshaj Features: Vataj/35 Pittaj/35 Kaphaj/35

Interpretation of Prakriti Type:

Predominant Dosha: 70% in one dosha / 40-60% in two doshas / 30-35% in three doshas

Interpretation of Doshaj Prakriti Type: Ekdoshaja Prakriti / Dwidoshaja Prakriti / Sama Prakriti



Teacher's Signature

7.2 PRAKRITI ASSESSMENT

Objective : To assess the Prakriti of an individual

Name: :

Age : **Gender:** M/F **Occupation:**

Body temperature : **Weight:**kg

Blood pressure : **Pulse rate:**...../min

Address :

Physical Traits		Prakriti assessment preform	Score		
Sr No	Domain	Feature	Vataj	Pittaj	Kaphaj
Anatomical Traits					
1	Built	अपचित: (Thin built)	1	0	0
		उपचित: (Well built) Decide with the help of BMI (<18.5 = Vata, >25 = Kapha) or Measure the width of the four fingers of the hand together to calculate one finger's breadth in cm. Calculate height as per using self-anguli (finger) measurement, which is 84 angul. Whether any confounder which is affecting the weight of the participant is present or not?	0	0	1
2	Height	ह्रस्वाकृति (Short height)	0	0	1
		दीर्घाकृति (Tall height) Short or tall decide as compared with others. (may considered below 5 feet & above 6 feet)	1	0	0
3	Appearance	बहु कण्डरा सिरा प्रतान: (Prominent tendons and veins)	1	0	0
		सुकुमार गात्र (Delicate body) Are you sensitive to excessive heat/summer season/hot food items especially spicy food items/ harsh medicine?	0	1	0

		सुकुमार गात्र (Delicate body) Whether the person has a delicate appearance?	0	0	1
		महाललाट (Broad forehead) (The forehead is broader than 4 anguli)	0	0	1
4	Skin	Colour			
		गौर (Lotus/straw/gold colour)	0	0	1
		गौर- पीतांग (Fair with yellowish tinge)	0	1	0
		धुसर/कृष्ण (Dusky/dark)	1	0	0
		Texture			
		रूक्षांग / स्फुटितांग: (Dry skin/cracking of skin) (Non lustrous skin)	1	0	0
		प्रसन्न स्निग्धांग: (Unctuous skin/ smooth clear skin without moles, freckles & dryness etc.)	0	0	1
		प्रभुततिलव्यंगपिप्पु: (Presence of moles, pimple, freckles)	0	1	0
		क्षिप्रवली/शीघ्रवली: (Early appearance of wrinkles, may be seen under the eyes first)	0	1	0
		Temperature			
		उष्माङ्गा (Warm feeling on touch at dorsal surface of the hand and forehead)	0	1	0
		अल्पसंताप (Body temperature towards the lower side of normalcy) Ask for the normal temperature of the body or Measure the temperature with the thermometer to decide the range of temperature.	0	0	1
		उष्णमुख: (Tendency for frequent ulceration of mouth) Do you feel hotness in mouth / suffer from ulceration of mouth?)	0	1	0
5	Joints	अनवस्थित सन्धिपाद (Undue frequent movement of different body parts like i.e joints, eyes, eyebrows, jaws, lips, tongue, head, shoulder, hands and legs (sandhi, akshi, bhru, hanu, otha, jihva, shira, skandha, paani and paad - physician's observations)	1	0	0

		सशब्द सन्धिगामि (Joints sound) Is there any sound (at times) on normal movement especially in knee/ ankle /toes or shoulder?	1	0	0
6	Muscle	शिथिल मृदु मांसः (Lax muscles) Observe the muscle tone while keeping in mind confounding factors like ageing.	0	1	0
		उद्भट्ट पिण्डक (Prominent/hard calf muscles)	1	0	0
7	Eyes	Size			
		तनु लोचनः (Small eyes)	1	0	0
		विशालाक्षः (Big eyes) (after opening the eyes, at the lower portion of the cornea a little part of the white sclera is also visible)	0	0	1
		Appearance/Texture			
		रूक्ष/खर नेत्रः (Dry/rough eyes) (often having feelings of dryness/roughness/irritation/eye fatigue)	1	0	0
		Colour of eyes			
		धुसर नेत्र (Dull white sclera)	1	0	0
		ताम्रनयन (Coppery, Reddish eyes)	0	1	0
		शुक्लाक्ष/व्यक्त सित असिताक्षः (milky white / clearly distinguished sclera) रक्तान्त नेत्र (Reddish canthus)	0	0	1
		Eye lashes			
		अल्पपक्ष्म/तनुपक्ष्मः (Less in number/thin eye lashes)	0	1	0
		पक्ष्मलाक्षः (Thick/dense eye lashes)	0	0	1
		Eye movements			
		चलदृष्टीः (Unsteady gaze) (Unable to hold attention/ frequent movement of eyes observed by assessor)	1	0	0
		Special feature of eyes			
		क्रोधेन/मद्येन/ रवेश्च रागमः (Eyes redden during anger/ after alcohol consumption/after exposure to sunlight)	0	1	0

8	Body hair	Texture			
		परुष/रूक्ष/स्फुटितकेश: (Hard/ dry/spilt hair ends)	1	0	0
		मृदु: (Soft)/ अल्प केश: (Scanty hair)/ कपिल केश: (Reddish brown hair)	0	1	0
		स्निग्ध: (Unctuous, oily hair)	0	0	1
		Appearance			
		कुटिल केश: (Curly hair)/ घन केश (Dense hair) श्याम केश:(Black hair)/	0	0	1
		कपिल केश: (Reddish brown hair)	0	1	0
		धूसर केश: (Dusky hair)	1	0	0
		Tendency of hair fall & Greying			
		अकाल पलित: (Premature greying) or शीघ्र खालित्य: (Early hair loss/baldness) Whether the person is having grey hair (approx. more than 20%) before the age of 35 years? Whether the person has baldness/is facing any significant hair fall (more than 50% before the age of 50 years)?	0	1	0
9	Teeth	Texture			
		रूक्ष दन्त/परुष दशन: (Dry teeth /rough teeth)	1	0	0
		Size			
		सूक्ष्म दन्त/अति दन्त: (Small teeth/big teeth)	1	0	0
10	Nails	Size			
		अल्प नख/तनु नख:(Small nails/thin nails)	1	0	0
		दीर्घ नख:(Long nails)	0	0	1
		Texture			
		रूक्ष नख/परुष नख/स्तब्ध नख (Dry nails/rough nails)	1	0	0
		Colour			
		ताम्र नख:(Coppery coloured nails)	0	1	0
		Nature			
		नख: वृद्धि (Excessive nail growth)	1	0	0

Physiological Traits

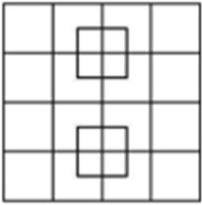
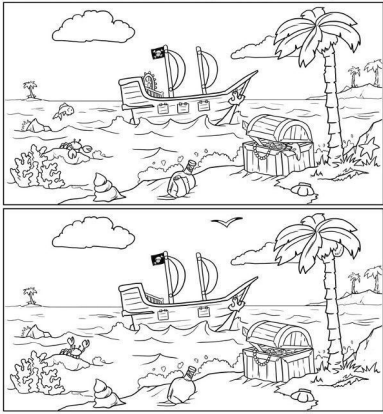
11	Locomotion	Gait and Locomotion			
		द्रुतगति/चपलगति/लघुगति: Speedy/ quick, Swift gait/Small steps The person walks very fast or he/she starts walking instantly/ immediately?	1	0	0
		सारगति/अधिष्ठित गति: (Consistent /straight unswerving way) Complete way i.e. heel strike first followed by toes touching the ground completely.	0	0	1
		Initiation of Action			
		लघु चेष्टा/शीघ्र समारम्भः:(Quick initiation of action) If a task is given to you do you have a tendency to initiate it immediately?	1	0	0
		मंद चेष्टा/अशीघ्रारम्भः:(Delayed initiation)	0	0	1
12	Speech/ Voice	Voice Quality			
		सन्न स्वर/रूक्ष स्वर/जर्जर स्वर/प्रतत स्वर/बद्ध स्वर/चलवाच/ स्तब्ध स्वर/तनुस्वर: The assessor has to analyse the voice quality as per Feeble voice/ Unpleasant/Shattered, Broken voice/ Unclear speech / Stammering/ Fast, shaking voice/Hoarseness of voice/Low speech voice. Whether any confounder which is affecting the voice of the participant is present or not?	1	0	0
		प्रसन्न स्वर/गंभीर स्निग्ध स्वर (Clear voice/Deep, Pleasant voice)	0	0	1
13	Appetite	Quantity & Frequency			
		शीघ्रक्षुत/दन्दशुक : (Frequent intake of more food) प्रभुताशन (Excessive intake/ Excess Quantity)	0	1	0
		अल्पक्षुत/अल्पअशन: (Less hunger/Less intake, less frequency)	0	0	1

		How many times in a day do you take main meals and how many times refreshments? 2 main meals and 3 or more refreshments or 3 main meals and 2 or more refreshments or None of the above			
		बहुभुज/अनल्पाशन (Frequent but less intake) How much quantity of food you take in a day? Less than average (4 chapati, little rice, 1 bowl dal etc.) or Average (6-10 chapattis, rice, dal, sabji, fruits, salads etc.) or More than average	1	0	0
		Tolerance to hunger			
		क्षुत सहिष्णु (Can tolerate hunger)	0	0	1
		तीक्ष्ण बुभुक्षा:(Intense feeling of hunger) Suppose you skip 1 meal do you tolerate it easily? Can tolerate (Kapha) or Cannot tolerate (Pitta) or It varies (None)	0	1	0
		Eating habit			
		लघु आहार/चपल आहार: (Fast eating habit)	1	0	0
		मंद आहार: (Slow eating habit) Suppose you are having lunch with your friends / family members. Usually you finish your food - Before your friends / family members (Vata) or At par / with your friends / family members (Kapha)	0	0	1
14	Thirst	Quantity & Intensity			
		शीघ्र पिपासा: (Frequently thirsty) How frequently you take water/fluids in a day? (<4 or 4-7 or >7 times/day)	0	1	0
		प्रभूतपान/तीक्ष्ण तृष्णा: (Drink more water/Intense thirst)	0	1	0
		अल्प पान/अल्पतृष्णा: (Drink less water/Less thirst) When you feel thirsty and water is not instantly available how you react? Immediately search for water or wait for sometime. How much water / liquids do you take per day approximately? (<1 or 1-2 or >2 lit/day)	0	0	1

15	Stool	प्रभूत-सृष्ट पुरीषः मूत्रः (Easy & satisfactory bowel evacuation) (Easy & adequate flow of urine) The quantity of stool passed by you in comparison to food intake is: Adequate/ Bulky or Less Do you usually have easy, quick and effortless bowel evacuation without constipation or use of any laxative?	0	1	0
16	Sweat	प्रभूत.सृष्ट स्वेदः (Profuse sweating)	0	1	0
		अल्प स्वेदः (Scanty sweating) How much do you sweat on exposure to sun rays / exercise / physical activity? Profuse (Pitta) Scanty (Kapha)	0	0	1
		कक्ष-आस्य-शिर प्रभूत पूतिगंध/स्वेदोदुर्गन्धः (Putrid, foul odour especially from axilla, scalp, mouth & head / Bad odour of sweat) Have any of your family members / friends ever told you about your strong body odour?	0	1	0
17	Sleep	जागरूक/अल्पनिद्रा (Easy awakening) (Less sleep) While sleeping are you awakened by a little noise? How many hours do you sleep in 24 hours? (<6 or 7-8 or > 8 hr.)	1	0	0
		निद्रा प्रियः (Like sleeping) Whether you like to sleep if given a chance or in free time?	0	0	1
18	Nature Dreams	आकाशचारी स्वप्नेषुः (Dreaming of Flying in the sky, walking mountains, dried water bodies, trees etc.)	1	0	0
		स्वप्नेषु ज्योतिषां दृष्टाः (Dreaming of falling of meteors, fire, Lightning, Gold etc.)	0	1	0
		स्वप्नेषु जलाशय लोकीः (Dreaming of water bodies, Lotus, water birds like swans, ducks etc.) Do you often see dreams? if ans is No – then skip the question	0	0	1

Psychological Traits

19	Indecisive-ness	अनवस्थित आत्मा/अव्यवस्थितमति: (Indecisiveness often change in decisions) After taking a decision how often you feel the need to change your decisions?	1	0	0
20	Comprehe-nsion	श्रुतग्राही: (Quick comprehension/ Grasping)	1	0	0
		चिरग्राही: (Delayed comprehension) If the participant is able to grasp new information quickly, he/ she may be considered as Shrutagrahi. Else, if his/her grasping is delayed he/she may be considered as Chiragrahi. Tell a short story or passage and ask the person to summarize it. The participant who answers 60 % or above (able to answer 6 or more out of 10) correctly may be taken as Shrutagrahi.	0	0	1
21	Memory	स्मृति मान: (Having good & long-term memory) To assess good memory, ask the person to recall details of a meaningful past event, like a childhood story and evaluate their ability to retrieve vivid, accurate and sequential details. Alternatively, give them a list of 5 single words and 5 paired words to memorize. Test recall after 5 minutes for short-term memory and after 1-2 hours for long-term memory. If they recall 4-5 single words and 3 or more pairs, they likely have good long-term memory.	0	0	1
		अल्प स्मृति/चल स्मृति: (Less memory/Unstable memory)	1	0	0
22	Relationship	मंद मित्र/अदृढ सौहृद्/शीघ्र सौहृद्: (Few friends/ Fickle friendship/ Make friends easily, quickly)	1	0	0
		स्थिर सौहृद् : (Stable cordial friendship)/ Do your relations with your friends / relatives / neighbours last for a long time?	0	0	1

23	Intelligence	निपुण मति (Learning ability)/ मेधावी (Intelligent) A picture puzzle may be presented to a person to identify a set number of differences (e.g., 15 differences) between two otherwise similar images within 3 minutes. If the person successfully identifies 12–15 differences within the time limit, they may be considered to have good learning ability or intelligence. Total triangles are 27, squares are 40 and 5 differences. Alternatively, another task could involve asking the person to count the number of triangles/squares/ image in a given figure. This test can also assess learning ability and attention to detail.   	0	1	0
24	Temperament	क्रोधी/क्षिप्र कोप-क्षिप्र प्रसाद (Short tempered/ Becomes angry quickly & gets pleased quickly) The assessor may ask the individual the following question: "What is your usual reaction if someone breaks a queue in front of you?" A. Ignore it, remain calm or politely request them to step back – overall calm demeanor. B. Start an argument, shout at them or lose your temper – overall short-tempered behaviour. C. Get angry or irritated quickly but calm down shortly afterward – overall volatile temper. (when an answer is B or C then score 1 for pitta) (when an answer is A then score 1 for Kapha) अल्प क्रोधः (Calm & patient, Less anger) Ask the question - What is the opinion of your family members/friends/relatives about you? (calm, short tempered, get angry)	0	1	0
			0	0	1

Behavioural Appearance

25	Nature of Speech/ Talking	वाचाल (talkative or talking irrelevantly) The Assessor will observe the talking style nature of speech of an individual. The nature of speech may be confirmed by friends or associates to check/observe talkativeness or way of manners. The assessor may be asked the following questions. What is your future plan? How much do you serve your parents? How do you judge your personality? etc. If the person seems to be talkative or talking irrelevant, may be considered as talkative & give a score 1 for vata	1	0	0
		मितवाक्, वदान्य/ विनित (Polite /humble) Do you often remain polite soft & submissive even in stressful/unwanted/anxious conditions? If a person speaks limited, mannered way, soft-spoken may be considered as polite/humble. If the answer is yes then score 1 for Kapha	0	0	1
		प्रगल्भ वक्ता (profound orator) Are you often able to talk effectively at length on a particular topic of your choice eg. Pollution, humanity, women power, agriculture, corruption etc? If the answer is yes then score 1 for pitta	0	1	0
		समितिषु दुर्निवारवीर्य (Dominant speaker) Are you often able to dominate during the conversation & establish your views If the answer is yes then score 1 for pitta			
		तीक्ष्ण पराक्रम/शुर/न भयात् प्रणम्याति: (Very brave, valour/ never surrender) Are you often able to stand for your faith /values/morals despite what others say & you face extreme adverse situations and challenges bravely till the goal is achieved? If the answer is yes then score 1 for pitta	0	1	0
26	Likings	शीताभिलाषी (like cold climate & food items)	0	1	0
		उष्ण अन्नपानाकांक्षा: (Desirous of hot food & drinks) What type of food items you like the most in moderate season?	1	0	0

		अतिस्वेदन /विमर्दन सौख्यम् गच्छति: (like sudation & massage)	1	0	0
		यात्राप्रियदयित माल्य विलेपन मंडन (Fond of cosmetics)	0	1	0
		शास्त्रप्रिय (Fond of reading/listening classical texts)	0	0	1
27	Tolerance	हीनसत्त्व/शीघ्रत्रास (Having less tolerances & get frightened easily) How do you usually react when you face a stressful situation?	1	0	0
28	Bravery	तीक्ष्ण पराक्रम/शूर न भयात् प्रणम्यति:। (Very brave/ never surrender in fear) Do you consider yourself someone who stands firm in the face of fear? Why or why not?	0	1	0
29	Affectionate	आश्रित वात्सल्य (affectionate to dependents) How do you typically show affection to those who depend on you?	0	1	0
30	Competitive spirit	भूरिईर्ष्या: (competitive spirit to friends/ relatives/ neighbours) What motivates you to compete with others? Is it the desire to win, improve or be ahead of others?	0	1	0
31	Jealous	मात्सर्यः (Jealous of friends/Relatives /Neighbours) How do you feel when someone close to you achieves something significant, like a promotion, a new car or a special recognition?	1	0	0
32	Egoist	अभिमानि (Having superiority complex) Have you ever dismissed someone's advice or opinions because you felt they were not qualified to give it?	0	1	0
33	Forgiveness & solemnity	गम्भीर व क्षमावान (Stable mind forgiveness to other) Do you believe in giving people second chances? Why or why not?	0	0	1
34	Rigid	अनतेषु/अमृदुः (Harsh towards non-loyal) How do you handle conflicts or disagreements with people you perceive as disloyal?	0	1	0
35	Respectfulness & Humanity	गुरुमानयिता व विनीत (Respect to teachers/ elders & polite/Humble) Scenario based questions- How is your attitude towards house help, waiters, etc?	0	0	1

36	Enmity	दृढवैरः (Having strong enmity) Do you find it difficult to forgive those who have wronged you in the past? Why or why not? Clue - Have you any friends /relatives/colleagues with whom you had differences & never patched up again because you never wanted to patch up or your friends, but you did not accept?	0	0	1
37	Compliment freak	स्तुतिप्रिय (Fond of exaggerated praise) How often do you seek feedback from others on your work? What kind of feedback do you value the most?	0	1	0
38	Modesty	सुलज्जौः (Feel shy while expressing) Do you feel awkward when someone praises you for your achievements?	0	0	1
30	Generosity	परिगन्य चिरात् प्रददातिबहुः (Denotes generosity & Judiciously) Do you feel comfortable saying no to donation requests when necessary?	0	0	1
40	Planner	अभियोगावानः (Systemic & ideal planner) How do you set goals for yourself, both short-term & long-term?	0	1	0
41	Physical activity	व्यायामशीलः (Doing regular physical exercise) How often do you exercise each week?	0	0	1

Result:

Predominance of Doshaj Features: Vataj/35 Pittaj/35 Kaphaj/35

Interpretation of Prakriti Type:

Predominant Dosha: 70% in one dosha / 40-60% in two doshas / 30-35% in three doshas

Interpretation of Doshaj Prakriti Type: Ekdoshaja Prakriti / Dwidoshaja Prakriti / Sama Prakriti



Teacher's Signature

7.3 PRAKRITI ASSESSMENT

Objective : To assess the Prakriti of an individual

Name: :

Age : **Gender:** M/F **Occupation:**

Body temperature : **Weight:**kg

Blood pressure : **Pulse rate:**...../min

Address :

Physical Traits		Prakriti assessment preform	Score		
Sr No	Domain	Feature	Vataj	Pittaj	Kaphaj
Anatomical Traits					
1	Built	अपचित: (Thin built)	1	0	0
		उपचित: (Well built) Decide with the help of BMI (<18.5 = Vata, >25 = Kapha) or Measure the width of the four fingers of the hand together to calculate one finger's breadth in cm. Calculate height as per using self-anguli (finger) measurement, which is 84 angul. Whether any confounder which is affecting the weight of the participant is present or not?	0	0	1
2	Height	ह्रस्वाकृति (Short height)	0	0	1
		दीर्घाकृति (Tall height) Short or tall decide as compared with others. (may considered below 5 feet & above 6 feet)	1	0	0
3	Appearance	बहु कण्डरा सिरा प्रतान: (Prominent tendons and veins)	1	0	0
		सुकुमार गात्र (Delicate body) Are you sensitive to excessive heat/summer season/hot food items especially spicy food items/ harsh medicine?	0	1	0

		सुकुमार गात्र (Delicate body) Whether the person has a delicate appearance?	0	0	1
		महाललाट (Broad forehead) (The forehead is broader than 4 anguli)	0	0	1
4	Skin	Colour			
		गौर (Lotus/straw/gold colour)	0	0	1
		गौर- पीतांग (Fair with yellowish tinge)	0	1	0
		धुसर/कृष्ण (Dusky/dark)	1	0	0
		Texture			
		रूक्षांग / स्फुटितांग: (Dry skin/cracking of skin) (Non lustrous skin)	1	0	0
		प्रसन्न स्निग्धांग: (Unctuous skin/ smooth clear skin without moles, freckles & dryness etc.)	0	0	1
		प्रभुततिलव्यंगपिप्पु: (Presence of moles, pimple, freckles)	0	1	0
		क्षिप्रवली/शीघ्रवली: (Early appearance of wrinkles, may be seen under the eyes first)	0	1	0
		Temperature			
		उष्माङ्गा (Warm feeling on touch at dorsal surface of the hand and forehead)	0	1	0
		अल्पसंताप (Body temperature towards the lower side of normalcy) Ask for the normal temperature of the body or Measure the temperature with the thermometer to decide the range of temperature.	0	0	1
		उष्णमुख: (Tendency for frequent ulceration of mouth) Do you feel hotness in mouth / suffer from ulceration of mouth?)	0	1	0
5	Joints	अनवस्थित सन्धिपाद (Undue frequent movement of different body parts like i.e joints, eyes, eyebrows, jaws, lips, tongue, head, shoulder, hands and legs (sandhi, akshi, bhru, hanu, osth, jihva, shira, skandha, paani and paad - physician's observations)	1	0	0

		सशब्द सन्धिगामि (Joints sound) Is there any sound (at times) on normal movement especially in knee/ ankle /toes or shoulder?	1	0	0
6	Muscle	शिथिल मृदु मांसः (Lax muscles) Observe the muscle tone while keeping in mind confounding factors like ageing.	0	1	0
		उद्भट्ट पिण्डिक (Prominent/hard calf muscles)	1	0	0
7	Eyes	Size			
		तनु लोचनः (Small eyes)	1	0	0
		विशालाक्षः (Big eyes) (after opening the eyes, at the lower portion of the cornea a little part of the white sclera is also visible)	0	0	1
		Appearance/Texture			
		रूक्ष/खर नेत्रः (Dry/rough eyes) (often having feelings of dryness/roughness/irritation/eye fatigue)	1	0	0
		Colour of eyes			
		धुसर नेत्र (Dull white sclera)	1	0	0
		ताम्रनयन (Coppery, Reddish eyes)	0	1	0
		शुक्लाक्ष/व्यक्त सित असिताक्षः (milky white / clearly distinguished sclera) रक्तान्त नेत्र (Reddish canthus)	0	0	1
		Eye lashes			
		अल्पपक्ष्म/तनुपक्ष्मः (Less in number/thin eye lashes)	0	1	0
		पक्ष्मलाक्षः (Thick/dense eye lashes)	0	0	1
		Eye movements			
		चलदृष्टीः (Unsteady gaze) (Unable to hold attention/ frequent movement of eyes observed by assessor)	1	0	0
		Special feature of eyes			
		क्रोधेन/मद्येन/ रवेश्च रागमः (Eyes redden during anger/ after alcohol consumption/after exposure to sunlight)	0	1	0

8	Body hair	Texture			
		परुष/रूक्ष/स्फुटितकेश: (Hard/ dry/spilt hair ends)	1	0	0
		मृदु: (Soft)/ अल्प केश: (Scanty hair)/ कपिल केश: (Reddish brown hair)	0	1	0
		स्निग्ध: (Unctuous, oily hair)	0	0	1
		Appearance			
		कुटिल केश: (Curly hair)/ घन केश (Dense hair) श्याम केश:(Black hair)/	0	0	1
		कपिल केश: (Reddish brown hair)	0	1	0
		धूसर केश: (Dusky hair)	1	0	0
		Tendency of hair fall & Greying			
		अकाल पलित: (Premature greying) or शीघ्र खालित्य: (Early hair loss/baldness) Whether the person is having grey hair (approx. more than 20%) before the age of 35 years? Whether the person has baldness/is facing any significant hair fall (more than 50% before the age of 50 years)?	0	1	0
9	Teeth	Texture			
		रूक्ष दन्त/परुष दशन: (Dry teeth /rough teeth)	1	0	0
		Size			
		सूक्ष्म दन्त/अति दन्त: (Small teeth/big teeth)	1	0	0
10	Nails	Size			
		अल्प नख/तनु नख:(Small nails/thin nails)	1	0	0
		दीर्घ नख:(Long nails)	0	0	1
		Texture			
		रूक्ष नख/परुष नख/स्तब्ध नख (Dry nails/rough nails)	1	0	0
		Colour			
		ताम्र नख:(Coppery coloured nails)	0	1	0
		Nature			
		नख: वृद्धि (Excessive nail growth)	1	0	0

Physiological Traits

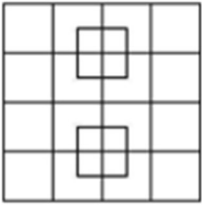
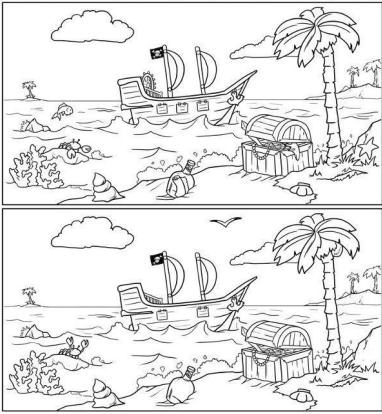
11	Locomotion	Gait and Locomotion			
		द्रुतगति/चपलगति/लघुगति: Speedy/ quick, Swift gait/Small steps The person walks very fast or he/she starts walking instantly/ immediately?	1	0	0
		सारगति/अधिष्ठित गति: (Consistent /straight unswerving way) Complete way i.e. heel strike first followed by toes touching the ground completely.	0	0	1
		Initiation of Action			
		लघु चेष्टा/शीघ्र समारम्भः:(Quick initiation of action) If a task is given to you do you have a tendency to initiate it immediately?	1	0	0
		मंद चेष्टा/अशीघ्रारम्भः:(Delayed initiation)	0	0	1
12	Speech/ Voice	Voice Quality			
		सन्न स्वर/रूक्ष स्वर/जर्जर स्वर/प्रतत स्वर/बद्ध स्वर/चलवाच/ स्तब्ध स्वर/तनुस्वर: The assessor has to analyse the voice quality as per Feeble voice/ Unpleasant/Shattered, Broken voice/ Unclear speech / Stammering/ Fast, shaking voice/Hoarseness of voice/Low speech voice. Whether any confounder which is affecting the voice of the participant is present or not?	1	0	0
		प्रसन्न स्वर/गंभीर स्निग्ध स्वर (Clear voice/Deep, Pleasant voice)	0	0	1
13	Appetite	Quantity & Frequency			
		शीघ्रक्षुत/दन्दशुक : (Frequent intake of more food) प्रभुताशन (Excessive intake/ Excess Quantity)	0	1	0
		अल्पक्षुत/अल्पअशन: (Less hunger/Less intake, less frequency)	0	0	1

		How many times in a day do you take main meals and how many times refreshments? 2 main meals and 3 or more refreshments or 3 main meals and 2 or more refreshments or None of the above			
		बहुभुज/अनल्पाशन (Frequent but less intake) How much quantity of food you take in a day? Less than average (4 chapati, little rice, 1 bowl dal etc.) or Average (6-10 chapattis, rice, dal, sabji, fruits, salads etc.) or More than average	1	0	0
		Tolerance to hunger			
		क्षुत सहिष्णु (Can tolerate hunger)	0	0	1
		तीक्ष्ण बुभुक्षा:(Intense feeling of hunger) Suppose you skip 1 meal do you tolerate it easily? Can tolerate (Kapha) or Cannot tolerate (Pitta) or It varies (None)	0	1	0
		Eating habit			
		लघु आहार/चपल आहार: (Fast eating habit)	1	0	0
14	Thirst	मंद आहार: (Slow eating habit) Suppose you are having lunch with your friends / family members. Usually you finish your food - Before your friends / family members (Vata) or At par / with your friends / family members (Kapha)	0	0	1
		Quantity & Intensity			
		शीघ्र पिपासा: (Frequently thirsty) How frequently you take water/fluids in a day? (<4 or 4-7 or >7 times/day)	0	1	0
		प्रभूतपान/तीक्ष्ण तृष्णा: (Drink more water/Intense thirst)	0	1	0
		अल्प पान/अल्पतृष्णा: (Drink less water/Less thirst) When you feel thirsty and water is not instantly available how you react? Immediately search for water or wait for sometime. How much water / liquids do you take per day approximately? (<1 or 1-2 or >2 lit/day)	0	0	1

15	Stool	प्रभूत-सृष्ट पुरीषः मूत्रः (Easy & satisfactory bowel evacuation) (Easy & adequate flow of urine) The quantity of stool passed by you in comparison to food intake is: Adequate/ Bulky or Less Do you usually have easy, quick and effortless bowel evacuation without constipation or use of any laxative?	0	1	0
16	Sweat	प्रभूत.सृष्ट स्वेदः (Profuse sweating)	0	1	0
		अल्प स्वेदः (Scanty sweating) How much do you sweat on exposure to sun rays / exercise / physical activity? Profuse (Pitta) Scanty (Kapha)	0	0	1
		कक्ष-आस्य-शिर प्रभूत पूतिगंध/स्वेदोदुर्गन्धः (Putrid, foul odour especially from axilla, scalp, mouth & head / Bad odour of sweat) Have any of your family members / friends ever told you about your strong body odour?	0	1	0
17	Sleep	जागरूक/अल्पनिद्रा (Easy awakening) (Less sleep) While sleeping are you awakened by a little noise? How many hours do you sleep in 24 hours? (<6 or 7-8 or > 8 hr.)	1	0	0
		निद्रा प्रियः (Like sleeping) Whether you like to sleep if given a chance or in free time?	0	0	1
18	Nature Dreams	आकाशचारी स्वप्नेषुः (Dreaming of Flying in the sky, walking mountains, dried water bodies, trees etc.)	1	0	0
		स्वप्नेषु ज्योतिषां दृष्टाः (Dreaming of falling of meteors, fire, Lightning, Gold etc.)	0	1	0
		स्वप्नेषु जलाशय लोकीः (Dreaming of water bodies, Lotus, water birds like swans, ducks etc.) Do you often see dreams? if ans is No – then skip the question	0	0	1

Psychological Traits

19	Indecisive-ness	अनवस्थित आत्मा/अव्यवस्थितमति: (Indecisiveness often change in decisions) After taking a decision how often you feel the need to change your decisions?	1	0	0
20	Comprehe-nsion	श्रुतग्राही: (Quick comprehension/ Grasping)	1	0	0
		चिरग्राही: (Delayed comprehension) If the participant is able to grasp new information quickly, he/ she may be considered as Shrutagrahi. Else, if his/her grasping is delayed he/she may be considered as Chiragrahi. Tell a short story or passage and ask the person to summarize it. The participant who answers 60 % or above (able to answer 6 or more out of 10) correctly may be taken as Shrutagrahi.	0	0	1
21	Memory	स्मृति मान: (Having good & long-term memory) To assess good memory, ask the person to recall details of a meaningful past event, like a childhood story and evaluate their ability to retrieve vivid, accurate and sequential details. Alternatively, give them a list of 5 single words and 5 paired words to memorize. Test recall after 5 minutes for short-term memory and after 1-2 hours for long-term memory. If they recall 4-5 single words and 3 or more pairs, they likely have good long-term memory.	0	0	1
		अल्प स्मृति/चल स्मृति: (Less memory/Unstable memory)	1	0	0
22	Relationship	मंद मित्र/अदृढ सौहृद्/शीघ्र सौहृद्: (Few friends/ Fickle friendship/ Make friends easily, quickly)	1	0	0
		स्थिर सौहृद् : (Stable cordial friendship)/ Do your relations with your friends / relatives / neighbours last for a long time?	0	0	1

23	Intelligence	निपुण मति (Learning ability)/ मेधावी (Intelligent) A picture puzzle may be presented to a person to identify a set number of differences (e.g., 15 differences) between two otherwise similar images within 3 minutes. If the person successfully identifies 12–15 differences within the time limit, they may be considered to have good learning ability or intelligence. Total triangles are 27, squares are 40 and 5 differences. Alternatively, another task could involve asking the person to count the number of triangles/squares/ image in a given figure. This test can also assess learning ability and attention to detail.   	0	1	0
24	Temperament	क्रोधी/क्षिप्र कोप-क्षिप्र प्रसाद (Short tempered/ Becomes angry quickly & gets pleased quickly) The assessor may ask the individual the following question: "What is your usual reaction if someone breaks a queue in front of you?" A. Ignore it, remain calm or politely request them to step back – overall calm demeanor. B. Start an argument, shout at them or lose your temper – overall short-tempered behaviour. C. Get angry or irritated quickly but calm down shortly afterward – overall volatile temper. (when an answer is B or C then score 1 for pitta) (when an answer is A then score 1 for Kapha) अल्प क्रोधः (Calm & patient, Less anger) Ask the question - What is the opinion of your family members/friends/relatives about you? (calm, short tempered, get angry)	0	1	0
			0	0	1

Behavioural Appearance

25	Nature of Speech/ Talking	वाचाल (talkative or talking irrelevantly) The Assessor will observe the talking style nature of speech of an individual. The nature of speech may be confirmed by friends or associates to check/observe talkativeness or way of manners. The assessor may be asked the following questions. What is your future plan? How much do you serve your parents? How do you judge your personality? etc. If the person seems to be talkative or talking irrelevant, may be considered as talkative & give a score 1 for vata	1	0	0
		मितवाक्, वदान्य/ विनित (Polite /humble) Do you often remain polite soft & submissive even in stressful/unwanted/anxious conditions? If a person speaks limited, mannered way, soft-spoken may be considered as polite/humble. If the answer is yes then score 1 for Kapha	0	0	1
		प्रगल्भ वक्ता (profound orator) Are you often able to talk effectively at length on a particular topic of your choice eg. Pollution, humanity, women power, agriculture, corruption etc? If the answer is yes then score 1 for pitta	0	1	0
		समितिषु दुर्निवारवीर्य (Dominant speaker) Are you often able to dominate during the conversation & establish your views If the answer is yes then score 1 for pitta			
		तीक्ष्ण पराक्रम/शुर/न भयात् प्रणम्याति: (Very brave, valour/ never surrender) Are you often able to stand for your faith /values/morals despite what others say & you face extreme adverse situations and challenges bravely till the goal is achieved? If the answer is yes then score 1 for pitta	0	1	0
26	Likings	शीताभिलाषी (like cold climate & food items)	0	1	0
		उष्ण अन्नपानाकांक्षा: (Desirous of hot food & drinks) What type of food items you like the most in moderate season?	1	0	0

		अतिस्वेदन /विमर्दनेन सौख्यम् गच्छति: (like sudation & massage)	1	0	0
		यात्राप्रियदयित माल्य विलेपन मंडन (Fond of cosmetics)	0	1	0
		शास्त्रप्रिय (Fond of reading/listening classical texts)	0	0	1
27	Tolerance	हीनसत्त्व/शीघ्रत्रास (Having less tolerances & get frightened easily) How do you usually react when you face a stressful situation?	1	0	0
28	Bravery	तीक्ष्ण पराक्रम/शूर न भयात् प्रणम्यति:। (Very brave/ never surrender in fear) Do you consider yourself someone who stands firm in the face of fear? Why or why not?	0	1	0
29	Affectionate	आश्रित वात्सल्य (affectionate to dependents) How do you typically show affection to those who depend on you?	0	1	0
30	Competitive spirit	भूरिईर्ष्या: (competitive spirit to friends/ relatives/ neighbours) What motivates you to compete with others? Is it the desire to win, improve or be ahead of others?	0	1	0
31	Jealous	मात्सर्यः (Jealous of friends/Relatives /Neighbours) How do you feel when someone close to you achieves something significant, like a promotion, a new car or a special recognition?	1	0	0
32	Egoist	अभिमानि (Having superiority complex) Have you ever dismissed someone's advice or opinions because you felt they were not qualified to give it?	0	1	0
33	Forgiveness & solemnity	गम्भीर व क्षमावान (Stable mind forgiveness to other) Do you believe in giving people second chances? Why or why not?	0	0	1
34	Rigid	अनतेषु/अमृदुः (Harsh towards non-loyal) How do you handle conflicts or disagreements with people you perceive as disloyal?	0	1	0
35	Respectfulness & Humanity	गुरुमानयिता व विनीत (Respect to teachers/ elders & polite/Humble) Scenario based questions- How is your attitude towards house help, waiters, etc?	0	0	1

36	Enmity	दृढवैरः (Having strong enmity) Do you find it difficult to forgive those who have wronged you in the past? Why or why not? Clue - Have you any friends /relatives/colleagues with whom you had differences & never patched up again because you never wanted to patch up or your friends, but you did not accept?	0	0	1
37	Compliment freak	स्तुतिप्रिय (Fond of exaggerated praise) How often do you seek feedback from others on your work? What kind of feedback do you value the most?	0	1	0
38	Modesty	सुलज्जौः (Feel shy while expressing) Do you feel awkward when someone praises you for your achievements?	0	0	1
30	Generosity	परिगन्य चिरात् प्रददातिबहुः (Denotes generosity & Judiciously) Do you feel comfortable saying no to donation requests when necessary?	0	0	1
40	Planner	अभियोगावानः (Systemic & ideal planner) How do you set goals for yourself, both short-term & long-term?	0	1	0
41	Physical activity	व्यायामशीलः (Doing regular physical exercise) How often do you exercise each week?	0	0	1

Result:

Predominance of Doshaj Features: Vataj/35 Pittaj/35 Kaphaj/35

Interpretation of Prakriti Type:

Predominant Dosha: 70% in one dosha / 40-60% in two doshas / 30-35% in three doshas

Interpretation of Doshaj Prakriti Type: Ekdoshaja Prakriti / Dwidoshaja Prakriti / Sama Prakriti



Teacher's Signature

EXERCISE

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OSPE-1

Title: Assessment of Prakriti Parikshana

Learning Objectives:

To understand the concept of Prakriti in Ayurveda.

To accurately assess an individual's Prakriti using a structured approach.

Station 1: Assessment of Physical Characteristics

Objective: Evaluate physical features related to Prakriti types (Vata, Pitta, Kapha).

Task: Observe and note physical traits such as body build, skin texture and hair quality.

Checklist: Body structure (Ectomorphic, Mesomorphic, Endomorphic), Skin texture (Dry, Oily, Normal), Hair type (Thin, Thick, Medium) etc.

Marks Allocation: 10 marks (2 marks for each parameter)

Station 2: Mental Characteristics Analysis

Objective: Identify mental traits linked to Prakriti.

Task: Analyze emotional responses and cognitive tendencies.

Checklist: Emotional tendencies (Calm, Irritable, Stable), Decision-making speed (Quick, Moderate, Slow), Memory retention capacity etc

Marks Allocation: 10 marks (2 marks for each parameter)

Station 3: Prakriti Determination Questionnaire

Objective: Assess Prakriti through structured questions.

Task: Fill out a questionnaire designed to identify dominant doshas.

Checklist: Completeness of responses, Accuracy of identified dominant dosha etc

Marks Allocation: 10 marks (5 marks for accuracy, 5 marks for completion)

Station 4: Pulse Examination (Nadi Pariksha)

Objective: Correlate Nadi Pariksha findings with Prakriti.

Task: Perform pulse diagnosis and interpret findings.

Checklist: Identification of pulse characteristics, Correlation with Prakriti type

Marks Allocation: 5 marks (for technique and interpretation)

Station 5: Integration and Report Preparation

Objective: Synthesize observations into a cohesive Prakriti assessment.

Task: Prepare a brief report summarizing findings from all stations.

Checklist: Accuracy of summary. Logical integration of findings, Clarity of the report

Marks Allocation: 5 marks

Total Marks: 40

Assessment Notes: Ensure each station is adequately equipped with necessary materials (e.g., questionnaires, pulse examination tools). Instructors should provide clear instructions at each station. Students must adhere to time limits set for each station.

(Note: The teacher may allocate different marks schemes as per different stations, questions and allowed time.)

CASE-BASED LEARNING

Case-Based Learning (CBL) Example: Prakriti Assessment (Parikshana)

Case Title: Understanding Prakriti Through Clinical Features

Case Scenario:

A 35-year-old male presents to the Ayurvedic clinic with complaints of excessive sweating, frequent acidity and irritability. He describes his body as moderately built with a warm complexion and reddish tinge to his skin. He prefers cool and bland foods, avoids spicy meals and has a sharp focus at work but tends to lose his temper easily. He sleeps lightly and wakes up early, feeling refreshed.

Learning Objectives:

1. To apply Ayurvedic principles in identifying the Prakriti of the patient.
2. To recognize the clinical features associated with Pitta, Vata and Kapha Doshas.
3. To recommend lifestyle modifications and dietary plans based on the patient's Prakriti.

Process:

1. Case Presentation to Participants: Key features:

- Warm body temperature, reddish skin tone.
- Excessive sweating, frequent acidity.
- Sharp focus, irritability and light sleep.
- Preference for cool, bland foods.

2. Prakriti Assessment:

Participants are provided with a Prakriti questionnaire to evaluate the patient's physical, physiological and psychological characteristics.

Physical Features:

Moderate body frame: Pitta characteristic. Warm skin with reddish hue: Pitta dominance. Excessive sweating: Pitta characteristic.

Physiological Features:

Frequent acidity: Pitta feature. Light sleep, refreshed awakening: Pitta feature with possible Vata influence.

Psychological Features:

Sharp focus, good decision-making: Pitta trait. Irritability: Pitta characteristic.

3. Discussion of Findings:

Dominant Pitta Prakriti: Explains the warm body, reddish skin tone, sweating, acidity, sharp focus and irritability.

Secondary Vata influence: Light sleep.

Minimal Kapha traits observed in the case.

4. Recommendations Based on Prakriti:

Dietary Suggestions:

Increase intake of cool, sweet and bland foods like milk, cucumbers and coconut water.

Avoid spicy, oily and fried foods.

Stay hydrated with room-temperature water.

Lifestyle Modifications:

Practice calming activities like meditation or walking in nature.

Avoid direct sun exposure for prolonged periods.

Schedule regular meals to prevent acidity.

Behavioural Suggestions:

Manage irritability through mindfulness and deep-breathing exercises.

Establish a consistent sleep schedule.

5. Questions for Participants:

1. How do the patient's features align with Pitta Prakriti?
2. What factors could aggravate Pitta in this individual?

3. How would lifestyle and dietary adjustments restore balance in this case?

Assessment of Learning:

- Participants articulate the relationship between the patient's symptoms and Pitta dominance.
- They provide practical, evidence-based recommendations for balance.
- They demonstrate a thorough understanding of Prakriti assessment principles.

Conclusion:

This case reinforces the importance of Prakriti examination in understanding individual health profiles and tailoring preventive and therapeutic measures. The participants learn to systematically apply Ayurvedic concepts in a real-world scenario.

PROBLEM-BASED LEARNING

Title: Exploring the Role of Prakriti in Personalized Healthcare

Scenario:

A new wellness clinic in a tribal area is facing challenges in tailoring treatments for its diverse population. The clinic's team includes a mix of modern medical practitioners and Ayurveda specialists. They are observing significant differences in how individuals respond to treatments, leading to questions about the role of Prakriti in health and disease.

Problem Statement:

You are a team of healthcare students tasked with developing a framework for personalized healthcare interventions based on Prakriti analysis to improve the health outcomes of the clinic's patients.

Learning Objectives:

1. Understand the concept of Prakriti in Ayurveda.
2. Differentiate between Vata, Pitta and Kapha Prakritis and their implications in health.

3. Identify modern scientific evidence linking Prakriti to genomics, metabolism and disease susceptibility.
4. Develop practical strategies for applying Prakriti analysis in clinical settings.

Steps in the PBL Process:

Step 1: Identify What You Know

Students discuss and list their current knowledge of Prakriti. Example:

- Prakriti is the individual constitution determined at birth.
- There are three primary Prakritis: Vata, Pitta and Kapha.
- Prakriti affects susceptibility to diseases and response to treatments.

Step 2: Define Learning Gaps

What more do you need to learn about Prakriti?

Possible gaps:

- Methods to assess Prakriti.
- The role of diet and lifestyle in maintaining balance for each Prakriti.
- Modern research validating Ayurvedic concepts of Prakriti.

Step 3: Research and Gather Information

Students divide into groups to research:

1. Classical Ayurvedic texts describing Prakriti.
2. Modern studies correlating Prakriti with genetics and metabolic profiles.
3. Case studies showing practical applications of Prakriti-based treatment.

Application Task:

Case Study: A 35-year-old male patient visits the clinic with chronic indigestion, irritability, & sleep disturbances. He works hours in a high-stress job.

His Ayurveda assessment reveals a predominant Pitta Prakriti.

Based on this, students are tasked with:

1. Identifying dietary and lifestyle recommendations for balancing Pitta.
2. Suggesting basic drugs or therapies for Pitta-related symptoms.
3. Justifying their recommendations using both traditional and modern scientific evidence.

Presentation and Feedback:

Each group presents their solutions to the clinic's challenges, highlighting:

- How they assessed Prakriti in the patient.
- The rationale behind their intervention plan.
- Potential challenges in implementation and strategies to overcome them.

Evaluation Criteria:

1. Accuracy of Prakriti analysis.
2. Integration of traditional Ayurvedic principles with modern scientific insights.
3. Creativity and feasibility of solutions for personalized healthcare.

Expected Outcomes:

- Enhanced understanding of Prakriti's role in individualized medicine.
- Practical experience in bridging Ayurveda with modern healthcare.
- Greater appreciation of the relevance of ancient knowledge in solving contemporary health problems.

CASE-BASED LEARNING

Case Study:

A 35-year-old male presents with complaints of frequent fatigue, dry skin and poor appetite. He mentions he often feels cold and has unsteady joints.

Problem:

Identify the patient's predominant Prakriti type and suggest suitable lifestyle and dietary modifications.

Guiding Questions:

- Which physical traits align with each dosha?
- How does unsteady movement indicate Vata dominance?
- What dietary recommendations would benefit a Vata-dominant individual?

Discussion and Resolution:

- Why is it important to balance Vata in this individual?
- What factors might exacerbate Vata in the patient's lifestyle?

Discussion and Resolution: Through guided discussion, students will identify Vata-dominant traits and recommend remedies such as warm, oily foods and regular routines to balance Vata.

Assessment: Evaluate students on:

- Ability to identify key traits of doshas.
- Formulating practical lifestyle and dietary advice.
- Overall understanding of Prakriti-based care.

8. DOSHA VRIDDHI-KSHAYA ASSESSMENT FORM

LEARNING OBJECTIVES

After completion of the practical, students should be able to:

- Understand the concept of Dosha Vriddhi (aggravation) and Kshaya (decrease) in Ayurveda.
- Learn to identify the features of Dosha imbalance through clinical observation and questioning.
- Develop skills in analyzing and scoring the severity of Dosha features based on responses.

PRINCIPLE

The concept of Dosha Vriddhi and Kshaya forms the cornerstone of Ayurvedic diagnosis, linking the physiological and pathological states of an individual with their Prakriti and Vikriti (current state). Accurate assessment aids in tailoring personalized dietary and therapeutic interventions.

PURPOSE

To Identify and score features of Dosha Vriddhi/Kshaya in individuals.

To Interpret the scores for clinical decision-making.

INSTRUCTIONS

For assessment of dosha vriddhi/kshaya, the following proforma contains 14 questions for each dosha vriddhi and 9 questions for each dosha kshaya. Questions were framed on the basis of features mentioned in the shlokas of respective dosha vriddhi/kshaya. Students have to ask all 14 questions to assess dosha vriddhi and 9 questions to assess dosha kshaya to the candidate. According to the severity of responses given by the candidate, students have to write the score in front of each question. For rating the severity of the assessed features, 4 categories are mentioned as Normal, Mild, Moderate and Severe.

Score 0 for no features present (Normal)

Score 1 for the presence of Mild features

Score 2 for the presence of Moderate features

Score 3 for the presence of Severe features

At the end of assessing each dosha vriddhi/kshaya, students have to calculate the scores.

PROCEDURE

1. Introduce yourself and explain the purpose of the assessment to the candidate.
2. Record basic details such as name, age, gender, occupation, body temperature, weight, etc
3. Ask the candidate the specific questions listed in the proforma & assessed by दर्शन (Inspection), स्पर्शन (Palpation, Percussion, Auscultation), प्रश्न (Interrogation) using advanced technology
4. Record their responses and Sum the scores for each Dosha to determine the level of Vriddhi or Kshaya.
5. Total the scores and categorize them as Normal, Mild, Moderate or Severe Vriddhi or Kshaya.

INTERPRETATION

दोष वृद्धि:	0–10 - Normal
	11–21 - Mild Dosha Kshaya
	22–32 - Moderate Dosha Kshaya
	33–42 - Severe Dosha Kshaya

दोष क्षय:	0–6 - Normal
	7–13 - Mild Dosha Vriddhi
	14–20 - Moderate Dosha Vriddhi
	21–27 - Severe Dosha Vriddhi

PRECAUTION

Ensure the patient is comfortable and understands the questions. Avoid leading questions that might bias the responses. Maintain a non-judgmental and professional demeanor.

वातदोष

■ वृद्धि

- वृद्धस्तु कुरुतेऽनिलः कार्श्यकाष्ण्योष्णकामित्व
कम्पाऽऽनाहशकृद्ग्रहान।बलनिद्रेन्द्रियभ्रंशप्रलापभ्रमदीनताः॥
- अ.ह.सू.11/6
- तत्र, वातवृद्धौ वाकपारुष्यं कार्श्यं काष्ण्यं गात्रस्फुरणं।
उष्णकामिता निद्रानाशोऽल्पबलत्वं गाढवर्चस्त्वं च॥
- सु.सू.15/13
- कार्श्यकाष्ण्यगात्रकम्पस्फुरणोष्णकामितासंज्ञानिद्रानाशबले-
न्द्रियोपघातास्थि शूल मज्जाशोष मलसङ्गाध्मानाटोपमोहदैत्य
भयशोक प्रलापादिभिर्वृद्धो वायुः पीडयति॥ - अ.सं.19/6

■ क्षय

- लिङ्गं क्षीणेऽनिलेऽङ्गस्य सादोऽल्पं भाषितेहितम्।
संज्ञामोहस्तथा श्लेष्मवृद्धयुक्तामयसम्भवः।
- अ.ह.सू.11/15
- वातक्षये मन्दचेष्टताऽल्पवाक्त्वमप्रहर्षो मूढसंज्ञता च॥
- सु.सू.15/7
- प्रसेकारुचिहृल्लासंज्ञामोहाल्पवाक्चेष्टता।
प्रहर्षाङ्गसादाग्रिवैषम्यादिभःक्षीणो वायुः पीडयति॥
- अ.सं.सू.11/6

पित्तदोष

■ वृद्धि

- पित्तवृद्धौ पीतावभासता सन्तापः शीतकामित्वमल्पनिद्रता मूर्च्छा
बलहानिरिन्द्रियदौर्बल्यं पीतविष्णुमूत्रनेत्रत्वं च।
- सु.सू.15/13
- पीतत्वग्लानिन्द्रियदौर्बल्यौजोविस्त्रंसशीताभिलाषदाहतिक्तास्यता
तृष्णमूर्च्छाऽल्प निद्राक्रोधादिभिः पित्तम्।
- अं.सं.सू. 19/6
- पीतविष्णुमूत्रनेत्रत्वक्क्षुत्तृड्दाहाऽल्पनिद्रताः पित्तम्।
- अ.ह.सू. 11/7

■ क्षय

- पित्तक्षये मन्दोष्माग्रिता निष्प्रभत्वं च। - सु.सू.15/7
स्तम्भ शैत्यानियततोददाहारोचकाविपाकाङ्गपारुष्य॥
- कंपगौरवनखनयन शौक्यादिभिः पित्तम् - अ.सं.सू.19/9
- पित्ते मन्दोऽनलः शीतं प्रभाहानिः। - अ.ह.सू.11/16

कफदोष

■ वृद्धि

- श्लेष्माऽग्निसदनप्रसेकालस्यगौरवम्।
श्वैत्यशैत्यश्लथाङ्गत्वं श्वासकासातिनिद्रताः।-अ.ह.सू.11/7
- श्लेष्मवृद्धौ शौक्ल्यं शैत्यं स्थैर्यं गौरवमवसादस्तन्द्रा निद्रा
सन्ध्यास्थिविश्लेषश्च। - सु.सू.15/13
- श्वैत्यशैत्यस्थौल्यालस्यगौरवाङ्गसादस्त्रोतःपिधानमूर्च्छा
निद्रातन्द्राश्वास कासप्रसेकहृल्लासाग्निसादसन्धिविश्लेषादिभिः
श्लेष्माः। - अ.सं.सू.19/6

■ क्षय

- श्लेष्मक्षये रुक्षताऽन्तर्दाह आमाशयेतरश्लेष्माशयशून्यता
सन्धिशैथिल्यं तृष्णा दौर्बल्यं प्रजागरणं च।
- सु.सू.15/7
- भ्रमोद्वेष्टनानिद्राङ्गमर्दपरिशोषतोददवदाहस्फोटनवेपनधूमायन
सन्धिशैथिल्यहृदयद्रवश्लेष्माशयशून्यतादिभिः श्लेष्मा।
- अ.सं.सू.19/9
- कफे भ्रमः। श्लेष्माशयानां शून्यत्वं हृद्द्रवः श्लथसन्धिता॥
- अ.ह.सू.11/16

GUIDELINES FOR DOSHA VRIDDHI-KSHAYA ASSESSMENT

To effectively assess Dosha Vriddhi (increase) and Dosha Kshaya (decrease), students should follow the structured steps provided in the proforma as a guideline. The proforma serves as a guideline for structured evaluation. This framework serves as a foundation and can be expanded beyond the given list by referring to research papers, classical texts and teacher insights. First-year BAMS students are advised to grade scores based on their current knowledge and understanding. Typically, simple examinations or question-based assessments are sufficient for their level. Where access to advanced instruments, equipment, blood tests or diagnostic tools is available, these can be utilized. Otherwise, students should focus on finding answers of questions using basic tools and techniques. This approach encourages practical learning while aligning with current resources and level of expertise.

1. Observations (दर्शन)

Observe the patient's physical appearance, posture, skin tone and expressions etc. Look for visible signs of dosha imbalance, such as dryness, discolouration, swelling etc. Document general physical traits, including gait, body language, facial cues etc.

2. Functional Assessment by Palpation, Percussion & Auscultation (स्पर्शन)

Examine areas for tenderness, stiffness, or swelling. Assess joint mobility and muscular strength. Feel for temperature variations such as heat or coldness that may indicate dosha imbalance through Palpation. Tap gently to detect abnormalities in underlying tissues or organs, such as fluid accumulation or structural changes through Percussion. Listen to heart sounds for irregularities or murmurs. Observe respiratory patterns and sounds. Assess for any abnormal sounds in the gastrointestinal tract through Auscultation.

3. Interrogation (प्रश्न)

Ask Detailed Questions: Onset, duration, and nature of symptoms (e.g., frequency, intensity). Associated symptoms like fatigue, pain, or digestive issues. Patient's dietary habits, lifestyle, and emotional state. Understand any pre-existing/current conditions, diseases, treatments or hereditary factors through medical history.

4. Exploring Features/Symptoms Using the Proforma

Guideline for assessment given for each feature or symptom listed in the proforma. Use the questionnaire to evaluate symptoms systematically:

5. Diagnostic Tools - Use modern tools to complement your assessment

Laboratory Tests: For blood, urine or stool analysis to identify systemic imbalances.

Imaging Studies: Ultrasound, X-ray, CT, MRI for structural or functional abnormalities.

Specialized Tools: Pulse diagnosis (Nadi Pariksha), Ayurvedic dosha-specific diagnostic methods.

Questionnaire based tool etc

Features of Aggravated Vata Dosha (वात वृद्धी लक्षण)

Sr. No	Features and Questions	Assessment by (दर्शन, स्पर्शन, प्रश्नै)	Score
1	कार्श्य (Muscle wasting) Observed weight loss or reduced muscle mass?	Muscle wasting is assessed by observing reduced muscle mass, weakness or asymmetry and palpating for atrophy or tenderness. Conditions like malnutrition, nerve damage or muscular dystrophy anorexia nervosa, starvation, LMN (Lower Motor Neuron) lesions or hyperthyroidism are diagnosed via imaging, EMG or blood tests.	
2	काष्ण्य (Hyperpigmentation) Darker or patchy skin?	Hyperpigmentation is assessed by observing skin discolouration patterns and distribution, checking for associated lesions and exploring symptom duration, triggers or sun exposure. It may indicate melasma, Addison's disease or post-inflammatory changes, diagnosed through skin biopsy and hormonal tests, spectrophotometry, tristimulus colourimetry, Wood's lamp Examination	
3	उष्ण कामित्व (Desire for heat) Do you feel a craving for warmth or aversion to cold?	Desire for heat is assessed by observing behaviours like seeking warmth, examining skin temperature or circulation issues. It may indicate hypothyroidism, anaemia laryngitis or pharyngitis or Raynaud's disease, diagnosed using blood tests, thermal imaging, body temperature, BSL, Thyroid and Hormone Test	
4	कंप (Tremor) Do you experience shaking or tremors?	Tremor seen in anxiety nervosa, cerebellar disease, parkinson's or neurological/psychological reasons, Anxiety, Cerebellar Diseases, hyperthyroidism. Examining muscle tone and coordination. Diagnosed through neurological exams, imaging and lab tests, EEG, Electromyography (EMG), LFT, KFT	
5	आनाह (Abdominal distension) Is there frequent bloating or fullness in the abdomen?	Abdominal distension is evaluated by observing visible swelling, assessing tenderness, rigidity. Often linked to gas or fluid accumulation, indigestion, lactose intolerance, gastroesophageal reflux disease, irritable bowel syndrome, ascites, obstruction, during menstrual cycles, grahani, ajirn, vataj gurm. Diagnosed using percussion, imaging, endoscopy and lab tests.	
6	शकृतग्रह (Constipation) Have you had trouble with bowel movements?	Constipation is evaluated by observing abdominal bloating or discomfort, assessing tenderness or hard stool presence. Constipation seen old age, due to bad dietary habits, irritable bowel syndrome, intestinal obstruction, intake of drugs like opium, cannabis, iron, hypothyroidism. Diagnosed by X-ray,	

		sigmoidoscopy, colonoscopy, anorectal manometry, blood tests.	
7	बलभ्रंश (Loss of strength) Feeling unusually fatigued or weak?	Evaluating muscle tone and reflexes. It may indicate conditions like neuropathy, myopathy, anaemia, electrolyte imbalance. Investigate for harward step test, muscle tone, muscle power, hand grip test, Fatigue index, electromyography, blood tests, imaging,	
8	निद्रा भ्रंश (Loss of sleep) Difficulty sleeping or insomnia?	Loss of sleep is assessed by observing fatigue, irritability or dark under-eye circles. Evaluating stress levels or physical discomfort and exploring triggers like lifestyle habits or emotional distress. It may indicate insomnia, anxiety or sleep apnea, diagnosed via polysomnography or psychological evaluation.	
9	इन्द्रिय भ्रंश (Debility of sense organs) Hearing, vision or other sensory impairments?	Debility of sense organs is assessed by observing functional impairments, testing sensory responses. Conditions like neuropathy, stroke or vitamin deficiencies are evaluated using neurological exams, imaging, blood tests, video nystagmography, rotary-chair testing, EEG, use questionnaire, polysomnography etc.	
10	प्रलाप (Delirium) Random, incoherent speech or confusion?	Delirium is assessed by observing confusion, agitation or disorientation and evaluating cognitive function. Questions focus on the onset, associated symptoms and triggers. It may occur in conditions like infections, drug withdrawal, electrolyte imbalances, metabolic imbalances, stroke, traumatic brain injury, degenerative neurological dementia, autism, vataj unmad etc. Clinical testing of indriyas/sense organs for general sensations like touch, proprioception, pain, temperature and imaging.	
11	भ्रम (Loss of consciousness) Have you fainted recently?	Evaluated by observing the patient's responsiveness and posture. It may indicate during vasovagal episodes. cervical spondylosis, head injury, labyrinthitis, cardiovascular issues, diagnosed through imaging, EEG, neurological assessments. electronystagmography (ENG) blood tests, videonystagmography (VNG), rotation test, MRI	
12	दिनता (Miserable feelings) Persistent feelings of sadness or depression?	Miserable feelings are assessed through observing mood changes, physical signs of distress and exploring emotional triggers. It may indicate conditions like depression, anxiety, stress, Cancer, iron deficiency anaemia. Diagnosis involves psychological evaluation, screenings for mental health conditions and possibly neuroimaging, Hb level, Vitamin B ₆ , B ₁₂ , ECG, CT Scan, MRI Scan, use depression/Anxiety rating Scale etc.	

13	अस्थिशूल (Bone pain) Pain in bones, especially during fever?	Common in. Bone pain is assessed by observing localized swelling, deformities or tenderness. Questioning focuses on onset, duration and associated symptoms like fever, dengue, chikungunya, weight loss. Conditions like osteoporosis, arthritis, infections may be diagnosed through imaging, blood tests, bone scans.	
14	भय (Fear or anxiety) Experiencing excessive fear or anxiety?	Fear or anxiety is assessed by observing signs like restlessness, sweating or rapid heart rate. It may indicate disorders such as generalized anxiety, electrolyte imbalance. Diagnosed using psychological assessments and screening tools like Questionnaires related with fear.	

Total Vata Vriddhi Score:.....

Features of Decreased Vata (वात क्षय लक्षण)

Sr. No.	Features and Questions	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	अंगसाद (Weakness of body parts) Reduced strength or weakness in any body parts?	Weakness of body parts is assessed by observing muscle tone, coordination and gait. Questioning focuses on symptoms like pain or numbness. It may indicate neurological disorders, autoimmune diseases or vitamin deficiencies, diagnosed through imaging, EMG and blood tests.	
2	अल्पभाषित्व (Reduced speech) Difficulty speaking or speaking less than usual?	Reduced speech is evaluated by observing speech clarity and volume, assessing for slurred or incomplete sentences and inquiring about associated symptoms like weakness or confusion. It may indicate neurological conditions such as stroke or Parkinson's disease, aphasia or aphonia, dysarthria, brain injury diagnosed through MRI, speech assessments and neurological exams.	
3	संज्ञानाश (Loss of memory) Do you feel forgetful or experience memory loss?	Loss of memory is assessed through observation of cognitive function, questioning about confusion or mood changes. It may indicate Alzheimer's, dementia or head injuries, diagnosed using neuroimaging, cognitive tests and blood work.	
4	प्रसेक (Increased salivation)	Increased salivation is assessed by observing drooling oral discomfort or swelling in the salivary glands. Questions focus on conditions like neurological disorders, infections or medications. Diagnosis is confirmed through lab tests,	

	Excessive saliva or drooling?	imaging or neurological evaluations, radioisotope scanning, cups strapped or the number of napkins used for the measurement of salivary flow.	
5	अरुची (Loss of appetite) Not feeling hungry?	Seen in anorexia nervosa. Loss of appetite is assessed through observation of physical condition and weight changes. It may indicate conditions like gastritis, infections, cancer.	
6	हृल्लास (Nausea) Feeling nauseous frequently?	Nausea is evaluated by observing signs of discomfort, assessing abdominal tenderness. It may indicate conditions like gastritis, infections, neurological disorders.	
7	अल्पचेष्टा (Reduced movement) Do you feel physically slower or have reduced movements?	Reduced movement is assessed by observing posture and gait, examining joint mobility. It may indicate conditions like arthritis, neurological disorders or muscle weakness, diagnosed using imaging, blood tests or electromyography.	
8	अप्रहर्ष (Reduced activity) Reduced interest and enthusiasm in activity?	Reduced enthusiasm is assessed by observing posture, facial expressions, behaviour, exploring fatigue, mood changes or sleep disturbances. It may indicate depression, hypothyroidism or chronic fatigue syndrome, diagnosed through psychological evaluations, hormone tests and imaging studies.	
9	अग्निवैषम्य (Irregular appetite) Do you have irregular eating habits or hunger?	Irregular appetite is assessed through observing eating patterns and weight changes, alongside questioning about onset, triggers and associated symptoms like nausea or fatigue. It may indicate gastrointestinal disorders, metabolic imbalances or psychological conditions, diagnosed through blood tests, imaging and endoscopy.	

Total Vata Kshaya Score:.....

Features of Aggravated Pitta (पित्त वृद्धी लक्षण)

Sr. No.	Features and Questions	Assessment by (दर्शन, स्पर्शन, प्रश्नै)	Score
1	पित्तविट् (Yellow-coloured stool)	Yellow-coloured stool is assessed by observing stool consistency and accompanying symptoms like abdominal pain or nausea. It may indicate conditions such as jaundice, liver disease, bile duct obstruction, malabsorption disorders,	

	Have you noticed more yellow discolouration in the stool?	diagnosed through stool analysis, liver function tests and imaging.	
2	पित्तमूत्र (Yellow-coloured urine) Is your urine yellow or darker than usual?	Yellow-coloured urine is assessed by observing urine colour and clarity. Questioning focuses on hydration, diet and medications. It may indicate dehydration, liver disorders or hematuria. Diagnosis involves urine analysis, blood tests for kidney function.	
3	पित्तेत्र (Yellow sclera) Are the whites of your eyes yellowish?	Yellow sclera is assessed by observing the eye's colour and checking for jaundice. Questioning focuses on the onset, associated symptoms like fatigue or dark urine and potential liver or bile duct issues. Conditions like hepatitis or cirrhosis are diagnosed through blood tests, imaging and liver function tests.	
4	पित्तवृक्क (Yellow skin discolouration) Do you observe yellowish skin tones?	Seen in neonatal jaundice or excess bilirubin. Yellow skin discolouration or jaundice is assessed by observing the skin and sclera for a yellow tint. Symptoms like fatigue, abdominal pain or dark urine may accompany it. Conditions such as jaundice, hepatitis, hemolysis are diagnosed through blood tests, liver function tests and imaging.	
5	क्षुधा (Increased hunger/appetite) Are you experiencing frequent hunger or overeating?	Increased hunger or appetite can be assessed by observing eating behaviour and weight changes. It may indicate conditions like hyperthyroidism, diabetes or stress, diagnosed through blood tests, glucose levels and hormone assessments.	
6	तृष्णा (Increased thirst) Do you feel excessively thirsty?	Increased thirst is evaluated by observing signs of dehydration and checking for dry skin or mucous membranes. Questioning explores onset, frequency and associated symptoms like frequent urination. Conditions like diabetes, hypercalcemia or kidney disease may be diagnosed using blood tests, urine analysis and imaging.	
7	दाह (Burning sensation) Do you often feel a burning sensation in your body or extremities?	Burning sensation is assessed by observing skin changes, tenderness or redness. It may indicate conditions like diabetic neuropathy, infections or inflammation, diagnosed through nerve conduction studies, blood tests or imaging.	
8	अल्प निद्रा (Insomnia)	Insomnia is assessed through observation of sleep patterns It may be linked to conditions such as depression, anxiety,	

	Are you having difficulty sleeping or experiencing restlessness?	sleep apnea, stress or hypertension, diagnosed with sleep studies and psychological evaluations.	
9	ओजविस्त्रंस (Lack of vitality) Do you feel a lack of energy or increased susceptibility to infections?	Lack of vitality is assessed through observations of fatigue, reduced activity, pallor. Questioning explores energy levels, sleep patterns and emotional health. It may indicate conditions like anaemia, depression or hypothyroidism, diagnosed with blood tests, thyroid function tests and psychological assessments.	
10	इन्द्रिय दौर्बल्य (Weakness of sense organs) Are you experiencing reduced vision, hearing or taste sensitivity?	Weakness of sense organs is assessed by observing the patient's response to stimuli, such as light or sound and asking about vision, hearing, smell changes. Conditions like neurological disorders or sensory nerve damage may be identified through neurological exams, imaging and tests.	
11	शीत अभिलाषा (Desire for cold) Do you frequently crave cold environments or substances?	Assessed by observing behaviour, checking for signs of heat intolerance or excessive sweating and exploring symptom onset, triggers and associated factors. It may indicate conditions like hyperthyroidism or fever, diagnosed through blood tests, thyroid function tests, heat-cold tolerance test	
12	तिक्त आस्यता (Bitter taste in mouth) Do you experience a bitter taste in your mouth?	A bitter taste in the mouth is assessed by observing the patient's oral hygiene and signs of infection. Often occurs during fever.	
13	मूर्च्छा (Fainting) Have you fainted recently or felt light-headed?	It may indicate conditions like hypotension, arrhythmias or neurological disorders, diagnosed through blood tests, ECG or imaging.	
14	क्रोध (Anger or irritability) Are you feeling more irritable or prone to anger?	Anger associated symptoms like fatigue or sleep disturbances., stress and sympathetic overactivity. It may indicate mood disorders, hormonal imbalances or neurological conditions.	

Total Pitta Vriddhi Score:.....

Features of Decreased Pitta Dosha (पित्त क्षय लक्षण)

Sr. No	Features and Questions	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	मन्दाग्नि Have you experienced a decrease in your appetite or digestive power?	Decreased digestive power or appetite is assessed by observing abdominal discomfort or bloating. It may indicate gastrointestinal disorders, infections, metabolic conditions.	
2	शित (Coldness) Do you often feel cold, particularly in your extremities?	Questioning explores onset, associated symptoms like shivering or numbness and potential triggers. It may indicate conditions like hypothyroidism, poor circulation, infections.	
3	प्रभाहानी (Loss of luster) Have you noticed a loss of luster or shine in your skin?	Questioning focuses on onset, associated symptoms and lifestyle factors. It may indicate nutritional deficiencies, hormonal imbalances or skin conditions, diagnosed through blood tests or dermatological examination.	
4	स्तंभ (Stiffness) Do you experience stiffness in your body?	Stiffness is assessed by observing joint movement limitations and muscle rigidity. Questioning focuses on onset, duration and associated pain. It may indicate conditions like arthritis, neurological disorders or fibromyalgia, diagnosed through physical exams, imaging, blood tests or electromyography.	
5	अनियत तोद् Do you experience general, non-specific pain in your body?	Physical assessments for tenderness and detailed questioning about onset, location and triggers. It may indicate conditions like fibromyalgia, infections or stress, diagnosed through clinical examination.	
6	कंप (Shivering) Do you experience shivering even without feeling cold?	Shivering is evaluated by observing body tremors and assessing for fever, cold intolerance or muscle rigidity. It may indicate infections, neurological disorders or endocrine imbalances, diagnosed through physical exams, blood tests, imaging and temperature monitoring.	
7	गौरव (Heaviness) Do you feel a general sense of heaviness in your body?	Heaviness can result from stagnation of blood and reduced metabolic activity, as seen in peripheral vascular diseases. Heaviness is assessed by observing posture, body movement and signs of discomfort. It may indicate conditions like obesity, heart disease or pelvic disorders.	
8	नख शुक्ल (Pallor of nails) Have you	Pallor may indicate anaemia, poor circulation or systemic conditions like hypothyroidism or heart failure. Associated	

	noticed pallor in your nails?	symptoms like fatigue or shortness of breath are explored, with diagnosis confirmed through blood tests.	
9	नयन शुक्ल (Pallor in eyes) Do your eyes appear pale or lack colour?	Questioning focuses on associated symptoms like fatigue or dizziness. It may indicate anaemia, vitamin deficiencies or systemic diseases, diagnosed through blood tests and clinical evaluation.	

Total Pitta Kshaya Score:.....

Features of Aggravated Kapha Dosha (कफ वृद्धि लक्षण)

Sr. No	Features and Questions	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	अग्निसाद (Decrease appetite/digestive power) Do you experience a loss of appetite?	It may observed in conditions like gastritis, hypothyroidism. Decreased appetite is assessed through a questionnaire regarding aahar pachan lakshan.	
2	प्रसेक (excessive salivation) Do you have increased salivation?	Excessive salivation is assessed by observing drooling oral discomfort and swelling of salivary glands. Questioning focuses on onset, associated symptoms and potential triggers. It may indicate conditions like stomach irritation, a prodromal stage of Nausea , vomiting, diseases of oesophagus, intestine, stomach, parkinson's disease, dental caries, infections or gastrointestinal issues, diagnosed through imaging, lab tests and clinical evaluation.	
3	आलस्य (Laziness) Do you often feel a lack of energy or lethargy?	Laziness is assessed by observing energy levels, physical movement and general alertness. Questioning explores onset, associated fatigue or psychological factors. It may indicate conditions like diabetes mellitus. shthoulya, medovridhi, depression, hypothyroidism or chronic fatigue syndrome, diagnosed through blood tests, imaging and psychological evaluation by international physical activity questionnaire.	
4	गौरव (Heaviness) Do you feel heaviness in your body?	Commonly observed in obesity. It may indicate conditions like edema, anaemia or circulatory issues, assessed through BMI, Thyroid profile, Lipid Profile blood tests, imaging and physical examination.	
5	शैत्य (Feeling of coldness / Cold Touch)	Coldness is assessed by temperature differences. Questioning focuses on possible triggers. It may find in circulatory disorders, hypothyroidism, anaemia, diabetes, drug/alcohol abuse, hypothyroidism, infection, kidney failure Liver failure,	

	Do you frequently feel cold?	sepsis, side effects of medications, shock, fasting, asthma and stress. Diagnosed through Liver function test, Kidney function test, T3, T4, Heat-Cold Test, imaging and thermal assessments.	
6	श्वेत्य (whitishness) Do you notice any pallor or pale discolouration in your body?	Whitishness is assessed through visual inspection for hypopigmentation or ischemic effects.	
7	श्लथंगता (Looseness of the body part) Do you feel laxity or looseness in your muscles or joints?	Looseness of a body part is assessed by observing poor posture, floppy muscles, muscular dystrophy, abnormal movement and joint instability. Questioning focuses on weakness. Conditions like ligament injuries, joint hypermobility or neurological disorders may be indicated, diagnosed through skin fold thickness, anthropometry measurement, physical tests, CT and MRI	
8	श्वास (Dyspnoea) Do you have difficulty breathing?	Dyspnoea is assessed by examining respiratory effort, signs of distress and oxygen saturation. It may indicate conditions such as asthma, pneumonia, laryngitis, bronchitis, heart failure, diagnosis using chest examination, chest X-ray, pulmonary function tests and blood tests.	
9	कास (Cough) Do you experience frequent coughing?	Cough is assessed by observing the frequency, type and triggers, alongside questioning about duration, associated symptoms like Laryngitis, Bronchitis, Asthma, Tuberculosis, Pneumonia, Pulmonary Oedema, Diagnosis using chest examination, chest X-rays, sputum analysis, pulmonary function tests, blood tests for ESR, IgE	
10	अतिनिद्रा (Excessive Sleep) Do you feel sleepy excessively?	As seen in, obesity. Excessive sleep is evaluated through observations of sleep patterns, fatigue levels and daytime drowsiness. Questioning explores onset, duration and associated symptoms like such as hypothyroidism, anaemia, sleep apnea, depression, diagnosed via polysomnography, EEG, blood tests and clinical evaluation.	
11	स्थूल्य (Obesity) Do you experience obesity or weight gain?	Due to a sedentary lifestyle or high-calorie intake. Obesity is assessed by observing body mass, measuring waist circumference and evaluating skinfold thickness. Questioning includes onset, diet, exercise and comorbidities like sedentary lifestyle, high-calorie intake diabetes and hypertension. Diagnostic tools like BMI calculation, blood tests and imaging help identify related conditions.	

12	स्रोतोपिधान (Obstruction/ covering of srotas) Do you feel a blockage in your body's channels (srotas)? e.g. <i>Nasa</i>	Obstruction of srotas is assessed by observing signs of obstruction, discomfort and retention of related system. It may indicate conditions like a common cold, edema, gastrointestinal blockages or infections.	
13	हल्लास (Nausea) Do you experience nausea or feel like vomiting?	Inquire if it occurs after eating, on an empty stomach or is triggered by specific smells or movements. Explore associated symptoms like vomiting, dizziness, abdominal pain, or headaches.	
14	मुच्छा (Fainting) Do you experience fainting, dizziness?	Fainting is observed signs of pallor, weakness or confusion. It may indicate conditions like dehydration, hypotension, hypovolemia, ischemic attack, diagnosed through blood tests, ECG and imaging studies to assess underlying causes.	

Total Kapha Vriddhi Score:.....

Features of Decreased Kapha Dosha (कफ क्षय लक्षण)

Sr. No	Features and Questions	Assessment by (दर्शन, स्पर्शन, प्रश्नै)	Score
1	भ्रम Have you experienced dizziness or moments of lightheadedness?	Loss of consciousness or dizziness is assessed by observing alertness, checking vitals and evaluating neurological signs. Questioning explores triggers, duration, and associated symptoms. It may indicate syncope, stroke or cardiac issues, diagnosed through ECG, imaging, and lab tests.	
2	श्लेष्माशय शून्यता Do you feel emptiness in areas like your chest or joints?	A feeling of emptiness in the chest or joints is assessed by observing posture and movement and questioning about Feeling of emptiness in chest or joints. It may indicate dehydration, anxiety, musculoskeletal disorders or heart conditions.	
3	हृद्रव (Palpitations) Do you experience palpitations or an abnormal heart rhythm?	Palpitations are assessed by observing the patient's regular or irregular rhythm, rate and strength of pulse, while questioning focuses on onset, triggers and associated symptoms. Measure for hypertension or hypotension. They may indicate arrhythmias, anxiety or thyroid disorders, diagnosed through ECG, Holter monitoring and blood tests.	

4	संधी शैथिल्य (Laxity of joints) Do you feel that your joints are loose or lack stability?	Laxity of joints may occur due to reduced synovial fluid, hypermobility, joint instability, joint pain, dislocations, osteoarthritis	
5	उद्वेष्टन् (Muscle cramps) Do you suffer from muscle cramps frequently?	Muscle cramps are evaluated by observing muscle tightness or spasms. Associated conditions like electrolyte imbalances, dehydration or neurological disorders may be present, diagnosed through blood tests, imaging and EMG studies.	
6	अनिद्रा (Sleeplessness or insomnia) Do you have difficulty sleeping, experience restlessness?	Sleeplessness or insomnia is assessed by observing signs of fatigue or irritability. It may indicate conditions such as depression, sleep apnea or anxiety disorders, diagnosed through sleep studies, psychological assessments or blood tests.	
7	अंगमर्द (Generalized body pain) Do you feel generalized body pain or discomfort?	Generalized body pain is assessed by observing posture, movement and muscle stiffness. Questioning focuses on pain onset, severity and associated symptoms. Conditions like malnutrition, vitamin deficiency, viral infections or autoimmune disorders may be involved, diagnosed through blood tests, imaging and physical examination.	
8	वेपन (Shivering or tremors) Do you experience shivering tremors, especially in response to stress or cold?	Shivering can occur as a protective mechanism in nutrient deficiency. Shivering or tremors are assessed by observing involuntary movements, muscle rigidity or weakness. Conditions like Parkinson's disease, anxiety or thyroid disorders may cause these symptoms.	
9	धूमायन (Burning sensation) Do you feel a burning sensation in your limbs or other parts of your body?	A burning sensation is assessed by observing redness or irritation, palpating for tenderness, temperature and Exploring triggers (specific activities, foods, or exposure). It may indicate conditions like diabetic neuropathy, infections, nerve damage or gastrointestinal issues, diagnosed through blood tests, imaging or endoscopy.	

Total Kapha Kshaya Score:.....



8.1 DOSHA VRIDDHI-KSHAYA ASSESSMENT FORM

Objective : To assess the Dosha Vriddhi-Kshaya lakshanas of an individual

Name: :

Age : **Gender:** M/F **Occupation:**

Body temperature : **Weight:**kg

Blood pressure : **Pulse rate:**...../min

Address :

Features of Aggravated Vata Dosha (वात वृद्धी लक्षण)

Sr. No.	Features and Questions	Score
1	काश्य (Muscle wasting) Observed weight loss or reduced muscle mass?	
2	काष्ण्य (Hyperpigmentation) Darker or patchy skin?	
3	उष्ण कामित्व (Desire for heat) Do you feel a craving for warmth or an aversion to cold?	
4	कंप (Tremor) Do you experience shaking or tremors?	
5	आनाह (Abdominal distension) Is there frequent bloating or fullness in the abdomen?	
6	शकृतग्रह (Constipation) Have you had trouble with bowel movements?	
7	बलभ्रंश (Loss of strength) Feeling unusually fatigued or weak?	
8	निद्रा भ्रंश (Loss of sleep) Difficulty sleeping or insomnia?	
9	इन्द्रिय भ्रंश (Debility of sense organs) Hearing, vision or other sensory impairments?	
10	प्रलाप (Delirium) Random, incoherent speech or confusion?	

11	भ्रम (Loss of consciousness) Have you fainted recently?	
12	दिनता (Miserable feelings) Persistent feelings of sadness or depression?	
13	अस्थिशूल (Bone pain) Pain in bones, especially during fever?	
14	भय (Fear or anxiety) Experiencing excessive fear or anxiety?	

Total Vata Vriddhi Score:.....

Features of Decreased Vata (वात क्षय लक्षण)

Sr. No.	Features and Questions	Score
1	अंगसाद (Weakness of body parts) Reduced strength or weakness in any body parts?	
2	अल्पभाषित्व (Reduced speech) Difficulty speaking or speaking less than usual?	
3	संज्ञानाश (Loss of memory) Do you feel forgetful or experience memory loss?	
4	प्रसेक (Increased salivation) Excessive saliva or drooling?	
5	अरुची (Loss of appetite) Not feeling hungry?	
6	हल्लास (Nausea) Feeling nauseous frequently?	
7	अल्पचेष्टा (Reduced movement) Do you feel physically slower or have reduced movements?	
8	अप्रहर्ष (Reduced activity) Reduced interest and enthusiasm in activity?	
9	अग्निवैषम्य (Irregular appetite) Do you have irregular eating habits or hunger?	

Total Vata Kshaya Score:.....

Features of Aggravated Pitta (पित्त वृद्धी लक्षण)

Sr. No.	Features and Questions	Score
1	पित्तविट् (Yellow-coloured stool) Have you noticed more yellow discolouration in the stool?	
2	पित्तमूत्र (Yellow-coloured urine) Is your urine yellow or darker than usual?	
3	पित्तनेत्र (Yellow sclera) Are the whites of your eyes yellowish?	
4	पित्तत्वक् (Yellow skin discolouration) Do you observe yellowish skin tones?	
5	क्षुधा (Increased hunger/appetite) Are you experiencing frequent hunger or overeating?	
6	तृष्णा (Increased thirst) Do you feel excessively thirsty?	
7	दाह (Burning sensation) Do you often feel a burning sensation in your body or extremities?	
8	अल्प निद्रा (Insomnia) Are you having difficulty sleeping or experiencing restlessness?	
9	ओजविस्त्रंस (Lack of vitality) Do you feel a lack of energy or increased susceptibility to infections?	
10	इन्द्रिय दौर्बल्य (Weakness of sense organs) Are you experiencing reduced vision, hearing or taste sensitivity?	
11	शीत अभिलाषा (Desire for cold) Do you frequently crave cold environments or substances?	
12	तिक्त आस्यता (Bitter taste in mouth) Do you experience a bitter taste in your mouth?	
13	मूर्च्छा (Fainting) Have you fainted recently or felt light-headed?	
14	क्रोध (Anger or irritability) Are you feeling more irritable or prone to anger?	

Total Pitta Vridhhi Score:.....

Features of Decreased Pitta Dosha (पित्त क्षय लक्षण)

Sr. No.	Features and Questions	Score
1	मन्दाग्नि (Decreased digestive power/appetite) Have you experienced a decrease in your appetite or digestive power?	
2	शित (Coldness) Do you often feel cold, particularly in your extremities?	
3	प्रभाहानी (Loss of luster) Have you noticed a loss of luster or shine in your skin?	
4	स्तम्भ (Stiffness) Do you experience stiffness in your body?	
5	अनियत तोद् (Non-specific pain) Do you experience general, non-specific pain in your body?	
6	कंप (Shivering) Do you experience shivering even without feeling cold?	
7	गौरव (Heaviness) Do you feel a general sense of heaviness in your body?	
8	नख शुक्ल (Pallor of nails) Have you noticed pallor in your nails?	
9	नयन शुक्ल (Pallor in eyes) Do your eyes appear pale or lack colour?	

Total Pitta Kshaya Score:.....

Features of Aggravated Kapha Dosha (कफ वृद्धि लक्षण)

Sr. No.	Features and Questions	Score
1	अग्निसाद (Decrease appetite/digestive power) Do you experience a loss of appetite?	
2	प्रसेक (excessive salivation) Do you have increased salivation?	
3	आलस्य (Laziness) Do you often feel a lack of energy or lethargy?	

4	गौरव (Heaviness) Do you feel heaviness in your body?	
5	शैत्य (Feeling of coldness / Cold Touch) Do you frequently feel cold?	
6	श्वैत्य (whitishness) Do you notice any pallor or pale discolouration in your body?	
7	श्लथान्गता (Looseness of the body part) Do you feel laxity or looseness in your muscles or joints?	
8	श्वास (Dyspnoea) Do you have difficulty breathing?	
9	कास (Cough) Do you experience frequent coughing?	
10	अतिनिद्रा (Excessive Sleep) Do you feel sleepy excessively?	
11	स्थूल्य (Obesity) Do you experience obesity or weight gain?	
12	स्रोतोपिधान (Obstruction/covering of srotas) Do you feel a blockage in your body's channels (srotas)? e.g. <i>Nasa etc</i>	
13	हल्लास (Nausea) Do you experience nausea or feel like vomiting?	
14	मुर्च्छा (Fainting) Do you experience fainting or dizziness?	

Total Kapha Vriddhi Score:.....

Features of Decreased Kapha Dosha (कफ क्षय लक्षण)

Sr. No.	Features and Questions	Score
1	भ्रम (Loss of consciousness or dizziness) Have you experienced dizziness or moments of lightheadedness?	
2	श्लेष्माशय शून्यता (Feeling of emptiness in chest or joints) Do you feel emptiness in areas like your chest or joints?	
3	हृद्रव (Palpitations) Do you experience palpitations or an abnormal heart rhythm?	
4	संधी शैथिल्य (Laxity of joints) Do you feel that your joints are loose or lack stability?	
5	उद्वेष्टन् (Muscle cramps) Do you suffer from muscle cramps frequently?	

6	अनिद्रा (Sleeplessness or insomnia) Do you have difficulty sleeping or experience restlessness?	
7	अंगमर्द (Generalized body pain) Do you feel generalized body pain or discomfort?	
8	वेपन (Shivering or tremors) Do you experience shivering or tremors, especially in response to stress or cold?	
9	धूमायन (Burning sensation) Do you feel a burning sensation in your limbs or other parts of your body?	

Total Kapha Kshaya Score:.....

Results:

Name of Dosha Vriddhi:

0–6 - Normal :.....
7–13 - Mild Dosha Vriddhi :.....
14–20 - Moderate Dosha Vriddhi :.....
21–27 - Severe Dosha Vriddhi :.....

Name of Dosha Kshaya:

0–10 - Normal :.....
11–21 - Mild Dosha Kshaya :.....
22–32 - Moderate Dosha Kshaya :.....
33–42 - Severe Dosha Kshaya :.....

Overall Dosha Vriddhi :.....

Overall Dosha Kshaya :.....



Teacher's Signature

8.2 DOSHA VRIDDHI-KSHAYA ASSESSMENT FORM

Objective : To assess the Dosha Vriddhi-Kshaya lakshanas of an individual

Name: :

Age : **Gender:** M/F **Occupation:**

Body temperature : **Weight:**kg

Blood pressure : **Pulse rate:**...../min

Address :

Features of Aggravated Vata Dosha (वात वृद्धी लक्षण)

Sr. No.	Features and Questions	Score
1	काश्य (Muscle wasting) Observed weight loss or reduced muscle mass?	
2	काष्ण्य (Hyperpigmentation) Darker or patchy skin?	
3	उष्ण कामित्व (Desire for heat) Do you feel a craving for warmth or an aversion to cold?	
4	कंप (Tremor) Do you experience shaking or tremors?	
5	आनाह (Abdominal distension) Is there frequent bloating or fullness in the abdomen?	
6	शकृतग्रह (Constipation) Have you had trouble with bowel movements?	
7	बलभ्रंश (Loss of strength) Feeling unusually fatigued or weak?	
8	निद्रा भ्रंश (Loss of sleep) Difficulty sleeping or insomnia?	
9	इन्द्रिय भ्रंश (Debility of sense organs) Hearing, vision or other sensory impairments?	
10	प्रलाप (Delirium) Random, incoherent speech or confusion?	

11	भ्रम (Loss of consciousness) Have you fainted recently?	
12	दिनता (Miserable feelings) Persistent feelings of sadness or depression?	
13	अस्थिशूल (Bone pain) Pain in bones, especially during fever?	
14	भय (Fear or anxiety) Experiencing excessive fear or anxiety?	

Total Vata Vriddhi Score:.....

Features of Decreased Vata (वात क्षय लक्षण)

Sr. No.	Features and Questions	Score
1	अंगसाद (Weakness of body parts) Reduced strength or weakness in any body parts?	
2	अल्पभाषित्व (Reduced speech) Difficulty speaking or speaking less than usual?	
3	संज्ञानाश (Loss of memory) Do you feel forgetful or experience memory loss?	
4	प्रसेक (Increased salivation) Excessive saliva or drooling?	
5	अरुची (Loss of appetite) Not feeling hungry?	
6	हल्लास (Nausea) Feeling nauseous frequently?	
7	अल्पचेष्टा (Reduced movement) Do you feel physically slower or have reduced movements?	
8	अप्रहर्ष (Reduced activity) Reduced interest and enthusiasm in activity?	
9	अग्निवैषम्य (Irregular appetite) Do you have irregular eating habits or hunger?	

Total Vata Kshaya Score:.....

Features of Aggravated Pitta (पित्त वृद्धी लक्षण)

Sr. No.	Features and Questions	Score
1	पित्तविट् (Yellow-coloured stool) Have you noticed more yellow discolouration in the stool?	
2	पित्तमूत्र (Yellow-coloured urine) Is your urine yellow or darker than usual?	
3	पित्तनेत्र (Yellow sclera) Are the whites of your eyes yellowish?	
4	पित्तत्वक् (Yellow skin discolouration) Do you observe yellowish skin tones?	
5	क्षुधा (Increased hunger/appetite) Are you experiencing frequent hunger or overeating?	
6	तृष्णा (Increased thirst) Do you feel excessively thirsty?	
7	दाह (Burning sensation) Do you often feel a burning sensation in your body or extremities?	
8	अल्प निद्रा (Insomnia) Are you having difficulty sleeping or experiencing restlessness?	
9	ओजविस्त्रंस (Lack of vitality) Do you feel a lack of energy or increased susceptibility to infections?	
10	इन्द्रिय दौर्बल्य (Weakness of sense organs) Are you experiencing reduced vision, hearing or taste sensitivity?	
11	शीत अभिलाषा (Desire for cold) Do you frequently crave cold environments or substances?	
12	तिक्त आस्यता (Bitter taste in mouth) Do you experience a bitter taste in your mouth?	
13	मूर्च्छा (Fainting) Have you fainted recently or felt light-headed?	
14	क्रोध (Anger or irritability) Are you feeling more irritable or prone to anger?	

Total Pitta Vridhhi Score:.....

Features of Decreased Pitta Dosha (पित्त क्षय लक्षण)

Sr. No.	Features and Questions	Score
1	मन्दाग्नि (Decreased digestive power/appetite) Have you experienced a decrease in your appetite or digestive power?	
2	शित (Coldness) Do you often feel cold, particularly in your extremities?	
3	प्रभाहानी (Loss of luster) Have you noticed a loss of luster or shine in your skin?	
4	स्तंभ (Stiffness) Do you experience stiffness in your body?	
5	अनियत तोद् (Non-specific pain) Do you experience general, non-specific pain in your body?	
6	कंप (Shivering) Do you experience shivering even without feeling cold?	
7	गौरव (Heaviness) Do you feel a general sense of heaviness in your body?	
8	नख शुक्ल (Pallor of nails) Have you noticed pallor in your nails?	
9	नयन शुक्ल (Pallor in eyes) Do your eyes appear pale or lack colour?	

Total Pitta Kshaya Score:.....

Features of Aggravated Kapha Dosha (कफ वृद्धी लक्षण)

Sr. No.	Features and Questions	Score
1	अग्निसाद (Decrease appetite/digestive power) Do you experience a loss of appetite?	
2	प्रसेक (excessive salivation) Do you have increased salivation?	
3	आलस्य (Laziness) Do you often feel a lack of energy or lethargy?	

4	गौरव (Heaviness) Do you feel heaviness in your body?	
5	शैत्य (Feeling of coldness / Cold Touch) Do you frequently feel cold?	
6	श्वैत्य (whitishness) Do you notice any pallor or pale discolouration in your body?	
7	श्लथान्गता (Looseness of the body part) Do you feel laxity or looseness in your muscles or joints?	
8	श्वास (Dyspnoea) Do you have difficulty breathing?	
9	कास (Cough) Do you experience frequent coughing?	
10	अतिनिद्रा (Excessive Sleep) Do you feel sleepy excessively?	
11	स्थूल्य (Obesity) Do you experience obesity or weight gain?	
12	स्रोतोपिधान (Obstruction/covering of srotas) Do you feel a blockage in your body's channels (srotas)? e.g. <i>Nasa etc</i>	
13	हल्लास (Nausea) Do you experience nausea or feel like vomiting?	
14	मुर्च्छा (Fainting) Do you experience fainting or dizziness?	

Total Kapha Vriddhi Score:.....

Features of Decreased Kapha Dosha (कफ क्षय लक्षण)

Sr. No.	Features and Questions	Score
1	भ्रम (Loss of consciousness or dizziness) Have you experienced dizziness or moments of lightheadedness?	
2	श्लेष्माशय शून्यता (Feeling of emptiness in chest or joints) Do you feel emptiness in areas like your chest or joints?	
3	हृद्रव (Palpitations) Do you experience palpitations or an abnormal heart rhythm?	
4	संधी शैथिल्य (Laxity of joints) Do you feel that your joints are loose or lack stability?	
5	उद्वेष्टन् (Muscle cramps) Do you suffer from muscle cramps frequently?	

6	अनिद्रा (Sleeplessness or insomnia) Do you have difficulty sleeping or experience restlessness?	
7	अंगमर्द (Generalized body pain) Do you feel generalized body pain or discomfort?	
8	वेपन (Shivering or tremors) Do you experience shivering or tremors, especially in response to stress or cold?	
9	धूमायन (Burning sensation) Do you feel a burning sensation in your limbs or other parts of your body?	

Total Kapha Kshaya Score:.....

Results:

Name of Dosha Vriddhi:

0–6 - Normal :.....
7–13 - Mild Dosha Vriddhi :.....
14–20 - Moderate Dosha Vriddhi :.....
21–27 - Severe Dosha Vriddhi :.....

Name of Dosha Kshaya:

0–10 - Normal :.....
11–21 - Mild Dosha Kshaya :.....
22–32 - Moderate Dosha Kshaya :.....
33–42 - Severe Dosha Kshaya :.....

Overall Dosha Vriddhi :.....

Overall Dosha Kshaya :.....



Teacher's Signature

8.3 DOSHA VRIDDHI-KSHAYA ASSESSMENT FORM

Objective : To assess the Dosha Vriddhi-Kshaya lakshanas of an individual

Name: :

Age : **Gender:** M/F **Occupation:**

Body temperature : **Weight:**kg

Blood pressure : **Pulse rate:**...../min

Address :

Features of Aggravated Vata Dosha (वात वृद्धी लक्षण)

Sr. No.	Features and Questions	Score
1	काश्य (Muscle wasting) Observed weight loss or reduced muscle mass?	
2	काष्ण्य (Hyperpigmentation) Darker or patchy skin?	
3	उष्ण कामित्व (Desire for heat) Do you feel a craving for warmth or an aversion to cold?	
4	कंप (Tremor) Do you experience shaking or tremors?	
5	आनाह (Abdominal distension) Is there frequent bloating or fullness in the abdomen?	
6	शकृतग्रह (Constipation) Have you had trouble with bowel movements?	
7	बलभ्रंश (Loss of strength) Feeling unusually fatigued or weak?	
8	निद्रा भ्रंश (Loss of sleep) Difficulty sleeping or insomnia?	
9	इन्द्रिय भ्रंश (Debility of sense organs) Hearing, vision or other sensory impairments?	
10	प्रलाप (Delirium) Random, incoherent speech or confusion?	

11	भ्रम (Loss of consciousness) Have you fainted recently?	
12	दिनता (Miserable feelings) Persistent feelings of sadness or depression?	
13	अस्थिशूल (Bone pain) Pain in bones, especially during fever?	
14	भय (Fear or anxiety) Experiencing excessive fear or anxiety?	

Total Vata Vriddhi Score:.....

Features of Decreased Vata (वात क्षय लक्षण)

Sr. No.	Features and Questions	Score
1	अंगसाद (Weakness of body parts) Reduced strength or weakness in any body parts?	
2	अल्पभाषित्व (Reduced speech) Difficulty speaking or speaking less than usual?	
3	संज्ञानाश (Loss of memory) Do you feel forgetful or experience memory loss?	
4	प्रसेक (Increased salivation) Excessive saliva or drooling?	
5	अरुची (Loss of appetite) Not feeling hungry?	
6	हल्लास (Nausea) Feeling nauseous frequently?	
7	अल्पचेष्टा (Reduced movement) Do you feel physically slower or have reduced movements?	
8	अप्रहर्ष (Reduced activity) Reduced interest and enthusiasm in activity?	
9	अग्निवैषम्य (Irregular appetite) Do you have irregular eating habits or hunger?	

Total Vata Kshaya Score:.....

Features of Aggravated Pitta (पित्त वृद्धी लक्षण)

Sr. No.	Features and Questions	Score
1	पित्तविट् (Yellow-coloured stool) Have you noticed more yellow discolouration in the stool?	
2	पित्तमूत्र (Yellow-coloured urine) Is your urine yellow or darker than usual?	
3	पित्तनेत्र (Yellow sclera) Are the whites of your eyes yellowish?	
4	पित्तत्वक् (Yellow skin discolouration) Do you observe yellowish skin tones?	
5	क्षुधा (Increased hunger/appetite) Are you experiencing frequent hunger or overeating?	
6	तृष्णा (Increased thirst) Do you feel excessively thirsty?	
7	दाह (Burning sensation) Do you often feel a burning sensation in your body or extremities?	
8	अल्प निद्रा (Insomnia) Are you having difficulty sleeping or experiencing restlessness?	
9	ओजविस्त्रंस (Lack of vitality) Do you feel a lack of energy or increased susceptibility to infections?	
10	इन्द्रिय दौर्बल्य (Weakness of sense organs) Are you experiencing reduced vision, hearing or taste sensitivity?	
11	शीत अभिलाषा (Desire for cold) Do you frequently crave cold environments or substances?	
12	तिक्त आस्यता (Bitter taste in mouth) Do you experience a bitter taste in your mouth?	
13	मूर्च्छा (Fainting) Have you fainted recently or felt light-headed?	
14	क्रोध (Anger or irritability) Are you feeling more irritable or prone to anger?	

Total Pitta Vridhhi Score:.....

Features of Decreased Pitta Dosha (पित्त क्षय लक्षण)

Sr. No.	Features and Questions	Score
1	मन्दाग्नि (Decreased digestive power/appetite) Have you experienced a decrease in your appetite or digestive power?	
2	शित (Coldness) Do you often feel cold, particularly in your extremities?	
3	प्रभाहानी (Loss of luster) Have you noticed a loss of luster or shine in your skin?	
4	स्तंभ (Stiffness) Do you experience stiffness in your body?	
5	अनियत तोद् (Non-specific pain) Do you experience general, non-specific pain in your body?	
6	कंप (Shivering) Do you experience shivering even without feeling cold?	
7	गौरव (Heaviness) Do you feel a general sense of heaviness in your body?	
8	नख शुक्ल (Pallor of nails) Have you noticed pallor in your nails?	
9	नयन शुक्ल (Pallor in eyes) Do your eyes appear pale or lack colour?	

Total Pitta Kshaya Score:.....

Features of Aggravated Kapha Dosha (कफ वृद्धि लक्षण)

Sr. No.	Features and Questions	Score
1	अग्निसाद (Decrease appetite/digestive power) Do you experience a loss of appetite?	
2	प्रसेक (excessive salivation) Do you have increased salivation?	
3	आलस्य (Laziness) Do you often feel a lack of energy or lethargy?	

4	गौरव (Heaviness) Do you feel heaviness in your body?	
5	शैत्य (Feeling of coldness / Cold Touch) Do you frequently feel cold?	
6	श्वैत्य (whitishness) Do you notice any pallor or pale discolouration in your body?	
7	श्लथांगता (Looseness of the body part) Do you feel laxity or looseness in your muscles or joints?	
8	श्वास (Dyspnoea) Do you have difficulty breathing?	
9	कास (Cough) Do you experience frequent coughing?	
10	अतिनिद्रा (Excessive Sleep) Do you feel sleepy excessively?	
11	स्थूल्य (Obesity) Do you experience obesity or weight gain?	
12	स्रोतोपिधान (Obstruction/covering of srotas) Do you feel a blockage in your body's channels (srotas)? e.g. <i>Nasa etc</i>	
13	हल्लास (Nausea) Do you experience nausea or feel like vomiting?	
14	मुर्च्छा (Fainting) Do you experience fainting or dizziness?	

Total Kapha Vriddhi Score:.....

Features of Decreased Kapha Dosha (कफ क्षय लक्षण)

Sr. No.	Features and Questions	Score
1	भ्रम (Loss of consciousness or dizziness) Have you experienced dizziness or moments of lightheadedness?	
2	श्लेष्माशय शून्यता (Feeling of emptiness in chest or joints) Do you feel emptiness in areas like your chest or joints?	
3	हृद्रव (Palpitations) Do you experience palpitations or an abnormal heart rhythm?	
4	संधी शैथिल्य (Laxity of joints) Do you feel that your joints are loose or lack stability?	
5	उद्वेष्टन् (Muscle cramps) Do you suffer from muscle cramps frequently?	

6	अनिद्रा (Sleeplessness or insomnia) Do you have difficulty sleeping or experience restlessness?	
7	अंगमर्द (Generalized body pain) Do you feel generalized body pain or discomfort?	
8	वेपन (Shivering or tremors) Do you experience shivering or tremors, especially in response to stress or cold?	
9	धूमायन (Burning sensation) Do you feel a burning sensation in your limbs or other parts of your body?	

Total Kapha Kshaya Score:.....

Results:

Name of Dosha Vriddhi:

0–6 - Normal :.....
7–13 - Mild Dosha Vriddhi :.....
14–20 - Moderate Dosha Vriddhi :.....
21–27 - Severe Dosha Vriddhi :.....

Name of Dosha Kshaya:

0–10 - Normal :.....
11–21 - Mild Dosha Kshaya :.....
22–32 - Moderate Dosha Kshaya :.....
33–42 - Severe Dosha Kshaya :.....

Overall Dosha Vriddhi :.....

Overall Dosha Kshaya :.....



Teacher's Signature

EXERCISE

Q.1 What is the significance of Dosha Vriddhi and Kshaya in disease?

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Q.2 What are the signs of Kapha Dosha Kshaya?

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Q.3 How can lifestyle modification balance aggravate Doshas? Give an example.

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Q.4 What are common symptoms (diseases) associated with Vata Dosha Vriddhi?

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Q.5 What are common symptoms (diseases) associated with Kapha Dosha Vriddhi?

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Q.6 How can modern diagnostic tools complement Dosha assessments?

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OSPE-1

Title: Assessment of Dosha Vriddhi-Kshaya

Learning Objectives:

To demonstrate the ability to assess Dosha imbalances using a structured proforma.

To interpret the scoring to identify Vriddhi or Kshaya levels.

Station Setup:

Table with the assessment form and scoring guide.

A simulated or real candidate for assessment.

Tasks:

- Collect demographic and clinical information from the candidate.
- Ask all the listed questions for Vata, Pitta and Kapha.
- Score the responses based on severity and interpret the total.

Checklist:

1. Collected all required demographic data.
2. Asked maximum relevant questions for each Dosha.
3. Correctly scored the responses.
4. Accurately interpreted the total scores.
5. Maintained professional communication throughout the interaction.

GUIDELINES FOR OTHER EXAMPLES OF OSPE

OSPE Stations Design: Each station targets specific competencies:

- Create interactive stations covering.
- History-taking and symptom analysis.
- Identification of Dosha features (e.g., signs of aggravated or decreased Doshas)
- Interpretation of results and diagnosis.

Station 1: Symptom Identification

Task: Identify Dosha imbalances from a set of symptoms and assign scores using the form.
Example: A scenario where the student recognizes symptoms like tremors and abdominal distension, calculates the aggravated Vata score and concludes it as "Moderate Aggravation."

Station 2: Practical Application

Task: Suggest remedies for the imbalanced Dosha based on Ayurveda principles.
Example: Recommend dietary modifications (e.g., warm, unctuous food for Vata imbalance).

Station 3: Communication Skills

Task: Explain the assessment process and results to a simulated patient. The evaluation focuses on clarity and patient-friendly language.

Assessment Checklist for Each Station:

Accuracy in identifying symptoms and calculating scores. Appropriateness of suggested remedies. Communication and patient education effectiveness.

1. Role of Facilitators:

Provide structured feedback at each station. Highlight common errors (e.g., misinterpreting symptoms, incorrect scoring).

Advantages of OSPE in Dosha Assessment

- **Structured Learning:** Enables a focused and stepwise approach to mastering Dosha evaluation.
- **Objective Evaluation:** Ensures fairness by evaluating specific skills at each station.
- **Skill Integration:** Combines theoretical knowledge, practical skills and communication abilities.

Learning Methodology

Dosha Vriddhi-Kshaya Assessment Form, which involves evaluating symptoms linked to the Ayurvedic doshas (Vata, Pitta and Kapha), the following teaching and learning methodologies paired with the Objective Structured Practical Examination (OSPE) would be the most effective:

Interactive Discussions: Conduct debriefing sessions post-OSPE to discuss key findings, Dosha interpretations and common misconceptions.

Reflection: Encourage students to reflect on their performance to reinforce learning.

Feedback: Provide individualized feedback, focusing on reasoning and scoring accuracy.

CASE-BASED LEARNING

Case-Based Learning (CBL) Example: Dosha Vriddhi Assessment

Case Title: Understanding Dosha Vriddhi Through Clinical Features

Case Scenario:

A 45-year-old male presents with fatigue, excessive sleepiness and heaviness in the body. He complains of a loss of appetite and frequent coughing.

Problem:

The patient's symptoms indicate a possible Kapha Dosha Vriddhi. Students need to assess the severity

of imbalance and recommend appropriate interventions.

Guiding Questions:

What features of Kapha Dosha Vriddhi are present in the patient?

How would you differentiate Kapha Vriddhi from other Dosha imbalances?

Guided Discussion Questions:

What dietary and lifestyle modifications would help balance Kapha Dosha?

How does modern medicine explain the physiological basis of these symptoms?

Expected Outcomes:

Identification of symptoms linked to Kapha Dosha Vriddhi.

Recommendation of practical interventions, including diet, exercise and Ayurved basic remedies as per dosha.

Discussion and Resolution:

Students assess the patient's symptoms using the proforma, score the severity and interpret the results. Based on the findings, they discuss interventions to reduce Kapha Dosha.

Assessment:

Students are evaluated on their ability to:

- Use the assessment form accurately.
- Interpret the scores correctly.
- Propose evidence-based interventions.
- Communicate their findings effectively.

ROLE PLAY

Role-Play Activity: Dosha Assessment

Title: Dosha Assessment Role-Play

Objective: To practice identifying dosha imbalances, communicating findings and recommending interventions using the Dosha Assessment Form.

Setup: Roles:

Vaidya: A student or teacher simulates the role of the vaidya.

Patient: A student simulates a patient experiencing symptoms of dosha imbalance.

Materials: Dosha Assessment Form. Case scenario details with symptoms representing dosha imbalances. Scoring guidelines for dosha severity levels.

Script Details

Vaidya:

Introduction: Greets the patient and introduces themselves. Verifies patient details (name, age and lifestyle habits).

History Taking: Ask the patient about their symptoms, such as appetite changes, digestion, sleep patterns and emotional well-being. Records observations and patient responses systematically.

Assessment: Refers to the Dosha Assessment Form to identify aggravated or diminished doshas. Calculates the scores based on patient symptoms.

Recommendation: Explain the findings to the patient in simple terms. Provides Ayurvedic advice, such as dietary changes, lifestyle modifications and Ayurveda basic remedies to balance the dosha(s).

Patient:

Describes symptoms such as bloating, weakness or fear, depending on the case scenario.

Ask questions or express concerns about the assessment and suggested remedies.

Scenario Example

Patient Case: A 35-year-old teacher presents with the following symptoms: Frequent bloating and abdominal distension. Tremors in the hands during stressful situations. Difficulty sleeping and feeling anxious at night.

Task for the Vaidya: Take the patient's history and record the symptoms. Use the Dosha Assessment Form to calculate the aggravated Vata Dosha score. Abdominal distension: 3 (severe). Tremors: 2 (moderate). Loss of sleep: 2 (moderate). Total Vata score: 7 (Mild Aggravation).

Recommend interventions:

Dietary: Warm, unctuous meals (e.g., soups, ghee, spices like ginger).

Lifestyle: Regular sleep routine, oil massage and yoga for relaxation.

Therapeutic: Basti procedure, Ashwagandha like drugs for calming Vata imbalance.

Debriefing Questions

What were the key symptoms that indicated Vata Dosha imbalance?

Highlight clear explanations, patient reassurance and practical advice.

What other recommendations could be made for this case?

Assessment

Use a checklist to evaluate Vaidya's performance: Accurate symptom identification and scoring.

Effective communication and patient engagement.

Appropriateness of the suggested interventions.

9. DHATU VRIDDHI-KSHAYA PARIKSHAN

LEARNING OBJECTIVES

After completion of the practical, students should be able to:

1. Understand the concept of Dhatu Vriddhi and Kshaya.
2. Learn the methods of assessing individual Dhatu states.
3. Correlate clinical findings of Ayurveda with modern interpretations.
4. Analyze the implications of Dhatu imbalance on overall health.

PRINCIPLE

The assessment of Dhatu Vriddhi and Kshaya involves evaluating clinical symptoms and physical signs related to the specific tissue. Each Dhatu's state reflects the individual's health status and helps in diagnosing imbalances.

PURPOSE

To examine and identify the symptoms and clinical signs associated with the aggravation and decreased of the seven Dhatus (Rasa, Rakta, Mamsa, Meda, Asthi, Majja, Shukra).

INSTRUCTIONS

Students must assess the seven dhatus for vriddhi (growth) and kshaya (depletion) using the provided questionnaire. The questions are based on features described in the shlokas of vriddhi and kshaya mentioned in the Samhitas. Assessments should be conducted through *darshana* (inspection), *sparshana* (palpation, percussion, auscultation) and *prashna* (interrogation) using advanced technology. Each symptom should be scored based on severity as reported by the participant: 0 - Not present, 1 - Mild, 2 - Moderate, 3 - Severe.

At the end of the questionnaire, students must calculate the percentage score for each dhatu. The

severity of the symptoms is categorized into four levels for easy interpretation: normal (dhatu samya), mild, moderate and severe dhatu vriddhi or kshaya.

Normal is considered when the score is 0-25%

Mild is considered when the score is 26-50%

Moderate is considered when the score is 51-75%

Severe is considered when the score is 76-100%

APPARATUS AND MATERIALS

For undergraduate students, we utilize the proforma for Dhatu Vriddhi/Kshaya Parikshan to ensure a structured approach. Depending on availability and necessity, advanced instruments or their reports can be incorporated for comprehensive assessment, including clinical tools like stethoscopes, sphygmomanometers and thermometers; laboratory investigations such as CBC, ESR, lipid profiles and serum calcium tests; imaging equipment like USG, X-ray and CT scans; Dhatu-specific assessment questionnaires; and patient case sheets with consent forms.

OBSERVATIONS AND RESULTS:

Document responses to each feature-related question.

Compute total scores for Vriddhi and Kshaya states of each Dhatu.

Calculate percentages and interpret results.

Detailed scoring for each Dhatu will be calculated based on symptoms and findings.

Compare scores with reference standards to determine imbalance severity.

Document and interpret findings in alignment with Ayurvedic principles.

PRECAUTIONS

Ensure patient comfort and informed consent before examinations. Maintain hygiene and proper handling of diagnostic equipment. Correlate clinical findings with laboratory

EXAMPLE - प्रसेक (Excessive salivation)

Students should understand assessment proforma on the basis of these examples. Excessive salivation can be examined and understood through a systematic approach involving *darshana* (inspection), *sparshana* (palpation, percussion, auscultation) and *prashna* (interrogation) using advanced technology.

1. Observation (Darshana)

Oral Examination: Check for mouth ulcers, gum inflammation or infections that may stimulate salivation.

Posture and Drooling: Observe for swelling, infection of salivary glands, Parkinson's disease

2. Palpation (Sparshana)

Examine swelling or tenderness, which may indicate infection or obstruction.

Assess the patient's ability to swallow and control oral secretions.

3. Interrogation (Prashna)

Onset and Duration: Ask when the excessive salivation started and whether it is continuous or episodic.

Associated Symptoms: Inquire about nausea, difficulty swallowing oral irritation or gastrointestinal symptoms like heartburn.

Triggers: Identify potential triggers such as specific foods, medications or exposure to toxins.

- **Medical History:** Assess for conditions such as pregnancy, GERD, Parkinson's disease or infections that may cause excessive salivation (sialorrhea).

4. Advanced Diagnostic Tools

Laboratory Tests: Check for infections, metabolic conditions or heavy metal poisoning.

Imaging: Ultrasound or MRI may be used to evaluate salivary gland abnormalities.

Neurological Assessment: Rule out neurological conditions affecting oral motor control.

EXAMPLE - रक्त मूत्र (Hematuria)

1. Observation (Darshana)

Urine Appearance: Observe the colour and clarity of urine. Presence of visible blood (gross hematuria) or pink/red tint (microscopic hematuria) can be noted.

General Appearance: Look for signs of systemic illness such as fever, swelling or weight loss.

2. Palpation (Sparshana)

Abdominal Palpation: Check for tenderness or swelling in the abdominal or lower back region, which could indicate a kidney or bladder issue.

Percussion: Percuss the kidneys to check for tenderness, which may suggest a kidney infection or stones.

3. Interrogation (Prashna)

Onset and Duration: Ask when blood in urine was first noticed and whether it is persistent or intermittent.

Associated Symptoms: Inquire about pain, burning sensation, fever, changes in urine colour.

Triggers: Identify potential causes such as recent trauma, physical activity or use of medications.

Medical History: Assess for past conditions like kidney stones, urinary tract infection (UTI) or blood disorders like hemophilia.

4. Advanced Diagnostic Tools

Laboratory Tests: Urinalysis to check for red blood cells and other abnormalities; urine culture to identify infections; blood tests RBC, WBC, HB, KFT to assess kidney function.

Imaging: Ultrasound, CT scan or MRI to evaluate for kidney stones, tumors or structural abnormalities.

Cystoscopy: A procedure to examine the bladder and urethra. **Urine Cytology:** To check for the presence of abnormal cells, indicating potential malignancy in the urinary tract.

SHLOKAS (VERSES) FOR RECITATION

■ धातु कर्म

- प्रीणनं जीवनं लेपः स्नेहो धारणपूरणे।
गर्भोत्पादश्च धातूनां श्रेष्ठं कर्म क्रमात् स्मृतम्॥ - अ.ह.सू.11/4
- रसस्तुष्टिं प्रीणनं रक्तपुष्टिं च करोति।
रक्तं वर्णप्रसादं मांसपुष्टिं जीवयति च॥
मांसं शरीरपुष्टिं मेदसश्च।
मेदः स्नेहस्वेदौ दृढत्वं पुष्टिमस्थानां च॥
अस्थिनि देहधारणं मज्जा; पुष्टिं च।
मज्जा प्रीतिं स्नेहं बलं शुक्रपुष्टिं पूरणमस्थानां च करोति॥
शुक्रं धैर्यं, च्यवनं, प्रीतिं देहबलं, हर्षं बीजार्थं च। - सु.सू.15
- तुष्टिप्रीणनरक्तपुष्टिभिस्तु रस।
जीवनं वर्णप्रसादनमांसं पोषणरसृक्॥
देहलेपमलमेदः पुष्टिभिर्मांसम्।
नेत्रगात्रस्निग्धता स्नेहदाढ्यास्थिपुष्टिभिर्मेदः॥
देहोर्ध्वताधारणमज्जापोषणाभ्यामस्थि।
स्नेहबलास्थिपूरणं शुक्रपुष्टिभिर्मज्जा॥
हर्षबलं गर्भोत्पादनैः शुक्रम्॥ - अ.सं.सू.19

■ धातु वृद्धि

■ अष्टांग हृदयं

- रसधातु-श्लेष्मा(वृद्धोऽग्निसदनप्रसेकालस्यगौरवम्।
श्वैत्यशैत्यश्लथाङ्गत्वं श्वासकासातिनिद्रताः। रसोऽपि
श्लेष्मवत्॥
- रक्तधातु-रक्तं विसर्पप्लीहविद्रधीन्। कुष्ठवातास्त्रपितास्त्रगुल्मोप
कुशकामलाः॥ व्यंगाग्निनाशसम्मोहरक्तवत्त्वग्नेत्रमूत्रताः॥
- मांसधातु-मांसं गण्डार्बुदग्रन्थिगण्डोरुदरवृद्धिताः॥
कण्ठादिष्वधिमांसं च॥
- मेदधातु-श्रमवासेऽल्पचेष्टितं श्वासं स्फिकस्तनउदरलम्बनम्।
अस्थिधातु-अस्थ्यध्यस्थ्यधिदन्ताश्च॥
- मज्जाधातु-मज्जानेत्राङ्गगौरवम्॥
पर्वसु स्थूलमूलानि कुर्यात्कृच्छ्राण्यरुंषि च॥
- शुक्रधातु-अतिस्त्रीकामतां वृद्धं शुक्रं शुक्राश्मरीमपि॥
- अ.ह.सू.11/7-12

■ सुश्रुत

- रसोऽतिवृद्धो हृदयोत्कलेदं प्रसेकं चापादयति॥
- रक्तं रक्ताङ्गाक्षितां सिरापूर्णत्वं चापादयति॥
- मांसं स्फिगण्डौष्ठोपस्थोरूबाहुजंघासु वृद्धिं गुरुगात्रतां च॥

- मेदः स्निग्धांगतामुदरपार्श्ववृद्धिं कासश्वासदीन् दौर्गन्ध्यं च॥
- अस्थ्यध्यस्थ्यधिदन्ताश्च॥
- मज्जा सर्वांगनेत्रगौरवं च॥
- शुक्रं शुक्राश्मरीमतिप्रादुर्भावं च॥

- सु.सू.15/14

■ धातु क्षय

■ चरक

- रसधातु- घट्टते सहते शब्दं नोच्चैर्द्रवति शूल्यते।
हृदयं ताम्यति स्वल्पचेष्टस्यापि रसक्षये॥
- रक्तधातु-परुषास्फुटिता म्लाना त्वग्रूक्षा रक्तसंक्षये।
मांसधातु-मांसक्षये विशेषणं स्फिग्रीवोदरशुष्कता॥
- मेदधातु-सन्धिनां स्फुटनं ग्लानिरक्षणोरायास एव च।
लक्षणं मेदसि क्षीणे तनुत्वं चोदरस्य च॥
- अस्थिधातु-केशलोमनखश्मश्रु द्विजप्रपतनं श्रमः।
ज्ञेयमस्थिक्षये लिंगसन्धिशैथिल्यमेव च॥
- मज्जाधातु-शीर्यन्त इव चास्थिनि दुर्बलानि लघुनि च।
प्रततं वातरोगीणि क्षीणे मज्जानि देहिनाम्॥
- शुक्रधातु-दौर्बल्यं मुखशोषश्च पाण्डुत्वं सदनं श्रमः।
क्लैब्यं शुक्राविसर्गश्च क्षीणशुक्रस्य लक्षणम्॥

- च.वि.17/62-69

■ अष्टांग हृदयं

- रसधातु-रसे (क्षीणे) रौक्ष्यं श्रमः शोषो ग्लानिः शब्दासहिष्णुता।
शोषो मुखादिनाम्। ग्लानिः क्लमः॥
- रक्तधातु-रक्तेऽम्लशिरशिरप्रीतिशिराशैथिल्यरूक्षताः॥
- मांसधातु-मांसेऽक्षग्लानिगण्डस्फिक्शुष्कतासन्धिवेदनाः॥
- मेदधातु-मेदसि स्वपनं कट्याः प्लीहोवृद्धीः कृशाङ्गता॥
- अस्थिधातु-अस्थनस्थितोदः सदनं दन्तकेशनखादिषु॥
- मज्जाधातु-अस्थानां मज्जानि सौषिर्यं भ्रमस्तिमिरदर्शनम्॥
- शुक्रधातु-शुक्ले चिरात् प्रसिच्येत शुक्रं शोणितमेव वा।
तोदोऽत्यर्थं वृषणयोमेदं धूमायतीव च॥

- अ.ह.सू.11/15-21

■ सुश्रुत

- रसक्षये-हृत्पीडाकम्पशून्यतास्तृष्णा च॥
- शोणितक्षये-त्वक्पारुष्यमम्लशीतप्रार्थना सिराशैथिल्यं च॥
- मांसक्षये-स्फिगण्डौष्ठोपस्थोरूवक्षःकक्षापिण्डिकोदरग्रीवा
शुष्कता रौक्ष्यतोदौ गात्राणां सदनं धमनीशैथिल्यं च॥
- मेदःक्षये-प्लीहाभिवृद्धिः सन्धिशून्यता रौक्ष्यं मेदूरमांसप्रार्थना च॥
- अस्थिक्षये-ऽस्थिशूलं दन्तनखभङ्गो रौक्ष्यं च॥
- मज्जाक्षये-ऽल्पशुक्रता पर्वभेदोऽस्थिनिस्तोदोऽस्थिशून्यता च॥
- शुक्रक्षये-मेदवृषणवेदनाऽशक्तिमैथुने चिराद्वा प्रसेकः प्रसेके
चाल्परक्तशुक्रदर्शनम्॥ - सु.सू.15/9

9.1 DHATU VRIDDHI-KSHAYA PARIKSHAN

Objective : To assess the Dhatu Vriddhi-Kshaya lakshanas of an individual

Name: :

Age : **Gender:** M/F **Occupation:**

Body temperature : **Weight:**kg

Blood pressure : **Pulse rate:**...../min

Address :

Rasa Dhatu Assessment

Features of Aggravated Rasa Dhatu (रस वृद्धि लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	प्रसेक (Excessive salivation)	Do you experience excessive salivation such as drooling or a constant watery mouth?	Excessive salivation is assessed by observing drooling or oral discomfort, examining salivary glands for swelling or tenderness and exploring symptom triggers, duration and associated conditions. Potential causes like infections or neurological issues can be confirmed through imaging or lab tests.	
2	अरोचक (Loss of appetite)	Have you lost your appetite or felt disinterested in eating?	Loss of appetite is evaluated by observing signs of malnutrition, assessing abdominal or systemic health and exploring dietary habits, psychological factors and associated symptoms like fever. Conditions like infections or gastrointestinal disorders can be identified through lab tests, imaging or endoscopy.	
3	आस्यवैरस्य (Tastelessness)	Have you noticed a loss of taste or change in how food tastes?	Tastelessness is evaluated by noting oral hygiene and tongue appearance and ask questions, assessing for nerve damage or infections and discussing associated symptoms. Conditions like nutritional deficiencies or neurological disorders.	
4	हल्लास (Nausea)	Do you feel nausea or a sense of	Nausea is evaluated by observing signs of discomfort, assessing abdominal tenderness or bloating and exploring symptom onset, triggers and associated factors through questioning. Clinical findings may suggest gastritis,	

		uneasiness in your stomach?	infections or neurological issues, confirmed through imaging, endoscopy or lab tests.	
5	स्रोतोरोध (Obstruction of channels)	Do you feel congestion phlegm or blockages that interfere with normal functions like breathing?	Blockage of channels by mucous or other secretions (as seen in bronchial asthma, Nasal congestion, COPD). Obstruction of channels is evaluated by observing swelling, discolouration or impaired movement, assessing affected areas for tenderness or rigidity and exploring symptom onset, triggers and progression. Conditions like blockages or inflammation are confirmed using imaging, lab tests CBC (complete blood count) , ESR, CRP etc. or functional studies.	
6	स्वादुद्वेष (Dislike food having sweet taste)	Have you developed an aversion to sweet tasting foods or drinks?	Dislike for sweet-tasting food is evaluated by observing eating behaviour, digestive discomfort and exploring symptom onset, triggers and related factors.	
7	अगमर्द (Body ache)	Do you experience generalized aches or pains in your body or muscles?	Generalized body pain (as seen in malnutrition, vitamin deficiency etc.). Body ache is assessed by observing movement limitations, palpating for muscle tenderness and exploring onset, severity and associated symptoms. It may indicate infections, inflammatory conditions or fibromyalgia, diagnosed through blood tests, imaging or specialized assessments like electromyography.	

Total Rasa Dhatu Vriddhi Score:/21 X 100 =%

Features of Decreased Rasa Dhatu (रस क्षय लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्नै)	Score
1	रौक्ष्यं (Dryness)	Do you experience dryness or roughness in your skin hair or mucous membranes?	Dryness is assessed by observing skin, mucous membranes and overall hydration, while exploring symptoms like thirst, irritation or difficulty swallowing. It may indicate dehydration, Sjögren's syndrome or diabetes, diagnosed through blood tests, salivary function or imaging.	
2	भ्रम	Have you felt dizziness,	The state of illusion is assessed by observing altered perceptions, questioning the onset, duration and evaluating associated symptoms like confusion or	

	(State of illusion)	fainting recently?	hallucinations. It may be linked to psychiatric disorders, neurological conditions or substance abuse, diagnosed using imaging, EEG and psychological evaluations.	
3	शोष (Wasting)	Have you unintentional weight loss, thinning or muscle wasting?	Wasting is evaluated through observation of muscle loss, questioning about weight changes, fatigue, appetite and assessing physical function. It may indicate conditions like malnutrition, malabsorption, cancer, HIV or endocrine disorders, diagnosed using blood tests, imaging, biopsies or electromyography.	
4	रलानी (Tiredness without work)	Do you frequently feel fatigue, unable to complete daily activities?	Tiredness without exertion is assessed by observing fatigue patterns, examine for high/low blood pressure, pulse rate, pulse volume, pulse character etc. Exploring associated symptoms like sleep disturbances or depression and conditions such as anaemia, hypothyroidism may be diagnosed through blood tests, thyroid function tests etc.	
5	शब्दअसहिष्णुता (Intolerance to noise)	Are you highly sensitive or intolerant to loud sudden sounds?	Intolerance to noise is assessed by observing signs and associated symptoms like dehydration, shock, headaches or dizziness, meningitis. It may indicate conditions such as migraine, hyperacusis or vestibular disorders, diagnosed using audiometry, MRI or neurological evaluations.	
6	हृद्रव, हृत्कम्प (Palpitation, Tachycardia)	Do you frequently feel Palpitation or Tachycardia?	Palpitation and tachycardia are assessed by observing heart rate, skin pallor or sweating, examining pulse and heart sounds. They may indicate arrhythmias, anemia or hyperthyroidism, diagnosed through ECG, blood tests or echocardiography.	

Total Rasa Dhatu Kshaya Score:/18 X 100 =%

Rakta Dhatu Assessment

Features of Aggravated Rakta Dhatu (रक्त वृद्धी लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	विसर्प	Do you frequently Suffer from skin diseases like inflammation?	Inflammatory conditions like cellulitis due to toxic effect in infections. Investigate for CBC, ESR, CRP	
2	विद्रुधी	Do you suffer from inflammatory conditions such as cellulitis or	An abscess inflammation with the collection of pus. Investigate for CBC, ESR	

		infections leading to swelling and redness?		
3	कुष्ठ	Do you frequently suffer from skin diseases like rashes, eczema or autoimmune diseases?	Skin diseases due to various causes. Examine the nature of lesions. Investigate for the presence of infective, allergic or autoimmune elements. Investigate for CBC, AEC, ESR, CRP, ASO.	
4	वातरक्त	Do you experience symptoms of arthritis or joint inflammation?	Rheumatoid Arthritis, Gout etc. due to inflammatory disorders of metabolic or autoimmune origin. Investigate for CBC, ESR, LFT, Serum uric acid, RA factor	
5	रक्तपित्त	Have you been diagnosed with bleeding disorders or prolonged bleeding?	Bleeding disorders due to defect in hemostasis mechanism-congenital & acquired. Investigate for Platelet count, BT, CT, LFT	
6	गुल्म	Do you experience abdominal distension or bloating regularly?	Abdominal distension due to pathological changes in intestine and surroundings. Investigate for USG, CT, CBC	
7	उपकुश	Have you noticed bleeding from the gums or Unexplained bruising?	Bleeding from the gums due to local causes like gingivitis or due to systemic pathology. Investigate for BT, CT, PT, LFT	
8	व्यंग	Do you experience hyperpigmentation of the skin or conditions like Addison's disease?	Hyperpigmentation is seen in endocrine disorders like Addison's disease, cushing's disease. Investigate for CBC, ESR, cortisol, TSH	
9	अग्निनाशः	Do you suffer from digestive disturbances like loss of appetite or acid-base imbalances?	Loss of digestive power is seen in conditions like acidosis, alkalosis etc. Investigate for CBC, LFT, Urine	
10	रक्त त्वक (Redness of skin)	Do you often observe skin redness or inflammation in response to minor triggers?	Increased redness of skin due to bleeding disorders or conditions like erythema nodosum, Raynaud's phenomenon etc. Investigate for ESR, BT, CT, PT, LFT	
11	रक्त नेत्र (Redness of eyes)	Do you notice increased redness or irritation in your eyes?	Increased redness of the eyes (due to bleeding or inflammation) It may indicate conditions like conjunctivitis, glaucoma or uveitis, diagnosed through slit-lamp examination or tonometry. Investigate for BT, CT, PT, LFT.	

12	रक्त मूत्र (Blood in urine)	Have you experienced red coloured urine without a known cause?	Red coloured urine due to bleeding or use of some drugs. Blood in urine is assessed by observing urine colour and clarity, palpating the abdomen for tenderness and asking about pain, trauma or related symptoms. It may indicate UTIs, kidney stones or tumors, diagnosed through urinalysis, imaging and cystoscopy.	
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Total Rakta Dhatu Vriddhi Score:/36 X 100 =%

Features of Decreased Rakta Dhatu (रक्त क्षय लक्षण)

Sr. No .	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	अम्लरस-प्रीति (Desire for food having sour taste)	Do you experience a strong desire for sour-tasting foods or drinks?	Desire for sour taste seen in conditions like anaemia. A desire for sour-tasting food is evaluated through observation of eating habits, questioning onset, frequency and related symptoms like acidity or digestive discomfort. It may indicate acid reflux, gastric ulcers or metabolic imbalances, confirmed through endoscopy, pH testing, CBC and PCV.	
2	शिशिर प्रीति (Desire for cold)	Do you often feel a desire for cold things, especially in not environments?	Desire for cold due to ischemic effect, as a mechanism to protect heat. Desire for cold can be assessed by observing behaviour and questioning about associated symptoms like fever, fatigue or digestive issues. It may indicate conditions like fever, thyroid disorders or anaemia, diagnosed through blood tests, temperature monitoring and thyroid function tests.	
3	सिराशैथिल्य (Hypovolemia or hypotension)	Have you noticed a loss of elasticity in your veins or Skin, particularly with aging?	Clinical examination for pulse volume, character, Blood pressure. Symptoms may indicate dehydration, blood loss or heart failure, diagnosed through blood tests, ECG and imaging studies like ultrasound.	
4	रूक्षता (Dryness)	Do you experience persistent dryness of the skin?	Dryness of skin due to ischemic effect or in conditions like DM, eczema, psoriasis. Investigate for CBC, ESR, CRP, RBS, Lipid profile.	

Total Rakta Dhatu Kshaya Score:/12 X 100 =%

Mamsa Dhatu Assessment

Features of Aggravated Mamsa Dhatu (मांस वृद्धि लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्नै)	Score
1	गण्डवृद्धी	Do you expect flabbiness in your cheeks?	Due to anaemia etc. Investigate for CBC, ESR	
2	उरूवृद्धी (hypertrophy of thigh muscle)	Do you experience flabbiness or Enlargement on your thighs?	Inflammation of low back due to lumbar paraspinal muscle spasm. Clinical examination by inspection, palpation, X-ray, CT, MRI	
3	उदरवृद्धी (abdominal muscles hypertrophy)	Do you experience enlargement in your Abdomen?	Inflammation of the abdomen due to abdominal muscle spasm. Clinical examination by inspection, palpation, USG	
4	अर्बुद, ग्रन्थि (Tumor)	Have you noticed any lumps or tumor growth in your body?	Tumor (benign & malignant). Clinical examination by inspection, palpation CBC, PSA (prostate-specific antigen)	
5	कण्ठादी अधिमास (over growth in the neck)	Do you experience excess Fleshy growth on your neck and other body parts?	Overgrowth in the neck in conditions like goiter or tonsillitis. Clinical examination by inspection, palpation, CBC, ESR, TFT (thyroid function test)	

Total Mamsa Dhatu Vriddhi Score:/15 X 100 =%

Features of Decreased Mamsa Dhatu (मांस क्षय लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्नै)	Score
1	अक्षग्लानी (Weakness in eyes)	Do you experience fatigue in your sensory organs (especially eyes) due to general weakness?	Weakness of body parts due to muscle fatigue. Investigate for Serum Calcium, Phosphorous, Total protein, CPK	
2	गंडस्फिकशुष्कता (Muscles wasting of related area)	Have you noticed a reduction in the size of your cheeks & buttocks?	Emaciation of neck and buttocks due to muscle wasting. Clinical examination by inspection, palpation (look for UMN, LMN lesion), X-ray, CT, MRI	

3	सन्धिवेदना (Arthralgia)	Do you have joint pain?	Joint pain is assessed by observing joint swelling or redness, palpating for tenderness. It may indicate conditions such as arthritis, infections or autoimmune diseases, diagnosed through imaging, blood	
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Total Mamsa Dhatu Kshaya Score:/9 X 100 =%

Meda Dhatu Assessment

Features of Aggravated Meda Dhatu (मेद वृद्धि लक्षण)

Sr. No	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्नै)	Score
1	श्रमः (Tiredness)	Do you experience palpitations and fatigue even during mild exercise or physical activity?	Examine physical fatigue, evaluating muscle weakness. It may indicate anaemia, thyroid disorders or chronic fatigue syndrome, diagnosed through blood tests, thyroid function tests and sleep studies.	
2	अल्पेऽपिचेष्टिते श्वासः (Feeling exhausted on even less exertion)	Have you ever noticed shortness of breath especially during physical activity or at rest?	Observing physical activity levels, assessing for weakness or pallor and inquiring about onset, triggers and associated symptoms. It may indicate anaemia, thyroid disorders. Diagnosed through blood tests, imaging or endocrine assessments.	
3	स्किक्स्तन उदर लंबनम् (Pendulous overgrowth of gluteal region, chest region and belly)	Have you noticed any signs of central obesity such as increased fat around your abdomen or waistline?	Central obesity due to increased deposition of fat. Investigate for lipid profile, RBS, BMI	
4.	अतिस्निग्धता (Very oily skin)	Have you ever noticed very oily skin in all season?	Skin is assessed excess shine, enlarged pores or acne. Questions focus on onset, triggers like diet or stress. It may indicate hormonal imbalances, PCOS or seborrheic dermatitis, diagnosed through skin tests.	

5.	प्रमेह पुरुरूप (Prodromal signs of diabetes)	Do you experience any prodromal signs of diabetes?	Prodromal signs of diabetes include fatigue, frequent urination, excessive thirst and unexplained weight loss. Blood glucose tests can help diagnose prediabetes or early diabetes mellitus.	
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Total Meda Dhatu Vriddhi Score:/15 X 100 =%

Features of Decreased Meda Dhatu (मेद क्षय लक्षण)

Sr. No	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	स्वपनम् कट्याः (Numbness in the lower back or lumbar region)	Have you ever felt numbness or loss of sensation in your Lower back or lumbar region?	Numbness in the lumbar region due to peripheral neuropathy or (IVDP) Intervertebral Disc Prolapse. Clinical examination by inspection, palpation, X-ray, CT, MRI	
2	प्लीहा वृद्धि (Splenomegaly)	Have you noticed any signs of discomfort or fullness in your left upper abdomen?	Splenomegaly is hypertrophy of the spleen due to overload in various conditions. Clinical examination by abdominal palpation, USG, RBC count, Platelet count	
3	कृशाङ्गता (Lean built)	Have you ever experienced extreme thinning or significant weight loss?	Emaciation in chronic diabetes. RBC, RBS	

Total Meda Dhatu Kshaya Score:/9 X 100 =%

Asthi Dhatu Assessment

Features of Aggravated Asthi Dhatu (अस्थि वृद्धि लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	अधिदंत (Hypertrophy of teeth/ Extra teeth)	Do you notice any extra teeth or calculus apart from the normal in your mouth?	Extra teeth in conditions like dental calculus. Serum Calcium, Phosphorus	

2	अध्यस्थि (Hypertrophy of bone)	Have you noticed any unusual bone growth or areas where your bones seem larger or thicker than normal?	Bone overgrowth due to hormonal imbalance. X-ray, Serum Calcium, Phosphorus, TFT, Plasma Calcitonin	
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Total Asthi Dhatu Vriddhi Score:/6 X 100 =%

Features of Decreased Asthi Dhatu (अस्थि क्षय लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	अस्थिशूल/तोद (Pain in bones/ Arthralgia)	Have you experienced pricking pain in bones?	Pricking pain due to calcium deficiency or fevers like Dengue. CBC, Serum Calcium	
2	दंतशातन (Cracking and breaking of teeth)	Have you noticed increased brittleness or fragility in teeth as breaking easily?	Increased brittleness of teeth due to deficiency of Calcium, phosphate etc. Serum Calcium	
3	केशशातन (Fall of hairs and beard)	Have you experienced any severe Hair loss recently?	Hair fall is evaluated by asking questions about hair fall and exploring onset, triggers factors. It is due to vitamin deficiency, infection, alopecia, hormonal imbalance, or nutritional deficiencies etc are diagnosed through comb test	
4	नखशातन (Brittleness of nails)	Have you noticed increased brittleness or breakage of nails?	Increased brittleness of nails due to vitamin deficiency, anaemia, infection, rheumatic disorders etc.	

Total Asthi Dhatu Kshaya Score:/12 X 100 =%

Majja Dhatu Assessment

Features of Aggravated Maja Dhatu (मज्जा वृद्धि लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	नेत्र गौरव (Heaviness in eyes)	Do you often experience the heaviness in your eyes?	Heaviness of eyes due to increased intraocular pressure. Tonometry can be used to measure IOP	

2	अंग गौरव (Heaviness in body)	Do you ever notice the heaviness in your body parts or any oedema on the body?	Heaviness of body parts due to some edematous or spastic conditions. Clinical examination by inspection and palpation. Investigate for CBC, Urine routine.	
3	पर्वसुस्थुलमुलानी Arunshi (Prominent-interphalangeal joints)	Have you ever seen the small vesicles in your joints?	Appearance of small vesicles with large bases in joints in conditions like osteomyelitis. Clinical examination by inspection and palpation. Investigate for CBC	

Total Majja Dhatu Vriddhi Score:/9 X 100 =%

Features of Decreased Majja Dhatu (मज्जा क्षय लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	अस्थि शौषिर्य (Osteoporosis)	Have you ever experienced frequent fractures or bone pain?	Osteoporosis is evaluated through observation for posture changes, questioning about fractures or back pain. It may be linked to aging, hormonal imbalances, vitamin D deficiency, diagnosed via bone density scans, X-rays, Serum Calcium, estrogen etc	
2	भ्रम (State of illusion)	Do you often feel like you are about to faint?	Fainting due to ischemic effect, reduced hemopoietic activity. Diagnosed by clinical examination and CBC	
3	तिमिर दर्शन (Blurred vision)	Have you had issues with vision loss recently?	Blurred vision, loss of vision due to neural defects with associated symptoms like headaches or dizziness. It may indicate refractive errors, cataracts or neurological conditions, diagnosed through eye exams, cranial nerve examination, imaging and visual field tests.	
4	पर्वभेद (pain in interphalangeal joints)	Do you often have pain in interphalangeal joints?	Interphalangeal pain is assessed by observing swelling, redness or deformity, palpating for tenderness or stiffness and exploring symptom onset and triggers. Conditions like arthritis, gout or trauma are diagnosed using imaging, blood tests.	

Total Majja Dhatu Kshaya Score:/12 X 100 =%

Shukra Dhatu Assessment

Features of Aggravated Shukra Dhatu (शुक्र वृद्धि लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	अतिस्त्रीकामता (Increased Libido)	Have you experienced increased sexual desire recently?	Increased sexual desire due to hormonal imbalance as seen in adrenal hyperplasia. Investigate by clinical examination, USG, ACTH	
2	शुक्राश्मरी (Spermolith)	Were you diagnosed with calculi in the genital tract in the recent past or do you feel obstruction during passing urine or semen?	Calculi in the genital tract like seminal vesicle calculi & prostatic calculi in males and ovarian calculi or uterine calculi in females. It can be associated with infections or obstructions. Diagnosed through USG, PSA, semen analysis.	

Total Shukra Dhatu Vriddhi Score:/6 X 100 =%

Features of Decreased Shukra Dhatu (शुक्रक्षय लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	चिरात् प्रसेक (Late ejaculation)	Do you experience delayed ejaculation, late sexual response? (Women expel fluids from the urethra during sexual arousal and orgasm)	Late ejaculation is evaluated by observing late sexual response, questioning about the onset, frequency and associated symptoms like erectile dysfunction or discomfort. It may be linked to psychological factors, hormonal imbalances or neurological conditions, diagnosed through blood tests, hormone levels and imaging.	
2	रक्तशोणित (Blood in semen)	Did you feel like your semen is mixed with blood after ejaculation?	Ejaculation mixed with blood due to bleeding from genital tract as seen in BPH, Carcinoma of prostate, TB etc. Clinical examination, USG, CBC, semen analysis, prostate-specific antigen (PSA) tests	
3	मेढ्रवृषण वेदना (Pain in penis and testes)	Do you experience pain in penis and testes?	Questioning about pain. It may indicate conditions like epididymitis, testicular torsion or varicocele. Diagnosis is	

			confirmed through ultrasound, urine tests and clinical evaluation.	
4	वृषण धुमायन (Burning sensation genital region)	Do you experience a burning sensation in the penis, testes, genital tract frequently after micturition or ejaculation?	Questioning focuses on onset, associated pain and urinary symptoms. It may indicate infections like UTIs, STIs or irritation, diagnosed through urine analysis, swabs and USG.	
5	मुखशोष (Xerostomia)	Do you experience dryness in your mouth?	Dryness in the mouth is assessed by observing oral moisture, tongue texture, and dental health. Questioning explores symptom, triggers. Conditions like Sjögren's syndrome, dehydration etc are diagnosed through saliva tests etc	

Total Shukra Dhatu Vriddhi Score:/15 X 100 =%

Result: Dhatu Vriddhi :.....

Dhatu Kshaya :.....



Teacher's Signature

9.2 DHATU VRIDDHI-KSHAYA PARIKSHAN

Objective : To assess the Dhatu Vriddhi-Kshaya lakshanas of an individual

Name: :

Age : **Gender:** M/F **Occupation:**

Body temperature : **Weight:**kg

Blood pressure : **Pulse rate:**...../min

Address :

Rasa Dhatu Assessment

Features of Aggravated Rasa Dhatu (रस वृद्धि लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्नै)	Score
1	प्रसेक (Excessive salivation)	Do you experience excessive salivation such as drooling or a constant watery mouth?	Excessive salivation is assessed by observing drooling or oral discomfort, examining salivary glands for swelling or tenderness and exploring symptom triggers, duration and associated conditions. Potential causes like infections or neurological issues can be confirmed through imaging or lab tests.	
2	अरोचक (Loss of appetite)	Have you lost your appetite or felt disinterested in eating?	Loss of appetite is evaluated by observing signs of malnutrition, assessing abdominal or systemic health and exploring dietary habits, psychological factors and associated symptoms like fever. Conditions like infections or gastrointestinal disorders can be identified through lab tests, imaging or endoscopy.	
3	आस्यवैरस्य (Tastelessness)	Have you noticed a loss of taste or change in how food tastes?	Tastelessness is evaluated by noting oral hygiene and tongue appearance and ask questions, assessing for nerve damage or infections and discussing associated symptoms. Conditions like nutritional deficiencies or neurological disorders.	
4	हल्लास (Nausea)	Do you feel nausea or a sense of	Nausea is evaluated by observing signs of discomfort, assessing abdominal tenderness or bloating and exploring symptom onset, triggers and associated factors through questioning. Clinical findings may suggest gastritis,	

		uneasiness in your stomach?	infections or neurological issues, confirmed through imaging, endoscopy or lab tests.	
5	स्रोतोरोध (Obstruction of channels)	Do you feel congestion phlegm or blockages that interfere with normal functions like breathing?	Blockage of channels by mucous or other secretions (as seen in bronchial asthma, Nasal congestion, COPD). Obstruction of channels is evaluated by observing swelling, discolouration or impaired movement, assessing affected areas for tenderness or rigidity and exploring symptom onset, triggers and progression. Conditions like blockages or inflammation are confirmed using imaging, lab tests CBC (complete blood count) , ESR, CRP etc. or functional studies.	
6	स्वादुद्वेष (Dislike food having sweet taste)	Have you developed an aversion to sweet tasting foods or drinks?	Dislike for sweet-tasting food is evaluated by observing eating behaviour, digestive discomfort and exploring symptom onset, triggers and related factors.	
7	अगमर्द (Body ache)	Do you experience generalized aches or pains in your body or muscles?	Generalized body pain (as seen in malnutrition, vitamin deficiency etc.). Body ache is assessed by observing movement limitations, palpating for muscle tenderness and exploring onset, severity and associated symptoms. It may indicate infections, inflammatory conditions or fibromyalgia, diagnosed through blood tests, imaging or specialized assessments like electromyography.	

Total Rasa Dhatu Vriddhi Score:/21 X 100 =%

Features of Decreased Rasa Dhatu (रस क्षय लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्नै)	Score
1	रौक्ष्यं (Dryness)	Do you experience dryness or roughness in your skin hair or mucous membranes?	Dryness is assessed by observing skin, mucous membranes and overall hydration, while exploring symptoms like thirst, irritation or difficulty swallowing. It may indicate dehydration, Sjögren's syndrome or diabetes, diagnosed through blood tests, salivary function or imaging.	
2	भ्रम	Have you felt dizziness,	The state of illusion is assessed by observing altered perceptions, questioning the onset, duration and evaluating associated symptoms like confusion or	

	(State of illusion)	fainting recently?	hallucinations. It may be linked to psychiatric disorders, neurological conditions or substance abuse, diagnosed using imaging, EEG and psychological evaluations.	
3	शोष (Wasting)	Have you unintentional weight loss, thinning or muscle wasting?	Wasting is evaluated through observation of muscle loss, questioning about weight changes, fatigue, appetite and assessing physical function. It may indicate conditions like malnutrition, malabsorption, cancer, HIV or endocrine disorders, diagnosed using blood tests, imaging, biopsies or electromyography.	
4	ग्लानी (Tiredness without work)	Do you frequently feel fatigue, unable to complete daily activities?	Tiredness without exertion is assessed by observing fatigue patterns, examine for high/low blood pressure, pulse rate, pulse volume, pulse character etc. Exploring associated symptoms like sleep disturbances or depression and conditions such as anaemia, hypothyroidism may be diagnosed through blood tests, thyroid function tests etc.	
5	शब्दअसहिष्णुता (Intolerance to noise)	Are you highly sensitive or intolerant to loud sudden sounds?	Intolerance to noise is assessed by observing signs and associated symptoms like dehydration, shock, headaches or dizziness, meningitis. It may indicate conditions such as migraine, hyperacusis or vestibular disorders, diagnosed using audiometry, MRI or neurological evaluations.	
6	हृद्रव, हृत्कम्प (Palpitation, Tachycardia)	Do you frequently feel Palpitation or Tachycardia?	Palpitation and tachycardia are assessed by observing heart rate, skin pallor or sweating, examining pulse and heart sounds. They may indicate arrhythmias, anemia or hyperthyroidism, diagnosed through ECG, blood tests or echocardiography.	

Total Rasa Dhatu Kshaya Score:/18 X 100 =%

Rakta Dhatu Assessment

Features of Aggravated Rakta Dhatu (रक्त वृद्धी लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	विसर्प	Do you frequently Suffer from skin diseases like inflammation?	Inflammatory conditions like cellulitis due to toxemic effect in infections. Investigate for CBC, ESR, CRP	
2	विद्रुधी	Do you suffer from inflammatory conditions such as cellulitis or	An abscess inflammation with the collection of pus. Investigate for CBC, ESR	

		infections leading to swelling and redness?		
3	कुष्ठ	Do you frequently suffer from skin diseases like rashes, eczema or autoimmune diseases?	Skin diseases due to various causes. Examine the nature of lesions. Investigate for the presence of infective, allergic or autoimmune elements. Investigate for CBC, AEC, ESR, CRP, ASO.	
4	वातरक्त	Do you experience symptoms of arthritis or joint inflammation?	Rheumatoid Arthritis, Gout etc. due to inflammatory disorders of metabolic or autoimmune origin. Investigate for CBC, ESR, LFT, Serum uric acid, RA factor	
5	रक्तपित्त	Have you been diagnosed with bleeding disorders or prolonged bleeding?	Bleeding disorders due to defect in hemostasis mechanism-congenital & acquired. Investigate for Platelet count, BT, CT, LFT	
6	गुल्म	Do you experience abdominal distension or bloating regularly?	Abdominal distension due to pathological changes in intestine and surroundings. Investigate for USG, CT, CBC	
7	उपकुश	Have you noticed bleeding from the gums or Unexplained bruising?	Bleeding from the gums due to local causes like gingivitis or due to systemic pathology. Investigate for BT, CT, PT, LFT	
8	व्यंग	Do you experience hyperpigmentation of the skin or conditions like Addison's disease?	Hyperpigmentation is seen in endocrine disorders like Addison's disease, cushing's disease. Investigate for CBC, ESR, cortisol, TSH	
9	अग्निनाशः	Do you suffer from digestive disturbances like loss of appetite or acid-base imbalances?	Loss of digestive power is seen in conditions like acidosis, alkalosis etc. Investigate for CBC, LFT, Urine	
10	रक्त त्वक (Redness of skin)	Do you often observe skin redness or inflammation in response to minor triggers?	Increased redness of skin due to bleeding disorders or conditions like erythema nodosum, Raynaud's phenomenon etc. Investigate for ESR, BT, CT, PT, LFT	
11	रक्त नेत्र (Redness of eyes)	Do you notice increased redness or irritation in your eyes?	Increased redness of the eyes (due to bleeding or inflammation) It may indicate conditions like conjunctivitis, glaucoma or uveitis, diagnosed through slit-lamp examination or tonometry. Investigate for BT, CT, PT, LFT.	

12	रक्त मूत्र (Blood in urine)	Have you experienced red coloured urine without a known cause?	Red coloured urine due to bleeding or use of some drugs. Blood in urine is assessed by observing urine colour and clarity, palpating the abdomen for tenderness and asking about pain, trauma or related symptoms. It may indicate UTIs, kidney stones or tumors, diagnosed through urinalysis, imaging and cystoscopy.	
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Total Rakta Dhatu Vriddhi Score:/36 X 100 =%

Features of Decreased Rakta Dhatu (रक्त क्षय लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	अम्लरस-प्रीति (Desire for food having sour taste)	Do you experience a strong desire for sour-tasting foods or drinks?	Desire for sour taste seen in conditions like anaemia. A desire for sour-tasting food is evaluated through observation of eating habits, questioning onset, frequency and related symptoms like acidity or digestive discomfort. It may indicate acid reflux, gastric ulcers or metabolic imbalances, confirmed through endoscopy, pH testing, CBC and PCV.	
2	शिशिर प्रीति (Desire for cold)	Do you often feel a desire for cold things, especially in not environments?	Desire for cold due to ischemic effect, as a mechanism to protect heat. Desire for cold can be assessed by observing behaviour and questioning about associated symptoms like fever, fatigue or digestive issues. It may indicate conditions like fever, thyroid disorders or anaemia, diagnosed through blood tests, temperature monitoring and thyroid function tests.	
3	सिराशैथिल्य (Hypovolemia or hypotension)	Have you noticed a loss of elasticity in your veins or Skin, particularly with aging?	Clinical examination for pulse volume, character, Blood pressure. Symptoms may indicate dehydration, blood loss or heart failure, diagnosed through blood tests, ECG and imaging studies like ultrasound.	
4	रूक्षता (Dryness)	Do you experience persistent dryness of the skin?	Dryness of skin due to ischemic effect or in conditions like DM, eczema, psoriasis. Investigate for CBC, ESR, CRP, RBS, Lipid profile.	

Total Rakta Dhatu Kshaya Score:/12 X 100 =%

Mamsa Dhatu Assessment

Features of Aggravated Mamsa Dhatu (मांस वृद्धि लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्नै)	Score
1	गण्डवृद्धि	Do you expect flabbiness in your cheeks?	Due to anaemia etc. Investigate for CBC, ESR	
2	उरुवृद्धि (hypertrophy of thigh muscle)	Do you experience flabbiness or Enlargement on your thighs?	Inflammation of low back due to lumbar paraspinal muscle spasm. Clinical examination by inspection, palpation, X-ray, CT, MRI	
3	उदरवृद्धि (abdominal muscles hypertrophy)	Do you experience enlargement in your Abdomen?	Inflammation of the abdomen due to abdominal muscle spasm. Clinical examination by inspection, palpation, USG	
4	अर्बुद, ग्रन्थि (Tumor)	Have you noticed any lumps or tumor growth in your body?	Tumor (benign & malignant). Clinical examination by inspection, palpation CBC, PSA (prostate-specific antigen)	
5	कण्ठादी अधिमास (over growth in the neck)	Do you experience excess Fleshy growth on your neck and other body parts?	Overgrowth in the neck in conditions like goiter or tonsillitis. Clinical examination by inspection, palpation, CBC, ESR, TFT (thyroid function test)	

Total Mamsa Dhatu Vriddhi Score:/15 X 100 =%

Features of Decreased Mamsa Dhatu (मांस क्षय लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्नै)	Score
1	अक्षग्लानी (Weakness in eyes)	Do you experience fatigue in your sensory organs (especially eyes) due to general weakness?	Weakness of body parts due to muscle fatigue. Investigate for Serum Calcium, Phosphorous, Total protein, CPK	
2	गंडस्फिकशुष्कता (Muscles wasting of related area)	Have you noticed a reduction in the size of your cheeks & buttocks?	Emaciation of neck and buttocks due to muscle wasting. Clinical examination by inspection, palpation (look for UMN, LMN lesion), X-ray, CT, MRI	

3	सन्धिवेदना (Arthralgia)	Do you have joint pain?	Joint pain is assessed by observing joint swelling or redness, palpating for tenderness. It may indicate conditions such as arthritis, infections or autoimmune diseases, diagnosed through imaging, blood	
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Total Mamsa Dhatu Kshaya Score:/9 X 100 =%

Meda Dhatu Assessment

Features of Aggravated Meda Dhatu (मेद वृद्धि लक्षण)

Sr. No	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्नै)	Score
1	श्रमः (Tiredness)	Do you experience palpitations and fatigue even during mild exercise or physical activity?	Examine physical fatigue, evaluating muscle weakness. It may indicate anaemia, thyroid disorders or chronic fatigue syndrome, diagnosed through blood tests, thyroid function tests and sleep studies.	
2	अल्पेऽपिचेष्टिते श्वासः (Feeling exhausted on even less exertion)	Have you ever noticed shortness of breath especially during physical activity or at rest?	Observing physical activity levels, assessing for weakness or pallor and inquiring about onset, triggers and associated symptoms. It may indicate anaemia, thyroid disorders. Diagnosed through blood tests, imaging or endocrine assessments.	
3	स्किक्स्तन उदर लंबनम् (Pendulous overgrowth of gluteal region, chest region and belly)	Have you noticed any signs of central obesity such as increased fat around your abdomen or waistline?	Central obesity due to increased deposition of fat. Investigate for lipid profile, RBS, BMI	
4.	अतिस्निग्धता (Very oily skin)	Have you ever noticed very oily skin in all season?	Skin is assessed excess shine, enlarged pores or acne. Questions focus on onset, triggers like diet or stress. It may indicate hormonal imbalances, PCOS or seborrheic dermatitis, diagnosed through skin tests.	

5.	प्रमेह पुरुरूप (Prodromal signs of diabetes)	Do you experience any prodromal signs of diabetes?	Prodromal signs of diabetes include fatigue, frequent urination, excessive thirst and unexplained weight loss. Blood glucose tests can help diagnose prediabetes or early diabetes mellitus.	
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Total Meda Dhatu Vriddhi Score:/15 X 100 =%

Features of Decreased Meda Dhatu (मेद क्षय लक्षण)

Sr. No	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	स्वपनम् कट्याः (Numbness in the lower back or lumbar region)	Have you ever felt numbness or loss of sensation in your Lower back or lumbar region?	Numbness in the lumbar region due to peripheral neuropathy or (IVDP) Intervertebral Disc Prolapse. Clinical examination by inspection, palpation, X-ray, CT, MRI	
2	प्लीहा वृद्धि (Splenomegaly)	Have you noticed any signs of discomfort or fullness in your left upper abdomen?	Splenomegaly is hypertrophy of the spleen due to overload in various conditions. Clinical examination by abdominal palpation, USG, RBC count, Platelet count	
3	कृशाङ्गता (Lean built)	Have you ever experienced extreme thinning or significant weight loss?	Emaciation in chronic diabetes. RBC, RBS	

Total Meda Dhatu Kshaya Score:/9 X 100 =%

Asthi Dhatu Assessment

Features of Aggravated Asthi Dhatu (अस्थि वृद्धि लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	अधिदंत (Hypertrophy of teeth/ Extra teeth)	Do you notice any extra teeth or calculus apart from the normal in your mouth?	Extra teeth in conditions like dental calculus. Serum Calcium, Phosphorus	

2	अध्यस्थि (Hypertrophy of bone)	Have you noticed any unusual bone growth or areas where your bones seem larger or thicker than normal?	Bone overgrowth due to hormonal imbalance. X-ray, Serum Calcium, Phosphorus, TFT, Plasma Calcitonin	
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Total Asthi Dhatu Vriddhi Score:/6 X 100 =%

Features of Decreased Asthi Dhatu (अस्थि क्षय लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	अस्थिशूल/तोद (Pain in bones/ Arthralgia)	Have you experienced pricking pain in bones?	Pricking pain due to calcium deficiency or fevers like Dengue. CBC, Serum Calcium	
2	दंतशातन (Cracking and breaking of teeth)	Have you noticed increased brittleness or fragility in teeth as breaking easily?	Increased brittleness of teeth due to deficiency of Calcium, phosphate etc. Serum Calcium	
3	केशशातन (Fall of hairs and beard)	Have you experienced any severe Hair loss recently?	Hair fall is evaluated by asking questions about hair fall and exploring onset, triggers factors. It is due to vitamin deficiency, infection, alopecia, hormonal imbalance, or nutritional deficiencies etc are diagnosed through comb test	
4	नखशातन (Brittleness of nails)	Have you noticed increased brittleness or breakage of nails?	Increased brittleness of nails due to vitamin deficiency, anaemia, infection, rheumatic disorders etc.	

Total Asthi Dhatu Kshaya Score:/12 X 100 =%

Majja Dhatu Assessment

Features of Aggravated Maja Dhatu (मज्जा वृद्धि लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	नेत्र गौरव (Heaviness in eyes)	Do you often experience the heaviness in your eyes?	Heaviness of eyes due to increased intraocular pressure. Tonometry can be used to measure IOP	

2	अंग गौरव (Heaviness in body)	Do you ever notice the heaviness in your body parts or any oedema on the body?	Heaviness of body parts due to some edematous or spastic conditions. Clinical examination by inspection and palpation. Investigate for CBC, Urine routine.	
3	पर्वसुस्थुलमुलानी Arunshi (Prominent-interphalangeal joints)	Have you ever seen the small vesicles in your joints?	Appearance of small vesicles with large bases in joints in conditions like osteomyelitis. Clinical examination by inspection and palpation. Investigate for CBC	

Total Majja Dhatu Vriddhi Score:/9 X 100 =%

Features of Decreased Majja Dhatu (मज्जा क्षय लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	अस्थि शौषिर्य (Osteoporosis)	Have you ever experienced frequent fractures or bone pain?	Osteoporosis is evaluated through observation for posture changes, questioning about fractures or back pain. It may be linked to aging, hormonal imbalances, vitamin D deficiency, diagnosed via bone density scans, X-rays, Serum Calcium, estrogen etc	
2	भ्रम (State of illusion)	Do you often feel like you are about to faint?	Fainting due to ischemic effect, reduced hemopoietic activity. Diagnosed by clinical examination and CBC	
3	तिमिर दर्शन (Blurred vision)	Have you had issues with vision loss recently?	Blurred vision, loss of vision due to neural defects with associated symptoms like headaches or dizziness. It may indicate refractive errors, cataracts or neurological conditions, diagnosed through eye exams, cranial nerve examination, imaging and visual field tests.	
4	पर्वभेद (pain in interphalangeal joints)	Do you often have pain in interphalangeal joints?	Interphalangeal pain is assessed by observing swelling, redness or deformity, palpating for tenderness or stiffness and exploring symptom onset and triggers. Conditions like arthritis, gout or trauma are diagnosed using imaging, blood tests.	

Total Majja Dhatu Kshaya Score:/12 X 100 =%

Shukra Dhatu Assessment

Features of Aggravated Shukra Dhatu (शुक्र वृद्धि लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	अतिस्त्रीकामता (Increased Libido)	Have you experienced increased sexual desire recently?	Increased sexual desire due to hormonal imbalance as seen in adrenal hyperplasia. Investigate by clinical examination, USG, ACTH	
2	शुक्राश्मरी (Spermolith)	Were you diagnosed with calculi in the genital tract in the recent past or do you feel obstruction during passing urine or semen?	Calculi in the genital tract like seminal vesicle calculi & prostatic calculi in males and ovarian calculi or uterine calculi in females. It can be associated with infections or obstructions. Diagnosed through USG, PSA, semen analysis.	

Total Shukra Dhatu Vriddhi Score:/6 X 100 =%

Features of Decreased Shukra Dhatu (शुक्रक्षय लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	चिरात् प्रसेक (Late ejaculation)	Do you experience delayed ejaculation, late sexual response? (Women expel fluids from the urethra during sexual arousal and orgasm)	Late ejaculation is evaluated by observing late sexual response, questioning about the onset, frequency and associated symptoms like erectile dysfunction or discomfort. It may be linked to psychological factors, hormonal imbalances or neurological conditions, diagnosed through blood tests, hormone levels and imaging.	
2	रक्तशोणित (Blood in semen)	Did you feel like your semen is mixed with blood after ejaculation?	Ejaculation mixed with blood due to bleeding from genital tract as seen in BPH, Carcinoma of prostate, TB etc. Clinical examination, USG, CBC, semen analysis, prostate-specific antigen (PSA) tests	
3	मेढ्रवृषण वेदना (Pain in penis and testes)	Do you experience pain in penis and testes?	Questioning about pain. It may indicate conditions like epididymitis, testicular torsion or varicocele. Diagnosis is	

			confirmed through ultrasound, urine tests and clinical evaluation.	
4	वृषण धुमायण (Burning sensation genital region)	Do you experience a burning sensation in the penis, testes, genital tract frequently after micturition or ejaculation?	Questioning focuses on onset, associated pain and urinary symptoms. It may indicate infections like UTIs, STIs or irritation, diagnosed through urine analysis, swabs and USG.	
5	मुखशोष (Xerostomia)	Do you experience dryness in your mouth?	Dryness in the mouth is assessed by observing oral moisture, tongue texture, and dental health. Questioning explores symptom, triggers. Conditions like Sjögren's syndrome, dehydration etc are diagnosed through saliva tests etc	

Total Shukra Dhatu Vriddhi Score:/15 X 100 =%

Result: Dhatu Vriddhi :.....

Dhatu Kshaya :.....



Teacher's Signature

9.3 DHATU VRIDDHI-KSHAYA PARIKSHAN

Objective : To assess the Dhatu Vriddhi-Kshaya lakshanas of an individual

Name: :

Age : **Gender:** M/F **Occupation:**

Body temperature : **Weight:**kg

Blood pressure : **Pulse rate:**...../min

Address :

Rasa Dhatu Assessment

Features of Aggravated Rasa Dhatu (रस वृद्धी लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	प्रसेक (Excessive salivation)	Do you experience excessive salivation such as drooling or a constant watery mouth?	Excessive salivation is assessed by observing drooling or oral discomfort, examining salivary glands for swelling or tenderness and exploring symptom triggers, duration and associated conditions. Potential causes like infections or neurological issues can be confirmed through imaging or lab tests.	
2	अरोचक (Loss of appetite)	Have you lost your appetite or felt disinterested in eating?	Loss of appetite is evaluated by observing signs of malnutrition, assessing abdominal or systemic health and exploring dietary habits, psychological factors and associated symptoms like fever. Conditions like infections or gastrointestinal disorders can be identified through lab tests, imaging or endoscopy.	
3	आस्यवैरस्य (Tastelessness)	Have you noticed a loss of taste or change in how food tastes?	Tastelessness is evaluated by noting oral hygiene and tongue appearance and ask questions, assessing for nerve damage or infections and discussing associated symptoms. Conditions like nutritional deficiencies or neurological disorders.	
4	हल्लास (Nausea)	Do you feel nausea or a sense of	Nausea is evaluated by observing signs of discomfort, assessing abdominal tenderness or bloating and exploring symptom onset, triggers and associated factors through questioning. Clinical findings may suggest gastritis,	

		uneasiness in your stomach?	infections or neurological issues, confirmed through imaging, endoscopy or lab tests.	
5	स्रोतोरोध (Obstruction of channels)	Do you feel congestion phlegm or blockages that interfere with normal functions like breathing?	Blockage of channels by mucous or other secretions (as seen in bronchial asthma, Nasal congestion, COPD). Obstruction of channels is evaluated by observing swelling, discolouration or impaired movement, assessing affected areas for tenderness or rigidity and exploring symptom onset, triggers and progression. Conditions like blockages or inflammation are confirmed using imaging, lab tests CBC (complete blood count) , ESR, CRP etc. or functional studies.	
6	स्वादुद्वेष (Dislike food having sweet taste)	Have you developed an aversion to sweet tasting foods or drinks?	Dislike for sweet-tasting food is evaluated by observing eating behaviour, digestive discomfort and exploring symptom onset, triggers and related factors.	
7	अगमर्द (Body ache)	Do you experience generalized aches or pains in your body or muscles?	Generalized body pain (as seen in malnutrition, vitamin deficiency etc.). Body ache is assessed by observing movement limitations, palpating for muscle tenderness and exploring onset, severity and associated symptoms. It may indicate infections, inflammatory conditions or fibromyalgia, diagnosed through blood tests, imaging or specialized assessments like electromyography.	

Total Rasa Dhatu Vriddhi Score:/21 X 100 =%

Features of Decreased Rasa Dhatu (रस क्षय लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्नै)	Score
1	रौक्ष्यं (Dryness)	Do you experience dryness or roughness in your skin hair or mucous membranes?	Dryness is assessed by observing skin, mucous membranes and overall hydration, while exploring symptoms like thirst, irritation or difficulty swallowing. It may indicate dehydration, Sjögren's syndrome or diabetes, diagnosed through blood tests, salivary function or imaging.	
2	भ्रम	Have you felt dizziness,	The state of illusion is assessed by observing altered perceptions, questioning the onset, duration and evaluating associated symptoms like confusion or	

	(State of illusion)	fainting recently?	hallucinations. It may be linked to psychiatric disorders, neurological conditions or substance abuse, diagnosed using imaging, EEG and psychological evaluations.	
3	शोष (Wasting)	Have you unintentional weight loss, thinning or muscle wasting?	Wasting is evaluated through observation of muscle loss, questioning about weight changes, fatigue, appetite and assessing physical function. It may indicate conditions like malnutrition, malabsorption, cancer, HIV or endocrine disorders, diagnosed using blood tests, imaging, biopsies or electromyography.	
4	ग्लानी (Tiredness without work)	Do you frequently feel fatigue, unable to complete daily activities?	Tiredness without exertion is assessed by observing fatigue patterns, examine for high/low blood pressure, pulse rate, pulse volume, pulse character etc. Exploring associated symptoms like sleep disturbances or depression and conditions such as anaemia, hypothyroidism may be diagnosed through blood tests, thyroid function tests etc.	
5	शब्दअसहिष्णुता (Intolerance to noise)	Are you highly sensitive or intolerant to loud sudden sounds?	Intolerance to noise is assessed by observing signs and associated symptoms like dehydration, shock, headaches or dizziness, meningitis. It may indicate conditions such as migraine, hyperacusis or vestibular disorders, diagnosed using audiometry, MRI or neurological evaluations.	
6	हृद्रव, हृत्कम्प (Palpitation, Tachycardia)	Do you frequently feel Palpitation or Tachycardia?	Palpitation and tachycardia are assessed by observing heart rate, skin pallor or sweating, examining pulse and heart sounds. They may indicate arrhythmias, anemia or hyperthyroidism, diagnosed through ECG, blood tests or echocardiography.	

Total Rasa Dhatu Kshaya Score:/18 X 100 =%

Rakta Dhatu Assessment

Features of Aggravated Rakta Dhatu (रक्त वृद्धी लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	विसर्प	Do you frequently Suffer from skin diseases like inflammation?	Inflammatory conditions like cellulitis due to toxemic effect in infections. Investigate for CBC, ESR, CRP	
2	विद्रुधी	Do you suffer from inflammatory conditions such as cellulitis or	An abscess inflammation with the collection of pus. Investigate for CBC, ESR	

		infections leading to swelling and redness?		
3	कुष्ठ	Do you frequently suffer from skin diseases like rashes, eczema or autoimmune diseases?	Skin diseases due to various causes. Examine the nature of lesions. Investigate for the presence of infective, allergic or autoimmune elements. Investigate for CBC, AEC, ESR, CRP, ASO.	
4	वातरक्त	Do you experience symptoms of arthritis or joint inflammation?	Rheumatoid Arthritis, Gout etc. due to inflammatory disorders of metabolic or autoimmune origin. Investigate for CBC, ESR, LFT, Serum uric acid, RA factor	
5	रक्तपित्त	Have you been diagnosed with bleeding disorders or prolonged bleeding?	Bleeding disorders due to defect in hemostasis mechanism-congenital & acquired. Investigate for Platelet count, BT, CT, LFT	
6	गुल्म	Do you experience abdominal distension or bloating regularly?	Abdominal distension due to pathological changes in intestine and surroundings. Investigate for USG, CT, CBC	
7	उपकुश	Have you noticed bleeding from the gums or Unexplained bruising?	Bleeding from the gums due to local causes like gingivitis or due to systemic pathology. Investigate for BT, CT, PT, LFT	
8	व्यंग	Do you experience hyperpigmentation of the skin or conditions like Addison's disease?	Hyperpigmentation is seen in endocrine disorders like Addison's disease, cushing's disease. Investigate for CBC, ESR, cortisol, TSH	
9	अग्निनाशः	Do you suffer from digestive disturbances like loss of appetite or acid-base imbalances?	Loss of digestive power is seen in conditions like acidosis, alkalosis etc. Investigate for CBC, LFT, Urine	
10	रक्त त्वक (Redness of skin)	Do you often observe skin redness or inflammation in response to minor triggers?	Increased redness of skin due to bleeding disorders or conditions like erythema nodosum, Raynaud's phenomenon etc. Investigate for ESR, BT, CT, PT, LFT	
11	रक्त नेत्र (Redness of eyes)	Do you notice increased redness or irritation in your eyes?	Increased redness of the eyes (due to bleeding or inflammation) It may indicate conditions like conjunctivitis, glaucoma or uveitis, diagnosed through slit-lamp examination or tonometry. Investigate for BT, CT, PT, LFT.	

12	रक्त मूत्र (Blood in urine)	Have you experienced red coloured urine without a known cause?	Red coloured urine due to bleeding or use of some drugs. Blood in urine is assessed by observing urine colour and clarity, palpating the abdomen for tenderness and asking about pain, trauma or related symptoms. It may indicate UTIs, kidney stones or tumors, diagnosed through urinalysis, imaging and cystoscopy.	
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Total Rakta Dhatu Vriddhi Score:/36 X 100 =%

Features of Decreased Rakta Dhatu (रक्त क्षय लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	अम्लरस-प्रीति (Desire for food having sour taste)	Do you experience a strong desire for sour-tasting foods or drinks?	Desire for sour taste seen in conditions like anaemia. A desire for sour-tasting food is evaluated through observation of eating habits, questioning onset, frequency and related symptoms like acidity or digestive discomfort. It may indicate acid reflux, gastric ulcers or metabolic imbalances, confirmed through endoscopy, pH testing, CBC and PCV.	
2	शिशिर प्रीति (Desire for cold)	Do you often feel a desire for cold things, especially in not environments?	Desire for cold due to ischemic effect, as a mechanism to protect heat. Desire for cold can be assessed by observing behaviour and questioning about associated symptoms like fever, fatigue or digestive issues. It may indicate conditions like fever, thyroid disorders or anaemia, diagnosed through blood tests, temperature monitoring and thyroid function tests.	
3	सिराशैथिल्य (Hypovolemia or hypotension)	Have you noticed a loss of elasticity in your veins or Skin, particularly with aging?	Clinical examination for pulse volume, character, Blood pressure. Symptoms may indicate dehydration, blood loss or heart failure, diagnosed through blood tests, ECG and imaging studies like ultrasound.	
4	रूक्षता (Dryness)	Do you experience persistent dryness of the skin?	Dryness of skin due to ischemic effect or in conditions like DM, eczema, psoriasis. Investigate for CBC, ESR, CRP, RBS, Lipid profile.	

Total Rakta Dhatu Kshaya Score:/12 X 100 =%

Mamsa Dhatu Assessment

Features of Aggravated Mamsa Dhatu (मांस वृद्धि लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्नै)	Score
1	गण्डवृद्धि	Do you expect flabbiness in your cheeks?	Due to anaemia etc. Investigate for CBC, ESR	
2	उरुवृद्धि (hypertrophy of thigh muscle)	Do you experience flabbiness or Enlargement on your thighs?	Inflammation of low back due to lumbar paraspinal muscle spasm. Clinical examination by inspection, palpation, X-ray, CT, MRI	
3	उदरवृद्धि (abdominal muscles hypertrophy)	Do you experience enlargement in your Abdomen?	Inflammation of the abdomen due to abdominal muscle spasm. Clinical examination by inspection, palpation, USG	
4	अर्बुद, ग्रन्थि (Tumor)	Have you noticed any lumps or tumor growth in your body?	Tumor (benign & malignant). Clinical examination by inspection, palpation CBC, PSA (prostate-specific antigen)	
5	कण्ठादी अधिमास (over growth in the neck)	Do you experience excess Fleshy growth on your neck and other body parts?	Overgrowth in the neck in conditions like goiter or tonsillitis. Clinical examination by inspection, palpation, CBC, ESR, TFT (thyroid function test)	

Total Mamsa Dhatu Vriddhi Score:/15 X 100 =%

Features of Decreased Mamsa Dhatu (मांस क्षय लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्नै)	Score
1	अक्षग्लानी (Weakness in eyes)	Do you experience fatigue in your sensory organs (especially eyes) due to general weakness?	Weakness of body parts due to muscle fatigue. Investigate for Serum Calcium, Phosphorous, Total protein, CPK	
2	गंडस्फिकशुष्कता (Muscles wasting of related area)	Have you noticed a reduction in the size of your cheeks & buttocks?	Emaciation of neck and buttocks due to muscle wasting. Clinical examination by inspection, palpation (look for UMN, LMN lesion), X-ray, CT, MRI	

3	सन्धिवेदना (Arthralgia)	Do you have joint pain?	Joint pain is assessed by observing joint swelling or redness, palpating for tenderness. It may indicate conditions such as arthritis, infections or autoimmune diseases, diagnosed through imaging, blood	
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Total Mamsa Dhatu Kshaya Score:/9 X 100 =%

Meda Dhatu Assessment

Features of Aggravated Meda Dhatu (मेद वृद्धि लक्षण)

Sr. No	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्नै)	Score
1	श्रमः (Tiredness)	Do you experience palpitations and fatigue even during mild exercise or physical activity?	Examine physical fatigue, evaluating muscle weakness. It may indicate anaemia, thyroid disorders or chronic fatigue syndrome, diagnosed through blood tests, thyroid function tests and sleep studies.	
2	अल्पेऽपिचेष्टिते श्वासः (Feeling exhausted on even less exertion)	Have you ever noticed shortness of breath especially during physical activity or at rest?	Observing physical activity levels, assessing for weakness or pallor and inquiring about onset, triggers and associated symptoms. It may indicate anaemia, thyroid disorders. Diagnosed through blood tests, imaging or endocrine assessments.	
3	स्किक्स्तन उदर लंबनम् (Pendulous overgrowth of gluteal region, chest region and belly)	Have you noticed any signs of central obesity such as increased fat around your abdomen or waistline?	Central obesity due to increased deposition of fat. Investigate for lipid profile, RBS, BMI	
4.	अतिस्निग्धता (Very oily skin)	Have you ever noticed very oily skin in all season?	Skin is assessed excess shine, enlarged pores or acne. Questions focus on onset, triggers like diet or stress. It may indicate hormonal imbalances, PCOS or seborrheic dermatitis, diagnosed through skin tests.	

5.	प्रमेह पुरुरूप (Prodromal signs of diabetes)	Do you experience any prodromal signs of diabetes?	Prodromal signs of diabetes include fatigue, frequent urination, excessive thirst and unexplained weight loss. Blood glucose tests can help diagnose prediabetes or early diabetes mellitus.	
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Total Meda Dhatu Vriddhi Score:/15 X 100 =%

Features of Decreased Meda Dhatu (मेद क्षय लक्षण)

Sr. No	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	स्वपनम् कट्याः (Numbness in the lower back or lumbar region)	Have you ever felt numbness or loss of sensation in your Lower back or lumbar region?	Numbness in the lumbar region due to peripheral neuropathy or (IVDP) Intervertebral Disc Prolapse. Clinical examination by inspection, palpation, X-ray, CT, MRI	
2	प्लीहा वृद्धि (Splenomegaly)	Have you noticed any signs of discomfort or fullness in your left upper abdomen?	Splenomegaly is hypertrophy of the spleen due to overload in various conditions. Clinical examination by abdominal palpation, USG, RBC count, Platelet count	
3	कृशाङ्गता (Lean built)	Have you ever experienced extreme thinning or significant weight loss?	Emaciation in chronic diabetes. RBC, RBS	

Total Meda Dhatu Kshaya Score:/9 X 100 =%

Asthi Dhatu Assessment

Features of Aggravated Asthi Dhatu (अस्थि वृद्धि लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	अधिदंत (Hypertrophy of teeth/ Extra teeth)	Do you notice any extra teeth or calculus apart from the normal in your mouth?	Extra teeth in conditions like dental calculus. Serum Calcium, Phosphorus	

2	अध्यस्थि (Hypertrophy of bone)	Have you noticed any unusual bone growth or areas where your bones seem larger or thicker than normal?	Bone overgrowth due to hormonal imbalance. X-ray, Serum Calcium, Phosphorus, TFT, Plasma Calcitonin	
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Total Asthi Dhatu Vriddhi Score:/6 X 100 =%

Features of Decreased Asthi Dhatu (अस्थि क्षय लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	अस्थिशूल/तोद (Pain in bones/ Arthralgia)	Have you experienced pricking pain in bones?	Pricking pain due to calcium deficiency or fevers like Dengue. CBC, Serum Calcium	
2	दंतशातन (Cracking and breaking of teeth)	Have you noticed increased brittleness or fragility in teeth as breaking easily?	Increased brittleness of teeth due to deficiency of Calcium, phosphate etc. Serum Calcium	
3	केशशातन (Fall of hairs and beard)	Have you experienced any severe Hair loss recently?	Hair fall is evaluated by asking questions about hair fall and exploring onset, triggers factors. It is due to vitamin deficiency, infection, alopecia, hormonal imbalance, or nutritional deficiencies etc are diagnosed through comb test	
4	नखशातन (Brittleness of nails)	Have you noticed increased brittleness or breakage of nails?	Increased brittleness of nails due to vitamin deficiency, anaemia, infection, rheumatic disorders etc.	

Total Asthi Dhatu Kshaya Score:/12 X 100 =%

Majja Dhatu Assessment

Features of Aggravated Maja Dhatu (मज्जा वृद्धि लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	नेत्र गौरव (Heaviness in eyes)	Do you often experience the heaviness in your eyes?	Heaviness of eyes due to increased intraocular pressure. Tonometry can be used to measure IOP	

2	अंग गौरव (Heaviness in body)	Do you ever notice the heaviness in your body parts or any oedema on the body?	Heaviness of body parts due to some edematous or spastic conditions. Clinical examination by inspection and palpation. Investigate for CBC, Urine routine.	
3	पर्वसुस्थुलमुलानी Arunshi (Prominent-interphalangeal joints)	Have you ever seen the small vesicles in your joints?	Appearance of small vesicles with large bases in joints in conditions like osteomyelitis. Clinical examination by inspection and palpation. Investigate for CBC	

Total Majja Dhatu Vriddhi Score:/9 X 100 =%

Features of Decreased Majja Dhatu (मज्जा क्षय लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	अस्थि शौषिर्य (Osteoporosis)	Have you ever experienced frequent fractures or bone pain?	Osteoporosis is evaluated through observation for posture changes, questioning about fractures or back pain. It may be linked to aging, hormonal imbalances, vitamin D deficiency, diagnosed via bone density scans, X-rays, Serum Calcium, estrogen etc	
2	भ्रम (State of illusion)	Do you often feel like you are about to faint?	Fainting due to ischemic effect, reduced hemopoietic activity. Diagnosed by clinical examination and CBC	
3	तिमिर दर्शन (Blurred vision)	Have you had issues with vision loss recently?	Blurred vision, loss of vision due to neural defects with associated symptoms like headaches or dizziness. It may indicate refractive errors, cataracts or neurological conditions, diagnosed through eye exams, cranial nerve examination, imaging and visual field tests.	
4	पर्वभेद (pain in interphalangeal joints)	Do you often have pain in interphalangeal joints?	Interphalangeal pain is assessed by observing swelling, redness or deformity, palpating for tenderness or stiffness and exploring symptom onset and triggers. Conditions like arthritis, gout or trauma are diagnosed using imaging, blood tests.	

Total Majja Dhatu Kshaya Score:/12 X 100 =%

Shukra Dhatu Assessment

Features of Aggravated Shukra Dhatu (शुक्र वृद्धि लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	अतिस्त्रीकामता (Increased Libido)	Have you experienced increased sexual desire recently?	Increased sexual desire due to hormonal imbalance as seen in adrenal hyperplasia. Investigate by clinical examination, USG, ACTH	
2	शुक्राश्मरी (Spermolith)	Were you diagnosed with calculi in the genital tract in the recent past or do you feel obstruction during passing urine or semen?	Calculi in the genital tract like seminal vesicle calculi & prostatic calculi in males and ovarian calculi or uterine calculi in females. It can be associated with infections or obstructions. Diagnosed through USG, PSA, semen analysis.	

Total Shukra Dhatu Vriddhi Score:/6 X 100 =%

Features of Decreased Shukra Dhatu (शुक्रक्षय लक्षण)

Sr. No.	Features	Question	Assessment by (दर्शन, स्पर्शन, प्रश्न)	Score
1	चिरात् प्रसेक (Late ejaculation)	Do you experience delayed ejaculation, late sexual response? (Women expel fluids from the urethra during sexual arousal and orgasm)	Late ejaculation is evaluated by observing late sexual response, questioning about the onset, frequency and associated symptoms like erectile dysfunction or discomfort. It may be linked to psychological factors, hormonal imbalances or neurological conditions, diagnosed through blood tests, hormone levels and imaging.	
2	रक्तशोणित (Blood in semen)	Did you feel like your semen is mixed with blood after ejaculation?	Ejaculation mixed with blood due to bleeding from genital tract as seen in BPH, Carcinoma of prostate, TB etc. Clinical examination, USG, CBC, semen analysis, prostate-specific antigen (PSA) tests	
3	मेढ्रवृषण वेदना (Pain in penis and testes)	Do you experience pain in penis and testes?	Questioning about pain. It may indicate conditions like epididymitis, testicular torsion or varicocele. Diagnosis is	

			confirmed through ultrasound, urine tests and clinical evaluation.	
4	वृषण धुमायण (Burning sensation genital region)	Do you experience a burning sensation in the penis, testes, genital tract frequently after micturition or ejaculation?	Questioning focuses on onset, associated pain and urinary symptoms. It may indicate infections like UTIs, STIs or irritation, diagnosed through urine analysis, swabs and USG.	
5	मुखशोष (Xerostomia)	Do you experience dryness in your mouth?	Dryness in the mouth is assessed by observing oral moisture, tongue texture, and dental health. Questioning explores symptom, triggers. Conditions like Sjögren's syndrome, dehydration etc are diagnosed through saliva tests etc	

Total Shukra Dhatu Vriddhi Score:/15 X 100 =%

Result: Dhatu Vriddhi :.....

Dhatu Kshaya :.....



Teacher's Signature

EXERCISE

Q.1 What are the symptoms of Rasa Dhatu Kshaya?

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Q.2 How do you assess Rakta Dhatu Vriddhi using clinical tools?

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Q.3 Which laboratory tests correlate with Mamsa Dhatu imbalances?

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Q.4 What is the significance of lipid profile tests in Meda Dhatu evaluation?

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Q.5 Which symptoms suggest Asthi Dhatu Kshaya?

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Q.6 Describe the pathophysiology of Majja Dhatu Vriddhi in neurological disorders.

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Q.7 How does Ayurvedic theory integrate with modern diagnostics in Shukra Dhatu analysis?

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Q.8 Collect and explore the assessment of additional vriddhi and kshaya lakshanas mentioned in Charak Samhita which are not included in the above proforma.

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Q.9 Collect and explore the assessment of additional vriddhi and kshaya lakshanas mentioned in Sushrut Samhita which are not included in the above proforma.

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Q.10 Find out similar and opposite pairs of different symptoms among Dhatu vriddhi-kshaya features.

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OSPE

Title: Assessment of Dhatu Vriddhi-Kshaya

Learning Objectives:

To apply theoretical knowledge in practical Dhatu assessment.

To enhance clinical diagnostic skills using Ayurvedic principles.

Station Setup:

Examination table, clinical tools and questionnaires.

A sample patient or simulated case scenario.

Score sheets and lab results for interpretation.

Tasks:

Record the patient's vitals and baseline data.
Conduct a Dhatu-specific assessment using structured questions. Perform physical examinations and correlate with laboratory findings.

Checklist:

Proper patient history documentation

Accurate recording of clinical symptoms

Utilization of appropriate diagnostic tests

Completion of Dhatu scoring and interpretation

CASE-BASED LEARNING

Case-Based Learning (CBL) Example: Dosha Vriddhi Assessment

Case Title: Understanding Dosha Vriddhi Through Clinical Features

Case Scenario: A 35-year-old male presents with complaints of generalized body pain, frequent fatigue and intolerance to loud noises. His blood work reveals mild anaemia and hypoproteinemia.

Problem: Identify the Dhatu imbalance causing these symptoms.

Guiding Questions: What symptoms point toward Rasa or Mamsa Dhatu involvement?

What investigations will confirm your diagnosis?

Guided Discussion Questions: How do the clinical symptoms correlate with Rasa Dhatu Kshaya?

What dietary and lifestyle changes could address these issues?

Expected Outcomes: Recognition of Dhatu-specific symptoms. Correlation of findings with Ayurvedic and modern medical interpretations.

Discussion and Resolution: Confirm Rasa and Mamsa Dhatu Kshaya through analysis. Discuss simple treatment principle/ strategies focusing on nutritional supplementation and Ayurvedic therapies.

Assessment:

Evaluate the student's ability to apply theoretical knowledge, conduct clinical assessments and propose appropriate interventions based on findings.

DOPS

DOPS (Direct Observation of Procedural Skills) for Meda Dhatu Assessment

Objective:

To assess and interpret the features of Meda Dhatu Vridhhi (Aggravation) and Kshaya (Depletion) through clinical observation and analysis of patient response in a 10-20 min session.

Step-wise DOPS Protocol for Meda Dhatu Assessment

1. Preparation

Material Required: Patient history sheet, clinical examination tools (stethoscope, BP monitor, BMI calculator, lipid profile requisition forms, ECG etc. as per availability).

Patient Preparation: Ensure privacy and comfort of patient and explain the purpose of the assessment.

2. Patient History Taking

Record demographic details (Name, Age, Gender, Occupation).

Note baseline vital signs: body temperature, weight, blood pressure and pulse rate etc.

Document complaints related to fatigue, breathlessness or obesity.

3. Rationale for Meda Dhatu Examination

Educate the patient about Meda Dhatu and its role in body structure and metabolism.

Explain the focus on symptoms like tiredness, central obesity and numbness in the lumbar region.

4. Clinical Examination

Assessment of Meda Dhatu Vridhhi Features:

Tiredness: Inquire about fatigue during mild activities. Perform auscultation for cardiac ischemic changes. Use Hardward step test or ECG or taking proper history to confirm findings.

Shortness of Breath: Ask about dyspnea during rest or exertion. Evaluate lung function and perform chest auscultation or take proper history to confirm findings.

Central Obesity: Measure waist-to-hip ratio and BMI. If possible, Check lipid profile to confirm excess fat deposition.

Assessment of Meda Dhatu Kshaya Features:

Numbness in Lumbar Region: Examine the spine for neurological deficits. Use palpation and assess sensory responses.

Splenomegaly: Conduct abdominal palpation to assess spleen enlargement. Request USG and platelet count if indicated.

Lean Build: Inquire about unintentional weight loss. Confirm with RBS and other metabolic assessments.

Skill Demonstration by the Examiner: Showcase the method of history taking and physical examination. Demonstrate interpreting diagnostic results such as lipid profile, ECG findings or BMI calculations.

5. Patient Feedback Collection

Engage the patient by asking for feedback on discomfort or specific challenges during the assessment.

6. Documentation

Record findings for both **Vridhhi** and **Kshaya**:

Assign scores for each observed feature.

Calculate percentages using the formula:

Score Percentage =

$(\text{Obtained Score} / \text{Maximum Score}) \times 100$

Ex. - Total Meda Dhatu Kshaya Score:/9 X 100
=%

Ex.- Total Meda Dhatu Vriddhi Score:/15 X 100
=%

7. Critical Analysis

Interpret results to diagnose Meda Dhatu imbalance. Suggest lifestyle, dietary or simple therapeutic interventions based on Ayurvedic principles.

Example Scenario with Solutions

Case

Patient Details:

- Name: Mr. Ravi Sharma
- Age: 45 years
- Gender: Male
- Occupation: Office Worker
- BMI: 32 (Obese)
- Blood Pressure: 140/90 mmHg

Findings

Meda Dhatu Vriddhi:

- Complaints of fatigue during mild physical activity (Score: 3/3).
- Dyspnea was observed during routine walking (Score: 3/3).
- Central obesity was noted with a waist-to-hip ratio >1.0 (Score: 3/3).

Total Vriddhi Score: $9/9 \times 100 = 100\%$

Meda Dhatu Kshaya:

No numbness (0/3), splenomegaly (0/3) or lean build (0/3) was observed (Total Score: 0/9).

Total Kshaya Score: $0/9 \times 100 = 0\%$

Solutions

Recommend lifestyle changes focusing on weight loss (e.g., diet low in fats and sugars, regular exercise).

Suggest Rasayana therapy to balance Meda Dhatu (e.g., Guggulu-based formulations).

Skill Assessment

- **History-Taking Skill:** Accurate questioning about lifestyle and symptoms.
- **Physical Examination Skill:** Proper use of tools like a stethoscope, BMI calculator and palpation techniques.
- **Interpretative Skill:** Correlation of clinical findings with diagnostic tests.
- **Communication Skill:** Explaining findings and recommendations to the patient effectively.

10. नाडी परीक्षण

सीखने का उद्देश्य

प्रायोगिक कार्य पूरा होने के बाद, छात्रों को निम्नलिखित में सक्षम होना चाहिए:

1. स्वास्थ्य की स्थिति का मूल्यांकन.
2. आधुनिक चिकित्सा पद्धतियों के साथ सामंजस्य.
3. व्यक्तिगत स्वास्थ्य सुधार.
4. प्राकृतिक और बिना दवाओं के उपचार की दिशा में मार्गदर्शन.

नाडी परीक्षा का प्रयोजन

- **गुणवत्ता और दक्षता में वृद्धि:** यह आयुर्वेदिक उपचार विधियों को अधिक प्रभावी और सटीक बनाता है।
- **वैयक्तिकृत चिकित्सा:** इस पद्धति से यह समझा जा सकता है कि किसी व्यक्ति को किस प्रकार की चिकित्सा, आहार की आवश्यकता है।
- **शरीर के अंदर के असंतुलन का पता लगाना:** नाड़ी परीक्षण व्यक्ति के शारीरिक, मानसिक और भावनात्मक स्वास्थ्य की स्थिति का व्यापक मूल्यांकन करता है।
- **शरीर के अंदर के असंतुलन का पता लगाना:** नाड़ी परीक्षण का प्रमुख उद्देश्य शरीर के भीतर के त्रिदोष (वात, पित्त, कफ) का संतुलन या असंतुलन पहचानना है।

नाड़ी परीक्षा

विधि

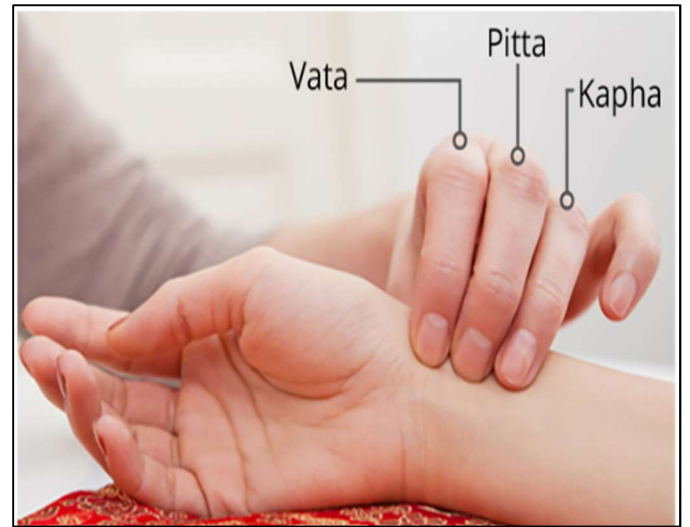
एकाङ्गुलं परित्यज्याधस्तादङ्गुष्ठमूलतः । यत्नवांस्तां परिक्षेत् ह्यभ्यासादेव लक्ष्यते ॥ करस्याङ्गुष्ठमूले या धमनी जीवसाक्षिणी । तच्चेष्ट्या सुखं दुःखं ज्ञेयं कायस्य पंडितैः ॥ स्त्रीणां भिषग्वामहस्ते वामपादे च यत्नतः । पुंसां दक्षिणमागे च नाडी विघात् विशेषतः ॥

गुल्फस्याधोऽङ्गुष्ठभागे पादे, त्वङ्गुष्ठमूलतः । एकाङ्गुलं परित्यज्य मणिबंधे परीक्षयेत् ॥

अंगूठे की जड़ के नीचे एक उंगली छोड़कर प्रयास करके नाड़ी की जांच करनी चाहिए।

अध्ययन द्वारा नाड़ी परीक्षण किया जाता है। नाड़ी को जीवसाक्षिणी कहा जाता है।

आमतौर पर एक चिकित्सक, किसी व्यक्ति के दाहिने हाथ की जांच करता है।



महिलाओं के बाएं हाथ की नाड़ी की जांच करते समय। चिकित्सक अपने दाहिने हाथ की तर्जनी, मध्यमा और अनामिका उंगलियों को रोगी के अंगूठे से एक उंगली छोड़कर मणिबंध पर रखता है।

प्रयास से पहले इन तीनों अंगुलियों से मणिबंध पर बराबर दबाव रखना चाहिए और नाड़ी का परीक्षण करना चाहिए।

नाड़ी एवं दोष संबंध

आदौवातवहा नाडी मध्ये वहति पितला ।

अन्ते श्लेष्मविकारेण नाडिकेति त्रिधा मता ।

उपरोक्तानुसार तीनों अंगुलियों से मणिबंध पर दबाव देने के बाद यदि नाड़ी पहले तर्जनी पर महसूस हो तो वात दोष,

यदि मध्यमा पर महसूस हो तो पित्त दोष, और यदि नाड़ी तर्जनी पर महसूस हो तो पित्त दोष अनामिका अंगुली के अंतिम भाग पर यह जानना चाहिए कि नाड़ी में कफ दोष की प्रधानता है।

नाड़ी व दोष गति

नाडी धतेमरुत्कोपे जलौकासर्पयोगतिम् ।

कुलिंगकालमण्डुकगति पित्तस्य कोपतः ।

हंसपारावतगति धते इलेष्मप्रकोपतः ॥

शा. सं. प्र. ख. ३/६४

- दोष गति का वर्णन करते समय, कई प्रकार की गति से तुलना करके दोष गति का वर्णन करता है।
- वातप्रधान में नाड़ी साँप या जोंक (जलौका) के समान होती है।
- पित्त में यह कुलिंग (लावा), काक या मंडुकमती के समान है।
- कफ में यह हंस या कबूतर की गति के समान है। इसका मतलब यह है कि साँप सरपट दौड़ता है, मेंढक उछलता है और हंस शान से चलता है। इस समय यही अपेक्षित है।

नाड़ी परीक्षण निषिद्ध काल

सद्यः स्नातस्य भुक्तस्य क्षुतृष्णातपशीलिनः ।

व्यायामश्रान्तदहस्य सम्यङ्गः नाडी न बुध्यते ॥

नहाने के तुरंत बाद, भोजन के तुरंत बाद, भूख-प्यास लगने के बाद, व्यायाम के बाद, धूप से चलने के बाद, क्रोध, भय, काम, क्रोध से ग्रस्त व्यक्ति की उपरोक्त स्थितियों में नाड़ी परीक्षण नहीं करना चाहिए।

सावधानियां

1. नाड़ी परीक्षण के दौरान व्यक्ति को आरामदायक और शांत स्थिति में होना चाहिए। अधिक सक्रिय या

तनावपूर्ण अवस्था में नाड़ी की सही जानकारी प्राप्त नहीं हो सकती।

2. नाड़ी परीक्षण से पहले चिकित्सक को स्वयं को मानसिक और शारीरिक रूप से शांत करना चाहिए। किसी भी प्रकार के बाहरी तनाव या उत्तेजना से नाड़ी में असमानताएँ उत्पन्न हो सकती हैं, जो निदान को गलत दिशा में ले जा सकती हैं।
3. नाड़ी परीक्षण करते समय चिकित्सक को अपने हाथों को स्वच्छ रखना चाहिए, ताकि कोई बाहरी संक्रमण या अशुद्धियाँ शरीर में प्रवेश न कर सकें।

EXERCISE

Q.1 नाड़ी परीक्षण करते समय कौन से लक्षण देखे जाते हैं?

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Q.2 नाड़ी परीक्षण के समय किस अंग की नाड़ी देखी जाती है?

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Q.3 नाड़ी परीक्षण में कितने प्रकार की नाड़ियाँ होती हैं?

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Q.4 वात दोष से संबंधित नाड़ी के लक्षणों का वर्णन करें।

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Q.5 नाड़ी निदान में लय और दर (rhythm and rate) का क्या महत्व है?

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OSPE-1

OSPE Station: नाड़ी परीक्षण

Learning Objectives

नाड़ी की पहचान और विभिन्न प्रकारों को समझना जैसे कि सामान्य नाड़ी, असामान्य नाड़ी (जैसे कि धीमी, तेज़, या अनियमित नाड़ी)।

नाड़ी की लय, रेट, और दबाव का मूल्यांकन करना। छात्रों को यह समझाना कि नाड़ी परीक्षण से कौन सी क्लिनिकल जानकारी प्राप्त की जा सकती है।

Station Setup

रोगी (नकली या पुतला) रक्त संग्रह के लिए हाथ बाहर निकालकर बैठा हुआ।

Tasks

प्रत्येक छात्र 15 मिनट में निम्नलिखित चरणों का प्रदर्शन और/या व्याख्या करेगा। प्रत्येक बिंदु के लिए 1 या 2 अंक होंगे।

Checklist:

नाड़ी के प्रकार और स्थितियों की समझ: (2 points):

नाड़ी की पहचान और विभिन्न प्रकारों को समझना जैसे कि सामान्य नाड़ी, असामान्य नाड़ी (जैसे कि धीमी, तेज़, या अनियमित नाड़ी)।(Y/N)

नाड़ी की लय, रेट, और दबाव का मूल्यांकन करना।(Y/N)

सही तकनीक और विधि का पालन (2 points):

नाड़ी जांचने के सही तरीके को सीखना, जैसे कि किस स्थान पर नाड़ी जांच करनी चाहिए. (Y/N)

नाड़ी की गुणवत्ता (जैसे कि मजबूत, कमजोर, धड़कन वाली नाड़ी) और उसका सही तरीके से आकलन करना। (Y/N)

संचार और पेशेवर नैतिकता (2 points):

मरीजों से संवाद करते समय सहानुभूति और पेशेवर दृष्टिकोण बनाए रखना।(Y/N)

नाड़ी परीक्षण के दौरान मरीज की गोपनीयता और सम्मान बनाए रखना। (Y/N)

अंतिम टिप्पणी

छात्र ने नाड़ी परीक्षा की सही तकनीक का पालन किया। सभी प्रमुख पहलुओं (जैसे रेट, लय, दबाव, गुणवत्ता) का सही मूल्यांकन किया गया।(Y/N)
किसी असामान्य स्थिति की पहचान कर, उचित निदान और उपचार की दिशा का सुझाव दिया गया।(Y/N)

ROLE PLAY ACTIVITY

Title: नाड़ी परीक्षण Role-Play

Objective: प्रभावी नाड़ी परीक्षण प्रक्रिया और संचार कौशल का अभ्यास और प्रदर्शन करना..

Roles: छात्र/परीक्षक: नाड़ी परीक्षण करने वाला व्यक्ति।
रोगी: रोगी की भूमिका निभाने वाला व्यक्ति (शिक्षक/लैब तकनीशियन)

Scenario: रोगी नियमित नाड़ी परीक्षण के लिए प्रयोगशाला में आता है।

Script: Student/Examiner

1. मरीज का अभिवादन करें और उनकी पहचान सत्यापित करें।
2. रोगी की समझ सुनिश्चित करने के लिए प्रक्रिया को समझाता है।
3. मरीज को अपनी आस्तीन ऊपर करने और हाथ को आराम देने के लिए कहें।
4. नाड़ी का पता लगाता है।
5. नाड़ी की दर, लय, आयतन समरूपता की जाँच करता है।

Patient:

1. प्रश्नों का उत्तर देता है और निर्देशों का पालन करता है।
2. प्रश्न पूछ सकते हैं या हल्की चिंता या बेचैनी दिखा सकते हैं (वैकल्पिक)।

Key Points:

रोगी की पहचान और सत्यापन। प्रक्रिया का स्पष्ट विवरण और सहमति। उचित हाथ स्वच्छता। रोगी के आराम के लिए संचार।

Debriefing Questions:

1. नाड़ी परीक्षण प्रक्रिया में मुख्य चरण क्या थे?
2. परीक्षक ने मरीज की सुविधा कैसे सुनिश्चित की?
3. आप ऐसे मरीज को कैसे संभाल सकते हैं जो चिंतित हैं?

PROBLEM-BASED LEARNING

Case Study:

जॉन नामक एक मरीज ने शिकायत प्रस्तुत की है: पिछले महीने से उसकी थकान और धड़कन बढ़ रही है। उसे दिन में कई बार धड़कन (दिल की धड़कन तेज होना) का अनुभव हो रहा है, खासकर शारीरिक गतिविधि के बाद या काम के दौरान, तनाव के दौरान।

Problem:

नाड़ी की जांच करें और नकली रोगी (छात्र / या प्रशिक्षक) पर गहन परीक्षण करें। नाड़ी की गति नाड़ी से संबंधित दोष नाड़ी परीक्षण के उचित समय का आकलन करें।

Guiding Questions:

नाड़ी परीक्षण करते समय कौन से लक्षण देखे जाते हैं?
उपरोक्त रुग्ण में नाड़ी की गति कैसी है?
नाड़ी परीक्षण के दौरान रुग्ण की नाड़ी में किस दोष का प्राधान्य है?

Discussion and Resolution:

इस सत्र में, छात्र धमनी नाड़ी की जांच करने के लिए एक साथ काम करते हैं और नाड़ी परीक्षण से प्राप्त मुख्य निष्कर्षों पर चर्चा करते हैं और कैसे इनसे विभेदक निदान हुआ। टैचीकार्डिया और इसके अंतर्निहित कारणों की पहचान करने में नाड़ी परीक्षण के महत्व को छात्र समझ सकते हैं।

नाडी परीक्षण (PULSE EXAMINATION)

By the end of this session, students will be able to:

- Perform a structured Naḍi examination.
- Interpret findings to assess tridoṣha (vata, pitta, kapha) imbalances.
- Apply principles of pulse diagnosis to clinical scenarios.
- Enhance diagnostic skills integrating traditional and modern methodologies.

PRINCIPLE

Pulse examination is a traditional diagnostic method in Ayurveda used to evaluate the balance or imbalance of tridoṣha. It reflects physiological and pathological states by analyzing rhythm, rate, volume and quality of the pulse.

PURPOSE OF PRACTICAL

Improve Diagnostic Efficiency: Enhance the precision of Ayurvedic treatment methods.

Personalized Medicine: Determine specific therapeutic approaches tailored to individual needs.

Identify Imbalances: Detect vata, pitta and kapha discrepancies in the body.

APPARATUS AND MATERIALS

A calm, quiet environment

Stopwatch

Clean towels

Hand sanitizer

Model patient (human subject or mannequin)

Ayurvedic diagnostic chart

PROCEDURE

Ensure the subject is relaxed, sitting or lying comfortably in a calm environment. Wash and sanitize hands before examination. Place the index, middle and ring fingers on the radial artery at the wrist, just below the thumb. Apply gentle pressure:

Index finger: Vata detection.

Middle finger: Pitta detection.

Ring finger: Kapha detection.

Note the following characteristics:

Rate: Beats per minute.

Rhythm: Regularity of the beats.

Volume: Strength of the pulse.

Quality: Type of movement (e.g., snake-like for vata, frog-like for pitta, swan-like for kapha).

Record findings and compare them against diagnostic criteria for tridoṣha.

OBSERVATIONS AND RESULTS

Parameter	Vata	Pitta	Kapha
Rate	Fast	Moderate	Slow
Rhythm	Irregular	Regular	Steady
Volume	Weak	Moderate	Strong
Quality	Snake-like	Frog-like	Swan-like

CONCLUSION

Based on the findings, the dominant doṣha is identified as

PRECAUTIONS

Conduct the examination in a calm and noise-free environment.

Ensure the subject has not eaten, exercised or undergone stress recently.

Maintain consistent pressure while palpating the pulse.

Avoid pulse assessment immediately after bathing or sun exposure.

EXERCISE

Describe the characteristics of the pulse associated with vata doṣha.

What are the contraindications for performing naḍi examination?

How does naḍi examination help in differential diagnosis?

List the factors affecting pulse characteristics.

Explain the significance of rhythm and rate in pulse diagnosis.

10.1 नाडी परीक्षण PULSE EXAMINATION

Objective: To assess the Nadi Parikshana of an individual

Name: :

Age: **Gender:** M/F **Occupation:**

Body temperature : **Weight:**kg

Blood pressure : **Pulse rate:**/min

Address:

Characteristics to Observe:

Rate: Beats per minute:.....

Fast [] Moderate [] Slow []

Rhythm: Regular [] Irregular [] Steady []

Volume: Weak [] Moderate [] Strong []

Quality: Snake-like (Vata predominant) []

Frog-like (Pitta predominant) []

Swan-like (Kapha predominant) []

Interpretation of Findings

Dominant Dosha: Vata [] Pitta [] Kapha []

Additional Observations:

.....
.....
.....

Clinical Correlations and Suggestions

Identify physiological and pathological conditions:

.....
Further diagnostic procedures if necessary:
.....

Recommendations for the Patient

Dietary Advice:

Warm, nourishing foods []

Avoid spicy/heavy foods []

Lifestyle Modifications:

Stress management techniques []

Physical activity:.....

Result:.....

10.2 नाडी परीक्षण PULSE EXAMINATION

Objective: To assess the Nadi Parikshana of an individual

Name: :

Age: **Gender:** M/F **Occupation:**

Body temperature : **Weight:**kg

Blood pressure : **Pulse rate:**/min

Address:

Characteristics to Observe:

Rate: Beats per minute:.....

Fast [] Moderate [] Slow []

Rhythm: Regular [] Irregular [] Steady []

Volume: Weak [] Moderate [] Strong []

Quality: Snake-like (Vata predominant) []

Frog-like (Pitta predominant) []

Swan-like (Kapha predominant) []

Interpretation of Findings

Dominant Dosha: Vata [] Pitta [] Kapha []

Additional Observations:

.....
.....
.....

Clinical Correlations and Suggestions

Identify physiological and pathological conditions:

.....
Further diagnostic procedures if necessary:
.....

Recommendations for the Patient

Dietary Advice:

Warm, nourishing foods []

Avoid spicy/heavy foods []

Lifestyle Modifications:

Stress management techniques []

Physical activity:.....

Result:.....

OSPE

OSPE Station: Naḍi Examination

Learning Objectives: Understand the pulse characteristics associated with vata, pitta and kapha. Accurately measure and interpret pulse parameters.

Station Setup: Simulated patient / live model.

Stopwatch, diagnostic chart and hand sanitizer.

Instruction sheet outlining the tasks.

Tasks:

Palpate the radial artery and identify the pulse type.

Record the rate, rhythm and volume.

Interpret findings and conclude the dominant dosha.

Checklist:

Criteria	Points
Proper hand hygiene	1
Correct finger placement	1
Accurate rate measurement	1
Identification of dosha	1
Professional communication	1

CASE-BASED LEARNING

Case Study: A 45-year-old patient complains of irregular bowel movements, fatigue and occasional palpitations. The physician suspects imbalances in vata and pitta.

Problem: Conduct a naḍi examination to determine the dominant dosha and provide dietary recommendations.

Guiding Questions:

What are the signs of vata imbalance in the pulse?

How do pulse characteristics differentiate between vata and pitta dominance?

What lifestyle modifications can address these imbalances?

Discussion and Resolution: Through collaborative discussion, students identify vata-pitta imbalance and suggest tailored interventions such as warm, unctuous foods and stress management techniques.

Assessment: Evaluate the student's ability to:

Perform a systematic pulse examination. Interpret findings accurately. Develop appropriate dietary and lifestyle recommendations.

DOPS

Skill: Naḍi Examination

Steps:

Preparation:

Wash hands and ensure proper equipment.

Explain the procedure to the patient.

Examination:

Place fingers on the radial artery.

Apply consistent pressure.

Note the rate, rhythm, volume and quality.

Interpretation:

Record findings.

Conclude the dominant dosha.

Communication:

Discuss findings with the patient.

Provide tailored recommendations.

Checklist:

Step	Y/N	Comments
Proper hand hygiene		
Correct positioning		
Accurate measurement		
Clear communication		

SKILL ASSESSMENT EXAMPLE

Scenario: A patient reports symptoms of digestive disturbances. Perform a naḍi examination to identify the imbalance.

Solution:

Perform pulse examination as per procedure.

Identify dominant dosha as vata.

Recommend a warm, nourishing diet and stress-reduction techniques.



Teacher's Signature

11. PULSE EXAMINATION

STUDENT OBJECTIVES

After completing this practical, students should be able to:

- Explain what is arterial pulse
- What are its clinical importance?

EXAMINATION OF RADIAL PULSE

- The radial artery is palpated with the tips of three fingers compressing the vessel against the head of radius bone.
- The subject's forearm should be slightly pronated and the wrist slightly flexed.
- The index finger (toward the heart) varies the pressure on the artery, the middle finger feels the pulse, while the distal finger prevents reflections of pulsations from the palmar arch of arteries.

The following observations are made:

1. Rate of pulse

The normal pulse rate at rest averages about 72/min. The rate is normally higher in children (90–110/min) and slower in old age (55–65/min). The pulse rate normally increases during deep inspiration and decreases during deep expiration. When this happens during quiet breathing, it is called sinus arrhythmia. The pulse rate should always be compared with the heart rate, as in some cases the pulse rate may be less than heart rate.

2. Rhythm

The normal pulse waves follow at regular intervals, i.e. the rhythm is regular. The common irregularities in the pulse, which may be occasional, regularly irregular or irregularly irregular, are extrasystoles (premature beats), atrial flutter and fibrillation and various degrees of heart block.

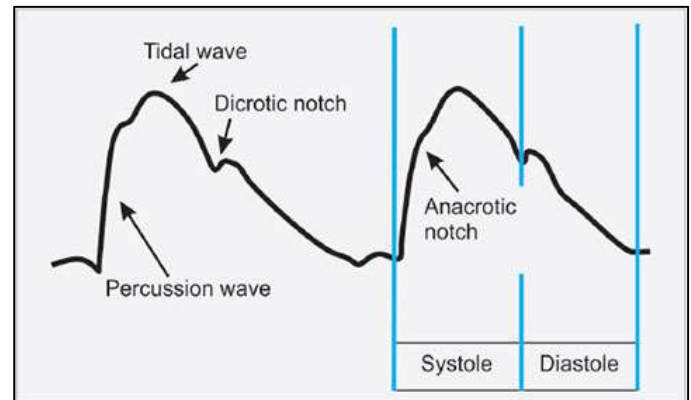
3. Character or form

By character or form is meant the waveform and volume of the pulse, i.e. whether the individual pulse wave has a normal rise, maintenance or fall (its contours) as the pulse is being palpated. The

character should be evaluated at the right carotid artery, i.e. the pulse closest to the heart and least subjected to distortion and damping in the arterial tree. Since the contours of the pulse waves cannot usually be clearly felt by palpation, one has to record the pulse with an electronic transducer. (Dudgeon's sphygmograph used to be employed in the past).

Normal Pulse Wave

The normal pulse wave shows the following components:



a. Percussion Wave or the Anacrotic Limb

This is the sharp upstroke. It is due to expansion of the artery due to ventricular systole and corresponds to the maximum ejection phase. The leisurely down stroke is called the catacrotic limb.

b. Tidal Wave

This predicrotic wave is due to elasticity of aorta. It is sometimes recorded soon after the peak of the tracing.

c. Dicrotic Notch and Wave

The descending limb shows a notch (negative wave) from aortic recoil, momentarily reversing blood flow. This reverse flow closes the aortic valve, causing a positive dicrotic wave. Arterial pulse tracing reflects systolic and diastolic phases. Maximum ejection lasts from the upstroke to percussion wave peak, and reduced ejection from peak to dicrotic notch. Systole spans the upstroke to the dicrotic notch.

4. Volume

The “volume” of the pulse refers to the amplitude of the movement or expansion of the artery during the passage of pulse wave. It is a rough guide to the pulse pressure. Experience is necessary before a student can distinguish a low volume pulse (thin, thready pulse) of low stroke output from a bounding pulse of hyperkinetic circulation, as in fever, pregnancy, anaemia, thyrotoxicosis, etc.

5. Tension

The amount of tension (or pressure) applied to best feel the artery can give only a very rough idea about the diastolic blood pressure; and the amount of pressure required to obliterate the artery gives a rough estimate of systolic pressure.

6. Condition of the vessel wall

The radial artery is emptied out by pressing on it with the finger toward the heart and an attempt is made to roll the vessel against the bone, by the other two fingers. In most young persons, the “empty” artery is so compliant that it cannot be felt as a separate structure. However, the vessel becomes palpable in middle age and is felt as a cord-like structure in old age due to atherosclerosis and calcification. (The surface may show irregularities and the vessel may be tortuous.) Another way to note the condition of the vessel wall (when emptied out of blood) is to compress the brachial artery with a thumb and then palpate the radial artery by rolling it against the bone. (A “full” radial artery is normally palpable in many thin individuals).

7. Delay

When the left femoral artery and the right radial artery are palpated simultaneously the two pulses normally beat together. A delay in the femoral artery is seen in coarctation of the aorta.

8. Equality on the two sides

The arterial pulse of one side is always compared with that of the other side for all of its features described above. Normally, there is no difference between the two.

Examine all Other Arterial Pulses

The examination of other pulses is important: brachial—at the elbow; carotids—in the neck; femoral—in the groin; posterior tibial—behind medial malleolus; and dorsalis pedis— on the dorsum of the foot at the midpoint between medial and lateral malleoli, at the base of the first metatarsal bone.

EXERCISE

Q.1 What is Dicrotic Pulse?

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Q.2 What is Corrigan’s, Water-hammer or Collapsing Pulse?

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Q.3 What is Alternating Pulse (Pulsus Alternans)?

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Q.4 What is Thready Pulse?

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Q.5 What is tachycardia and what are its causes?

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Q.6 What is bradycardia and what are its causes?

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Q.7 How can stress or emotional states impact pulse findings?

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OSPE-1

Aim: To examine the radial artery of the subject provided.

Checklist:

1. Stands on the subject's right side and explains the procedure. (Y/N)
2. Holds the subject's right hand in a semipronated and slightly flexed position. Then places her three middle fingers on the radial artery and compresses it slightly against the bone. (Y/N)
3. Notes the rhythm, volume and character of the pulse. Counts the rate for one minute and notes the result. (Y/N)
4. Compresses the artery with the proximal finger and tries to roll the artery against the bone with the other two fingers. (Y/N)
5. Compares the equality of pulses in both arms. Counts the heart rate to see if there is any pulse deficit. (Y/N)

CASE-BASED QUESTION

Case Study: A 60-year-old female patient, Mrs. Singh, Complaints of fatigue, Shortness of breath, Dizziness with Clinical Findings of Pulse: 120 bpm, Blood Pressure: 180/100 mmHg. Respiratory Rate: 24/min. Oxygen Saturation: 92% on room air

Problem: Examine Mrs. Singh's pulse and interpret your findings.

Guiding Questions:

Describe the technique for examining pulse.
What are the characteristics of a normal pulse?
Interpret Mrs. Singh's pulse rate and rhythm.
Relate your findings to her clinical presentation.

Guided Discussion Questions:

1. What are the differences between radial, carotid and apical pulses?
2. How does pulse rate and rhythm relate to cardiac function?
3. What are the clinical implications of tachycardia or bradycardia?

Expected Outcomes:

Accurate examination of pulse. Understanding of pulse characteristics. Ability to interpret pulse findings. Critical thinking & problem-solving skills.

Assessment:

Written report on pulse examination.
Group presentation on pulse characteristics.
Class discussion participation.

Tips for Facilitators: Encourage hands-on practice. Foster group discussion. Provide feedback. Monitor progress.



Teacher's Signature

12. WBC ESTIMATION

LEARNING OBJECTIVES

After completion of the practical, students should be able to:

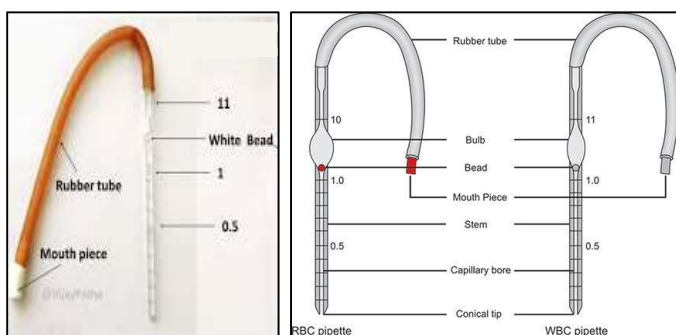
- Perform the Procedure Accurately
- Count and calculate the total number of WBCs in a blood sample
- Interpret the clinical significance of WBC counts and relate them to common physiological conditions.

PRINCIPLE

A sample of blood is diluted with a diluting fluid which destroys the red cells and stains the nuclei of the leukocytes. The cells are then counted in a counting chamber and their number in undiluted blood reported as leukocytes/mm³.

APPARATUS AND MATERIALS

1. Microscope.
2. Counting chamber with a heavy coverslip.
3. Blood lancet/pricking needle.
4. Sterile cotton/gauze swabs.
5. 70% alcohol (spirit)
6. WBC pipettes: white bead in bulb and markings 0.5, 1.0 and 11. Two such, clean and dry pipettes, with free-rolling beads are required.



7. **Turk's fluid (WBC diluting fluid)** - This fluid is used for diluting the blood.

Glacial acetic acid = 1.5 ml (hemolyzes RBCs without affecting WBCs).

Gentian violet (1% solution) = 1.5 ml (it stains the nuclei of leukocytes).

Distilled water to 100 ml.

PROCEDURES

1. Take 1 ml of Turk's fluid in a watch glass.
2. Place the counting chamber on the microscope stage. Adjust the illumination and focus the right upper group of 16 WBC squares. You will see all the squares in one field.
3. Observing all the aseptic precautions, get a finger-prick, discard the first 2 drops of blood and let a good-sized drop to form.
4. **Filling the pipette:** Dip the tip of the pipette in the edge of the drop, draw blood to the mark 0.5 and suck Turk's fluid to the mark 11. Mix the contents of the bulb thoroughly for 2-3 minutes.
5. **Charging the chamber:** Discard the first 2 drops of fluid from the pipette and charge the chamber with 1 in 20 dilution. The chamber should neither be over-charged nor under-charged.
6. Allow the cells to settle for 2-3 minutes, then carefully transfer the chamber to the microscope. Use the fine adjustment again and try to identify the WBCs.

Important

Under low magnification: The leukocytes appear as round, shiny (refractile), darkish dots, with a halo around them. These 'dots' represent the nuclei, which have been stained by gentian violet. The cytoplasm is not stained. Do not confuse with dust particles which have varying sizes and shapes, often angular. They are usually opaque, with no 'halo' around them. They may be brown, black or yellow in colour.

7. **Switch to high magnification** and study the leukocytes. By racking the microscope, you should be able to make out the morphology, of these cells, their round shapes, the clear unstained cytoplasm and the deep blue-violet nuclei which appear lobed in some cells and single in others. You will also see the remnants of the red cell membranes; these are called 'ghost' cells since they are faintly visible.

8. Counting the cells:

Count the cells under high power lens. You may count the WBCs in 16 squares under low power and then under high power and compare the results. Count the cells in the 4 groups of 16 squares each, i.e., in a total of 64 squares. Draw appropriate squares in your work-book for entering the counts.

Calculations: The leukocytes were counted in 64 squares, the volume of one square being =

$$1/4 (L) \times 1/4 (B) \times \text{Depth } 10 (H) = 1/160 \text{ mm}^3.$$

$$\text{Volume of 64 squares} = 1/160 \times 64 = 2/5 \text{ mm}^3.$$

Thus, the total volume of diluted blood in which WBCs were counted = $2/5 \text{ mm}^3$.

Let the count in $2/5 \text{ mm}^3$ be = N

Then 1 mm^3 of diluted blood will contain
= $N \times 5/2$ white cells.

Since the dilution employed is 20 times 1 mm^3 of undiluted blood will contain = $N \times 5/2 \times 20$

$$= N \times 100/2$$

$$= N \times 50$$

This means multiplying the number of cells N in 64 squares with 50 will give the total leukocyte count.

RESULTS

In the given sample, the following results are observed

Result: WBC =/Cumm

PRECAUTIONS

1. Observe all precautions for finger-prick, filling the pipette, and charging the counting chamber.
2. Keep equipment ready before pricking.
3. Allow sufficient time for hemolysis when mixing blood with Turk's fluid.
4. Continuously "rack" the microscope while identifying and counting cells.
5. Although the charged chamber remains stable for 80-90 minutes, the count is reliable for 30-40 minutes. Afterward, drying causes the count to decrease, so avoid delays.

EXERCISE

Q.1 A patient has a WBC count of $15,000/\text{mm}^3$. What conditions could this indicate, and what further tests should be done?

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Q.2 What do the three markings on the pipette indicate? How do you get a dilution of 1 in 20?

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Q.3 What is the volume of the bulb in the WBC pipette? Why is its bulb smaller than that of the RBC pipette?

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Q.4 What is the function of the bead in the bulb?

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Q.5 What are the other uses of WBC pipette?

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Q.6 What is the normal total leukocyte count as per age?

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Q.7 Compare the manual and automated methods of WBC estimation. What are the advantages and limitations of each?

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Q.8 What is meant by the terms leukocytosis and granulocytosis? Name the physiological and pathological conditions which cause leukocytosis.

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Q.9 What is the purpose of using Turk's fluid in WBC estimation?

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Q.10 How can abnormal WBC counts indicate specific diseases or conditions?

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OSPE

Aim: To dilute the given sample of blood 20 times for TLC by using diluent and a pipette provided.

Checklist:

1. Checks that the pipette is clean, dry and patent. (Y/N)
2. Takes sufficient diluting fluid in a watch glass and sucks blood exactly to the mark 0.5 and confirms that there is no air bubble. (Y/N)
3. Wipes off blood sticking to the outside of the pipette tip. (Y/N)
4. Sucks diluting fluid to the mark 11. (Y/N)
5. Holds the pipette horizontally between the palms and rolls it gently. (Y/N)

OSPE-2

Aim: To charge the counting chamber for TLC with diluted blood provided in a Pipette.

Checklist:

6. Checks that the chamber and the pipette are clean and dry. (Y/N)
7. Places the coverslip on the floor piece and trenches. (Y/N)
8. Mixes the contents of the pipette by rolling it between her palms and discards the first two. (Y/N)
9. Charges the chamber by slow and controlled release of diluted blood at the edge of the coverslip. (Y/N)
10. Allows the diluted blood to spread under the coverslip to cover the ruled area. Allows the cells to settle down. (Y/N)

CASE-BASED QUESTION

Case Study : A 35-year-old male patient, Mr. Patel, presents to the emergency department with: Severe sore throat, - Fever (103°F), Swollen lymph nodes

Laboratory Results: - Hemoglobin (Hb): 14 g/dL
- Platelet count: 250,000/ μ L
- Differential count: Neutrophils-80%, Lymphocytes -15% Monocytes: 5%

Problem: Estimate Mr. Patel's Total WBC count using the dilution method.

Questions:

1. Describe the principle of the dilution method for WBC counting.
2. Explain the procedure for preparing the WBC dilution.
3. Calculate the Total WBC count using the dilution method.
4. Interpret Mr. Patel's WBC count in relation to his clinical presentation.

Guided Discussion Questions:

What are the types of WBCs and their functions?

How does infection affect WBC count?

How would you correlate WBC count with differential count?

Expected Outcomes:

Accurate estimation of Total WBC count.

Understanding of WBC counting principles.

Ability to interpret WBC count.

Critical thinking and problem-solving skills.

Assessment:

Written report on WBC counting procedure.

Group presentation on WBC types and functions.

Class discussion participation.

Modified Scenario for Simulated Learning:

- Use simulated blood samples.
- Use electronic counters or software.
- Incorporate variations in WBC count and differential.

REFERENCES

- A textbook of practical physiology, CL. Ghai, 8th edition.
- Essentials of medical physiology, K. Sembulingam and P. Sembulingam, 6th edition.



Teacher's Signature

13. RBC ESTIMATION

LEARNING OBJECTIVES

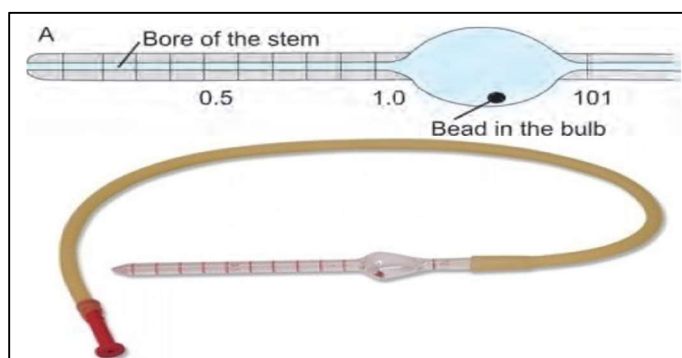
After completion of the practical, students should be able to:

- Perform the Procedure Accurately
- Count and calculate the total number of RBCs in a blood sample
- Interpret the clinical significance of RBC counts and relate them to common physiological conditions.

PRINCIPLE

The blood is diluted 200 times in a red cell pipette and the cells are counted in the counting chamber. RBC estimation involves diluting blood with an isotonic solution, loading it into a hemocytometer, and counting cells under a microscope within a defined grid. The RBC count is calculated using a formula that accounts for dilution, grid area, and chamber depth.

APPARATUS AND MATERIALS



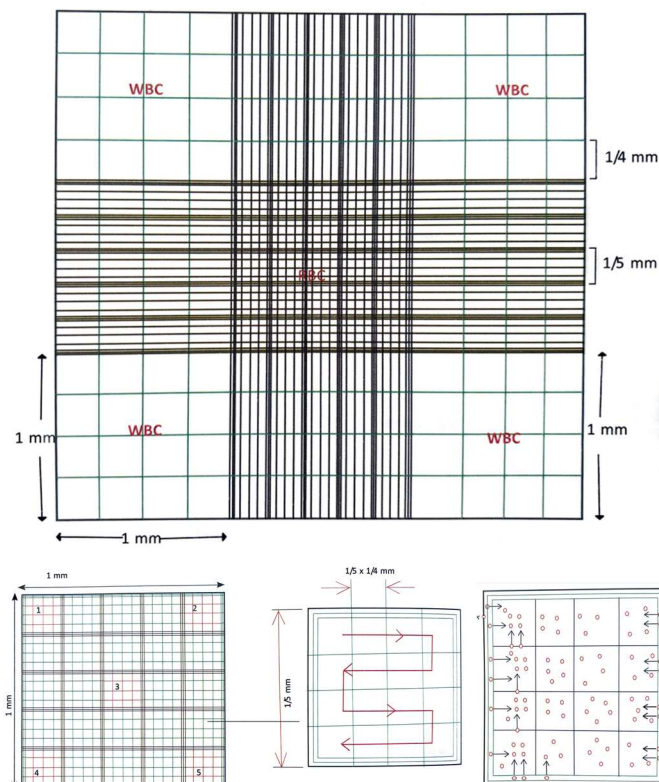
1. Improved Neubauer chamber with coverslip.
2. Microscope with LP and HP objectives and 10 x eyepiece.
3. Disposable blood lancet/pricking needle.
4. Sterile cotton/gauze swabs.
5. 70% alcohol/methylated spirit.
6. Hayem's fluid (RBC diluting fluid)

Composition of Hayem's fluid:

Sodium chloride (NaCl) - 0.50 g
Sodium sulfate (Na₂SO₄) - 2.50 g
Mercuric chloride (Hg Cl₂) - 0.25 g
Distilled water - 100 ml

PROCEDURES

1. Place about 2 ml of Hayem's fluid in a watch glass.
2. Examine the chamber, with the coverslip 'centred' on it, under low magnification. Adjust the illumination and focus the central 1 mm square (RBC square on the counting grid) containing 25 groups of 16 smallest squares each.
3. Charge the chamber with diluted blood.
4. **Filling the pipette with blood and diluting it:** Get a finger-prick. Wipe the first 2 drops of blood and fill the pipette from a fresh drop of blood up to the mark 0.5. Suck Hayem's fluid to the mark 101 and mix the contents of the bulb for 2-3 minutes.
5. Since the RBC pipette is a slow-speed pipette, it will need to be kept at an angle of 70-80° while charging the chamber.
6. Move the chamber to the microscope and focus the grid once again to see the central 1 mm square with the red cells distributed all over.
7. Wait for 2-3 minutes for the cells to settle down.
8. **Counting the cells:** Switch over to high magnification (HP lens) and check the distribution of cells. If they are unevenly distributed, i.e., bunched at some places and scanty at others, the chamber has to be washed, dried and recharged.
9. Move the chamber carefully and bring the left upper corner block of 16 smallest squares in the field of view. (There are no smallest squares above and to its left).
10. **Rules for Counting:** Cells lying within a square are to be counted with that square. Cells lying on or touching its upper horizontal and left vertical lines are to be counted with that particular square. Cells lying on or touching its lower horizontal and right vertical lines are to be omitted from that square because they will be counted with the adjacent squares. In this way you will avoid counting a cell twice.



11. First count the cells in the upper 4 horizontal squares from left to right, then come down to the next row and count the cells in each square from right to left. Then count the cells in the 3rd row from left to right and in the 4th row, from right to left. As the counts are made, enter your results in the appropriate squares drawn in your workbook, showing the count in each square.

OBSERVATIONS

Add up the number of cells in each of the 5 blocks of 16 smallest squares. A difference of more than 20 between any 2 blocks indicates uneven distribution.

A. Dilution factor

Blood is taken to mark 0.5 for the dilution with pipette is 1 in 200.

B. Calculation of volume of fluid examined

Volume of the fluid in 1 medium square
 $= 1/5 (L) \times 1/5 (B) \times \text{Depth of the chamber (H)}: 1/10$
 $= 1/250 \text{ mm}^3$

Volume of the fluid in 5 medium square:
 $= 1/250 \times 5 = 1/50 \text{ mm}^3$

Let N number of RBC be the number of cells in $1/50 \text{ mm}^3$ of diluted blood.

Cells in 1 mm^3 of diluted blood $= N \times 50$

Dilution employed was $= 1 \text{ in } 200$

Number of cells in 1 mm^3 is $= N \times 50 \times 200$
 $= N \times 10000$

RESULTS

In the given sample, the following results are observed

Result: RBC =Mill/Cumm

PRECAUTIONS

1. Observe all precautions described for getting a finger-prick, filling the pipette with blood and diluent and charging the chamber.
2. Clean the chamber with a lint-free piece of cloth. Once cleaned, do not touch the central part. Any oils left from your fingers on the coverslip or the chamber are fatal to cell counting.
3. If there is over-charging or under-charging, wash, clean, dry and refill the chamber.
4. Continuously rack the fine adjustment while counting the cells.

EXERCISE

Q.1 Why is blood diluted 200 times for red cell count?

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Q.2 What are the normal ranges of RBCs in different age groups??

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Q.3 What is the function of the coverslip?

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Q.4 What is the life span and fate of red cells?

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Q.5 How will you differentiate a WBC pipette from a RBC pipette?

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Q.5 What are the normal ranges of RBCs in different age groups?

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Q.6 Which physiological condition causes a decrease in RBC count?

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Q.7 What is polycythemia and what are its causes?

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Q.8 Why should you discard the first two drops from the pipette before charging the chamber?

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Q.9 What is the function of the bead in the bulb?

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Q.10 How will you differentiate red cells from dust particles?

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OSPE

Aim:

To charge the Neubauer's chamber with diluted blood provided to you in RBC pipette.

Checklist:

1. Ensures that the chamber and cover slip are clean and dry. (Y/N)
2. Mixes the contents of the bulb between her palms. (Y/N)
3. Places the cover slip on the central plateau of the chamber to cover both the ruled areas. (Y/N)
4. Discards the first two drops from the pipette and allows a suitable-sized drop to form. (Y/N)
5. Places the tip of pipette on the chamber, touching the edge of the coverslip and allows the fluid to spread evenly over the counting grid without over- or under-charging. Charges the other side also. (Y/N)

PROBLEM-BASED LEARNING

Case Study : A 40-year-old male patient, Mr. Kumar, presents to the outpatient department with: Fatigue, Shortness of breath, Pallor

Laboratory Results:

- Hemoglobin (Hb): 10 g/dL
- Hematocrit (Hct): 30%
- MCV: 80 fL
- MCH: 25 pg

Problem:

Estimate Mr. Kumar's RBC count using the hemocytometer method and comment on disease type he has with probable cause.

Given Data:

- Dilution factor: 1:200
- RBCs counted in 80 small squares: 250

Questions:

1. Describe the principle of RBC counting using a hemocytometer.

2. Explain the procedure for preparing the RBC dilution.
3. Calculate the RBC count using the hemocytometer method.
4. Interpret Mr. Kumar's RBC count in relation to his clinical presentation.

Guided Discussion Questions:

What are the factors affecting RBC count?

How does anaemia affect RBC count?

How would you correlate RBC count with Hb and Hct?

Expected Outcomes:

Accurate estimation of RBC count.

Understanding of RBC counting principles.

Ability to interpret RBC count.

Assessment:

Written report on RBC counting procedure.

Group presentation on RBC functions.

Class discussion participation.

Modified Scenario for Simulated Learning:

Use simulated blood samples.

Use electronic counters or software.

Incorporate variations in RBC count and morphology.

REFERENCES

- A textbook of practical physiology, CL. Ghai, 8th edition.
- Essentials of medical physiology, K. Sembulingam and P. Sembulingam, 6th edition.



Teacher's Signature

14. DLC ESTIMATION

STUDENT OBJECTIVES

After completion of the practical, students should be able to:

1. Prepare satisfactory blood films, fix and stain them and describe the features of a well-stained film.
2. Identify different blood cells in a film and indicate the identifying features of each type of leukocyte.
3. Differentiate between neutrophils, eosinophils and basophils and between a large lymphocyte and a monocyte.

PRINCIPLE

A blood film is stained with Leishman's stain and scanned under oil immersion, from one end to the other. As each WBC is encountered, it is identified until 100 leukocytes have been examined. The percentage distribution of each type of WBC is then calculated. Knowing the TLC and the differential count, it is easy to determine the number of each type of cell per mm³.

APPARATUS AND MATERIALS

1. Microscope.
2. Counting chamber with a heavy coverslip.
3. Blood lancet/pricking needle.
4. Sterile cotton/gauze swabs.
5. 70% alcohol (spirit)
6. A drop bottle containing Leishman's stain.
7. A wash bottle of distilled water (or buffered water, if available).
8. Fluff-free blotting paper.

Leishman's stain. This stain is a simplification of Romanowsky group of stains. It is probably one of the simplest and most precise method of staining blood for diagnostic purposes. It contains a compound dye-eosinate of methylene-blue dissolved in acetone-free methyl alcohol.

i. Eosin. It is an acidic dye (negatively charged) and stains basic (positive) particles/granules of eosinophils and RBCs a pink colour.

ii. Methylene-blue. It is a basic dye (positively charged) and stains acidic (negatively charged) granules in the cytoplasm, nuclei of leukocytes, especially the granules of basophils, a blue-violet colour.

PROCEDURES

A. Preparing the Blood Films

1. Prepare 4 or 5 blood films.
2. Use fresh capillary blood from a finger prick or EDTA anticoagulated venous blood.
3. Take a clean, grease-free glass slide.
4. Place a small drop of blood (approximately 2-3 mm in diameter) near one end of the slide using a pipette or applicator stick.
5. Hold a spreader slide (another glass slide) at a 30-45° angle to the slide with the blood drop.
6. Back the spreader slide into the drop until the blood spreads along its edge.
7. In one smooth motion, push the spreader slide forward, spreading the blood over the slide to create a thin, even smear.
8. Air dry the slides immediately by waving them in the air.
9. After fixing and staining, examine slides under low and then under high magnifications/

B. Fixing and Staining of Blood Films

Fixation is the process that makes the blood film and its cells adhere to the glass slide. It also preserves the shape and chemistry of blood cells as near living cells as possible.

Staining is the process that stains (colours) the nuclei and cytoplasm of the cells. Both these purposes are achieved by the Leishman's stain.

Fixing the Blood Films

1. Place the slides, smear side up, on a 'staining rack' assembled over a sink (two glass rods placed across the sink, with the ends fitted into

short pieces of rubber tubing). Ensure that they are horizontal.

2. Pour 8–10 drops of the distilled water diluted Leishman stain (1:2) on each unfixed slide by dripping it from a drop bottle or use a dropper. This amount of stain usually covers the entire surface and “stands up” from the edges of the slides without running off. Note the time. Leishman stain contains methanol, which acts as a fixative to preserve cell morphology.
3. Allow the stain to remain undisturbed for 1–2 minutes, as advised.
4. During this time, watch the stain carefully, especially during hot weather and see that it does not become syrupy (thick) due to evaporation of alcohol. If the stain dries, it will precipitate on the blood film and appear as round, blue granules. This can be prevented by pouring more stains on the slides as required.

Staining the blood film

1. After the fixing time is over, add an equal number of drops of distilled water (or buffered water, if available) to the stain.
2. Mix the stain and water by gently blowing at different places on the slides through a dropper, without scratching the smear. A glossy greenish layer (scum) soon appears on the surface of the diluted stain. Allow the diluted stain to remain on the slide for 7-10 minutes or as advised.
3. Flush off the diluted stain in a gentle stream of distilled water for about 30 seconds and leave the slides on the rack for about a minute with the last wash of water covering them. Drain the slides and put them in an inclined position against a support, stained sides facing downwards (to prevent dust particles settling on them) to drain and dry. The under sides of the slides may be blotted with filter paper.

In short - Fixation (2 min), Staining (8 min), Washing (5 Sec), Drying (5 min)

OBSERVATIONS

1. Use a low-power lens (10x) and (40x) to scan the slide and a high-power lens (100x, oil

immersion) to count the cells and study cell morphology in detail.

2. Draw 100 squares in your workbook for recording various WBCs as they are encountered and identified one after another.
3. Enter these cells by using the letters ‘N’ for neutrophils, ‘M’ for monocytes, ‘LL’ for large lymphocytes, ‘SL’ for small lymphocytes, ‘E’ for eosinophils and ‘B’ for basophils.
4. Place a drop of cedar wood oil on the right upper corner of the film, a few mm away from the head end. Bring the oil immersion lens into position till it enters the oil drop. Adjust the focus. Do not flood the entire surface of the slide with oil; as you move the slide, the oil will move with the objective lens.
5. Move the slide slowly to the right (the image will move to the left) and as you encounter a leukocyte, identify it and enter it in your workbook. As you approach the end of the smear, move 2 fields down and scan the film in the opposite direction. As you near the head, again move 2 fields down and scan the film towards the tail. Traverse the film in this to and fro fashion till you have examined 100 cells.

RESULTS

In the given sample, the following results are observed

Neutrophils	:...../Cumm
Lymphocytes	:...../Cumm
Monocytes	:...../Cumm
Eosinophils	:...../Cumm
Basophils	:...../Cumm

PRECAUTIONS

1. Absolutely clean, grease and dust-free slides must be used.
2. When taking blood on to the slide before spreading it, do not touch the skin with the slide but only the periphery (top) of the blood drop.
3. Do not allow Leishman stain to become syrupy (thick) or dry up on the slide as this is likely to cause precipitation of the stain.

- 4. Avoid tap water for diluting the stain and for washing off the diluted stain after ‘staining’, because methylene-blue components may not stain the cells properly.
- 5. When counting the cells, use the “battlement” method. This is to avoid counting a cell twice.
- 6. Avoid counting the leukocytes in the extreme ends of the head or tail and along the edges of the blood film.

EXERCISE

Q.1 Why is cedar wood oil used with oil-immersion objective?

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Q.2 How are leukocytes classified?

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Q.3 What are the functions of various leukocytes?

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Q.4 How would you differentiate between neutrophils, eosinophils and basophils?

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Q.5 What is the clinical importance of doing DLC?

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Q.6 What are the normal ranges of different leukocytes in a DLC?

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Q.7 What is absolute leukocyte count and what is its importance?

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Q.8 What is meant by the terms neutrophilia and neutropenia and what are their causes?

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Q.9 Enumerate the conditions that cause eosinophilia and eosinopenia

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Q.10 Name the conditions where basophils increase and decrease in number.

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Q 11. What are the causes of monocytosis and monocytopenia?

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Q.12 What are the causes of lymphocytosis and lymphocytopenia?

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Q.13 Describe the staining techniques used in DLC (e.g., Leishman stain).

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Q.14 How can DLC help in diagnosing bacterial vs. viral infections?

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Q 15. What are the DLC findings in allergic reactions?

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OSPE 1

OSPE Station: OSPE (on Blood Smear Preparation and Staining (Leishman Stain))

The following **OSPE checklist** includes **15 points** to evaluate the student's skills in making a blood smear and staining using Leishman stain:

1. Sample Collection (1 point) (Y/N)
2. Choose a Proper clean, dry glass slide, free from grease or contamination. (1 point) (Y/N)
3. Proper blood drop Placement (1 point) (Y/N)
4. Smear Technique: 30-45 degree angle. Push the spreader slide smoothly to the other end to form an even, thin smear. (1 point) (Y/N)
5. Smear Quality - Check for a feathered edge & evenly spread at the end of the smear. (1 point) (Y/N)
6. Air-drying Allow the stained smear to air dry for 5 minutes before examination. (1 point) (Y/N)
7. Dilute Leishman stain with distilled water in the ratio of 1:2 or as instructed. (1 point) (Y/N)
8. Fixation - Cover the smear with Leishman stain and allow it to fix for 1-2 minutes. (1 point) (Y/N)
9. After fixation, add an appropriate amount of distilled water (7-10 drops) for 7-10 minutes.
10. Washing the Slide with distilled water Ensure that water flows off the slide (1 point) (Y/N)
11. Air Drying After Staining (1 point) (Y/N)
12. Use oil immersion lens (100x) for detailed examination of cells. (1 point) (Y/N)
13. Identification of some Cell Types (1 points) (Y/N)
14. Identification of all Cell Types (1 points) (Y/N)
15. Reporting Results: (1 point) (Y/N)

PROBLEM-BASED LEARNING

Case Study :

A 50-year-old female patient, Mrs. Sharma, presents to the outpatient department with: Fever (101°F) with infected wound

Laboratory Results:

Total WBC count: 15,000/ μ L, Neutrophils- 85, Monocytes – 40

Problem:

Complete the DLC report and interpret the results in relation to Mrs. Sharma's clinical presentation.

Questions:

1. Explain the clinical significance of DLC.
2. Interpret Mrs. Sharma's DLC results.

Guided Discussion Questions:

What are the functions of different WBC types?
How do infections affect DLC?

Expected Outcomes:

Understanding of DLC principles.
Ability to interpret DLC.
Critical thinking and problem-solving skills.

Assessment:

1. Written report on DLC procedure.
2. Group presentation on WBC functions.
3. Class discussion participation.

REFERENCES

- A textbook of practical physiology, CL. Ghai, 8th edition.
- Essentials of medical physiology, K. Sembulingam and P. Sembulingam, 6th edition.



Teacher's Signature

15. RECORDING OF SYSTEMIC ARTERIAL BLOOD PRESSURE

STUDENT OBJECTIVES

After completion of the practical, students should be able to:

1. Define systolic, diastolic, pulse and mean arterial pressures. Indicate their significance.
2. Determine blood pressure by palpatory, oscillometric and auscultatory methods. Name the advantages and disadvantages of each method.
3. Indicate the precautions that must be taken before and during recording of blood pressure.

DEFINITIONS AND NORMAL VALUES

Arterial blood pressure is defined as the lateral pressure exerted by the column of blood on wall of arteries. Arterial blood pressure is expressed in four different terms:

Systolic Blood Pressure: Systolic blood pressure (systolic pressure) is defined as the maximum pressure exerted in the arteries during systole of heart. Normal systolic pressure: 120 mm Hg (110 mm Hg to 140 mm Hg).

Diastolic Blood Pressure: Diastolic blood pressure (diastolic pressure) is defined as the minimum pressure exerted in the arteries during diastole of heart. Normal diastolic pressure: 80 mm Hg (60 mm Hg to 80 mm Hg).

Pulse Pressure: Pulse pressure is the difference between the systolic pressure and diastolic pressure. Normal pulse pressure: 40 mm Hg ($120 - 80 = 40$).

Mean Arterial Blood Pressure: Mean arterial blood pressure is the average pressure existing in the arteries. This is because, the diastolic Period of the cardiac cycle is longer (0.53 second) than the systolic period (0.27 second).

Normal mean arterial pressure: 93 mm Hg. ($80 + 13 = 93$).

Mean arterial blood pressure = Diastolic pressure + $\frac{1}{3}$ of pulse pressure

METHODS OF MEASUREMENT OF BP

Direct Method

The direct method of recording blood pressure (BP), in which an artery is punctured with a cannula connected to a manometer,

Indirect Methods

Recording blood pressure (BP), with sphygmomanometer.

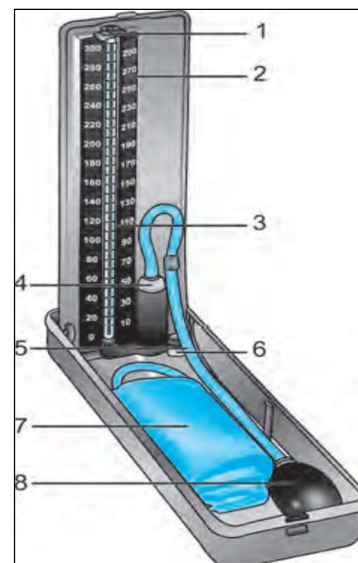
PRINCIPLE

The brachial artery is first compressed by inflating a rubber bag (connected to a manometer) placed around the arm to stop the blood flow through the occluded section of the artery. The pressure is then slowly released and the flow of blood through the obstructed segment of the artery is studied by:

1. Feeling the pulse - the palpatory method.
2. Observing the oscillations of the mercury column - the oscillometric method and
3. Listening to the sounds produced in the part of the artery just below the obstructed segment - the auscultatory method.

APPARATUSES

1. Stethoscope.
2. Sphygmomanometer.



1. Sphygmomanometer (the “BP apparatus”). 1: Spring-loaded clip; 2: Lid of the apparatus; 3: Graduated glass tube; 4: One-way valve; 5: Mercury reservoir; 6: Stop cock; 7: Armlet; 8: Air pump (rubber bulb, with leak valve.) (source- A Textbook of Practical Physiology, CL Ghai)

PROCEDURES

A. Palpatory Method (Riva Rocci 1896)

1. Have the subject sit or lie down, relaxed for 5 minutes.
2. Ensure the mercury is at zero by releasing the reservoir lock.
3. Wrap the cuff on the upper arm, with its center over the brachial artery and the lower edge 3 cm above the elbow.
4. Palpate the radial artery at the wrist with your fingertips, hold the air bulb in the other hand, and tighten the leak valve.
5. Inflate the cuff until the pulsations disappear; note the reading and inflate 30-40 mmHg further.
6. Gradually deflate the cuff (2-3 mmHg per step) using the leak valve.
7. Note the pressure when the pulse reappears- this is the systolic pressure. Deflate quickly to zero.
8. Repeat for the other arm, taking three readings per arm with short breaks between measurements.

B. Oscillatory Method

1. As the cuff pressure was raised and then lowered, oscillations appeared which became maximum and then disappeared.
2. Some workers took the midpoint of maximum oscillations as the DP while others considered the lower level of these oscillations as the DP (oscillations are best seen with an aneroid manometer).
3. The students must have seen these oscillations in the mercury column.

C. Auscultatory Method (Korotkoff, 1905)

1. Place the cuff on the upper arm and measure BP using the palpatory method.
2. Locate the bifurcation of the brachial artery just medial to the biceps tendon.
3. Position the stethoscope on this point and hold it with your left hand (if right-handed).

4. Inflate the cuff 40-50 mm Hg above systolic BP as per the palpatory method.
5. Gradually release pressure until a sharp tapping sound is heard, known as Korotkoff sounds.
6. Take 3 readings on both arms using the auscultatory method.

DIGITAL BLOOD PRESSURE MONITOR.

Fig 1: It is a compact, battery-operated palm-top unit with an LCD screen and memory function. The recorder uses a 2-cuff "oscillometric measuring" principle to automatically convert pulse wave oscillations into SP and DP. This method is easy for laypeople to use. The pressure range is -280 mm Hg, and the heart rate range is 40-180/min. The unit has 2 input sockets, one on each side. Connect the rubber bulb to one socket and the cuff (on the upper arm) to the other. The procedure is the same as the auscultatory method, except no Korotkoff sounds are auscultated. As pressure is raised and lowered, the pressure and pulse readings display on the screen, remaining until the unit is switched off. Fig 2 - This innovative, compact, battery-operated BP monitor also works on the oscillometric principle. It fits over the wrist with a Velcro cuff. The controls include: Mode/set: for date and time; Start/stop and Recall buttons.



Fig 1



Fig 2

OBSERVATIONS AND RESULTS

Record the observations in your workbook. Take the average of 3 readings as shown below and report your result as:

A. Palpatory method (mm Hg)

	Right Arm	Left Arm
1 st reading		
2 nd reading		
3 rd reading		

Result: Systolic BPmmHg

B. Auscultatory method (mm Hg)

	Systolic	Diastolic
1 st reading		
2 nd reading		
3 rd reading		

Result: Systolic BPmmHg

Diastolic BPmmHg

PRECAUTIONS

1. The subject should be relaxed, free from tension, and assured for 5 minutes to prevent 'white coat hypertension.' Compare BP in both arms on the first measurement.
2. Keep the arm with the cuff at heart level to avoid gravity's influence.
3. Position the cuff tubing anterolaterally to the cubital fossa to avoid interference with the stethoscope.
4. Ensure the cuff is neither too tight nor too loose.
5. Avoid leaving the cuff inflated at high pressures to prevent false high readings due to discomfort and reflex spasm.
6. Do not apply pressure on the artery with the stethoscope chest-piece to prevent a false low reading.
7. Always use the palpatory method before the auscultatory method.
8. In suspected or known hypertension, raise the pressure above 200 mm Hg or the level estimated by palpation.

EXERCISE

Q.1 Define blood pressure and its clinical importance.

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Q.2 What is systolic blood pressure and its clinical importance?

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Q.3 What is diastolic pressure and its clinical importance?

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Q.4 What is mean arterial pressure and what is its significance?

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Q.5 What is pulse pressure and what is its significance?

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Q.6 Why should the cuff be positioned at the level of the heart when measuring blood pressure?

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Q.7 What are the physiological variations in blood pressure?

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Q.8. How does the palpatory method help estimate systolic pressure?

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Q.9 What is hypertension and what are its causes and complications?

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Q.10 What is hypotension and what are its effects?

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Q.11 What is considered high blood pressure (hypertension) according to current guidelines?

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Q.12 How can "white coat hypertension" affect blood pressure readings?

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Q.13 What are the two main measurements in a blood pressure reading?

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Q.14 What is the difference between systolic and diastolic blood pressure?

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Q.15 Why is it important to compare blood pressure in both arms during the first measurement?

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Q.16 What are the potential risks of leaving the blood pressure cuff inflated for too long?

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OSPE-II

Aim: To record the blood pressure of the subject provided by the auscultatory method.

Checklist: (15 points): Each correctly performed step = 1 mark. Total = 15 marks.

1. Checks the BP apparatus and stethoscope for functionality and exposes the upper arm. (Y/N)
2. Ensures the subject is seated comfortably with their arm at heart level. (Y/N)
3. Selects the appropriate cuff size based on the arm circumference of the subject. (Y/N)
4. Wraps the cuff snugly around the upper arm with the tubing positioned anterolateral to the cubital fossa. (Y/N)
5. Confirms that the cuff is neither too tight nor too loose. (Y/N)
6. Records the systolic pressure by the palpatory method before proceeding. (Y/N)
7. Correctly locates the brachial artery at the cubital fossa using palpation. (Y/N)
8. Positions the stethoscope chest-piece directly over the brachial artery and applies the earpieces. (Y/N)
9. Inflates the cuff rapidly to 20–30 mm Hg above the palpated systolic pressure. (Y/N)
10. Deflates the cuff slowly at a rate of 2–3 mm Hg per second while listening for Korotkoff sounds. (Y/N)
11. Identifies the systolic pressure as the point where the first Korotkoff sound is heard. (Y/N)
12. Records the diastolic pressure as the point where the Korotkoff sounds disappear. (Y/N)
13. Removes the cuff after recording the readings and ensures the subject is comfortable. (Y/N)
14. Reports the systolic and diastolic readings accurately. (Y/N)
15. Cleans and stores the equipment properly after the procedure. (Y/N)

OSPE-I

Aim: To record the blood pressure of the subject provided by the palpatory method.

Checklist (5 points) Each correctly performed step = 3 mark. Total = 15 marks.:

1. Checks the zero reading of the manometer. Explains the procedure. Exposes the arm upto the shoulder. (Y/N)
2. Wraps the cuff firmly around the upper arm, keeping its lower edge about 3 cm above the elbow, its middle lying over the brachial artery and the tubes lying anterolaterally. (Y/N)
3. Keeps the BP apparatus at the level of the heart. Opens the stopcock between the two limbs of the mercury manometer. (Y/N)
4. Palpates the brachial artery and marks its position. Then holds the rubber bulb in her right hand, closes the leak-valve screw and inflates the cuff slowly until the radial pulse disappears. (Y/N)
5. Releases the pressure in the cuff slowly till the radial pulse reappears. Notes the reading. (Y/N)

ROLE PLAY ACTIVITY

Objective:

Students will practice taking blood pressure readings, understand the steps involved and learn how to communicate effectively with patients during the procedure.

Scenario:

Preparation:

- The Patient sits comfortably in a chair with their arm resting on a table at heart level.
- Student prepares the sphygmomanometer and stethoscope, ensuring that the equipment is functioning properly.

Role-Playing the Procedure:

STUDENT:

- a) Student introduces themselves and explains the procedure to the Patient.

- b) Ensures the Patient is relaxed and seated with their arm supported.
- c) Selects the appropriate cuff size and wraps it snugly around the Patient's upper arm, just above the elbow.
- d) Palpates the brachial artery to locate it.
- e) Places the stethoscope's diaphragm over the brachial artery.
- f) Inflates the cuff to approximately 20-30 mmHg above the estimated systolic pressure.
- g) Slowly deflates the cuff while listening for the first Korotkoff sound (systolic pressure) and the point where the sound disappears (diastolic pressure).
- h) Records the blood pressure reading and notes any observations.

PATIENT

- a) Responds to questions and follows instructions, such as sitting still and remaining calm.
- b) Provides feedback on comfort and any symptoms if asked.

OBSERVER:

- a) Takes notes on the student's technique.
- b) Provides constructive feedback on procedure adherence, communication and patient interaction.

Debriefing:

- After each role play, the group discusses what went well and what could be improved.
- The Observer shares their feedback with both the Patient and student.
- The student and Patient switch roles and the process is repeated to ensure that each student gets a chance to practice.

Discussion Points:

- The importance of proper cuff placement and size.
- Techniques for accurately listening sounds.
- How to communicate effectively with patients to reduce anxiety.
- Common errors and how to avoid them.

CASE-BASED LEARNING

1. Case Presentation

Case Scenario:

Dolly, a 40-year-old woman, comes into the clinic for a routine physical exam. She is concerned about her recent symptoms of occasional headaches and fatigue. Her primary care physician decides to assess her blood pressure to rule out hypertension.

2. Case Analysis

a) History Taking:

- **Symptom Inquiry:** Ask Dolly about the frequency, duration and nature of her headaches and fatigue.
- **Medical History:** Review any past issues with blood pressure, heart disease or other relevant conditions.
- **Lifestyle Factors:** Discuss her diet, physical activity, stress levels and caffeine or alcohol consumption.
- **Family History:** Ask if there is a family history of hypertension or cardiovascular diseases.

b) Physical Examination and Preparation:

- **Proper Setting:** Ensure Dolly is seated comfortably in a quiet, relaxed environment for at least 5 minutes before measuring her blood pressure.
- **Proper Positioning:** Dolly should be seated with her back supported, feet flat on the floor and arm supported at heart level.
- **Selection of Cuff Size:** Choose the correct cuff size based on Emily's arm circumference to ensure accurate readings.

c) Measurement Technique: As per procedure.

Discussion

a) Accuracy and Technique:

- **Factors Affecting Accuracy:** Discuss potential sources of error, such as cuff size, patient positioning and technique.

- **Repeat Measurements:** Emphasize the importance of taking multiple readings to confirm accuracy.

b) Interpretation of Results:

- **Normal Ranges:** Explain the normal blood pressure ranges.
- **Clinical Implications:** Discuss what different readings might indicate regarding Dolly's health.

c) Patient Communication:

- **Explaining Results:** Practice how to explain the blood pressure results to Emily, including any necessary follow-up or lifestyle changes.
- **Addressing Concerns:** Teach strategies to address patient concerns or questions about their blood pressure readings.

Reflection and Learning Objectives

Key Takeaways: Understand the importance of proper blood pressure measurement techniques and accurate interpretation of results.

Critical Thinking: Reflect on how different factors can influence blood pressure readings and the importance of a standardized approach.

Patient Engagement: Develop skills in patient education and communication regarding blood pressure and overall health management.

REFERENCES

- A textbook of practical physiology, CL. Ghai, 8th edition.
- Essentials of medical physiology, K. Sembulingam and P. Sembulingam, 6th edition.



Teacher's Signature

16. PERFORM PROCEDURE INSPECTION OF RESPIRATORY SYSTEM

STUDENT OBJECTIVES

After completion of the practical, students should be able to:

1. Carry out a systematic examination of the respiratory system.
2. List the abnormal forms of the chest.
3. Describe the differences between vesicular and bronchial breath sounds.
4. Describe the importance of vocal fremitus and resonance.

INSPECTION

Form of the chest

- The normal chest is bilaterally symmetrical and there are no large bulges or hollows.
- It is elliptical in shape, the normal ratio of transverse to anteroposterior diameter (Hutchinson's index) being 7:5, (the chest becomes barrel-shaped in emphysema).
- Observe carefully the positions of trachea and apex beat and note whether engorged veins are present over the chest.
- Abnormal forms of chest include: alar and flat chests due to poor posture, rachitic chest in rickets, pigeon breast chest; and barrel-shaped chest in emphysema.

Respiratory movements

- The rate, depth, rhythm and type (manner) of breathing should be noted.
- The rate should be counted surreptitiously, while keeping the fingers on the radial pulse, because a nervous patient may breathe rapidly and irregularly.
- The normal rate of respiration is 14-20 per minute, one inspiration and one expiration making up one cycle.
- Type of breathing - This depends on age and sex. Males: Breathing is predominantly abdominal, as

the diaphragm is used more freely than the intercostal muscles. The downward movement of the diaphragm causes noticeable outward movement of the abdomen. Females: Breathing is predominantly thoracic, with greater chest movements compared to the abdomen. Children: Breathing is primarily abdominal, similar to that of males.

- A particular note should be taken of the equality of expansion on the two sides. Both sides move equally, symmetrically and simultaneously. Asymmetric expansion of the lungs may be seen when the underlying lung is diseased. Fibrosis, consolidation, collapse or pleural effusion can all decrease chest expansion on the affected side though other physical signs will also be present.
- Flat chest: pulmonary TB and fibrothorax. (Fig1)
Pigeon chest (Pectus carinatum): in childhood asthma and rickets. (Fig 2)



Fig:1

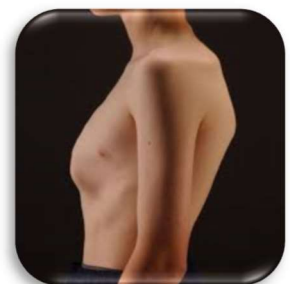


Fig:2

PALPATION

Position of trachea.

- Feel the rings of the trachea in the suprasternal notch with the tip of your index finger and try to judge the space between it and the insertion of sternomastoid muscle on either side of it.
- Normally, trachea is in the midline or slightly to one side. It may be pulled to the affected side (fibrosis, lung collapse) or pushed away from the affected side (pneumothorax, pleural effusion).

For effective chest palpation, the hands should be warm and used gently. The chest is palpated in the upper, middle, and lower regions on both the front and back. In the upper zones, the hands are placed over the apices from behind with the thumbs meeting at the midline. The movement of the thumbs away from the midline during deep breathing helps assess equal or unequal lung expansion. For the middle and lower zones, the hands are positioned on either side of the chest with the thumbs touching at the midline. The extent of thumb movement away from the midline indicates the degree of lung expansion. Additionally, palpation can detect subcutaneous emphysema, which feels spongy and may result from rib fractures.

Assessment of Posterior thoracic expansion (Fig 3)
Assessment of Anterior thoracic expansion (Fig 4)



Fig. 3



Fig. 4

Vocal Fremitus

Vocal fremitus refers to vibrations transmitted from the larynx through the bronchi, lungs, and chest wall during phonation. To assess it, the palm or ulnar border of the hand is placed on the intercostal spaces while the patient repeats “ninety-nine,” “one-two-three,” or “ek-do-teen.” The vibrations are compared at identical points from top to bottom on the front, axillary region, and back. Fremitus is diminished if the voice is weak, a bronchus is blocked, or fluid/air in the pleural cavity dampens vibrations. It is increased when vibrations are better conducted, as in lung consolidation (e.g., pneumonia).

PERCUSSION

Percussion creates artificial vibrations in tissues using a sharp tap. The middle finger of the left hand (pleximeter finger) is placed firmly on the skin and struck on its middle phalanx with the tip of the right middle finger 2–3 times. The striking finger should be parallel to the pleximeter finger, relaxed, and lifted only 2–3 inches. The movement should originate from the wrist, not the elbow or shoulder.

The following things are to be noted while percussing:

- Air-containing organs, such as lungs, produce a note (sound) called **resonance**.
- The opposite of resonance, i.e. lack of note, called dullness, is found over solid viscera like heart and liver or when the lung becomes solidified as in pneumonia, growth or fibrosis.
- An extreme form of **dullness** is called **stony dullness**, in which a feeling of resistance is felt by the tapping finger along with a dull note; such dullness is found by percussing over the thigh and is encountered in pleural effusion.

AUSCULTATION

Auscultation and Breath Sounds

Auscultation is performed over the lungs (front, axillary regions, and back), comparing sounds at corresponding points on both sides. The patient breathes deeply through an open mouth for better sound clarity. Key observations include:

Type of breath sounds: Vesicular or bronchial.

Intensity: Normal, diminished, or absent.

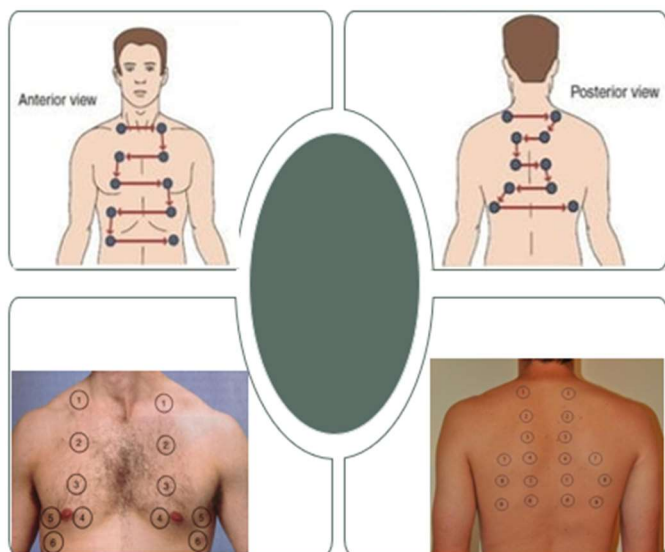
Added sounds: Crepitations, rhonchi, or pleural rub.

Vocal resonance: The sounds heard during phonation, assessed by having the patient say “ninety-nine” or “ek-do-teen,” comparing regions on both sides.

Added Sounds

- Rhonchi (Wheezes): Dry sounds caused by air passing through narrowed airways.
- Crepitations (Crackles): Bubbling sounds from air moving through fluid in small airways or alveoli; can be fine or coarse.
- Pleural Rub: A creaking sound from friction between inflamed pleural layers, heard during deep inspiration, disappearing if fluid accumulates in the pleural cavity.

Lungs Anterior & posterior Auscultation sites



Lungs Anterior & posterior Percussion sites

RESULT

Write key findings of the respiratory system, based on examination/observations on student, patient or simulated patient.

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EXERCISE

Q.1 What are the common causes of abnormal size and shape of chest?

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Q.2 Name and explain types of breathing in female and male?

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Q.3 What is the importance of vocal fremitus?

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Q.4 What is the importance of vocal resonance?

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Q.5 Explain different percussion sound in different condition?

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Q.6 Name auscultatory areas where bronchial breath sounds can normally be heard.

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Q.7 Scan the first QR Code of lung sounds and write your reflection on learning by Simulation.

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OSPE 1

Aim:

To assess the expansion of lower part of chest on the back.

Checklist:

1. Gives proper instructions to the subject exposes the chest. (Y/N)
2. Places both hands on either side of lower chest, on the back, with fingers stretched out on either side and thumbs just touching in the midline. (Y/N)
3. Asks the subject to take two or three deep breaths. (Y/N)
4. Observes the expansion of chest by noting the movement of each thumb away from the midline. (Y/N)
5. Repeats the maneuver once again. (Y/N)

REFERENCES

- A textbook of practical physiology, CL. Ghai, 8th edition.
- Essentials of medical physiology, K. Sembulingam and P. Sembulingam, 6th edition.



Teacher's Signature

17. PERFORM THE PROCEDURE INSPECTION OF HEART SOUND

STUDENT OBJECTIVES

After completion of the practical, students should be able to:

Listen to the heart sounds.

Find the apex beat

INTRODUCTION

Heart sounds are the sounds produced by the mechanical activities of the heart during each cardiac cycle.

Heart sounds are produced by:

1. The flow of blood through cardiac chambers
2. Contraction of cardiac muscle
3. Closure of valves of the heart.

Four heart sounds are produced during each cardiac cycle: First, Second, Third and Fourth heart sound

- First and second heart sounds are called classical heart sounds and are heard by using the stethoscope.
- Third heart sound is a mild sound and it is not heard by using stethoscope in normal conditions. But it can be heard by using a microphone.
- Fourth heart sound is an inaudible sound. It becomes audible in pathological conditions only. This sound is studied only by graphic registration, i.e. the phonocardiogram.

Features	1 st heart sound	2 nd heart sound	3 rd heart sound	4 th heart sound
Occurs during	Isometric contraction & part of ejection period	Protodiastole and part of isometric relaxation	Rapid filling phase	Atrial systole
Characteristics	Long, soft and low pitched Resembles the word 'LUBB'	Short, sharp and high pitched Resembles the word 'DUP'	Low pitched	Inaudible sound
Cause	Closure of atrioventricular valves	Closure of semilunar valves	Rushing of blood into ventricle	Contraction of atrial musculature
Relation with ECG	Coincides with peak of 'R' wave	Precedes or appears 0.09 second after peak of 'T' wave	Between 'T' wave and 'P' wave	Between 'P' wave and 'Q' wave

AUSCULTATION AREAS

1. **Mitral area (Bicuspid area):** Mitral area is in the left 5th intercostal space, about 10 cm away from the midline (midclavicular line).
2. **Tricuspid area:** Tricuspid area is on the xiphoid process. Sound produced by the closure of tricuspid valve (first heart sound) is transmitted well into this area.
3. **Pulmonary area:** Pulmonary area is on the left 2nd intercostal space, close to sternum. Sound produced by the closure of pulmonary valve (second heart sound) is heard well on this area.
4. **Aortic area:** Aortic area is over the right 2nd intercostal space, close to the sternum. On this area, the sound produced by the closure of aortic valve (second heart sound) is heard well.

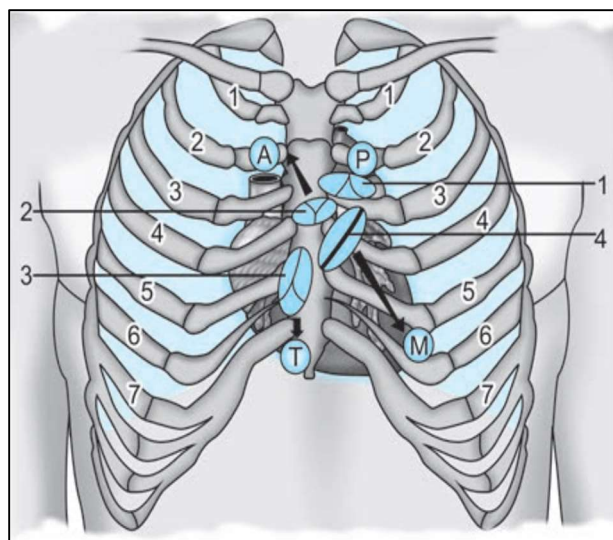


Diagram showing the projection of heart valves and the auscultatory areas. (1) Pulmonary artery valve, P— Pulmonary area, (2) Aortic valve, A—Aortic area, (3) Tricuspid valve, T—Tricuspid area, (4) Mitral valve, M—Mitral area. The ribs are numbered from 1 to 7 on each side. (source- A Textbook of Practical Physiology, CL Ghai)

EXERCISE

Q.1 Describe the normal heart sounds and their physiological correlates.

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Q.2 Explain types of murmur?

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Q.3 What is the clinical significance of heart sound?

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Q.4 Describe the difference between S1 and S2.

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Q.5 What are the common causes of murmur?

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Q.6 How to find the apex beat?

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OSPE

Aim:

Examine the heart sounds of a patient and identify any abnormalities.

Checklist:

1. Did the candidate introduce themselves and explain the procedure? (Y/N)
2. Was the patient positioned correctly? (Y/N)
3. Were heart sounds accurately identified? (Y/N)
4. Were findings correctly interpreted? (Y/N)
5. Was communication effective? (Y/N)



Teacher's Signature

18 ASSESSMENT OF AGNI

LEARNING OBJECTIVES

After completion of the practical, students should be able to:

- Perform an Agni assessment using the standardized proforma.
- Analyze the results to determine the patient's Agni type.
- Provide basic dietary recommendations based on the Agni type.

PRINCIPLE

Agni governs digestion, absorption and metabolism. Its assessment evaluates the strength and balance of Agni essential for breaking down food, assimilating nutrients and eliminating waste. Observations of appetite, digestion, stool characteristics and metabolic symptoms help identify whether Agni is balanced (Sama) or imbalanced (Manda, Tikshna, or Vishama).

PURPOSE

The primary aim is to evaluate digestive health, detect irregularities and prevent diseases caused by impaired digestion or toxin buildup (Ama). It also guides personalized dietary and therapeutic recommendations to enhance vitality, immunity and overall well-being.

PROCEDURE

1. Introduce the assessment purpose to the individual.
2. Record basic information: name, age, gender, occupation, body temperature, weight, blood pressure, pulse rate and address.
3. Ask the individual each of the 10 questions from the proforma.

4. Mark the appropriate response under the four Agni types: Mandagni, Vishamagni, Samagni or Teekshnagni.
5. Calculate the score and percentage for each Agni type using the provided formula.
6. Determine the predominant Agni type based on the highest percentage.
7. Provide dietary recommendations based on the result.

INSTRUCTIONS & INTERPRETATION

In total, there are 10 points for the assessment which questions are given below in the proforma. Students have to ask each of the 10 questions to the person whose Agni assessment he/she has to undergo. For each question, there is a statement given under each of the four states of Agni satisfying their respective characteristic. Students must put a tick mark '✓', in the box given before each option satisfied by the person undergoing assessment. After asking all the 10 questions, count the ticked boxes in each state of Agni and calculate the percentage by the formula given below at the end of the proforma.

For example, if a person gets the maximum number of boxes ticked under *Vishamagni*, say 7 out of 10, then place the scored value in the numerator box and calculate the percentage. The Agni type with the highest score will be considered the individual's Agni. In case of a tie, where the same percentage is obtained in more than one Agni type due to improper history taking, *Vaidya* should resolve it by re-evaluating the responses and obtaining a more detailed and accurate history.

PRECAUTION

Ensure the patient is comfortable and understands the questions. Avoid leading questions that might bias the responses. Maintain a non-judgmental and professional demeanor.

EXERCISE

1. Q.1 What are the four types of Agni?

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Q.2 Why is it important to assess Agni?

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Q.3 What role does lifestyle play in determining Agni?

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Q.4 How can Agni assessment aid in personalized healthcare?

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Q.5 Explain the significance of satiety in Agni assessment.

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Q.6 What are the limitations of the Agni assessment method?

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Q.7 Can the concept of Agni be related to enzymatic or metabolic activities in modern medicine?

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Q.8 How do stress and environmental factors affect Agni, according to both Ayurveda and contemporary physiology?

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Method for Agni Parikshan (Assessment of Agni)**Station Setup:****1. Station 1: History Taking**

- Objective: Gather relevant information about the patient's digestion, appetite, bowel habits and tolerance to food types.
- Task: Students interact with a simulated patient (actor) or a case sheet to note details of the digestive pattern.

2. Station 2: Clinical Analysis

- Objective: Identify the type of *Agni* based on gathered data.
- Task: Using the proforma provided, analyze the answers and calculate the percentage score for each *Agni* type.

3. Station 3: Interpretation and Conclusion

- Objective: Determine the predominant *Agni* type (Sama, Vishama, Tikshna, Manda).
- Task: Students write a conclusion based on their findings and provide general dietary or lifestyle recommendations.

4. Station 4: Feedback and Education

- Objective: Educate the patient (simulated) on their *Agni* type and appropriate adjustments in diet or habits.
- Task: Students explain their diagnosis and give suggestions in simple language, ensuring patient understanding.

Assessment Parameters: 15 marks

- **Station 1:** Clarity and relevance of questions asked (4 marks).
- **Station 2:** Accuracy in proforma completion and calculations (4 marks).
- **Station 3:** Correct identification of *Agni* type (4 marks).
- **Station 4:** Communication skills and appropriateness of advice (3 marks).

Evaluation Checklist:

- Did the student ask all relevant questions for *Agni* assessment? (Y/N)
- Was the data interpreted correctly using the proforma? (Y/N)
- Did the student accurately identify the *Agni* type? (Y/N)
- Was the explanation clear, patient-friendly and culturally sensitive? (Y/N)

CASE BASED LEARNING. (SCENARIO)

Case 1: A 30-year-old individual reports inconsistent digestion. Some days, they experience indigestion after small meals, while on other days, they can eat heavy meals without discomfort. They often skip meals due to irregular appetite.

Question: Identify the state of *Agni* and suggest dietary modifications.

Case 2: A 45-year-old reports feeling hungry within two hours of eating and can easily digest oily and spicy foods. They rarely experience bloating or heaviness after meals.

Question: Based on their symptoms, determine their *Agni* type and discuss potential risks of imbalanced *Tikshna Agni*.

Case 3: A 60-year-old feels sluggish and often complains of bloating and heaviness, even after consuming small meals. It takes 5–6 hours to feel hungry again.

Question: Evaluate their *Agni* type and recommend therapeutic interventions to balance it.

PROBLEM-BASED LEARNING

Case Study

A 35-year-old office worker presents with complaints of irregular digestion, occasional bloating and inconsistent appetite. They experience sluggish digestion after heavy meals and feel hungry at irregular intervals.

Problem

What is the individual's Agni type and how can their digestion be improved?

Guiding Questions

1. What are the primary symptoms presented?
2. How do the symptoms align with the Agni types?

Guided Discussion Questions

Why is the digestion pattern important in determining Agni?

How can lifestyle disturbances impact digestion?

Expected Outcomes

1. Identifying the Agni type (likely Vishamagni).
2. Providing targeted dietary and lifestyle recommendations.
3. Understanding the relationship between Agni and overall health.

Discussion and Resolution

- Analyze the individual's responses.
- Compare symptoms with the characteristics of each Agni type.
- Develop a personalized plan to improve digestion through diet and lifestyle.

Assessment

Evaluate the student's ability to:

1. Conduct the Agni assessment accurately.
2. Interpret the results to identify the Agni type.

DOPS

DOPS for Assessment of Agni

Step 1: Introduction and Explanation

Greet the individual and introduce yourself.

Explain the purpose of the Agni assessment.

Obtain consent to proceed.

Step 2: Data Collection

Record the following demographic and physical parameters:

Name: Age:

Gender: Male/Female Occupation:

Body Temperature:Weight:kg

Blood Pressure:Pulse Rate:/min

Address:

Step 3: Administering the Questionnaire

1. Explain the questionnaire:

Inform the individual that you will ask 10 questions about their digestion and dietary habits.

2. Ask each question one at a time:

Read the question aloud and provide the four options. Allow the individual to select the option that best describes their situation.

Tick (✓) the box corresponding to their answer.

Step 4: Scoring

After completing the questionnaire, count the tick marks under each Agni type:

Mandagni (Slow digestion)

Vishamagni (Irregular digestion)

Samagni (Balanced digestion)

Teekshnagni (Intense digestion)

Step 5: Calculate Percentages

Use the formula to calculate the percentage for each Agni type:

Percentage of Agni Type=(Score Obtained/10)×100

Record the results:

Mandagni: %

Vishamagni: %

Samagni: %

Teekshnagni: %

Step 6: Interpret Results

Identify the Agni type with the highest percentage as the individual's dominant Agni.

Step 7: Conclusion

Fill in the conclusion:

"The Agni of the given person is"

Step 8: Provide Recommendations

Based on the identified Agni type, give general dietary and lifestyle suggestions:

- **Mandagni:** Light, warm and easily digestible foods. Avoid cold and heavy foods.
- **Vishamagni:** Regular meal timings. Avoid irregular eating and excessive spices.
- **Samagni:** Maintain a balanced diet and healthy lifestyle to preserve balance.
- **Teekshnagni:** Cooling foods and drinks. Avoid overly spicy or hot foods.

Step 9: Feedback and Documentation

Discuss findings with the individual.

Record observations in their health records.

Example Solution - Case Details

Name: Mr. Rahul Shine

Age: 32

Gender: Male

Occupation: Teacher

Body Temperature: 98.6°F

Weight: 70 kg

Blood Pressure: 120/80 mmHg

Pulse Rate: 72/min

Assessment Results

Mandagni: 2/10 → $2/10 \times 100 = 20\%$

Vishamagni: 3/10 → $3/10 \times 100 = 30\%$

Samagni: 4/10 → $4/10 \times 100 = 40\%$

Teekshnagni: 1/10 → $1/10 \times 100 = 10\%$

Conclusion- Agni of the given person is **Samagni**.

Recommendations - Maintain balanced meals with proper portions. Avoid skipping meals or overeating. Include fresh fruits and vegetables, whole grains and moderate spices.

SHLOKAS (VERSES) FOR RECITATION

अग्नि

- **व्याख्या:** तेजोगुण उत्कर्षात् क्षपितं सोमगुण त्यक्तद्रवत्वं स्वभावं लब्धाग्नि शब्द तत् पाचकमइत्युच्यते। - अ.सं.20/5
- **स्थान:** षष्ठी पित्तधरा नाम या कला परिकीर्तिता। पक्वामाशमध्यस्था ग्रहणी या प्रकीर्तिता। - सु.उ.40
- **लाभ:** आयुवर्णो बलं स्वस्थमुत्साहोपचयो प्रभा। ओजस्तेजोअग्रयः प्राणाश्चोक्ता देहान्निहेतुकाः॥ शान्तेऽअग्रो प्रियते, युक्ते चिरम् जीवत्यनामयः। - च.चि.15/3
- प्राणापानसमानैस्तु सर्वतः पवनैस्त्रिभिः ध्मायते पाल्यते चापि स्वे स्वे स्थाने व्यवस्थिते। - सु.सु.21
- **आहार-परीक्षा**
- उदगार शुद्धी उत्साही वेगोत्सर्गो यथोचितः। लघुता क्षुत्पिपासा जीर्णाहारस्य लक्षणम्॥ सं.शा.11/45
- **जाठराग्नि के प्रकार - 4**
- (कुल अग्निप्रकार $1+5+7=13$)
- समाग्नि** - सम समाग्रेरशिता मात्रा सम्याग्विपच्यते
- विषमाग्नि** - विषमाग्रेस्तु देहीनः कदाचित् पच्यते सम्यक् कदाचित् न विपच्यते
- तीक्ष्णाग्नि** - मात्राअतिमात्राअप्यशितां सुखं यस्य विपच्यते तीक्ष्णाग्निरिती
- मन्दाग्नि** - तं विद्यात् स्वल्पापि नैव मन्दाग्रे। - मा.नि.
- **भुताग्नि (5)**
- भौमआप्यअग्रेयवायव्याः पंचउष्णानः सनाभसाः। - संग्रह
- **धात्वाग्नि (7)**
- स्वस्थानस्थस्य कायाग्रेरंशा धातुषु संस्थिताः। तेषांसादतिदीप्तिभ्यां धातुववृद्धिक्षयोद्भवः॥ - अ.ह.सु.11/34

कोष्ठ

- क्रूरकोष्ठ - स्निह्यति क्रूरकोष्ठस्तु सप्तरात्रेण मानवः।
- मृदुकोष्ठ - मृदुकोष्ठस्त्रिरात्रेण स्निह्यत्यच्छोपसेवया। बहुपित्तो मृदुः स्यात् सः दुग्धोनापि विरिच्यते। - सुश्रुत
- **जीर्णआहार लक्षणे**
- आहारशक्तिश्चेति आहारशक्तिर्भ्यवहरणशक्त्या जरणशक्त्या च परीक्ष्या बलायुषी ह्याहारायतके ॥120॥
- सम्यक् स्निग्ध लक्षणे
- वातनुलोम्यं दीप्तोऽग्निर्वचः स्निग्धसंहतम्। मार्दवं स्निग्धाता च अंगे स्निग्धनामुपजायते॥ च.सु.27/13

18.1 ASSESSMENT OF AGNI

Objective : To assess the Agni Parikshana of an individual

Name: :

Age : **Gender:** M/F **Occupation:**

Body temperature : **Weight:**kg

Blood pressure : **Pulse rate:**/min

Address :

Sr. No.	Characteristic feature	Question	Mandagni मंदग्नि	Vishamagni विषमग्नि	Samagni समग्नि	Teekshnagni तिक्ष्णग्नि
1	Pattern of digestion	Which statement describes your pattern of digestion?	☐ Unable To digest even small quantity of food.	☐ Ability to digest food is variable.	☐ Able to digest food only if consumed in appropriate quantity.	☐ Able to digest almost all food items.
2	Appetite	What time do you need to feel like eating again, after having your meal?	☐ Only after about 6 hours of having meal.	☐ Not consistent, keeps varying.	☐ 3 to 5 hours after having meal.	☐ Before 3 hours of having meal.
3	Frequency	How frequently do you have your meals in a day	☐ 1 to 2 meals per day.	☐ 1 to 5 meals per day.	☐ 3 to 4 meals per day.	☐ more than 5 meals per day.
4	Time duration of hunger tolerance	When you are hungry, how much time you can tolerate it?	☐ Tolerate longer (> 2 hours).	☐ Sometimes upto 2 hours and sometimes <1 hour).	☐ Upto 1 to 2 hours.	☐ Very difficult to tolerate (not even half hour).

5	Quantity of food	What quantity of mid-day meals do you quite often have in a day?	☐ Small meals.	☐ Sometimes small meal or large meal.	☐ Neither too small nor too large (regular meal).	☐ Large meals.
6	Quality of food	How do you describe your capacity to digest heavy food stuffs (oily food/ fast food/ junk food)?	☐ Takes longer than average time (more than 5 hours).	☐ Keeps varying.	☐ Normal average time (3 to 4 hours).	☐ Less than 3 hours.
7	Bowel habits	How do you describe your bowel habits?	☐ Slow and sluggish bowel movement.	☐ Irregular bowel habit (constipation/ diarrhoea).	☐ Regular bowel habit with well formed stool.	☐ Often soft or loose stool.
8	Lifestyle disturbances	What effects do you observe in your digestion due to disturbances in your lifestyle?	☐ Digestion gets disturbed easily.	☐ Gets disturbed sometimes only due to appreciable disturbances.	☐ Digestion is not affected much due to disturbances.	☐ Digestion is not affected at all.
9	Symptoms of digestion	How do you feel after complete digestion of Meals? Do you feel any heaviness in abdomen and body?	☐ Frequently.	☐ Occasionally.	☐ Feeling of lightness in abdomen and body.	☐ Feeling of lightness in Abdomen and body quite early.
10	Satiety	How do you express your feelings that you develop after looking at your favourite food	☐ Do not feel like eating it.	☐ Sometimes.	☐ Feel like eating favourite food item but	☐ Feel like eating my favourite food item and can

		item, just after having a regular meal?			nothing other than it.	have any other food item as well.
	Total score		/10	/10	/10	/10

Percentage of Agni (%) = $\frac{\text{Score obtained for individual type of Agni}}{\text{Total score allotted to that type of Agni}} \times 100$

$$1) \text{ Percentage of Mandagni (\%)} = \frac{\boxed{}}{10} \times 100 = \boxed{} \%$$

$$2) \text{ Percentage of Vishamagni (\%)} = \frac{\boxed{}}{10} \times 100 = \boxed{} \%$$

$$3) \text{ Percentage of Samagni (\%)} = \frac{\boxed{}}{10} \times 100 = \boxed{} \%$$

$$4) \text{ Percentage of Teekshnagni (\%)} = \frac{\boxed{}}{10} \times 100 = \boxed{} \%$$

The type of Agni with the maximum % is

Conclusion: The type of Agni in the given individual is.....



Teacher's Signature

18.2 ASSESSMENT OF AGNI

Objective : To assess the Agni Parikshana of an individual

Name: :

Age : **Gender:** M/F **Occupation:**

Body temperature : **Weight:**kg

Blood pressure : **Pulse rate:**/min

Address :

Sr. No.	Characteristic feature	Question	Mandagni मंदग्नि	Vishamagni विषमग्नि	Samagni समग्नि	Teekshnagni तिक्ष्णग्नि
1	Pattern of digestion	Which statement describes your pattern of digestion?	☐ Unable To digest even small quantity of food.	☐ Ability to digest food is variable.	☐ Able to digest food only if consumed in appropriate quantity.	☐ Able to digest almost all food items.
2	Appetite	What time do you need to feel like eating again, after having your meal?	☐ Only after about 6 hours of having meal.	☐ Not consistent, keeps varying.	☐ 3 to 5 hours after having meal.	☐ Before 3 hours of having meal.
3	Frequency	How frequently do you have your meals in a day	☐ 1 to 2 meals per day.	☐ 1 to 5 meals per day.	☐ 3 to 4 meals per day.	☐ more than 5 meals per day.
4	Time duration of hunger tolerance	When you are hungry, how much time you can tolerate it?	☐ Tolerate longer (> 2 hours).	☐ Sometimes upto 2 hours and sometimes <1 hour).	☐ Upto 1 to 2 hours.	☐ Very difficult to tolerate (not even half hour).

5	Quantity of food	What quantity of mid-day meals do you quite often have in a day?	☐ Small meals.	☐ Sometimes small meal or large meal.	☐ Neither too small nor too large (regular meal).	☐ Large meals.
6	Quality of food	How do you describe your capacity to digest heavy food stuffs (oily food/ fast food/ junk food)?	☐ Takes longer than average time (more than 5 hours).	☐ Keeps varying.	☐ Normal average time (3 to 4 hours).	☐ Less than 3 hours.
7	Bowel habits	How do you describe your bowel habits?	☐ Slow and sluggish bowel movement.	☐ Irregular bowel habit (constipation/ diarrhoea).	☐ Regular bowel habit with well formed stool.	☐ Often soft or loose stool.
8	Lifestyle disturbances	What effects do you observe in your digestion due to disturbances in your lifestyle?	☐ Digestion gets disturbed easily.	☐ Gets disturbed sometimes only due to appreciable disturbances.	☐ Digestion is not affected much due to disturbances.	☐ Digestion is not affected at all.
9	Symptoms of digestion	How do you feel after complete digestion of Meals? Do you feel any heaviness in abdomen and body?	☐ Frequently.	☐ Occasionally.	☐ Feeling of lightness in abdomen and body.	☐ Feeling of lightness in Abdomen and body quite early.
10	Satiety	How do you express your feelings that you develop after looking at your favourite food	☐ Do not feel like eating it.	☐ Sometimes.	☐ Feel like eating favourite food item but	☐ Feel like eating my favourite food item and can

		item, just after having a regular meal?			nothing other than it.	have any other food item as well.
	Total score		/10	/10	/10	/10

Percentage of Agni (%) = $\frac{\text{Score obtained for individual type of Agni}}{\text{Total score allotted to that type of Agni}} \times 100$

$$1) \text{ Percentage of Mandagni (\%)} = \frac{\boxed{}}{10} \times 100 = \boxed{} \%$$

$$2) \text{ Percentage of Vishamagni (\%)} = \frac{\boxed{}}{10} \times 100 = \boxed{} \%$$

$$3) \text{ Percentage of Samagni (\%)} = \frac{\boxed{}}{10} \times 100 = \boxed{} \%$$

$$4) \text{ Percentage of Teekshnagni (\%)} = \frac{\boxed{}}{10} \times 100 = \boxed{} \%$$

The type of Agni with the maximum % is

Conclusion: The type of Agni in the given individual is.....



Teacher's Signature

18.3 ASSESSMENT OF AGNI

Objective : To assess the Agni Parikshana of an individual

Name: :

Age : **Gender:** M/F **Occupation:**

Body temperature : **Weight:**kg

Blood pressure : **Pulse rate:**/min

Address :

Sr. No.	Characteristic feature	Question	Mandagni मंदग्नि	Vishamagni विषमग्नि	Samagni समग्नि	Teekshnagni तिक्ष्णग्नि
1	Pattern of digestion	Which statement describes your pattern of digestion?	☐ Unable To digest even small quantity of food.	☐ Ability to digest food is variable.	☐ Able to digest food only if consumed in appropriate quantity.	☐ Able to digest almost all food items.
2	Appetite	What time do you need to feel like eating again, after having your meal?	☐ Only after about 6 hours of having meal.	☐ Not consistent, keeps varying.	☐ 3 to 5 hours after having meal.	☐ Before 3 hours of having meal.
3	Frequency	How frequently do you have your meals in a day	☐ 1 to 2 meals per day.	☐ 1 to 5 meals per day.	☐ 3 to 4 meals per day.	☐ more than 5 meals per day.
4	Time duration of hunger tolerance	When you are hungry, how much time you can tolerate it?	☐ Tolerate longer (> 2 hours).	☐ Sometimes upto 2 hours and sometimes <1 hour).	☐ Upto 1 to 2 hours.	☐ Very difficult to tolerate (not even half hour).

5	Quantity of food	What quantity of mid-day meals do you quite often have in a day?	☐ Small meals.	☐ Sometimes small meal or large meal.	☐ Neither too small nor too large (regular meal).	☐ Large meals.
6	Quality of food	How do you describe your capacity to digest heavy food stuffs (oily food/ fast food/ junk food)?	☐ Takes longer than average time (more than 5 hours).	☐ Keeps varying.	☐ Normal average time (3 to 4 hours).	☐ Less than 3 hours.
7	Bowel habits	How do you describe your bowel habits?	☐ Slow and sluggish bowel movement.	☐ Irregular bowel habit (constipation/ diarrhoea).	☐ Regular bowel habit with well formed stool.	☐ Often soft or loose stool.
8	Lifestyle disturbances	What effects do you observe in your digestion due to disturbances in your lifestyle?	☐ Digestion gets disturbed easily.	☐ Gets disturbed sometimes only due to appreciable disturbances.	☐ Digestion is not affected much due to disturbances.	☐ Digestion is not affected at all.
9	Symptoms of digestion	How do you feel after complete digestion of Meals? Do you feel any heaviness in abdomen and body?	☐ Frequently.	☐ Occasionally.	☐ Feeling of lightness in abdomen and body.	☐ Feeling of lightness in Abdomen and body quite early.
10	Satiety	How do you express your feelings that you develop after looking at your favourite food	☐ Do not feel like eating it.	☐ Sometimes.	☐ Feel like eating favourite food item but	☐ Feel like eating my favourite food item and can

		item, just after having a regular meal?			nothing other than it.	have any other food item as well.
	Total score		/10	/10	/10	/10

Percentage of Agni (%) = $\frac{\text{Score obtained for individual type of Agni}}{\text{Total score allotted to that type of Agni}} \times 100$

$$1) \text{ Percentage of Mandagni (\%)} = \frac{\boxed{}}{10} \times 100 = \boxed{} \%$$

$$2) \text{ Percentage of Vishamagni (\%)} = \frac{\boxed{}}{10} \times 100 = \boxed{} \%$$

$$3) \text{ Percentage of Samagni (\%)} = \frac{\boxed{}}{10} \times 100 = \boxed{} \%$$

$$4) \text{ Percentage of Teekshnagni (\%)} = \frac{\boxed{}}{10} \times 100 = \boxed{} \%$$

The type of Agni with the maximum % is

Conclusion: The type of Agni in the given individual is.....



Teacher's Signature

19 ASSESSMENT OF KOSHHA

कोष्ठ परिक्षण

LEARNING OBJECTIVES

After completion of the practical, students should be able to:

- Assess Koshtha using a standardized proforma.
- Analyze the findings to identify the type of Koshtha (Mridu, Madhyam or Krura).
- Recommend dietary and therapeutic measures based on Koshtha type.

PRINCIPLE

The assessment of Koshtha helps in determining bowel habits, digestion and evacuation characteristics, which guide dietary and therapeutic interventions in Ayurveda. It reflects the prakriti (constitution) and state of health.

PURPOSE OF PRACTICAL

To understand and assess an individual's bowel habits and digestive tendencies to classify their Koshtha type and provide suitable lifestyle and dietary advice.

APPARATUS AND MATERIALS

- Proforma for Koshtha assessment.
- Pen and clipboard.
- Weighing scale

PROCEDURE

1. Obtain informed consent from the individual undergoing the assessment.
2. Use the proforma with the eight characteristic features of Koshtha assessment.
3. Ask the individual the relevant questions and record responses by ticking the appropriate box under Mridu, Madhyam or Krura Koshtha.
4. Calculate the total score for each Koshtha type.
5. Determine the individual's Koshtha type based on the maximum score.

INSTRUCTIONS FOR ASSESSMENT:

There are 8 characteristic features given in the proforma. Based on that feature, it is described that feature according to the type of koshtha under the respective headings. Students have to put a tick mark '✓', in the box given before each option satisfied by the person undergoing assessment. After asking all the 8 points against 3 types of koshtha, just calculate the total ticked boxes and sum it up out of 8. The maximum scoring koshtha type will be the Koshtha of that individual.

For example, if a person gets the maximum number of boxes ticked under *Krura Koshtha*, say 6 out of 8, then place the scored value in the numerator box and calculate the percentage. The *Koshtha* type with the highest score will be considered the individual's *Koshtha*. In case of a tie, where the same percentage is obtained in more than one *Koshtha* type due to improper history taking, *Vaidya* should resolve it by re-evaluating the responses and obtaining a more detailed and accurate history.

OBSERVATIONS AND RESULTS

Tally the scores under each Koshtha category. Calculate the total and determine the maximum scoring Koshtha type.
Example: If Mridu Koshtha scores 6/8, Madhyam 2/8 and Krura 1/8, the person has Mridu Koshtha.

PRECAUTIONS

Ensure the individual understands the questions clearly.
Avoid leading questions to ensure unbiased responses.
Maintain confidentiality of the individual's information.

19.1 ASSESSMENT OF KOSHITA

कोष्ठ परिक्षण

Objective : To assess the Agni Parikshana of an individual

Name: :

Age : **Gender:** M/F **Occupation:**

Body temperature : **Weight:**kg

Blood pressure : **Pulse rate:**/min

Address :

Sr. No	Characteristic feature	मृदु कोष्ठ (Mridu Koshtha)	मध्यम कोष्ठ (Madhyam koshtha)	क्रूर कोष्ठ (Krura koshtha)
1	Nature of Bowel Movement	☐ Passes stool daily once/ twice or even mor	☐ passes stool daily once or twice.	☐ Irregularly passing stool.
2	Consistency	☐ Semi-formed stools/ loose stools.	☐ Well-formed stools.	☐ Hard and dry stools.
3	Time required for defecation	☐ Requires less time (less than 2-5 minutes)	☐ Requires longer than mridu Koshtha (5-10 minutes)	☐ Requires long time (more than 10 minutes)
4	Amount of stool evacuation	More (approximately >200 gm/day)	Moderate (approximately 150-200 gm/day)	Less (approximately <150 gm/day)
5	Evacuation	☐ Easy defecation	☐ Defecation with less pressure	☐ Defecation with more pressure.

6	Associated Symptoms	☐ Satisfactory and quite easy bowel clearance with increase frequency/ loose stools	☐ Satisfactory bowel clearance	☐ Bowel clearance is unsatisfactory
7	Experience regarding purgatives/laxatives if any.	☐ Even mild laxative can lead to passing of Stools	☐ Require medium dose of laxative or may not require.	☐ Strong purgatives required to pass stools..
8	Experience regarding diarrhoea and constipation if any.	☐ Experience diarrhoea (watery stools) due to hot drinks (milk, tea etc.)	☐ Doesn't often experience diarrhoea & constipation	☐ Rarely experience diarrhoea and frequently experience constipation
	Total score	/8	/8	/8

The type of Koshta with the maximum score is

Conclusion: The type of Koshta in the given individual is.....



Teacher's Signature

19.2 ASSESSMENT OF KOSHITA

कोष्ठ परिक्षण

Objective : To assess the Agni Parikshana of an individual

Name: :

Age : **Gender:** M/F **Occupation:**

Body temperature : **Weight:**kg

Blood pressure : **Pulse rate:**/min

Address :

Sr. No	Characteristic feature	मृदु कोष्ठ (Mridu Koshtha)	मध्यम कोष्ठ (Madhyam koshtha)	क्रूर कोष्ठ (Krura koshtha)
1	Nature of Bowel Movement	☐ Passes stool daily once/ twice or even mor	☐ passes stool daily once or twice.	☐ Irregularly passing stool.
2	Consistency	☐ Semi-formed stools/ loose stools.	☐ Well-formed stools.	☐ Hard and dry stools.
3	Time required for defecation	☐ Requires less time (less than 2-5 minutes)	☐ Requires longer than mridu Koshtha (5-10 minutes)	☐ Requires long time (more than 10 minutes)
4	Amount of stool evacuation	More (approximately >200 gm/day)	Moderate (approximately 150-200 gm/day)	Less (approximately <150 gm/day)
5	Evacuation	☐ Easy defecation	☐ Defecation with less pressure	☐ Defecation with more pressure.

6	Associated Symptoms	☐ Satisfactory and quite easy bowel clearance with increase frequency/ loose stools	☐ Satisfactory bowel clearance	☐ Bowel clearance is unsatisfactory
7	Experience regarding purgatives/laxatives if any.	☐ Even mild laxative can lead to passing of Stools	☐ Require medium dose of laxative or may not require.	☐ Strong purgatives required to pass stools..
8	Experience regarding diarrhoea and constipation if any.	☐ Experience diarrhoea (watery stools) due to hot drinks (milk, tea etc.)	☐ Doesn't often experience diarrhoea & constipation	☐ Rarely experience diarrhoea and frequently experience constipation
	Total score	/8	/8	/8

The type of Koshta with the maximum score is

Conclusion: The type of Koshta in the given individual is.....



Teacher's Signature

19.3 ASSESSMENT OF KOSHITA

कोष्ठ परिक्षण

Objective : To assess the Agni Parikshana of an individual

Name: :

Age : **Gender:** M/F **Occupation:**

Body temperature : **Weight:**kg

Blood pressure : **Pulse rate:**/min

Address :

Sr. No	Characteristic feature	मृदु कोष्ठ (Mridu Koshtha)	मध्यम कोष्ठ (Madhyam koshtha)	क्रूर कोष्ठ (Krura koshtha)
1	Nature of Bowel Movement	☐ Passes stool daily once/ twice or even mor	☐ passes stool daily once or twice.	☐ Irregularly passing stool.
2	Consistency	☐ Semi-formed stools/ loose stools.	☐ Well-formed stools.	☐ Hard and dry stools.
3	Time required for defecation	☐ Requires less time (less than 2-5 minutes)	☐ Requires longer than mridu Koshtha (5-10 minutes)	☐ Requires long time (more than 10 minutes)
4	Amount of stool evacuation	More (approximately >200 gm/day)	Moderate (approximately 150-200 gm/day)	Less (approximately <150 gm/day)
5	Evacuation	☐ Easy defecation	☐ Defecation with less pressure	☐ Defecation with more pressure.

6	Associated Symptoms	☐ Satisfactory and quite easy bowel clearance with increase frequency/ loose stools	☐ Satisfactory bowel clearance	☐ Bowel clearance is unsatisfactory
7	Experience regarding purgatives/laxatives if any.	☐ Even mild laxative can lead to passing of Stools	☐ Require medium dose of laxative or may not require.	☐ Strong purgatives required to pass stools..
8	Experience regarding diarrhoea and constipation if any.	☐ Experience diarrhoea (watery stools) due to hot drinks (milk, tea etc.)	☐ Doesn't often experience diarrhoea & constipation	☐ Rarely experience diarrhoea and frequently experience constipation
	Total score	/8	/8	/8

The type of Koshta with the maximum score is

Conclusion: The type of Koshta in the given individual is.....



Teacher's Signature

EXERCISE

Q.1 What are the characteristic features of Mridu Koshtha?

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Q.2 How does Madhyam Koshtha differ from Krura Koshtha?

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Q.3 Why is it essential to assess Koshtha in Ayurveda?

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Q.4 What dietary interventions can be recommended for Krura Koshtha?

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Q.5 Which factors influence changes in an individual's Koshtha over time?

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Q.6 How do purgatives affect the three types of Koshtha?

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Q.7 What are the implications of untreated Krura Koshtha?

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Q.8 Discuss therapeutic recommendations for each Koshtha type.

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Key for OSPE

OSPE Method for Koshta Parikshan

Evaluation Checklist:

- Did the student ask all relevant questions about bowel habits, stool consistency and related symptoms? (Y/N)
- Was the proforma completed correctly and were calculations accurate? (Y/N)
- Was the correct *Koshta* type identified and linked to *dosha* dominance? (Y/N)
- Did the student communicate findings effectively and suggest appropriate interventions? (Y/N)

SCENARIO-BASED KOSHHA ASSESSMENT

1. **Case 1:** A patient defecates easily once daily, with stools that are soft and semi-formed. They experience no discomfort or need for laxatives.
 - Question: Classify their *Koshta* (Mridu, Madhyama or Krura) and explain its relationship with *Kapha dosha*.
2. **Case 2:** A person experiences irregular bowel movements, sometimes with constipation and other times diarrhea. Their stools are hard during constipation episodes.
 - Question: Determine their *Koshta* type and outline management strategies for improving bowel regularity.
3. **Case 3:** A 50-year-old individual reports requiring strong laxatives to evacuate bowels, which are hard and dry. They pass stools irregularly, every 2–3 days, with significant effort.
 - Question: Analyze their *Koshta* type and propose treatment approaches, including dietary and lifestyle changes.

OSPE

Station: Koshta Assessment

Station Setup:

Proforma, pen, clipboard.

Dummy patient or case scenario.

Tasks:

1. Ask the patient the provided eight questions and tick the corresponding responses.
2. Calculate scores for each *Koshtha* type.
3. Conclude the dominant *Koshtha* type.

Checklist:

- Proper communication with the individual: ____/2
- Accuracy in recording responses: ____/4
- Correct calculation and interpretation: ____/4

PROBLEM-BASED LEARNING (PBL)

Case Study:

A 35-year-old male presents with irregular bowel habits, often constipated with hard stools, requiring significant effort during defecation. He occasionally takes strong purgatives to pass stools.

Problem:

Identify his *Koshtha* type and recommend suitable therapeutic measures.

Guiding Questions:

Which *Koshtha* characteristics match the patient's symptoms?

What dietary changes can benefit Krura *Koshtha*?

Guided Discussion Questions:

1. How do lifestyle habits affect *Koshtha*?
2. What are the implications of prolonged Krura *Koshtha*?

Expected Outcomes:

Patient identified with Krura *Koshtha*.

Recommended fiber-rich diet, hydration and Ayurvedic medicines like Triphala.

Discussion and Resolution:

Students analyze the symptoms, identify Krura *Koshtha* and propose suitable interventions.

Assessment:

Evaluate through participation in discussion and practical identification.

20. URINE EXAMINATION

LEARNING OBJECTIVES

- After completion of practical, students should be able to:
- Follow guidelines for the proper collection of urine
- Assess urine colour, clarity and odour.
- Identify common abnormalities in urine appearance
- Conduct Chemical Testing of Urine.

PURPOSE OF URINE SAMPLE COLLECTION

Urine analysis is a simple yet essential test used to assess the key features of urine. It is commonly done as part of routine health exams, during hospital admissions, or before surgeries. It's also helpful for confirming abnormal results from a quick urine test. The test includes:

Colour, Cloudiness, and Concentration: Observing the physical appearance of urine helps detect hydration levels and possible health issues.

Chemical Composition: Using a test strip to analyze the chemical makeup of urine.

Microscopic Examination: Looking at urine under a microscope to find any bacteria, cells, or cell fragments, which could indicate infections or other conditions.

COLLECTION OF URINE SAMPLE

To ensure the accuracy of urine tests and avoid contamination, it is essential to collect the sample properly:

Preparation: Clean the genital area with water (without soap) to remove any bacteria or debris. This prevents contamination of the urine sample.

Midstream Sample: For a clean sample, collect only the middle portion of the urine stream. Start urinating, pause after a few seconds, and then collect the urine in a cup. This is known as the "midstream urine" method, which helps ensure the sample is free from external contaminants.

24-Hour Urine Collection

A 24-hour urine collection test involves gathering all urine over a 24-hour period to assess kidney function or to measure specific substances:

First Void: The very first urine after waking up is not used. The time of this first void should be noted.

Collection Process: For the next 24 hours, collect all urine in a provided container. Once the 24 hours are over, the final sample is taken (this includes the last urine void).

Storage: The collected urine must be kept in a refrigerator to prevent bacterial growth. The container will usually have a preservative to help with this.

Laboratory Testing: Once the collection period is complete, the sample is sent to a lab for analysis. Urine specimen collection is used for a variety of purposes, including:

- **Screening for disease**

Urine tests can help detect diseases of the urinary system, as well as metabolic diseases like diabetes or liver disease.

- **Diagnosing medical conditions**

Urine tests can help diagnose the cause of symptoms like abdominal pain, back pain, frequent or painful urination or blood in urine.

- **Monitoring medical conditions**

Urine tests can help monitor the condition and treatment of a medical condition, such as kidney disease or a urinary tract infection

I. PHYSICAL EXAMINATION OF URINE

The physical urine examination consists of assessment of urine colour, its smell, foam and turbidity; and also measurements of its pH, relative specific gravity and osmolality. For functional examinations, estimation of volume of urine per defined period of time is necessary.

1. Test for Volume

Name: Volume Test

Principle: Measuring the volume of urine helps assess kidney function and fluid balance in the body.

Procedure: Collect all urine voided during a specified period (e.g., 24 hours). Measure the total volume using a graduated container.

Observation:

Normal volume: 800 to 2,000 mL per day (for an average adult).

Oliguria: Urine output less than 400 mL per day.

Polyuria: Urine output greater than 2,500–3,000 mL per day.

Physiological Condition: Normal urine output is around 1-2 liters per day in healthy adults.

Pathological Condition: Oliguria (kidney disease, dehydration), polyuria (diabetes, diuretics).

Result:

Normal volume (800-2,000 mL/day): Healthy kidney function.

Oliguria or polyuria: Possible indication of renal or endocrine disorders.

2. Test for Colour

Name: Colour Test

Principle: The colour of urine can provide valuable information about the body's hydration status, metabolism, and the presence of certain substances.

Procedure: Observe the urine sample directly against a white background. Note the colour of the urine.

Observation:

Pale yellow: Normal, well-hydrated urine.

Dark yellow/amber: Dehydration or high concentration of urine.

Red or brown: Presence of blood or hemoglobin (hematuria/hemoglobinuria).

Orange: Presence of certain medications (e.g., rifampicin) or bile pigments.

Green/blue: Presence of certain drugs, pseudomonas infection.

Physiological Condition: Normal colour ranges from pale yellow to amber, depending on hydration.

Pathological Condition: Hematuria (red/brown), jaundice (yellow/orange), infections (green/blue).

Result:

Normal colour (yellow): Healthy hydration status.

Abnormal colours (red/brown/green): Possible pathological conditions such as bleeding or infection.

3. Test for Transparency (Turbidity)

Name: Transparency Test

Principle: Normal urine is clear or slightly cloudy. Cloudy urine may indicate the presence of various substances such as cells, bacteria, or crystals.

Procedure: Observe the urine sample in a clear container against a white background. Compare the clarity of the urine to a reference sample (e.g., distilled water or normal urine).

Observation:

Clear: Normal urine.

Cloudy or turbid: Presence of mucus, pus, bacteria, or crystals.

Physiological Condition: Mild cloudiness may occur after standing, due to crystal formation.

Pathological Condition: Infections (bacteria, pus), kidney stones, crystals (uric acid, calcium phosphate).

Result:

Clear urine: Normal result.

Cloudy urine: May indicate infection, crystals, or other pathological conditions.

4. Test for Odour

Name: Odour Test

Principle: Urine odour can provide insight into metabolic and pathological conditions, as certain substances or bacteria can change the urine's odour.

Procedure: Smell the urine sample after opening the container. Note the presence of any unusual odour.

Observation:

Normal odour: Mild, characteristic urine smell.

Sweet/fruity odour: Presence of ketones (diabetic ketoacidosis).

Foul odour: Infection (e.g., due to bacteria like *E. coli*).

Ammoniacal odour: Bacterial infection (e.g., UTI).

Mousy odour: Phenylketonuria (PKU).

Physiological Condition: Normal urine has a mild characteristic odour.

Pathological Condition: Diabetic ketoacidosis (fruity), UTI (foul), PKU (mousy).

Result:

Normal odour: No abnormal findings.

Abnormal odour: Potential indication of metabolic or infectious conditions.

5. Test for Specific Gravity

Name: Specific Gravity Test

Principle: Specific gravity indicates the concentration of urine, reflecting the kidney's ability to concentrate or dilute urine.

Procedure: Fill a graduated cylinder with 5-10 mL of urine. Insert a urinometer into the sample and allow it to float freely. Record the specific gravity at the point where the meniscus touches the scale.

Observation:

1.005 to 1.030: Normal range for healthy individuals.

>1.030: Dehydration or concentrated urine.

<1.005: Diluted urine, possible renal failure.

Physiological Condition: Normal specific gravity varies depending on hydration, but usually falls between 1.005 and 1.030.

Pathological Condition: Dehydration (higher values), renal disease (lower values).

Result:

1.005–1.030: Normal specific gravity.

<1.005 or >1.030: Possible pathological conditions.

6. Test for pH of Urine

Name: pH Test of Urine

Principle: The pH of urine indicates its acidity or alkalinity, which reflects the body's metabolic and dietary conditions. It is used to assess kidney function and overall health.

Procedure: Collect a fresh urine sample in a clean container. Dip a pH strip into the urine sample for a few seconds. Compare the colour change on the strip to the pH chart provided with the strips. Record the pH value that corresponds to the closest colour match.

Observation:

Normal range: 4.5 to 8.0.

Acidic urine (pH < 5.0) may indicate dehydration, high-protein diet, or metabolic acidosis.

Alkaline urine (pH > 7.0) can be a result of a vegetarian diet, urinary tract infection (UTI) caused by urease-producing bacteria, or renal tubular acidosis.

Physiological Condition: Normal urine pH typically ranges from 5.5 to 7.0, depending on diet and hydration. Acidic urine is more common in individuals consuming high-protein foods or fasting. Alkaline urine is usually seen in those consuming a vegetarian diet or when the body is in an alkaline state due to certain medical conditions.

Pathological Condition: Acidic pH (< 5.0): Can be associated with metabolic acidosis, dehydration, diarrhea, or starvation. Alkaline pH (> 7.0): Can be linked to urinary tract infections (UTIs), renal tubular acidosis, or the use of certain medications like diuretics.

Result: Normal pH (4.5–8.0): Typically indicates a healthy metabolic state. Abnormal pH (< 5.0 or > 7.0): May suggest an underlying health condition requiring further investigation.

CHEMICAL EXAMINATION OF URINE

1. Test for Glucose (Benedict's Test)

Name: Benedict's Test

Principle: Reducing sugars, such as glucose, reduce copper sulfate in Benedict's reagent to form a coloured precipitate of cuprous oxide.

Procedure: Add 2 mL of Benedict's reagent to a test tube. Add 5 drops of urine to the test tube. Heat the mixture gently in a water bath or over a flame for 2–3 minutes.

Observation:

Blue: No glucose (no reaction).

Green: Low concentration of glucose.

Yellow: Moderate concentration of glucose.

Orange or Brick Red: High concentration of glucose.

Physiological Condition: Mild glucosuria may occur after heavy meals or during pregnancy.

Pathological Condition: Diabetes mellitus, renal glucosuria, stress.

Result:

No colour change: Negative result for glucose.

Green to red precipitate: Positive result indicating glucose.

2. Test for Protein (Heat Coagulation Test)

Name: Heat Coagulation Test

Principle: Proteins in urine coagulate when heated in an acidic or neutral environment.

Procedure: Take 5 mL of urine in a test tube. Heat the test tube gently over a flame for 2–3 minutes. Add 2–3 drops of acetic acid to confirm coagulation.

Observation: No change: No protein present. White precipitate or coagulum: Presence of protein.

Physiological Condition: Post-exercise proteinuria, fever.

Pathological Condition: Nephrotic syndrome, glomerulonephritis, kidney disease.

Result:

No precipitate: Negative result for protein.

White precipitate: Positive result for protein.

3. Test for Ketone Bodies (Rothera's Test)

Name: Rothera's Test

Principle: Ketone bodies react with sodium nitroprusside in an alkaline medium to form a purple colour.

Procedure: Add a pinch of sodium nitroprusside to 5 mL of urine in a test tube. Add a few drops of ammonia solution by tube side. Observe for colour change at the interface of the urine and reagent.

Observation:

No colour change: No ketones present.

Purple or violet ring: Presence of ketone bodies.

Physiological Condition: Prolonged fasting, low carbohydrate diet.

Pathological Condition: Diabetes ketoacidosis, starvation, alcoholism.

Result:

No ring: Negative result for ketones.

Purple ring: Positive result indicating ketones.

4. Test for Bile Salts (Hay's Test)

Name: Hay's Test

Principle: Bile salts decrease the surface tension of urine, causing sulfur powder to sink.

Procedure: Sprinkle sulfur powder on the surface of urine in a test tube. Observe the behavior of the sulfur powder.

Observation: Sulfur powder sinks: Presence of bile salts. Sulfur powder floats or remains suspended: Absence of bile salts.

Physiological Condition: None.

Pathological Condition: Obstructive jaundice, hepatocellular jaundice, cholestasis.

Result:

Sulfur powder sinks: Positive result for bile salts.

Sulfur powder floats: Negative result for bile salts.

5. Test for Bile Pigments (Gmelin's Test)

Name: Gmelin's Test

Principle: Bile pigments oxidize in the presence of nitric acid, forming coloured rings.

Procedure: Add 1 mL of concentrated nitric acid to a test tube. Slowly add 2–3 mL of urine, forming two layers. Observe the colour at the junction of the two layers.

Observation:

Yellow or green ring: Presence of bile pigments.

No colour change: Absence of bile pigments.

Physiological Condition: None.

Pathological Condition: Jaundice (hepatocellular, obstructive).

Result:

Colour ring (green, blue, or yellow): Positive for bile pigments.

No colour ring: Negative result.

6. Test for Blood (Benzidine Test)

Name: Benzidine Test

Principle: Hemoglobin in urine reacts with hydrogen peroxide and benzidine to produce a blue or green colour.

Procedure: Take 2 mL of urine in a test tube. Add a few drops of hydrogen peroxide and a pinch of benzidine powder. Observe any colour change.

Observation:

No colour change: No blood present.

Blue or green colour: Presence of blood or hemoglobin.

Physiological Condition: None.

Pathological Condition: Hematuria (trauma, stones, infections). Hemoglobinuria (hemolysis, burns).

Result:

No colour: Negative result for blood.

Blue/Green: Positive result for blood.

7. Test for Urobilinogen (Ehrlich's Test)

Name: Ehrlich's Test

Principle: Urobilinogen reacts with Ehrlich's reagent (p-dimethylaminobenzaldehyde) in an acidic medium to form a pink or red colour.

Procedure: Add 2 mL of Ehrlich's reagent to 5 mL of urine. Shake and observe the colour change.

Observation:

No colour change: No urobilinogen present.

Pink or red colour: Presence of urobilinogen.

Physiological Condition: Normal small amounts of urobilinogen.

Pathological Condition: Liver diseases, hemolytic anemia.

Result:

No colour change: Negative result for urobilinogen.

Pink/red colour: Positive result indicating urobilinogen.

8. Test for Ammonia (Litmus Paper Test)

Name: Litmus Paper Test

Principle: Ammonia in urine changes the colour of red litmus paper to blue due to its basic nature.

Procedure: Take a fresh urine sample in a test tube. Hold a strip of red litmus paper above the open test tube without touching the urine. Observe for any colour change in the litmus paper.

Observation:

Red litmus turns blue: Ammonia is present.

No colour change: Ammonia is absent.

Physiological Condition: Normal small amounts of ammonia.

Pathological Condition: Infections, dehydration, metabolic alkalosis.

Result: Blue litmus: Positive for ammonia.

No change: Negative for ammonia.

MICROSCOPIC EXAMINATION:

Sample preparation

The most suitable sample for microscopic examination of urinary sediment is the middle flow of the first or second morning urine. The second morning urine is recommended because in the first morning sample the cellular elements can be damaged or decomposed. For examination of urinary sediment, it is necessary to process a fresh urine sample within one hour from the collection.

Red Blood Cells (RBCs):

Appearance: Small, circular, biconcave discs that may appear colourless or slightly yellow under the microscope.

Significance: The presence of RBCs is indicative of hematuria, which can result from various causes like trauma, infection, kidney disease (such as glomerulonephritis), or stones.

White Blood Cells (WBCs):

Appearance: Larger than RBCs, with a prominent nucleus that may appear slightly granular.

Significance: An increase in WBCs (pyuria) is indicative of inflammation and infection, commonly found in urinary tract infections (UTIs), pyelonephritis, or glomerulonephritis.

Neutrophils are most common, but the presence of eosinophils can indicate drug-induced interstitial nephritis.

Epithelial Cells:

Appearance: Larger, with a visible nucleus; may be squamous, transitional, or renal tubular cells.

Significance: Squamous epithelial cells: Found in urine samples that may have been contaminated by skin or genital tract; generally not clinically significant.

Renal tubular epithelial cells: May indicate acute tubular necrosis (ATN) or kidney injury.

Transitional epithelial cells: May indicate bladder infection or bladder cancer.

Casts:

Appearance: Cylindrical structures that are often transparent and may contain cellular components.

Types and Significance:

Hyaline Casts: Colourless and transparent; often seen in healthy individuals but can be increased in dehydration, fever, or renal diseases.

Granular Casts: Have a granular appearance and indicate acute tubular necrosis or other kidney injuries.

Waxy Casts: Broad and shiny; suggest chronic renal disease or end-stage kidney failure.

RBC Casts: Contain RBCs and are indicative of glomerulonephritis or severe renal injury.

WBC Casts: Contain WBCs and are found in cases of pyelonephritis or kidney infection.

Epithelial Casts: Contain renal tubular epithelial cells, indicating acute tubular necrosis.

Crystals:

Appearance: Vary in shape, size, and colour depending on the substance.

Common Types and Significance:

Calcium Oxalate Crystals: Envelope-shaped or octahedral; common in urine and may indicate kidney stones or hypercalciuria.

Uric Acid Crystals: Diamond or needle-like; can be associated with gout or uric acid kidney stones.

Cystine Crystals: Hexagonal; can indicate cystinuria, a hereditary condition that can lead to kidney stones.

Triple Phosphate Crystals: Found in alkaline urine; associated with UTIs and may form struvite stones.

Bacteria:

Appearance: Small rod-shaped or cocci forms.

Significance: The presence of bacteria suggests a urinary tract infection (UTI). Gram stain and culture are needed to identify the organism.

Yeast Cells:

Appearance: Oval-shaped with a budding appearance.

Significance: The presence of yeast suggests a fungal infection, commonly *Candida* species, especially in immunocompromised individuals.

Mucus Threads:

Appearance: Long, thin, wavy structures, often colourless.

Significance: Mucus is often present in small amounts and is typically not of clinical concern. Excessive mucus may indicate inflammation.

OSPE

OSPE Example 1: Physical Examination of Urine – pH Test

Station Title: pH Examination of Urine

Required Materials: Urine sample (provided), pH paper strips, Small container for urine, Colour chart for pH paper

Instructions for Students:

1. Preparation:

Take 2–3 mL of the urine sample in a clean container.

2. Testing pH:

Dip a pH paper strip into the urine sample.

Remove the strip and compare the colour change on the pH paper with the colour chart provided.

Record the pH value (normal pH range: 4.5–8.0).

3. Interpretation:

If the pH is acidic (below 5.0), discuss possible conditions such as dehydration, high-protein diet, or respiratory acidosis.

If the pH is alkaline (above 7.5), consider conditions like urinary tract infections (UTIs), renal tubular acidosis, or a vegetarian diet.

Explain the physiological and pathological implications of the observed pH value.

Time Allocation: 5 minutes

OSPE

OSPE Example 2: Chemical Examination of Urine – Test for Glucose (Benedict's Test)

Station Title: Test for Glucose in Urine

Required Materials: Urine sample (provided), Benedict's reagent, Test tubes, Water bath or Bunsen burner, Test tube rack

Instructions for Students:

1. Procedure:

Add 2 mL of Benedict's reagent into a clean test tube.

Add 5–6 drops of urine sample to the test tube.

Place the test tube in a water bath or gently heat it using a Bunsen burner.

Observe the colour change during heating.

2. Interpretation:

No colour change (blue): No glucose present (negative result).

Green/Yellow/Orange/Brick red: The presence of glucose is indicated by the intensity of the colour change. Green to brick red indicates increasing amounts of glucose.

3. Clinical Relevance:

A positive result (yellow, orange, or red) suggests the presence of glucose in the urine, which is commonly seen in diabetes mellitus.

Discuss potential causes of glycosuria, such as uncontrolled diabetes, stress, or renal glucosuria.

Time Allocation: 5 minutes

URINE REPORT

Objective : To assess the Urine Examination of an individual

Name: :

Age: **Gender:** M/F **Occupation:**

Date of Collection:..... **Time of Collection:**

Based on the given sample, provide the following urine report

I. PHYSICAL EXAMINATION OF URINE

1. Test for Volume :.....
2. Test for Colour :.....
3. Test for Transparency :.....
4. Test for Odour :.....
5. Test for Specific Gravity :.....
6. pH of Urine :.....

II. CHEMICAL EXAMINATION OF URINE

1. Glucose :.....
2. Protein :.....
3. Ketone Bodies :.....
4. Bile Salts :.....
5. Bile Pigments :.....
6. Blood in urine :.....
7. Urobilinogen :.....
8. Ammonia :.....

III. MICROSCOPIC EXAMINATION

- Red Blood Cells (RBCs) :.....
- White Blood Cells (WBCs): :.....
- Epithelial Cells: :.....
- Casts :.....
- Crystals: :.....
- Mucus Threads: :.....

EXERCISE

Q.1 What is the normal pH range of urine, and what factors can influence it?

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Q.2 What are some examples of abnormal odours?

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Q.3 Explain the principle of the Benedict's test

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Q.4 How does the Benzidine test detect the presence of blood in urine, and what conditions can cause hematuria or hemoglobinuria?

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Q.5 What are the normal and abnormal results for bile salts in urine, and what does it indicate?

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Q.6 How would you differentiate between squamous epithelial cells and renal tubular cells in urine sediment?

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Q.7 What types of casts can be found in urine, and what do they indicate about kidney function?

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Teacher's Signature

21.(A) DEMONSTRATE ESR

STUDENT OBJECTIVES

After completion of practical, students should be able to:

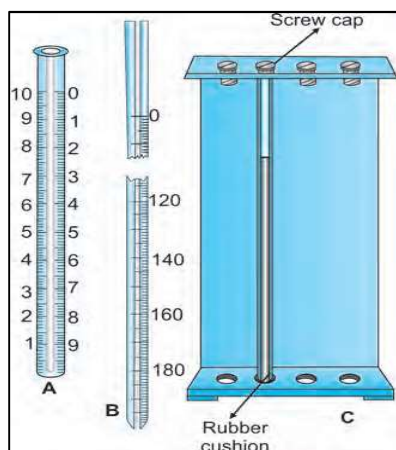
1. Perform practical for estimation of ESR.
2. Describe the clinical importance of doing ESR.

PRINCIPLE

In the circulating blood the red cells remain uniformly suspended in the plasma. However, when a sample of blood, to which an anticoagulant has been added, is allowed to stand in a narrow vertical tube, the red cells (specific gravity = 1.095) being heavier (denser) than the colloid plasma (specific gravity = 1.032), settle or sediment gradually towards the bottom of the tube. The rate, in mm, at which the red cells sediment, called ESR, is recorded at the end of one hour.

WINTROBE'S METHOD

APPARATUS AND MATERIALS



Determination of ESR.
(A) Wintrobe tube; (B) Westergren tube; (C) Westergren stand with the Westergren tube in position.
(source- A Textbook of Practical Physiology, CL Ghai)

APPARATUS AND MATERIALS

Disposable syringe and needle.
Sterile swabs moist with 70% alcohol.
Container (discarded penicillin bottle, etc.) with double oxalate mixture.
Pasteur pipette with a long thin nozzle.
Wintrobe tube and stand.

Normal values

Males : 2–8 mm 1st hour

Females : 4–10 mm 1st hour.

PROCEDURES

1. Draw 2.0 ml of venous blood and transfer it to a container of anticoagulant. Mix the contents gently but well by inverting the vial a few times or by swirling it. Do not shake, as it will cause frothing.
2. Using the Pasteur pipette, fill the Wintrobe tube from below upwards.
3. Transfer the tube to its stand and adjust the screws so that it will remain vertical. Leave the tube undisturbed in this position for one hour, at the end of which read the mm of clear plasma above the red cells.

RESULTS

The given blood sample ESR in 1st hour (Wintrobe) is mm

WESTERGREN'S METHOD

APPARATUS AND MATERIALS

2 ml disposable syringe with needle.
Sterile cotton/gauze swabs moist with alcohol.
Container (discarded penicillin bottle).
Sterile solution of 3.8 percent sodium citrate as the anticoagulant.
Westergren pipette (tube) and stand.

Normal values

Males : 3–9 mm 1st hour

Females : 5–12 mm 1st hour

PROCEDURES

1. Draw 2.0 ml of venous blood and transfer it into a vial containing 0.5 ml of 3.8% sodium citrate solution. This will give a blood: citrate ratio of 4:1.
2. Mix the contents by inverting or swirling the vial. Do not shake, as it will cause frothing.
3. Fill the Westergren's pipette with blood-citrate mixture by sucking, after placing the tip of your finger over the top of the pipette to control the

flow of blood into and out of it or with a rubber bulb. Bring the blood column to exact zero mark.

4. Keeping your finger (or the rubber bulb) over the pipette, transfer it to the Westergren stand by firmly pressing its lower end into the rubber cushion. Now slip the upper end of the pipette under the screw cap. Confirm that there is no leakage of blood and that the pipette will remain vertical.
5. Leave the pipette undisturbed for one hour at the end of which read the mm of clear plasma above the red cells.

RESULTS

The given blood sample ESR in 1st hour (Westergren) is mm

PRECAUTIONS

Proper anticoagulant should be used for each method. The blood should be collected in the fasting state. The test should preferably be done within 2–3 hours of collecting the blood sample and at a room temperature of 20–35°C. Clotted or hemolysed blood must be discarded. The Wintrobe tube or Westergren pipette should not be disturbed from the vertical position for the duration of the test.

EXERCISE

Q.1 Why do the red cells settle down in a sample of anticoagulated blood?

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Q.2 What are the disadvantages of Wintrobe method?

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Q.3 Why is ESR reading taken after one hour?

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Q.4 What are the factors on which the rate of sedimentation of red cells depends?

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Q.5 Name the physiological and pathological variations in ESR.

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Q.6 What is the clinical significance of ESR?

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PROBLEM-BASED LEARNING (PBL)

Case Study: A 45-year-old female patient, Mrs. Kumar, presents to the outpatient department with: Fever, Fatigue, Joint pain, Swelling in hands and feet

Laboratory Results:

Hemoglobin (Hb): 12 g/dL

Total WBC count: 15,000/ μ L

Count: Neutrophils 70%

Lymphocytes 20%, ESR: 40 mm/hour

Problem:

Interpret Mrs. Kumar's ESR result and correlate it with her clinical presentation.

Questions:

1. What is the principle of ESR?
2. What are the normal values for ESR?
3. What are the possible causes of elevated ESR?
4. How does ESR relate to Mrs. Kumar's symptoms?

Guided Discussion Questions:

1. What are the differences between ESR and CRP?
2. How does ESR relate to inflammation?
3. What are the limitations of ESR?
4. How would you manage Mrs. Kumar's condition based on her ESR result?

Expected Outcomes:

Accurate interpretation of ESR result.

Understanding of ESR principles.

Ability to correlate ESR with clinical presentation.

Critical thinking and problem-solving skills.

Assessment:

1. Written report on ESR interpretation.
2. Group presentation on ESR principles.
3. Class discussion participation.

Tips for Facilitators:

1. Encourage critical thinking.
2. Foster group discussion.
3. Provide feedback.
4. Monitor progress.

OSPE-1

Aim: To load the Wintrobe tube with the supplied anticoagulated blood.

Checklist:

1. Selects the tube and sees that it is clean and dry. (Y/N)
2. Mixes the blood sample and draws blood into a Pasteur pipette or a dropper with a long nozzle. (Y/N)
3. Fills the tube with blood, starting from its bottom and withdrawing the pipette till blood column reaches the zero mark. (Y/N)
4. Checks that there are no bubbles; if there are, she removes them with a filter paper strip. (Y/N)
5. Places the tube vertically in the Wintrobe stand and notes the time. (Y/N)

OSPE-2

Aim: To load the Westergren tube with the supplied anticoagulated blood.

Checklist:

1. Selects the Westergren tube and checks that it is dry and clean. (Y/N)
2. Mixes the blood sample. (Y/N)
3. Sucks blood into the tube and takes it to the zero mark, taking care to avoid any bubbles. (Y/N)
4. Presses the lower end of the tube into the rubber cushion of the stand and the upper end under the screw cap. (Y/N)
5. Confirms that it is vertical and notes the time. (Y/N)



Teacher's Signature

21.(B) DEMONSTRATE PCV

STUDENT OBJECTIVES

After completion of this experiment, you should be able to:

1. Define hematocrit and its clinical importance.
2. Identify Wintrobe tube and explain how to fill it with blood.

APPARATUS AND MATERIALS

1. Sterile swabs,
2. Alcohol,
3. Syringe and needle.
4. Container with anticoagulant.
5. Wintrobe tube (hematocrit tube).
6. Pasteur pipette.
7. Centrifuge machine.

PROCEDURES

1. Draw 5 ml of venous blood and transfer it to a container of anticoagulant. Rotate the bulb between your palms.
2. Fill the pasteur pipette with blood and take its nozzle to the bottom of the Wintrobe tube.
3. Expel the blood gently by pressing the rubber teat and fill the tube from below upwards while withdrawing the pipette but always keeping its tip below the level of blood.
(Do not try to fill the tube from its top as blood will not flow down to its bottom because of air present in the tube.)
4. Bring the blood column exactly to the mark 10 (or the mark 0 on the other side of the scale) at the top. There should not be any bubbles at the top of blood.
5. Close the mouth of the tube with its rubber cap and centrifuge it at 3000 rpm for 30 minutes. Balance this tube with another tube filled with water or another sample of blood placed in the opposite tube holder.
6. At the end of 30 minutes, take the reading of upper level of packed red cells on the side of the scale where zero is at the bottom. Replace the

tube in the machine and centrifuge it again for 15 minutes.

7. Read the packed cell height again; it should be the same as before. If the height is reduced, centrifuge it again for 5 minutes.
8. To be reliable, at least 3 successive readings, at intervals of 5 minutes, should be the same.

OBSERVATIONS AND RESULTS

The percentage of the volume of blood occupied by the red cells constitutes hematocrit or packed cell volume, i.e., the percentage of whole blood that is red cells Hematocrit (Hct) =

$$\frac{\text{Height of packed red cells mm}}{\text{Height of packed RBCs and plasma (ie height of blood column)}} \times 100$$

Hematocrit Reading 1:%

Hematocrit Reading 2:%

Hematocrit Reading 3:%

The given blood sample Hematocrit is%

PRECAUTIONS

1. Use the recommended amount of EDTA as the anticoagulant. There should be no clotting or hemolysis of the blood.
2. The test should be done within 6–8 hours of collection of the sample.
3. The Wintrobe tube should be filled from below upwards so that no air bubble is trapped in the tube.

EXERCISE

Q.1 Give normal values of PCV.

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Q.2 What is the effect of a high hematocrit?

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Q.3 Name the conditions where the PCV is increased and those where it is decreased?

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Q.4 What is the importance of determining hematocrit?

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Q.5 What is the relationship between PCV and blood viscosity, and how can this influence circulation?

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Q.6 Given a PCV value of 25%, how would you interpret this result in a clinical setting?

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Q.7 How would you calculate the percentage of hematocrit (PCV) from a given blood sample, and what does this percentage represent?

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OSPE

Aim: Determine the PCV of a blood sample.

Checklist:

1. Did the candidate label the tube correctly? (Y/N)
2. Was the tube filled to the marked line? (Y/N)
3. Was the centrifugation time and speed correct? (Y/N)
4. Was the PCV measurement accurate? (Y/N)
5. Was the result recorded correctly? (Y/N)

REFERENCES

- A textbook of practical physiology, CL. Ghai, 8th edition.
- Essentials of medical physiology, K. Sembulingam and P. Sembulingam, 6th edition.



Teacher's Signature

22. OBSERVED THE PROCEDURE OF ECG

STUDENT OBJECTIVES

After completing this practical, students should be able to:

1. Define ECG and describe the equipment needed for recording ECG.
2. Handle the equipment properly to record ECG.
3. Indicate the clinical uses of ECG.
4. Interpret changes in various waves and intervals.

APPARATUS

1. The ECG, Machine.
2. Electrodes.
3. ECG Paper.
4. Electrocardiographic Leads

ELECTRODE POSITIONS

Bipolar limb leads

- Lead I-** Lead I is obtained by connecting right arm and left arm. Right arm is connected to the negative terminal of the instrument and the left arm is connected to the positive terminal.
- Lead II-** Lead II is obtained by connecting right arm and left leg. Right arm is connected to the negative terminal of the instrument and the left leg is connected to the positive terminal.
- Lead III-** Lead III is obtained by connecting left arm and left leg. Left arm is connected to the negative terminal of the instrument and the left leg is connected to the positive terminal.

Unipolar limb leads

- aVR lead-** Active electrode is from right arm.
- aVL lead-** Active electrode is from left arm.
- aVF lead-** Active electrode is from left leg (foot).

Unipolar chest leads (unipolar precordial leads)

V1- in the 4th intercostal space (ICS), just to the right of the sternum.

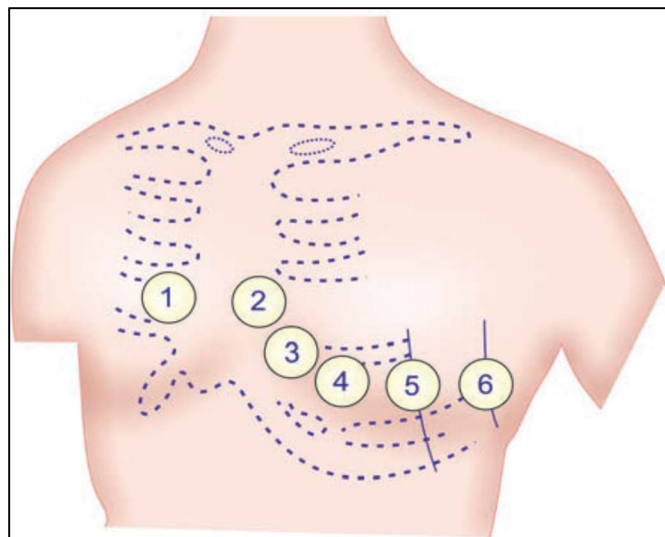
V2 - in the 4th ICS, just to the left of the sternum.

V3- halfway between V2 and V4.

V4 - at midclavicular line in the 5th intercostal space.

V5- in anterior axillary line at the same level as V4.

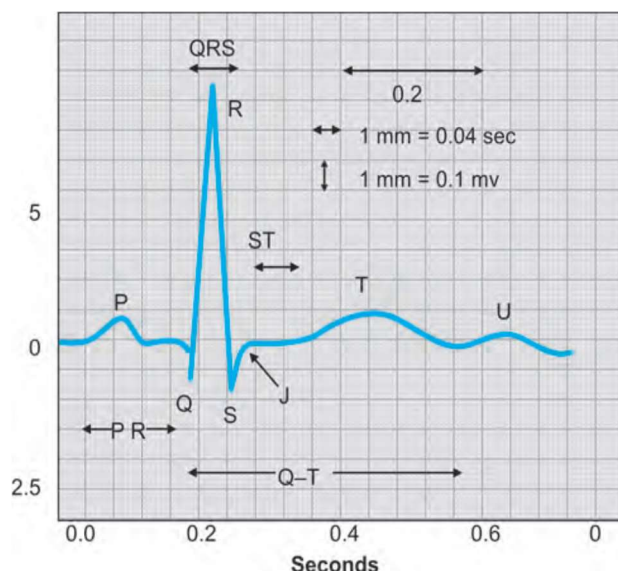
V6 - in mid-axillary line in the 5th intercostal space.



Source- K Sembulingam, Essentials of Medical Physiology, Sixth Edition

PROCEDURES

1. Ask the subject to lie down supine on the bed and be comfortable and relaxed.
2. Rub small amounts of electrolyte jelly on the fronts of wrists and just above the ankles.
3. Apply the limb electrodes firmly on these points and fix them in place with rubber straps.
4. Fix the lead wires, identified with the letters- RA, LA, LL and RL electrodes. Connect the connector cable to the machine.
5. Apply the electrode jelly on the chest positions for V1 to V6.
6. Switch on the machine.



SYSTEMATIC ANALYSIS OF ECG

- 1. Heart rate-** It can be determined by any of the following 2 methods:

- By dividing 1500 by the number of small squares between 2 successive R waves. (1500 small squares represent 1 minute) For example, number of small squares between 2 R waves = 21
Heart rate = $1500/21 = 70$ per min.
- By dividing 60 by the RR interval in seconds. For example, number of small squares between 2 R waves = 20
RR interval = $20 \times 0.4 = 0.80$
Heart rate = $60/0.80 = 75$ per min.

2. Rhythm- In normal sinus rhythm, P waves precede each QRS complex. Atrial, junctional and ventricular arrhythmias are detected by in-hospital and ambulatory ECG monitoring.

3. Mean cardiac vector- Evaluation of the frontal plane QRS axis provides the information.

4. Morphology of various waves, intervals and segments are carefully studied.

OBSERVATIONS AND RESULTS

After completion of demonstration of ECG, Interpret the heart rate from a given ECG strip using the big block method.

Heart rate=/min

EXERCISE

Q.1 What is ECG and what is its basis?

[illegible]

Q.2 What do the lines on ECG paper indicate?

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Q.3 What are the different waves recorded in normal ECG and what do they represent?

[illegible]

Q.4 Which wave represents atrial repolarization?

[illegible]

Q.5 What is complete heart block?

[illegible]

[illegible]

This image shows a full page of primary-ruled paper. It contains ten identical horizontal rows. Each row is defined by three lines: a solid top line, a dashed midline, and a solid bottom line. The entire page is white, and there are no margins or additional markings.

[illegible][illegible][illegible]

Aim: Demonstrate and interpret a 12-lead ECG on a patient.

1. Did the candidate introduce themselves and explain the procedure? (Y/N)
2. Were ECG electrodes applied correctly? (Y/N)
3. Was the ECG recording technique correct? (Y/N)
4. Was the ECG interpretation accurate? (Y/N)
5. Were findings documented clearly? (Y/N)

- A textbook of practical physiology, CL. Ghai, 8th edition.
- Essentials of medical physiology, K. Sembulingam and P. Sembulingam, 6th edition.



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23. PERFORM THE PROCEDURE OF EXAMINING THE CRANIAL NERVES AND REFLEXES

STUDENT OBJECTIVES

After completing this clinical examination, you should be able to:

1. Explain the various cranial nerves, their functions and the subjective and objective features of their lesions.

PROCEDURES OF EXAMINATION OF CRANIAL NERVE

The 1st or Olfactory Nerve (Sensory)

1. Ask the subject to close his eyes and occlude one of his nostrils. Then have him smell and distinguish the odours of each of the test substances, one by one, in each nostril, separately.
2. Ask him to occlude one nostril and have him smell the oil of cloves until the odour can no longer be detected. Immediately after this, ask the subject to try to distinguish, with the adapted nostril, between turpentine and alcohol.

Result- Normal/Abnormal.

The 2nd or Optic Nerve (Sensory)

A. VISUAL ACUITY (VA)

Test for distant vision with Snellen's chart

1. The subject is seated at a distance of 6 meters (20 feet) from a well-lighted chart and is asked to read the letters down the chart as far as she can read. Each eye is tested separately.
2. The VA is recorded according to the formula $VA = d/D$ where d is the distance at which the letters are read and D is the distance at which the letter should be read by a person with normal vision.
3. Thus if only the top letter can be read, the VA is 6/60. If the subject can read only the first 4 lines, then the VA is 6/18 and so on. A normal person should be able to read at least the 7th line, i.e. have a VA of 6/6 (20/20 in terms of feet, commonly pronounced "twenty-twenty" vision.)
4. If the VA is less than 6/60, the subject is moved towards the chart until she can read the top letter.

The VA is 2/60 if she can read it at 2 meters. The VA less than 1/60 is recorded as "counting fingers" (CF; if she correctly counts fingers when held in front of her face; "hand movements" (HM); "perception of light" (PL) or as "no perception of light" (no PL).

V/A of subject-

Test for near vision with Jaeger's chart

1. The subject is asked to read Jaeger's chart held at the ordinary reading distance of 15 inches. This chart is made up of reading material of various sizes with the smallest size at the bottom.
2. The subject reads the material down the chart and VA is recorded as the smallest type that can be read comfortably.
3. The result used to be expressed as J1, J2, J3, J4 etc. J1-- (the smallest size) indicates normal vision.
4. These days a modification of the original Jaeger system is used and VA is expressed in terms of printers' point system—N36, N18, N8, N6 and N5 instead of J1, J2 etc; N5 being the smallest type instead of J1.

V/A of subject-

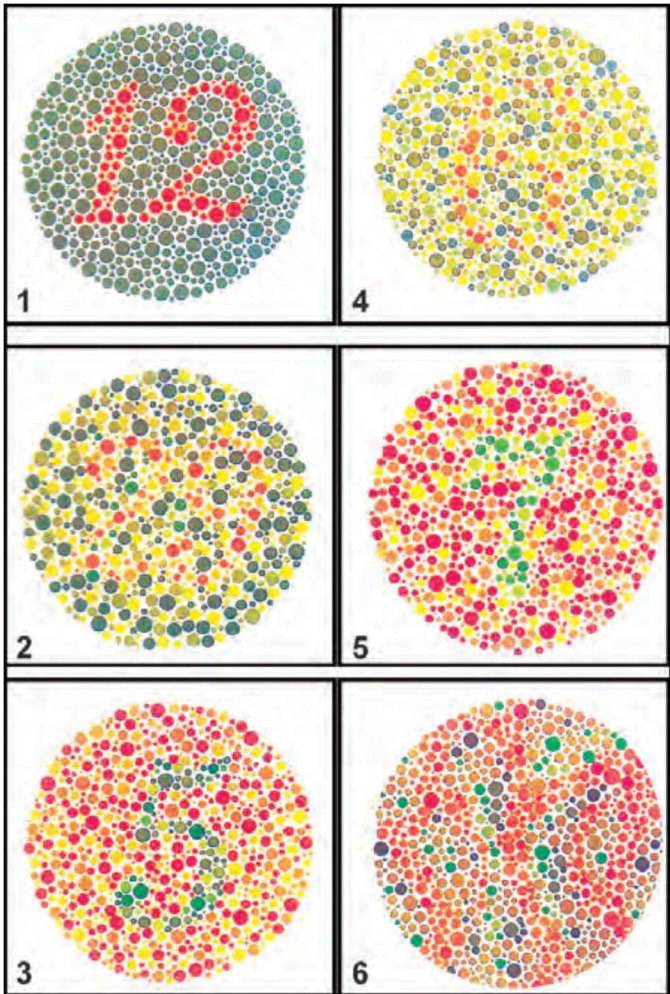
B. TESTING OF COLOUR VISION

Ishihara charts. These charts consist of lithographic colour plates available in book form. The plates are so constructed that numbers and wavy lines made up of spots of confusing colours, are printed against backgrounds of differently coloured spots of identical size.

Procedure

1. Seat the subject in a room adequately lit by day light. Direct sunlight or electric light may cause discrepancies in the result due to changes in the shades of the colours.
2. Ask the subject to read the numbers or trace the wavy lines on successive plates. The correct answers are given at the back of the book to

exclude the possibility of examiner being colour blind.



Result- Normal/Abnormal.

C. FIELD OF VISION (Confrontation test)

- It is a rough test to compare a person's visual fields with the examiner's own (presuming his own to be normal).
- The subject and the examiner sit facing each other about 3 feet apart.
- When testing the subject's left eye, he places his cupped right hand over his right eye and with the left eye he fixes his gaze on the examiner's right eye, while the examiner closes his left eye.
- The subject is instructed not to move his left eye in any direction.
- The examiner then holds out his right arm to its full extent, midway between himself and the subject and asks the subject to say "yes" when he sees any movement of the examiner's finger.

- If no movement is perceived, the hand is moved in, kept still and the finger moved once again.
- In this way, the examiner compares his own *first sighting* of the movement with that of the subject.
- Using this procedure, the peripheral field is tested in all the four quadrants—temporal, upper, lower and nasal.
- The subject's right eye is tested in a similar manner.
- The normal peripheral field of vision extends beyond 90° on the temporal side, about 50° in the vertical direction, about 55° on the nasal side and about 65° downwards.
- Only gross changes in the field of vision can be detected with this method.
- Scotomas (blind areas within the field of vision) are impossible to locate, for which a perimeter is employed

Result- Normal/Abnormal.

Oculomotor and Pupillary Innervation— The 3rd (Oculomotor), 4th (Trochlear), 6th (Abducent) and Sympathetic Nerves

The 3rd, 4th and 6th cranial nerves are usually considered together because they function as a physiological unit in the control of the eye movements.

A. Testing the conjugate movements of the eyes

- To test the eye movements, the head of the patient must be fixed with the left hand and he/she must be asked to follow the examiner's index finger to the right, to the left, upwards and downwards as far as possible in each direction.
- Normally, the eyes move 50° outwards, 50° downwards, 50° inwards and 33° upwards. The rotatory movements should also be tested. It is observed if there is any limitation of movement in any direction.

Result- Normal/Abnormal.

B. Testing light reflex

Direct light reflex:

- Each eye is tested separately in a shady place.
- The subject is asked to look at a distance.
- A bright light from a torch, brought from the side of the eye, is shined into the eye—the result is a prompt constriction of the pupil.
- When the light is switched off, the pupil quickly dilates to its previous size.

Indirect or consensual light reflex:

- A hand is placed between the two eyes and light is shined into one eye, observing the effect on the pupil of the unstimulated side.
- There is a constriction of the pupil in the other eye—a response called the indirect or consensual light reflex. Thus, the pupils of both eyes constrict when light is thrown into any eye.

Result- Normal/Abnormal.

C. Reaction of the pupil to accommodation for near vision.

- The subject is asked to look at the far wall of the room.
- The observer then suddenly brings his finger, holding it vertically, about 15 cm in front of the subject's nose and the subject is asked to look at it.
- The response is convergence of the eyes and pupillary constriction as he accommodates for the finger.
- The pupils dilate as the finger is moved away.

Result- Normal/Abnormal.

The 5th or Trigeminal Nerve (Sensory & Motor)

A. SENSORY FUNCTIONS

The Corneal reflex

Light wisp of absorbent cotton is twisted to a fine hair. The subject is asked to look at the far wall and, approaching from the side, the lateral edge of the cornea is lightly touched with the cotton. The response is bilateral blinking; and the two sides should be compared.

The conjunctival reflex

- Touching the conjunctiva with a wisp of cotton causes bilateral blinking. (The conjunctiva of the

lower lid is supplied by maxillary division of the 5th nerve).

The nasal or sneeze reflex

- Sneezing when the nasal mucosa is irritated, also employs 5th nerve as its afferent path, while the motor path employs motor components of 5th to 10th cranial and upper cervical nerves.

Result- Normal/Abnormal.

B. Motor Functions

The motor fibers of the 5th nerve, innervate the muscles of mastication—masseter, temporalis and medial and lateral pterygoids—and the **tensor** tympani of middle ear.

- The subject is asked to open his mouth and show the teeth. Normally, the jaw is symmetrical. If there is paralysis on one side, the jaw deviates to the side of paralysis, the healthy pterygoids pushing it to that side.
- The subject is asked to clench his teeth—the temporalis and masseter muscles contract and become equally prominent on the two sides. The muscles can be palpated to note if there is any difference in the strength of contraction.
- The subject is asked to open his mouth and move the mandible from side to side.
- The jaw jerk (maxillary reflex) is tested by placing a finger on the chin below the lower lip, with the mouth open and striking it with a percussion hammer. The response is closure of the mouth. Normally this jerk is hardly detectable, but it is exaggerated in upper motor neuron lesions (as are other deep reflexes). Both the afferent and efferent paths are along 5th nerve and the center is in the pons.

Result- Normal/Abnormal.

The 7th or Facial Nerve (Almost Purely Motor)

Testing the upper face

1. The subject is asked to look up and wrinkle the skin on his forehead (occipitofrontalis muscle tested). Normally, the wrinkling of the skin is symmetrical on the two sides. By asking the

subject to frown, the corrugator supercilii can be tested.

2. He is asked to shut his eyes as tightly as possible. (The corners of the mouth also get drawn up). The examiner then tries to open one and then the other eye. Normally, it is impossible to do so against the subject's wishes. (Orbicularis oculi muscles tested).

Result- Normal/Abnormal.

Testing the lower face

1. The nasolabial folds on both sides are observed, which are normally symmetrical. Paralysis on one side causes flattening of the folds on that side; it also affects facial symmetry at rest or during voluntary facial movements.
2. The subject is asked to smile or show his upper teeth or to whistle. Normally, the face remains symmetrical (Levator angularis muscle tested). Paralysis of one side causes the angle of the mouth to be drawn toward the healthy side, while that on the paralyzed side remains stationary. The buccinator is also involved in whistling.
3. The subject is asked to inflate his mouth with air and blow out his cheeks (Buccinator tested). Each inflated cheek is then tapped with a finger. If there is paralysis, the air escapes easily through the angle of the mouth on the paralyzed side.
4. The subject is asked to depress the lower lip (Depressor labii inferioris and quadratus labii inferioris tested). In case of paralysis, the asymmetry is obvious.

Result- Normal/Abnormal.

The 8th or Vestibulocochlear Nerve

Tuning-Fork Tests

I. Rinne's Test

This test compares the subject's AC hearing with his BC hearing, in each ear separately.

1. Hold the stem of the tuning fork with the thumb and finger and set it into vibration by striking one of its prongs on the heel of your hand. (The other prong will also start to vibrate).

2. Place its base on the mastoid process (the bony prominence behind the ear). The subject will hear a sound. Ask him to raise his hand when the sound stops. Note the time for which the sound is heard.
3. When the sound stops, bring the prongs in front of the ear—the sound will become audible once again. Note the time for which it lasts (E.g., for another 10 seconds; total= 45 seconds).

Results

In normals. For example, sound heard on mastoid process = 35 seconds.

Sound heard in front of ear = $35 + 10 = 45$ seconds

Thus, $AC > BC$ (Rinne positive).

In conduction deafness. BC sound remains normal at 35 seconds, but AC sound not heard after BC sound stops.

Thus, $AC < BC$ (Rinne negative).

In nerve deafness. Hearing will be impaired in both BC and AC sounds.

For example, BC becomes 15 seconds, AC becomes 20 seconds.

Thus, $AC > BC$, if nerve deafness is partial.

4. Test the other ear and record the timings for BC and AC sounds.

II. Weber's Test

This test compares the bone conduction of the subject in his two ears.

1. Set the tuning fork into vibration and place its base in the midline on the top of the subject's head or on his forehead. Ask the subject if he hears the sound equally well in both the ears or louder on one side.

In a normal subject. Bone conducted sounds are heard equally well on the two sides.

In conduction deafness. Sound is louder/better heard in deaf or deafer ear because of masking effect of environmental noise is absent on diseased side.

In nerve deafness. The sound is louder /better heard on the healthy side i.e. the patient lateralizes the sound to healthy side.

2. Conduction deafness can be artificially created if you close one ear (say, left) of the subject with your finger. Set the tuning fork into vibration once again and place it on his head or forehead.

In a normal person. Sound is better heard in the closed ear due to bone conduction. Closing his ear produces a situation of conduction deafness and demonstrates the masking effect of environmental noise.

The 9th or Glossopharyngeal Nerve (Mixed Nerve)

1. The sensation of taste over the posterior third of the tongue is tested.
2. Each side of the pharynx is touched lightly with a wooden spatula. The response is constriction of the pharynx. The afferent path is 9th nerve; the center is in medulla; and the efferent path is 10th nerve. Thus, this pharyngeal (or gag) reflex tests vagus as well.
3. A soft touch is applied on the soft palate; the response is elevation of the soft palate. The reflex arc is the same as in the gag reflex described above.

Result- Normal/Abnormal.

The 10th or Vagus Nerve (Mixed Nerve)

1. The *pharyngeal* and *palate reflexes* are tested as described for 9th nerve.
2. Using a tongue depressor, the subject is asked to open his mouth wide and say “ah”. The response is constriction of posterior pharyngeal wall (Vernet’s rideau phenomenon) and movement of the uvula backwards in the midline. But in vagal paralysis, the uvula is deflected to the normal side.
3. The subject is asked for history of regurgitation of food through the nose, which is due to total paralysis of vagus; a nasal voice may also be noted.
4. Laryngoscopy is done to note the position and movement of the true vocal cords.

Result- Normal/Abnormal.

The 11th or Accessory Nerve (Motor Nerve)

1. The examiner presses on the shoulders from behind and asks the subject to shrug his shoulders (this tests the upper part of trapezius). If the 11th nerve is damaged, shrugging is weaker on that side; the shoulder also droops. The subject is asked to approximate his shoulder blades against examiner’s resistance (this tests the lower part of the muscle).
2. A hand is placed against the right side of the subject’s face and he is asked to rotate the head to the right. The left sternomastoid is seen to become prominent. The procedure is repeated on the left side also. In case of a unilateral lesion, the head cannot be rotated to the healthy side.
3. The examiner places a hand on the subject’s forehead and asks him to bend his head forwards against resistance. Normally, both sternomastoids become prominent.

Result- Normal/Abnormal.

The 12th or Hypoglossal Nerve (Motor Nerve)

1. The subject is asked to push out his tongue as far as possible. Normally, it remains in the midline (genioglossus tested). If the 12th nerve is paralyzed, the tongue is pushed over to the side of the lesion by the healthy muscles on the opposite side. The affected side is also wasted, wrinkled and may show fasciculation, which indicates lower motor neuron lesion.
2. The subject is asked to move the tongue from side to side over the lips and against the walls of the cheeks (extrinsic and intrinsic muscles of the tongue tested). A finger is placed against the cheek while the subject is asked to press against it with his tongue through the wall of the cheek. The strength of contraction is compared on the two sides.
3. The subject is asked to touch the tongue to the palate (palatoglossus tested) and to depress the tongue in the floor of the mouth (hypoglossus tested).

EXERCISE

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15. Describe the clinical features of vagus nerve palsy.

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16. Describe the motor functions of the accessory nerve.

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16. Describe the motor functions of the hypoglossal nerve.

OSPE-1

Aim:

To test the 5th cranial nerve in the subject provided.

Checklist:

1. Makes the subject sit on a stool and explains the procedure. (Y/N)
2. Asks the subject to look at a distance and touches his/her conjunctiva with a wisp of cotton and notes the response. (Y/N)
3. Tests the sensations of touch and pain with a wisp of cotton and a pin on identical points on the two sides of the face. (Y/N)
4. Asks the subject to show his/her teeth and then to clench his/her teeth. Watches and feels the masseter and temporalis muscles contracting. (Y/N)
5. Asks the subject to open his/her mouth and move the mandible from side to side Then tests the mandibular reflex. (Y/N)

OSPE-2

Aim:

To test the 7th cranial nerve in the subject provided.

Checklist:

1. Explains the procedure to the subject. Looks for facial symmetry, furrows on the forehead and the width of the palpebral fissure. (Y/N)
2. Asks the subject to look up and wrinkle his/her forehead and then to shut his/her eyes as tightly as possible against the examiner's resistance. (Y/N)
3. Asks the subject to show his/her upper teeth and to smile. (Y/N)
4. Asks the subject to inflate his/her mouth with air and to blow out his/her cheeks. Then taps his/her cheeks, on either side with his/her finger, to see if air escapes from the angle of the mouth. (Y/N)
5. Asks the subject to depress his/her lower lip. (Y/N)

OSPE-3

Aim:

To demonstrate Rinne test on the subject provided.

Checklist:

1. Explains the test procedure and gives suitable instructions. (Y/N)
2. Selects either 216 Hz or 250 Hz tuning fork and strikes one of the prongs on the heel of her hand. (Y/N)
3. Closes the external auditory meatus of the subject's other ear with her finger. (Y/N)
4. Places the base of the tuning fork on the subject's mastoid process; when he raises his finger to indicate that the sound can no longer be heard. (Y/N)
5. Quickly transfers the vibrating tuning fork close to the subject's ear. Notes if he can hear the sound once again. Notes down whether air or bone conduction is better. (Y/N)

OSPE-4

Aim:

To test the 12th cranial nerve in the subject provides.

Checklist:

1. Explains the procedure to the subject (Y/N)
2. Asks the subject to push out his/her tongue as far as possible, Then inspects its position, evidence of wasting and fasciculation. (Y/N)
3. Asks the subject to move his/her tongue from side to side over the lips and against the walls of the cheeks. (Y/N)
4. Places his/her finger over the subject's cheek and asks him/her to push against it. Repeats on the opposite side. (Y/N).
5. Asks the subject to touch the tongue to the palate and then to depress it into the floor of the mouth. (Y/N)

SPE-5

Aim:

To test the 11th cranial nerve of the subject provided.

Checklist:

1. Asks the subject to sit comfortably and explains the procedure. (Y/N)
2. Stands behind the subject and places his/her hands on his/her shoulders. Then asks him/her to shrug his/her shoulders against his/her resistance. (Y/N)
3. Places his/her hand on the right side of the subjects face and asks him/her to rotate his/ her head to the opposite side. Watches the left sternomastoid. (Y/N)
4. Repeats the procedure on the left side and asks him/her to rotate his/her head to the left and watches the right sternomastoid muscle. (Y/N)
5. Places his/her hand on the subject's forehead and asks him/her to bend his/her head forwards against resistance. (Y/N)

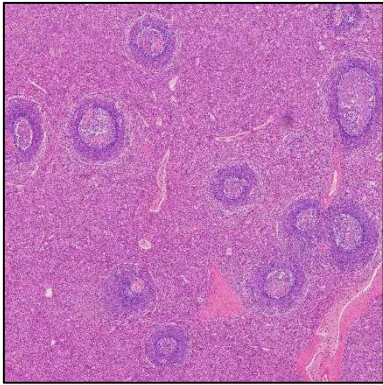
REFERENCES

- A textbook of practical physiology, CL. Ghai, 8th edition.
- Essentials of medical physiology, K. Sembulingam and P. Sembulingam, 6th edition.
- If you have any suggestion, query kindly share on kriyasharirpractical@gmail.com



Teacher's Signature

24. HISTOPATHOLOGY SLIDES



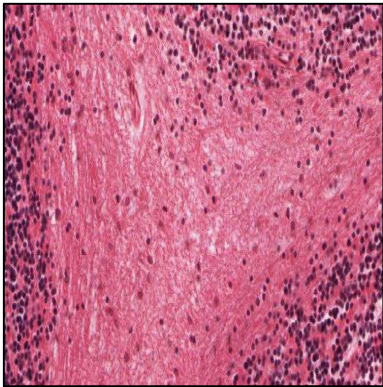
Spleen



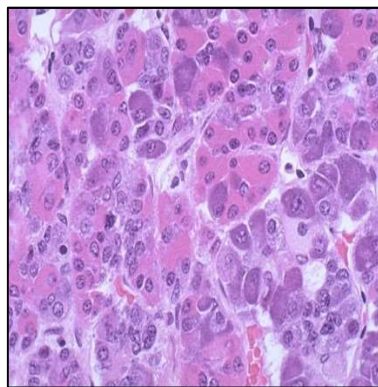
Liver



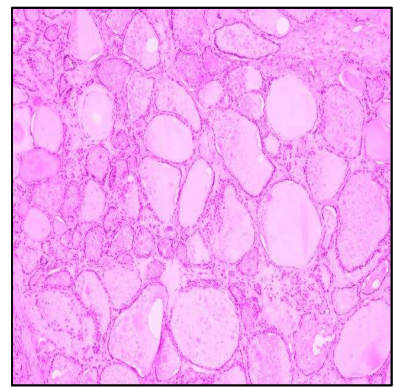
Testes



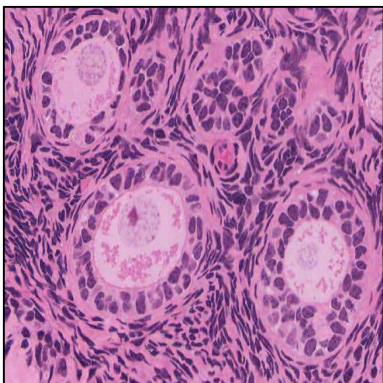
Brain



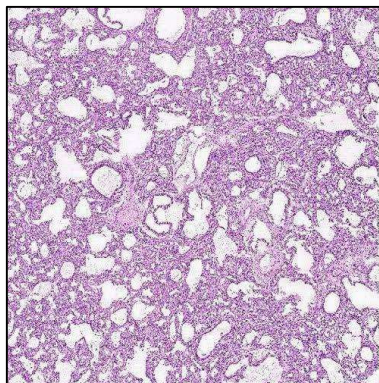
Anterior pituitary



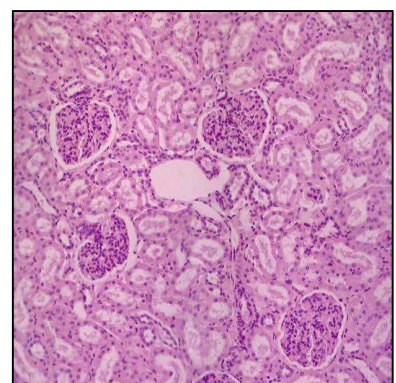
Thyroid gland



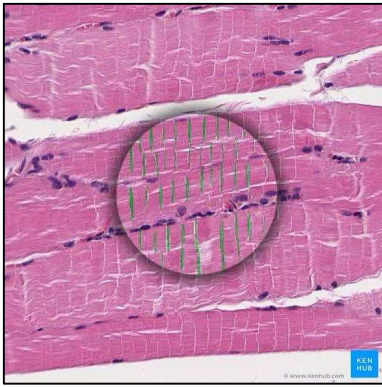
Ovary



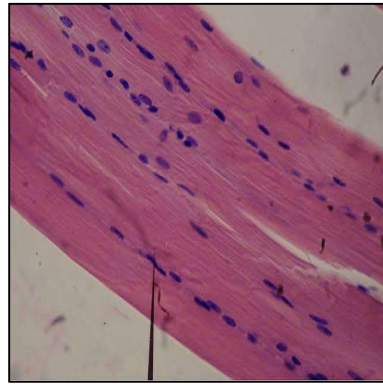
Lungs



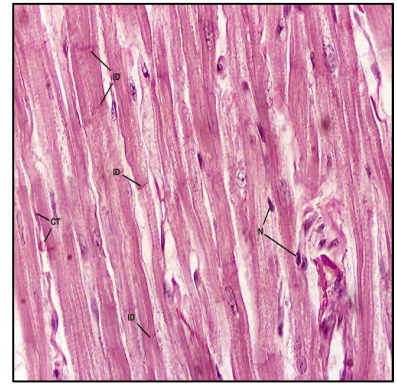
Kidney



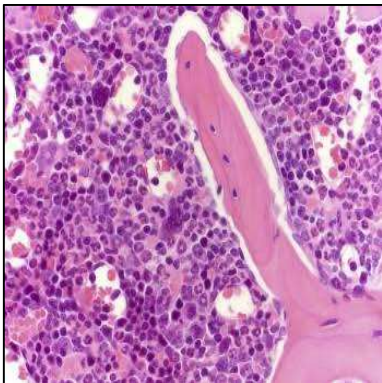
Muscle



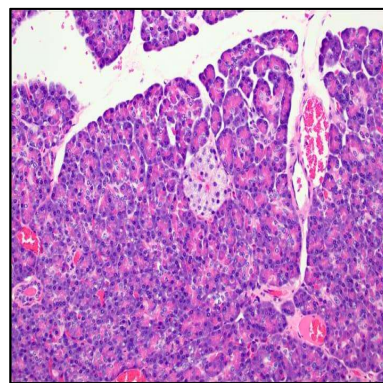
Smooth muscle



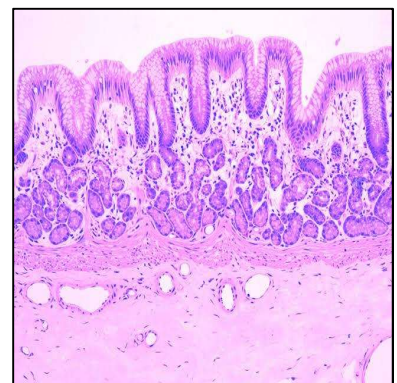
Cardiac muscle



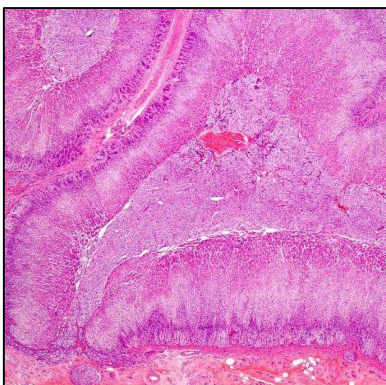
Bone Marrow



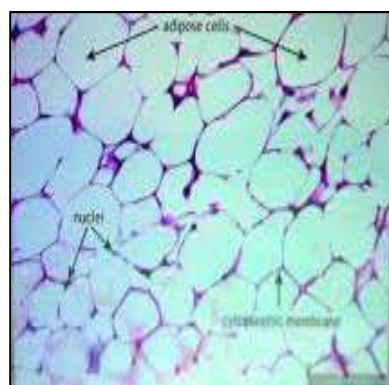
Pancreas



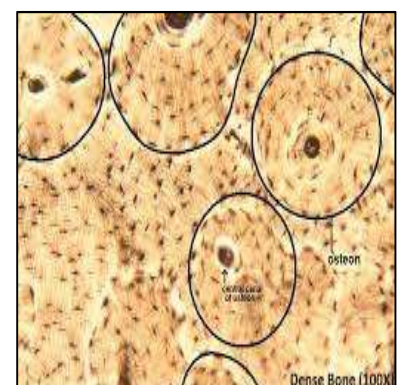
Stomach



Adrenal Gland



Adipose Tissue



Bone Cell



Books



HSET



New